

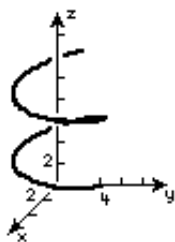
9

Vector Calculus

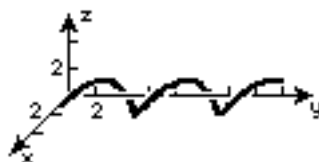
EXERCISES 9.1

Vector Functions

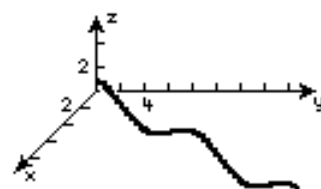
1.



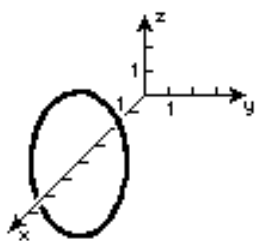
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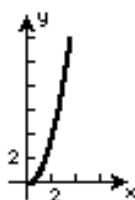
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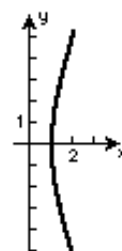
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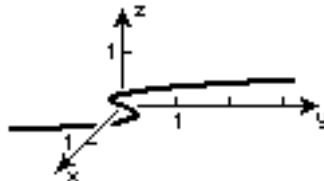
6.



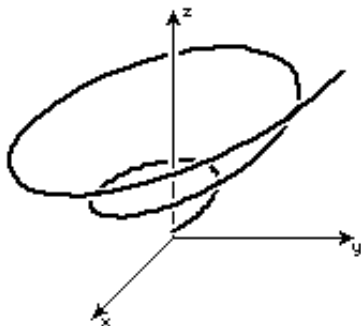
7.



8.

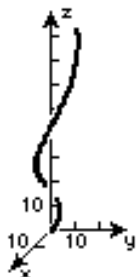
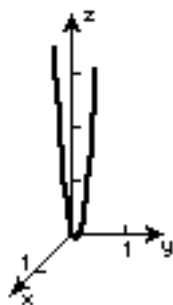
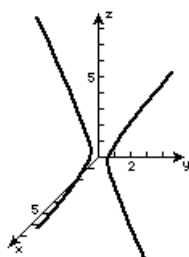
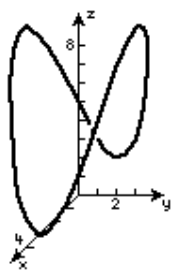
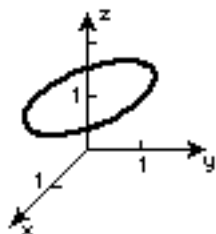


9.



Note: the scale is distorted in this graph. For $t = 0$, the graph starts at $(1,0,1)$. The upper loop shown intersects the xz -plane at about $(286751, 0, 286751)$.

10.


 11. $x = t, y = t, z = t^2 + t^2 = 2t^2; \mathbf{r}(t) = t\mathbf{i} + t\mathbf{j} + 2t^2\mathbf{k}$

 12. $x = t, y = 2t, z = \pm\sqrt{t^2 + 4t^2 + 1} = \pm\sqrt{5t^2 + 1}; \mathbf{r}(t) = t\mathbf{i} + 2t\mathbf{j} \pm \sqrt{5t^2 + 1}\mathbf{k}$

 13. $x = 3 \cos t, z = 9 - 9 \cos^2 t = 9 \sin^2 t, y = 3 \sin t; \mathbf{r}(t) = 3 \cos t\mathbf{i} + 3 \sin t\mathbf{j} + 9 \sin^2 t\mathbf{k}$

 14. $x = \sin t, z = 1, y = \cos t; \mathbf{r}(t) = \sin t\mathbf{i} + \cos t\mathbf{j} + \mathbf{k}$


9.1 Vector Functions

15. $\mathbf{r}(t) = \frac{\sin 2t}{t} \mathbf{i} + (t-2)^5 \mathbf{j} + \frac{\ln t}{1/t} \mathbf{k}$. Using L'Hôpital's Rule,

$$\lim_{t \rightarrow 0^+} \mathbf{r}(t) = \left[\frac{2 \cos 2t}{1} \mathbf{i} + (t-2)^5 \mathbf{j} + \frac{1/t}{-1/t^2} \mathbf{k} \right] = 2\mathbf{i} - 32\mathbf{j}.$$

16. (a) $\lim_{t \rightarrow \alpha} [-4\mathbf{r}_1(t) + 3\mathbf{r}_2(t)] = -4(\mathbf{i} - 2\mathbf{j} + \mathbf{k}) + 3(2\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}) = 2\mathbf{i} + 23\mathbf{j} + 17\mathbf{k}$

(b) $\lim_{t \rightarrow \alpha} \mathbf{r}_1(t) \cdot \mathbf{r}_2(t) = (\mathbf{i} - 2\mathbf{j} + \mathbf{k}) \cdot (2\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}) = -1$

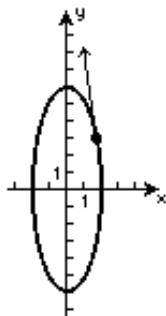
17. $\mathbf{r}'(t) = \frac{1}{t} \mathbf{i} - \frac{1}{t^2} \mathbf{j}$; $\mathbf{r}''(t) = -\frac{1}{t^2} \mathbf{i} + \frac{2}{t^3} \mathbf{j}$

18. $\mathbf{r}'(t) = \langle -t \sin t, 1 - \sin t \rangle$; $\mathbf{r}''(t) = \langle -t \cos t - \sin t, -\cos t \rangle$

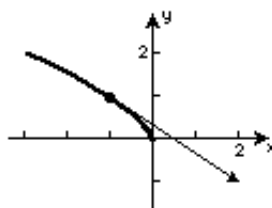
19. $\mathbf{r}'(t) = \langle 2te^{2t} + e^{2t}, 3t^2, 8t - 1 \rangle$; $\mathbf{r}''(t) = \langle 4te^{2t} + 4e^{2t}, 6t, 8 \rangle$

20. $\mathbf{r}'(t) = 2t\mathbf{i} + 3t^2\mathbf{j} + \frac{1}{1+t^2} \mathbf{k}$; $\mathbf{r}''(t) = 2\mathbf{i} + 6t\mathbf{j} - \frac{2t}{(1+t^2)^2} \mathbf{k}$

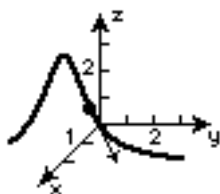
21. $\mathbf{r}'(t) = -2 \sin t \mathbf{i} + 6 \cos t \mathbf{j}$
 $\mathbf{r}'(\pi/6) = -\mathbf{i} + 3\sqrt{3}\mathbf{j}$



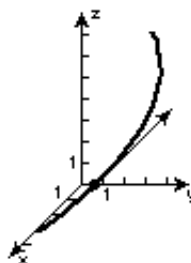
22. $\mathbf{r}'(t) = 3t^2\mathbf{i} + 2t\mathbf{j}$
 $\mathbf{r}'(-1) = 3\mathbf{i} - 2\mathbf{j}$



23. $\mathbf{r}'(t) = \mathbf{j} - \frac{8t}{(1+t^2)^2} \mathbf{k}$
 $\mathbf{r}'(1) = \mathbf{j} - 2\mathbf{k}$



24. $\mathbf{r}'(t) = -3 \sin t \mathbf{i} + 3 \cos t \mathbf{j} + 2\mathbf{k}$
 $\mathbf{r}'(\pi/4) = \frac{-3\sqrt{2}}{2} \mathbf{i} + \frac{3\sqrt{2}}{2} \mathbf{j} + 2\mathbf{k}$



25. $\mathbf{r}(t) = t\mathbf{i} + \frac{1}{2}t^2\mathbf{j} + \frac{1}{3}t^3\mathbf{k}$; $\mathbf{r}(2) = 2\mathbf{i} + 2\mathbf{j} + \frac{8}{3}\mathbf{k}$; $\mathbf{r}'(t) = \mathbf{i} + t\mathbf{j} + t^2\mathbf{k}$; $\mathbf{r}'(2) = \mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$

Using the point $(2, 2, 8/3)$ and the direction vector $\mathbf{r}'(2)$, we have $x = 2 + t$, $y = 2 + 2t$, $z = 8/3 + 4t$.

26. $\mathbf{r}(t) = (t^3 - t)\mathbf{i} + \frac{6t}{t+1}\mathbf{j} + (2t+1)^2\mathbf{k}$; $\mathbf{r}(1) = 3\mathbf{j} + 9\mathbf{k}$; $\mathbf{r}'(t) = (3t^2 - 1)\mathbf{i} + \frac{6}{(t+1)^2}\mathbf{j} + (8t+4)\mathbf{k}$; $\mathbf{r}'(1) = 2\mathbf{i} + \frac{3}{2}\mathbf{j} + 12\mathbf{k}$.

Using the point $(0, 3, 9)$ and the direction vector $\mathbf{r}'(1)$, we have $x = 2t$, $y = 3 + \frac{3}{2}t$, $z = 9 + 12t$.

27. $\frac{d}{dt}[\mathbf{r}(t) \times \mathbf{r}'(t)] = \mathbf{r}(t) \times \mathbf{r}''(t) + \mathbf{r}'(t) \times \mathbf{r}'(t) = \mathbf{r}(t) \times \mathbf{r}''(t)$

28. $\frac{d}{dt}[\mathbf{r}(t) \cdot (t\mathbf{r}(t))] = \mathbf{r}(t) \cdot \frac{d}{dt}(t\mathbf{r}(t)) + \mathbf{r}'(t) \cdot (t\mathbf{r}(t)) = \mathbf{r}(t) \cdot (t\mathbf{r}'(t) + \mathbf{r}(t)) + \mathbf{r}'(t) \cdot (t\mathbf{r}(t))$
 $= \mathbf{r}(t) \cdot (t\mathbf{r}'(t)) + \mathbf{r}(t) \cdot \mathbf{r}(t) + \mathbf{r}'(t) \cdot (t\mathbf{r}(t)) = 2t(\mathbf{r}(t) \cdot \mathbf{r}'(t)) + \mathbf{r}(t) \cdot \mathbf{r}(t)$

29. $\frac{d}{dt}[\mathbf{r}(t) \cdot (\mathbf{r}'(t) \times \mathbf{r}''(t))] = \mathbf{r}(t) \cdot \frac{d}{dt}(\mathbf{r}'(t) \times \mathbf{r}''(t)) + \mathbf{r}'(t) \cdot (\mathbf{r}'(t) \times \mathbf{r}''(t))$
 $= \mathbf{r}(t) \cdot (\mathbf{r}'(t) \times \mathbf{r}'''(t) + \mathbf{r}''(t) \times \mathbf{r}''(t)) + \mathbf{r}'(t) \cdot (\mathbf{r}'(t) \times \mathbf{r}''(t))$
 $= \mathbf{r}(t) \cdot (\mathbf{r}'(t) \times \mathbf{r}'''(t))$
30. $\frac{d}{dt}[\mathbf{r}_1(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3(t))] = \mathbf{r}_1(t) \times \frac{d}{dt}(\mathbf{r}_2(t) \times \mathbf{r}_3(t)) + \mathbf{r}_1'(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3(t))$
 $= \mathbf{r}_1(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3'(t) + \mathbf{r}_2'(t) \times \mathbf{r}_3(t)) + \mathbf{r}_1'(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3(t))$
 $= \mathbf{r}_1(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3'(t)) + \mathbf{r}_1(t) \times (\mathbf{r}_2'(t) \times \mathbf{r}_3(t)) + \mathbf{r}_1'(t) \times (\mathbf{r}_2(t) \times \mathbf{r}_3(t))$
31. $\frac{d}{dt}\left[\mathbf{r}_1(2t) + \mathbf{r}_2\left(\frac{1}{t}\right)\right] = 2\mathbf{r}_1'(2t) - \frac{1}{t^2}\mathbf{r}_2'\left(\frac{1}{t}\right)$
32. $\frac{d}{dt}[t^3\mathbf{r}(t^2)] = t^3(2t)\mathbf{r}'(t^2) + 3t^2\mathbf{r}(t^2) = 2t^4\mathbf{r}'(t^2) + 3t^2\mathbf{r}(t^2)$
33. $\int_{-1}^2 \mathbf{r}(t) dt = \left[\int_{-1}^2 t dt\right] \mathbf{i} + \left[\int_{-1}^2 3t^2 dt\right] \mathbf{j} + \left[\int_{-1}^2 4t^3 dt\right] \mathbf{k} = \frac{1}{2}t^2 \Big|_{-1}^2 \mathbf{i} + t^3 \Big|_{-1}^2 \mathbf{j} + t^4 \Big|_{-1}^2 \mathbf{k} = \frac{3}{2}\mathbf{i} + 9\mathbf{j} + 15\mathbf{k}$
34. $\int_0^4 \mathbf{r}(t) dt = \left[\int_0^4 \sqrt{2t+1} dt\right] \mathbf{i} + \left[\int_0^4 -\sqrt{t} dt\right] \mathbf{j} + \left[\int_0^4 \sin \pi t dt\right] \mathbf{k}$
 $= \frac{1}{3}(2t+1)^{3/2} \Big|_0^4 \mathbf{i} - \frac{2}{3}t^{3/2} \Big|_0^4 \mathbf{j} - \frac{1}{\pi} \cos \pi t \Big|_0^4 \mathbf{k} = \frac{26}{3}\mathbf{i} - \frac{16}{3}\mathbf{j}$
35. $\int \mathbf{r}(t) dt = \left[\int te^t dt\right] \mathbf{i} + \left[\int -e^{-2t} dt\right] \mathbf{j} + \left[\int te^{t^2} dt\right] \mathbf{k}$
 $= [te^t - e^t + c_1]\mathbf{i} + \left[\frac{1}{2}e^{-2t} + c_2\right]\mathbf{j} + \left[\frac{1}{2}e^{t^2} + c_3\right]\mathbf{k} = e^t(t-1)\mathbf{i} + \frac{1}{2}e^{-2t}\mathbf{j} + \frac{1}{2}e^{t^2}\mathbf{k} + \mathbf{c},$
 where $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$.
36. $\int \mathbf{r}(t) dt = \left[\int \frac{1}{1+t^2} dt\right] \mathbf{i} + \left[\int \frac{t}{1+t^2} dt\right] \mathbf{j} + \left[\int \frac{t^2}{1+t^2} dt\right] \mathbf{k}$
 $= [\tan^{-1} t + c_1]\mathbf{i} + \left[\frac{1}{2} \ln(1+t^2) + c_2\right]\mathbf{j} + \left[\int \left(1 - \frac{1}{1+t^2}\right) dt\right] \mathbf{k}$
 $= [\tan^{-1} t + c_1]\mathbf{i} + \left[\frac{1}{2} \ln(1+t^2) + c_2\right]\mathbf{j} + [t - \tan^{-1} t + c_3]\mathbf{k}$
 $= \tan^{-1} t \mathbf{i} + \frac{1}{2} \ln(1+t^2)\mathbf{j} + (t - \tan^{-1} t)\mathbf{k} + \mathbf{c},$
 where $\mathbf{c} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$.
37. $\mathbf{r}(t) = \int \mathbf{r}'(t) dt = \left[\int 6 dt\right] \mathbf{i} + \left[\int 6t dt\right] \mathbf{j} + \left[\int 3t^2 dt\right] \mathbf{k} = [6t + c_1]\mathbf{i} + [3t^2 + c_2]\mathbf{j} + [t^3 + c_3]\mathbf{k}$
 Since $\mathbf{r}(0) = \mathbf{i} - 2\mathbf{j} + \mathbf{k} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$, $c_1 = 1$, $c_2 = -2$, and $c_3 = 1$. Thus,
 $\mathbf{r}(t) = (6t + 1)\mathbf{i} + (3t^2 - 2)\mathbf{j} + (t^3 + 1)\mathbf{k}.$
38. $\mathbf{r}(t) = \int \mathbf{r}'(t) dt = \left[\int t \sin t^2 dt\right] \mathbf{i} + \left[\int -\cos 2t dt\right] \mathbf{j} = \left[-\frac{1}{2} \cos t^2 + c_1\right]\mathbf{i} + \left[-\frac{1}{2} \sin 2t + c_2\right]\mathbf{j}$
 Since $\mathbf{r}(0) = \frac{3}{2}\mathbf{i} = \left(-\frac{1}{2} + c_1\right)\mathbf{i} + c_2\mathbf{j}$, $c_1 = 2$ and $c_2 = 0$. Thus,
 $\mathbf{r}(t) = \left(-\frac{1}{2} \cos t^2 + 2\right)\mathbf{i} - \frac{1}{2} \sin 2t\mathbf{j}.$
39. $\mathbf{r}'(t) = \int \mathbf{r}''(t) dt = \left[\int 12t dt\right] \mathbf{i} + \left[\int -3t^{-1/2} dt\right] \mathbf{j} + \left[\int 2 dt\right] \mathbf{k} = [6t^2 + c_1]\mathbf{i} + [-6t^{1/2} + c_2]\mathbf{j} + [2t + c_3]\mathbf{k}$

9.1 Vector Functions

Since $\mathbf{r}'(1) = \mathbf{j} = (6 + c_1)\mathbf{i} + (-6 + c_2)\mathbf{j} + (2 + c_3)\mathbf{k}$, $c_1 = -6$, $c_2 = 7$, and $c_3 = -2$. Thus,

$$\mathbf{r}'(t) = (6t^2 - 6)\mathbf{i} + (-6t^{1/2} + 7)\mathbf{j} + (2t - 2)\mathbf{k}.$$

$$\begin{aligned}\mathbf{r}(t) &= \int \mathbf{r}'(t) dt = \left[\int (6t^2 - 6) dt \right] \mathbf{i} + \left[\int (-6t^{1/2} + 7) dt \right] \mathbf{j} + \left[\int (2t - 2) dt \right] \mathbf{k} \\ &= [2t^3 - 6t + c_4]\mathbf{i} + [-4t^{3/2} + 7t + c_5]\mathbf{j} + [t^2 - 2t + c_6]\mathbf{k}.\end{aligned}$$

Since

$$\mathbf{r}(1) = 2\mathbf{i} - \mathbf{k} = (-4 + c_4)\mathbf{i} + (3 + c_5)\mathbf{j} + (-1 + c_6)\mathbf{k},$$

$c_4 = 6$, $c_5 = -3$, and $c_6 = 0$. Thus,

$$\mathbf{r}(t) = (2t^3 - 6t + 6)\mathbf{i} + (-4t^{3/2} + 7t - 3)\mathbf{j} + (t^2 - 2t)\mathbf{k}.$$

$$\begin{aligned}40. \quad \mathbf{r}'(t) &= \int \mathbf{r}''(t) dt = \left[\int \sec^2 t dt \right] \mathbf{i} + \left[\int \cos t dt \right] \mathbf{j} + \left[\int -\sin t dt \right] \mathbf{k} \\ &= [\tan t + c_1]\mathbf{i} + [\sin t + c_2]\mathbf{j} + [\cos t + c_3]\mathbf{k}\end{aligned}$$

Since $\mathbf{r}'(0) = \mathbf{i} + \mathbf{j} + \mathbf{k} = c_1\mathbf{i} + c_2\mathbf{j} + (1 + c_3)\mathbf{k}$, $c_1 = 1$, $c_2 = 1$, and $c_3 = 0$. Thus,

$$\mathbf{r}'(t) = (\tan t + 1)\mathbf{i} + (\sin t + 1)\mathbf{j} + \cos t\mathbf{k}.$$

$$\begin{aligned}\mathbf{r}(t) &= \int \mathbf{r}'(t) dt = \left[\int (\tan t + 1) dt \right] \mathbf{i} + \left[\int (\sin t + 1) dt \right] \mathbf{j} + \left[\int \cos t dt \right] \mathbf{k} \\ &= [\ln |\sec t| + t + c_4]\mathbf{i} + [-\cos t + t + c_5]\mathbf{j} + [\sin t + c_6]\mathbf{k}.\end{aligned}$$

Since $\mathbf{r}(0) = -\mathbf{j} + 5\mathbf{k} = c_4\mathbf{i} + (-1 + c_5)\mathbf{j} + c_6\mathbf{k}$, $c_4 = 0$, $c_5 = 0$, and $c_6 = 5$. Thus,

$$\mathbf{r}(t) = (\ln |\sec t| + t)\mathbf{i} + (-\cos t + t)\mathbf{j} + (\sin t + 5)\mathbf{k}.$$

$$41. \quad \mathbf{r}'(t) = -a \sin t \mathbf{i} + a \cos t \mathbf{j} + c \mathbf{k}; \quad \|\mathbf{r}'(t)\| = \sqrt{(-a \sin t)^2 + (a \cos t)^2 + c^2} = \sqrt{a^2 + c^2}$$

$$s = \int_0^{2\pi} \sqrt{a^2 + c^2} dt = \sqrt{a^2 + c^2} t \Big|_0^{2\pi} = 2\pi \sqrt{a^2 + c^2}$$

$$42. \quad \mathbf{r}'(t) = \mathbf{i} + (\cos t - t \sin t)\mathbf{j} + (\sin t + t \cos t)\mathbf{k}$$

$$\|\mathbf{r}'(t)\| = \sqrt{1^2 + (\cos t - t \sin t)^2 + (\sin t + t \cos t)^2} = \sqrt{2 + t^2}$$

$$s = \int_0^\pi \sqrt{2 + t^2} dt = \left(\frac{t}{2} \sqrt{2 + t^2} + \ln \left| t + \sqrt{2 + t^2} \right| \right) \Big|_0^\pi = \frac{\pi}{2} \sqrt{2 + \pi^2} + \ln(\pi + \sqrt{2 + \pi^2}) - \ln \sqrt{2}$$

$$43. \quad \mathbf{r}'(t) = (-2e^t \sin 2t + e^t \cos 2t)\mathbf{i} + (2e^t \cos 2t + e^t \sin 2t)\mathbf{j} + e^t \mathbf{k}$$

$$\|\mathbf{r}'(t)\| = \sqrt{5e^{2t} \cos^2 2t + 5e^{2t} \sin^2 2t + e^{2t}} = \sqrt{6e^{2t}} = \sqrt{6} e^t$$

$$s = \int_0^{3\pi} \sqrt{6} e^t dt = \sqrt{6} e^t \Big|_0^{3\pi} = \sqrt{6} (e^{3\pi} - 1)$$

$$44. \quad \mathbf{r}'(t) = 3\mathbf{i} + 2\sqrt{3}t\mathbf{j} + 2t^2\mathbf{k}; \quad \|\mathbf{r}'(t)\| = \sqrt{3^2 + (2\sqrt{3}t)^2 + (2t^2)^2} = \sqrt{9 + 12t^2 + 4t^4} = 3 + 2t^2$$

$$s = \int_0^1 (3 + 2t^2) dt = \left(3t + \frac{2}{3}t^3 \right) \Big|_0^1 = 3 + \frac{2}{3} = \frac{11}{3}$$

$$45. \quad \mathbf{r}'(t) = -a \sin t \mathbf{i} + a \cos t \mathbf{j}; \quad \|\mathbf{r}'(t)\| = \sqrt{a^2 \sin^2 t + a^2 \cos^2 t} = a, \quad a > 0; \quad s = \int_0^t a du = at$$

$$\mathbf{r}(s) = a \cos(s/a)\mathbf{i} + a \sin(s/a)\mathbf{j}; \quad \mathbf{r}'(s) = -\sin(s/a)\mathbf{i} + \cos(s/a)\mathbf{j}$$

$$\|\mathbf{r}'(s)\| = \sqrt{\sin^2(s/a) + \cos^2(s/a)} = 1$$

$$46. \mathbf{r}'(s) = -\frac{2}{\sqrt{5}} \sin(s/\sqrt{5})\mathbf{i} + \frac{2}{\sqrt{5}} \cos(s/\sqrt{5})\mathbf{j} + \frac{1}{\sqrt{5}} \mathbf{k}$$

$$\|\mathbf{r}'(s)\| = \sqrt{\frac{4}{5} \sin^2(s/\sqrt{5}) + \frac{4}{5} \cos^2(s/\sqrt{5}) + \frac{1}{5}} = \sqrt{\frac{4}{5} + \frac{1}{5}} = 1$$

47. Since $\frac{d}{dt}(\mathbf{r} \cdot \mathbf{r}) = \frac{d}{dt}\|\mathbf{r}\|^2 = \frac{d}{dt}c^2 = 0$ and $\frac{d}{dt}(\mathbf{r} \cdot \mathbf{r}) = \mathbf{r} \cdot \mathbf{r}' + \mathbf{r}' \cdot \mathbf{r} = 2\mathbf{r} \cdot \mathbf{r}'$, we have $\mathbf{r} \cdot \mathbf{r}' = 0$. Thus, $\mathbf{r}'$ is perpendicular to $\mathbf{r}$.

48. Since $\|\mathbf{r}(t)\|$ is the length of $\mathbf{r}(t)$, $\|\mathbf{r}(t)\| = c$ represents a curve lying on a sphere of radius c centered at the origin.

49. Let $\mathbf{r}_1(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$. Then

$$\begin{aligned} \frac{d}{dt}[u(t)\mathbf{r}_1(t)] &= \frac{d}{dt}[u(t)x(t)\mathbf{i} + u(t)y(t)\mathbf{j}] = [u(t)x'(t) + u'(t)x(t)]\mathbf{i} + [u(t)y'(t) + u'(t)y(t)]\mathbf{j} \\ &= u(t)[x'(t)\mathbf{i} + y'(t)\mathbf{j}] + u'(t)[x(t)\mathbf{i} + y(t)\mathbf{j}] = u(t)\mathbf{r}_1'(t) + u'(t)\mathbf{r}_1(t). \end{aligned}$$

50. Let $\mathbf{r}_1(t) = x_1(t)\mathbf{i} + y_1(t)\mathbf{j}$ and $\mathbf{r}_2(t) = x_2(t)\mathbf{i} + y_2(t)\mathbf{j}$. Then

$$\begin{aligned} \frac{d}{dt}[\mathbf{r}_1(t) \cdot \mathbf{r}_2(t)] &= \frac{d}{dt}[x_1(t)x_2(t) + y_1(t)y_2(t)] = x_1(t)x_2'(t) + x_1'(t)x_2(t) + y_1(t)y_2'(t) + y_1'(t)y_2(t) \\ &= [x_1(t)x_2'(t) + y_1(t)y_2'(t)] + [x_1'(t)x_2(t) + y_1'(t)y_2(t)] = \mathbf{r}_1(t) \cdot \mathbf{r}_2'(t) + \mathbf{r}_1'(t) \cdot \mathbf{r}_2(t). \end{aligned}$$

$$\begin{aligned} 51. \frac{d}{dt}[\mathbf{r}_1(t) \times \mathbf{r}_2(t)] &= \lim_{h \rightarrow 0} \frac{\mathbf{r}_1(t+h) \times \mathbf{r}_2(t+h) - \mathbf{r}_1(t) \times \mathbf{r}_2(t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\mathbf{r}_1(t+h) \times \mathbf{r}_2(t+h) - \mathbf{r}_1(t+h) \times \mathbf{r}_2(t) + \mathbf{r}_1(t+h) \times \mathbf{r}_2(t) - \mathbf{r}_1(t) \times \mathbf{r}_2(t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\mathbf{r}_1(t+h) \times [\mathbf{r}_2(t+h) - \mathbf{r}_2(t)]}{h} + \lim_{h \rightarrow 0} \frac{[\mathbf{r}_1(t+h) - \mathbf{r}_1(t)] \times \mathbf{r}_2(t)}{h} \\ &= \mathbf{r}_1(t) \times \left(\lim_{h \rightarrow 0} \frac{\mathbf{r}_2(t+h) - \mathbf{r}_2(t)}{h} \right) + \left(\lim_{h \rightarrow 0} \frac{\mathbf{r}_1(t+h) - \mathbf{r}_1(t)}{h} \right) \times \mathbf{r}_2(t) \\ &= \mathbf{r}_1(t) \times \mathbf{r}_2'(t) + \mathbf{r}_1'(t) \times \mathbf{r}_2(t) \end{aligned}$$

52. Let $\mathbf{v} = a\mathbf{i} + b\mathbf{j}$ and $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$. Then

$$\int_a^b \mathbf{v} \cdot \mathbf{r}(t) dt = \int_a^b [ax(t) + by(t)] dt = a \int_a^b x(t) dt + b \int_a^b y(t) dt = \mathbf{v} \cdot \int_a^b \mathbf{r}(t) dt.$$

EXERCISES 9.2

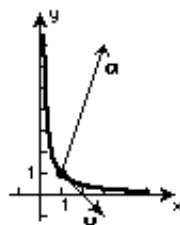
Motion on a Curve

1. $\mathbf{v}(t) = 2t\mathbf{i} + t^3\mathbf{j}$; $\mathbf{v}(1) = 2\mathbf{i} + \mathbf{j}$; $\|\mathbf{v}(1)\| = \sqrt{4+1} = \sqrt{5}$;
 $\mathbf{a}(t) = 2\mathbf{i} + 3t^2\mathbf{j}$; $\mathbf{a}(1) = 2\mathbf{i} + 3\mathbf{j}$

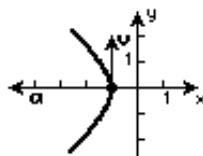


9.2 Motion on a Curve

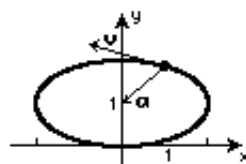
2. $\mathbf{v}(t) = 2t\mathbf{i} - \frac{2}{t^3}\mathbf{j}$; $\mathbf{v}(1) = 2\mathbf{i} - 2\mathbf{j}$; $\|\mathbf{v}(1)\| = \sqrt{4+4} = 2\sqrt{2}$;
 $\mathbf{a}(t) = 2\mathbf{i} + \frac{6}{t^4}\mathbf{j}$; $\mathbf{a}(1) = 2\mathbf{i} + 6\mathbf{j}$



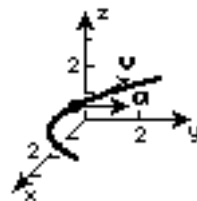
3. $\mathbf{v}(t) = -2\sinh 2t\mathbf{i} + 2\cosh 2t\mathbf{j}$; $\mathbf{v}(0) = 2\mathbf{j}$; $\|\mathbf{v}(0)\| = 2$;
 $\mathbf{a}(t) = -4\cosh 2t\mathbf{i} + 4\sinh 2t\mathbf{j}$; $\mathbf{a}(0) = -4\mathbf{i}$



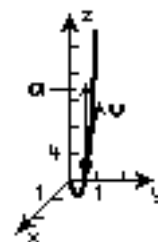
4. $\mathbf{v}(t) = -2\sin t\mathbf{i} + \cos t\mathbf{j}$; $\mathbf{v}(\pi/3) = -\sqrt{3}\mathbf{i} + \frac{1}{2}\mathbf{j}$; $\|\mathbf{v}(\pi/3)\| = \sqrt{3+1/4} = \sqrt{13}/2$;
 $\mathbf{a}(t) = -2\cos t\mathbf{i} - \sin t\mathbf{j}$; $\mathbf{a}(\pi/3) = -\mathbf{i} - \frac{\sqrt{3}}{2}\mathbf{j}$



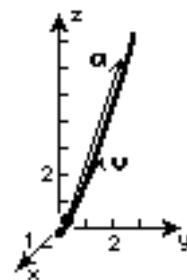
5. $\mathbf{v}(t) = (2t-2)\mathbf{j} + \mathbf{k}$; $\mathbf{v}(2) = 2\mathbf{j} + \mathbf{k}$; $\|\mathbf{v}(2)\| = \sqrt{4+1} = \sqrt{5}$;
 $\mathbf{a}(t) = 2\mathbf{j}$; $\mathbf{a}(2) = 2\mathbf{j}$



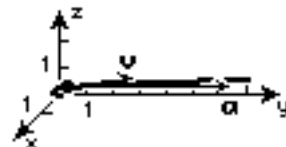
6. $\mathbf{v}(t) = \mathbf{i} + \mathbf{j} + 3t^2\mathbf{k}$; $\mathbf{v}(2) = \mathbf{i} + \mathbf{j} + 12\mathbf{k}$; $\|\mathbf{v}(2)\| = \sqrt{1+1+144} = \sqrt{146}$; $\mathbf{a}(t) = 6t\mathbf{k}$;
 $\mathbf{a}(2) = 12\mathbf{k}$



7. $\mathbf{v}(t) = \mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k}$; $\mathbf{v}(1) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$; $\|\mathbf{v}(1)\| = \sqrt{1+4+9} = \sqrt{14}$;
 $\mathbf{a}(t) = 2\mathbf{j} + 6t\mathbf{k}$; $\mathbf{a}(1) = 2\mathbf{j} + 6\mathbf{k}$



8. $\mathbf{v}(t) = \mathbf{i} + 3t^2\mathbf{j} + \mathbf{k}$; $\mathbf{v}(1) = \mathbf{i} + 3\mathbf{j} + \mathbf{k}$; $\|\mathbf{v}(1)\| = \sqrt{1+9+1} = \sqrt{11}$;
 $\mathbf{a}(t) = 6t\mathbf{j}$; $\mathbf{a}(1) = 6\mathbf{j}$



9. The particle passes through the xy -plane when $z(t) = t^2 - 5t = 0$ or $t = 0, 5$ which gives us the points $(0, 0, 0)$ and $(25, 115, 0)$. $\mathbf{v}(t) = 2t\mathbf{i} + (3t^2 - 2)\mathbf{j} + (2t - 5)\mathbf{k}$; $\mathbf{v}(0) = -2\mathbf{j} - 5\mathbf{k}$, $\mathbf{v}(5) = 10\mathbf{i} + 73\mathbf{j} + 5\mathbf{k}$; $\mathbf{a}(t) = 2\mathbf{i} + 6t\mathbf{j} + 2\mathbf{k}$;
 $\mathbf{a}(0) = 2\mathbf{i} + 2\mathbf{k}$, $\mathbf{a}(5) = 2\mathbf{i} + 30\mathbf{j} + 2\mathbf{k}$

10. If $\mathbf{a}(t) = \mathbf{0}$, then $\mathbf{v}(t) = \mathbf{c}_1$ and $\mathbf{r}(t) = \mathbf{c}_1 t + \mathbf{c}_2$. The graph of this equation is a straight line.

11. Initially we are given $\mathbf{s}_0 = \mathbf{0}$ and $\mathbf{v}_0 = (480 \cos 30^\circ)\mathbf{i} + (480 \sin 30^\circ)\mathbf{j} = 240\sqrt{3}\mathbf{i} + 240\mathbf{j}$. Using $\mathbf{a}(t) = -32\mathbf{j}$ we find

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c} \\ 240\sqrt{3}\mathbf{i} + 240\mathbf{j} &= \mathbf{v}(0) = \mathbf{c} \\ \mathbf{v}(t) &= -32t\mathbf{j} + 240\sqrt{3}\mathbf{i} + 240\mathbf{j} = 240\sqrt{3}\mathbf{i} + (240 - 32t)\mathbf{j} \\ \mathbf{r}(t) &= \int \mathbf{v}(t) dt = 240\sqrt{3}t\mathbf{i} + (240t - 16t^2)\mathbf{j} + \mathbf{b} \\ \mathbf{0} &= \mathbf{r}(0) = \mathbf{b}.\end{aligned}$$

- (a) The shell's trajectory is given by $\mathbf{r}(t) = 240\sqrt{3}t\mathbf{i} + (240t - 16t^2)\mathbf{j}$ or $x = 240\sqrt{3}t$, $y = 240t - 16t^2$.
 (b) Solving $dy/dt = 240 - 32t = 0$, we see that y is maximum when $t = 15/2$. The maximum altitude is $y(15/2) = 900$ ft.
 (c) Solving $y(t) = 240t - 16t^2 = 16t(15 - t) = 0$, we see that the shell is at ground level when $t = 0$ and $t = 15$. The range of the shell is $x(15) = 3600\sqrt{3} \approx 6235$ ft.
 (d) From (c), impact is when $t = 15$. The speed at impact is

$$\|\mathbf{v}(15)\| = |240\sqrt{3}\mathbf{i} + (240 - 32 \cdot 15)\mathbf{j}| = \sqrt{240^2 \cdot 3 + (-240)^2} = 480 \text{ ft/s}.$$

12. Initially we are given $\mathbf{s}_0 = 1600\mathbf{j}$ and $\mathbf{v}_0 = (480 \cos 30^\circ)\mathbf{i} + (480 \sin 30^\circ)\mathbf{j} = 240\sqrt{3}\mathbf{i} + 240\mathbf{j}$. Using $\mathbf{a}(t) = -32\mathbf{j}$ we find

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c} \\ 240\sqrt{3}\mathbf{i} + 240\mathbf{j} &= \mathbf{v}(0) = \mathbf{c} \\ \mathbf{v}(t) &= -32t\mathbf{j} + 240\sqrt{3}\mathbf{i} + 240\mathbf{j} = 240\sqrt{3}\mathbf{i} + (240 - 32t)\mathbf{j} \\ \mathbf{r}(t) &= \int \mathbf{v}(t) dt = 240\sqrt{3}t\mathbf{i} + (240t - 16t^2)\mathbf{j} + \mathbf{b} \\ 1600\mathbf{j} &= \mathbf{r}(0) = \mathbf{b}.\end{aligned}$$

- (a) The shell's trajectory is given by $\mathbf{r}(t) = 240\sqrt{3}t\mathbf{i} + (240t - 16t^2 + 1600)\mathbf{j}$ or $x = 240\sqrt{3}t$, $y = 240t - 16t^2 + 1600$.
 (b) Solving $dy/dt = 240 - 32t = 0$, we see that y is maximum when $t = 15/2$. The maximum altitude is $y(15/2) = 2500$ ft.
 (c) Solving $y(t) = -16t^2 + 240t + 1600 = -16(t - 20)(t + 5) = 0$, we see that the shell hits the ground when $t = 20$. The range of the shell is $x(20) = 4800\sqrt{3} \approx 8314$ ft.
 (d) From (c), impact is when $t = 20$. The speed at impact is

$$\|\mathbf{v}(20)\| = |240\sqrt{3}\mathbf{i} + (240 - 32 \cdot 20)\mathbf{j}| = \sqrt{240^2 \cdot 3 + (-400)^2} = 160\sqrt{13} \approx 577 \text{ ft/s}.$$

13. We are given $\mathbf{s}_0 = 81\mathbf{j}$ and $\mathbf{v}_0 = 4\mathbf{i}$. Using $\mathbf{a}(t) = -32\mathbf{j}$, we have

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c} \\ 4\mathbf{i} &= \mathbf{v}(0) = \mathbf{c} \\ \mathbf{v}(t) &= 4\mathbf{i} - 32t\mathbf{j} \\ \mathbf{r}(t) &= \int \mathbf{v}(t) dt = 4t\mathbf{i} - 16t^2\mathbf{j} + \mathbf{b} \\ 81\mathbf{j} &= \mathbf{r}(0) = \mathbf{b} \\ \mathbf{r}(t) &= 4t\mathbf{i} + (81 - 16t^2)\mathbf{j}.\end{aligned}$$

9.2 Motion on a Curve

Solving $y(t) = 81 - 16t^2 = 0$, we see that the car hits the water when $t = 9/4$. Then

$$\|\mathbf{v}(9/4)\| = |4\mathbf{i} - 32(9/4)\mathbf{j}| = \sqrt{4^2 + 72^2} = 20\sqrt{13} \approx 72.11 \text{ ft/s.}$$

14. Let θ be the angle of elevation. Then $\mathbf{v}(0) = 98 \cos \theta \mathbf{i} + 98 \sin \theta \mathbf{j}$. Using $\mathbf{a}(t) = -9.8\mathbf{j}$, we have

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = -9.8t\mathbf{j} + \mathbf{c} \\ 98 \cos \theta \mathbf{i} + 98 \sin \theta \mathbf{j} &= \mathbf{v}(0) = \mathbf{c} \\ \mathbf{v}(t) &= 98 \cos \theta \mathbf{i} + (98 \sin \theta - 9.8t)\mathbf{j} \\ \mathbf{r}(t) &= 98t \cos \theta \mathbf{i} + (98t \sin \theta - 4.9t^2)\mathbf{j} + \mathbf{b}.\end{aligned}$$

Since $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{b} = \mathbf{0}$ and $\mathbf{r}(t) = 98t \cos \theta \mathbf{i} + (98t \sin \theta - 4.9t^2)\mathbf{j}$. Setting $y(t) = 98t \sin \theta - 4.9t^2 = t(98 \sin \theta - 4.9t) = 0$, we see that the projectile hits the ground when $t = 20 \sin \theta$. Thus, using $x(t) = 98t \cos \theta$, $490 = x(t) = 98(20 \sin \theta) \cos \theta$ or $\sin 2\theta = 0.5$. Then $2\theta = 30^\circ$ or 150° . The angles of elevation are 15° and 75° .

15. Let s be the initial speed. Then $\mathbf{v}(0) = s \cos 45^\circ \mathbf{i} + s \sin 45^\circ \mathbf{j} = \frac{s\sqrt{2}}{2} \mathbf{i} + \frac{s\sqrt{2}}{2} \mathbf{j}$. Using $\mathbf{a}(t) = -32\mathbf{j}$, we have

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c} \\ \frac{s\sqrt{2}}{2} \mathbf{i} + \frac{s\sqrt{2}}{2} \mathbf{j} &= \mathbf{v}(0) = \mathbf{c} \\ \mathbf{v}(t) &= \frac{s\sqrt{2}}{2} \mathbf{i} + \left(\frac{s\sqrt{2}}{2} - 32t \right) \mathbf{j} \\ \mathbf{r}(t) &= \frac{s\sqrt{2}}{2} t \mathbf{i} + \left(\frac{s\sqrt{2}}{2} t - 16t^2 \right) \mathbf{j} + \mathbf{b}.\end{aligned}$$

Since $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{b} = \mathbf{0}$ and

$$\mathbf{r}(t) = \frac{s\sqrt{2}}{2} t \mathbf{i} + \left(\frac{s\sqrt{2}}{2} t - 16t^2 \right) \mathbf{j}.$$

Setting $y(t) = s\sqrt{2}t/2 - 16t^2 = t(s\sqrt{2}/2 - 16t) = 0$ we see that the ball hits the ground when $t = \sqrt{2}s/32$. Thus, using $x(t) = s\sqrt{2}t/2$ and the fact that 100 yd = 300 ft, $300 = x(t) = \frac{s\sqrt{2}}{2}(\sqrt{2}s/32) = \frac{s^2}{32}$ and $s = \sqrt{9600} \approx 97.98 \text{ ft/s}$.

16. Let s be the initial speed and θ the initial angle. Then $\mathbf{v}(0) = s \cos \theta \mathbf{i} + s \sin \theta \mathbf{j}$. Using $\mathbf{a}(t) = -32\mathbf{j}$, we have

$$\begin{aligned}\mathbf{v}(t) &= \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c} \\ s \cos \theta \mathbf{i} + s \sin \theta \mathbf{j} &= \mathbf{v}(0) = \mathbf{c} \\ \mathbf{v}(t) &= s \cos \theta \mathbf{i} + (s \sin \theta - 32t)\mathbf{j} \\ \mathbf{r}(t) &= st \cos \theta \mathbf{i} + (st \sin \theta - 16t^2)\mathbf{j} + \mathbf{b}.\end{aligned}$$

Since $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{b} = \mathbf{0}$ and $\mathbf{r}(t) = st \cos \theta \mathbf{i} + (st \sin \theta - 16t^2)\mathbf{j}$. Setting $y(t) = st \sin \theta - 16t^2 = t(s \sin \theta - 16t) = 0$, we see that the ball hits the ground when $t = (s \sin \theta)/16$. Using $x(t) = st \cos \theta \mathbf{i}$, we see that the range of the ball is

$$x\left(\frac{s \sin \theta}{16}\right) = \frac{s^2 \sin \theta \cos \theta}{16} = \frac{s^2 \sin 2\theta}{32}.$$

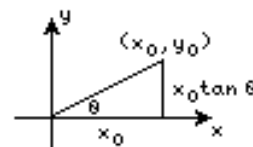
9.2 Motion on a Curve

For $\theta = 30^\circ$, the range is $s^2 \sin 60^\circ / 32 = \sqrt{3} s^2 / 64$ and for $\theta = 60^\circ$ the range is $s^2 \sin 120^\circ / 32 = \sqrt{3} s^2 / 64$. In general, when the angle is $90^\circ - \theta$ the range is

$$[s^2 \sin 2(90^\circ - \theta)] / 32 = s^2 [\sin(180^\circ - 2\theta)] / 32 = s^2 (\sin 2\theta) / 32.$$

Thus, for angles θ and $90^\circ - \theta$, the range is the same.

17. Let the initial speed of the projectile be s and let the target be at (x_0, y_0) . Then $\mathbf{v}_p(0) = s \cos \theta \mathbf{i} + s \sin \theta \mathbf{j}$ and $\mathbf{v}_t(0) = \mathbf{0}$. Using $\mathbf{a}(t) = -32\mathbf{j}$, we have



$$\mathbf{v}_p(t) = \int \mathbf{a}(t) dt = -32t\mathbf{j} + \mathbf{c}$$

$$s \cos \theta \mathbf{i} + s \sin \theta \mathbf{j} = \mathbf{v}_p(0) = \mathbf{c}$$

$$\mathbf{v}_p(t) = s \cos \theta \mathbf{i} + (s \sin \theta - 32t)\mathbf{j}$$

$$\mathbf{r}_p(t) = st \cos \theta \mathbf{i} + (st \sin \theta - 16t^2)\mathbf{j} + \mathbf{b}.$$

Since $\mathbf{r}_p(0) = \mathbf{0}$, $\mathbf{b} = \mathbf{0}$ and $\mathbf{r}_p(t) = st \cos \theta \mathbf{i} + (st \sin \theta - 16t^2)\mathbf{j}$. Also, $\mathbf{v}_t(t) = -32t\mathbf{j} + \mathbf{c}$ and since $\mathbf{v}_t(0) = \mathbf{0}$, $\mathbf{c} = \mathbf{0}$ and $\mathbf{v}_t(t) = -32t\mathbf{j}$. Then $\mathbf{r}_t(t) = -16t^2\mathbf{j} + \mathbf{b}$. Since $\mathbf{r}_t(0) = x_0\mathbf{i} + y_0\mathbf{j}$, $\mathbf{b} = x_0\mathbf{i} + y_0\mathbf{j}$ and $\mathbf{r}_t(t) = x_0\mathbf{i} + (y_0 - 16t^2)\mathbf{j}$. Now, the horizontal component of $\mathbf{r}_p(t)$ will be x_0 when $t = x_0 / s \cos \theta$ at which time the vertical component of $\mathbf{r}_p(t)$ will be

$$(sx_0 / s \cos \theta) \sin \theta - 16(x_0 / s \cos \theta)^2 = x_0 \tan \theta - 16(x_0 / s \cos \theta)^2 = y_0 - 16(x_0 / s \cos \theta)^2.$$

Thus, $\mathbf{r}_p(x_0 / s \cos \theta) = \mathbf{r}_t(x_0 / s \cos \theta)$ and the projectile will strike the target as it falls.

18. The initial angle is $\theta = 0$, the initial height is 1024 ft, and the initial speed is $s = 180(5280) / 3600 = 264$ ft/s. Then $x(t) = 264t$ and $y(t) = -16t^2 + 1024$. Solving $y(t) = 0$ we see that the pack hits the ground at $t = 8$ seconds. The horizontal distance travelled is $x(8) = 2112$ feet. From the figure in the text, $\tan \alpha = 1024 / 2112 = 16 / 33$ and $\alpha \approx 0.45$ radian or 25.87° .

19. $\mathbf{r}'(t) = \mathbf{v}(t) = -r_0\omega \sin \omega t \mathbf{i} + r_0\omega \cos \omega t \mathbf{j}$; $v = \|\mathbf{v}(t)\| = \sqrt{r_0^2\omega^2 \sin^2 \omega t + r_0^2\omega^2 \cos^2 \omega t} = r_0\omega$
 $\omega = v / r_0$; $\mathbf{a}(t) = \mathbf{r}''(t) = -r_0\omega^2 \cos \omega t \mathbf{i} - r_0\omega^2 \sin \omega t \mathbf{j}$
 $a = \|\mathbf{a}(t)\| = \sqrt{r_0^2\omega^4 \cos^2 \omega t + r_0^2\omega^4 \sin^2 \omega t} = r_0\omega^2 = r_0(v / r_0)^2 = v^2 / r_0.$

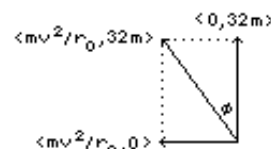
20. (a) $\mathbf{v}(t) = -b \sin t \mathbf{i} + b \cos t \mathbf{j} + c\mathbf{k}$; $\|\mathbf{v}(t)\| = \sqrt{b^2 \sin^2 t + b^2 \cos^2 t + c^2} = \sqrt{b^2 + c^2}$

(b) $s = \int_0^t \|\mathbf{v}(u)\| du = \int_0^t \sqrt{b^2 + c^2} du = t\sqrt{b^2 + c^2}$; $\frac{ds}{dt} = \sqrt{b^2 + c^2}$

(c) $\frac{d^2s}{dt^2} = 0$; $\mathbf{a}(t) = -b \cos t \mathbf{i} - b \sin t \mathbf{j}$; $\|\mathbf{a}(t)\| = \sqrt{b^2 \cos^2 t + b^2 \sin^2 t} = |b|$. Thus, $d^2s / dt^2 \neq \|\mathbf{a}(t)\|$.

21. By Problem 19, $a = v^2 / r_0 = 1530^2 / (4000 \cdot 5280) \approx 0.1108$. We are given $mg = 192$, so $m = 192 / 32$ and $w_e = 192 - (192 / 32)(0.1108) \approx 191.33$ lb.

22. By Problem 19, the centripetal acceleration is v^2 / r_0 . Then the horizontal force is mv^2 / r_0 . The vertical force is $32m$. The resultant force is $\mathbf{U} = (mv^2 / r_0)\mathbf{i} + 32m\mathbf{j}$. From the figure, we see that $\tan \phi = (mv^2 / r_0) / 32m = v^2 / 32r_0$. Using $r_0 = 60$ and $v = 44$ we obtain $\tan \phi = 44^2 / 32(60) \approx 1.0083$ and $\phi \approx 45.24^\circ$.



23. Solving $x(t) = (v_0 \cos \theta)t$ for t and substituting into $y(t) = -\frac{1}{2}gt^2 + (v_0 \sin \theta)t + s_0$ we obtain

$$y = -\frac{1}{2}g \left(\frac{x}{v_0 \cos \theta} \right)^2 + (v_0 \sin \theta) \frac{x}{v_0 \cos \theta} + s_0 = -\frac{g}{2v_0^2 \cos^2 \theta} x^2 + (\tan \theta)x + s_0,$$

9.2 Motion on a Curve

which is the equation of a parabola.

24. Since the projectile is launched from ground level, $s_0 = 0$. To find the maximum height we maximize $y(t) = -\frac{1}{2}gt^2 + (v_0 \sin \theta)t$. Solving $y'(t) = -gt + v_0 \sin \theta = 0$, we see that $t = (v_0/g) \sin \theta$ is a critical point. Since $y''(t) = -g < 0$,

$$H = y\left(\frac{v_0 \sin \theta}{g}\right) = -\frac{1}{2}g \frac{v_0^2 \sin^2 \theta}{g^2} + v_0 \sin \theta \frac{v_0 \sin \theta}{g} = \frac{v_0^2 \sin^2 \theta}{2g}$$

is the maximum height. To find the range we solve $y(t) = -\frac{1}{2}gt^2 + (v_0 \sin \theta)t = t(v_0 \sin \theta - \frac{1}{2}gt) = 0$. The positive solution of this equation is $t = (2v_0 \sin \theta)/g$. The range is thus

$$x(t) = (v_0 \cos \theta) \frac{2v_0 \sin \theta}{g} = \frac{v_0^2 \sin 2\theta}{g}.$$

25. Letting $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, the equation $d\mathbf{r}/dt = \mathbf{v}$ is equivalent to $dx/dt = 6t^2x$, $dy/dt = -4ty^2$, $dz/dt = 2t(z+1)$. Separating variables and integrating, we obtain $dx/x = 6t^2 dt$, $dy/y^2 = -4t dt$, $dz/(z+1) = 2t dt$, and $\ln x = 2t^3 + c_1$, $-1/y = -2t^2 + c_2$, $\ln(z+1) = t^2 + c_3$. Thus,

$$\mathbf{r}(t) = k_1 e^{2t^3} \mathbf{i} + \frac{1}{2t^2 + k_2} \mathbf{j} + (k_3 e^{t^2} - 1) \mathbf{k}.$$

26. We require the fact that $d\mathbf{r}/dt = \mathbf{v}$. Then

$$\frac{d\mathbf{L}}{dt} = \frac{d}{dt}(\mathbf{r} \times \mathbf{p}) = \mathbf{r} \times \frac{d\mathbf{p}}{dt} + \frac{d\mathbf{r}}{dt} \times \mathbf{p} = \boldsymbol{\tau} + \mathbf{v} \times \mathbf{p} = \boldsymbol{\tau} + \mathbf{v} \times m\mathbf{v} = \boldsymbol{\tau} + m(\mathbf{v} \times \mathbf{v}) = \boldsymbol{\tau} + \mathbf{0} = \boldsymbol{\tau}.$$

27. (a) Since $\mathbf{F}$ is directed along $\mathbf{r}$ we have $\mathbf{F} = c\mathbf{r}$ for some constant c . Then

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times (c\mathbf{r}) = c(\mathbf{r} \times \mathbf{r}) = \mathbf{0}.$$

(b) If $\boldsymbol{\tau} = \mathbf{0}$ then $d\mathbf{L}/dt = \mathbf{0}$ and $\mathbf{L}$ is constant.

28. (a) Using Problem 27, $\mathbf{F} = -k(Mm/r^2)\mathbf{u} = m\mathbf{a}$. Then $\mathbf{a} = d^2\mathbf{r}/dt^2 = -k(M/r^2)\mathbf{u}$.

(b) Using $\mathbf{u} = \mathbf{r}/r$ we have

$$\mathbf{r} \times \mathbf{r}'' = \mathbf{r} \times \left(-k \frac{M}{r^2} \mathbf{u}\right) = -\frac{kM}{r^2} \left[\mathbf{r} \times \left(\frac{1}{r} \mathbf{r}\right)\right] = -\frac{kM}{r^3} (\mathbf{r} \times \mathbf{r}) = \mathbf{0}.$$

(c) From Theorem 9.4 (iv) we have

$$\frac{d}{dt}(\mathbf{r} \times \mathbf{v}) = \mathbf{r} \times \frac{d\mathbf{v}}{dt} + \frac{d\mathbf{r}}{dt} \times \mathbf{v} = \mathbf{r} \times \mathbf{r}'' + \mathbf{v} \times \mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

(d) Since $\mathbf{r} = r\mathbf{u}$ we have $\mathbf{c} = \mathbf{r} \times \mathbf{v} = r\mathbf{u} \times r\mathbf{u}' = r^2(\mathbf{u} \times \mathbf{u}')$.

(e) Since $\mathbf{u} = (1/r)\mathbf{r}$ is a unit vector, $\mathbf{u} \cdot \mathbf{u} = 1$ and

$$\frac{d}{dt}(\mathbf{u} \cdot \mathbf{u}) = \mathbf{u} \cdot \frac{d\mathbf{u}}{dt} + \frac{d\mathbf{u}}{dt} \cdot \mathbf{u} = 2\mathbf{u} \cdot \frac{d\mathbf{u}}{dt} = \frac{d}{dt}(1) = 0.$$

Thus, $\mathbf{u} \cdot \mathbf{u}' = 0$.

- (f)
$$\begin{aligned} \frac{d}{dt}(\mathbf{v} \times \mathbf{c}) &= \mathbf{v} \times \frac{d\mathbf{c}}{dt} + \frac{d\mathbf{v}}{dt} \times \mathbf{c} = \mathbf{v} \times \mathbf{0} + \mathbf{a} \times \mathbf{c} = -\frac{kM}{r^2} \mathbf{u} \times \mathbf{c} = -\frac{kM}{r^2} \mathbf{u} \times [r^2(\mathbf{u} \times \mathbf{u}')] \\ &= -kM[\mathbf{u} \times (\mathbf{u} \times \mathbf{u}')] = -kM = -kM[(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - (\mathbf{u} \cdot \mathbf{u})\mathbf{u}'] \quad \boxed{\text{by (10) of 7.4}} \\ &= -kM[\mathbf{0} - \mathbf{u}'] = kM\mathbf{u}' = kM \frac{d\mathbf{u}}{dt} \end{aligned}$$

(g) Since

$$\begin{aligned}\mathbf{r} \cdot (\mathbf{v} \times \mathbf{c}) &= (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{c} && \boxed{\text{by Problem 61 in 7.4}} \\ &= \mathbf{c} \cdot \mathbf{c} = c^2 && \boxed{\text{where } c = \|\mathbf{c}\|}\end{aligned}$$

and

$$\begin{aligned}(kM\mathbf{u} + \mathbf{d}) \cdot \mathbf{r} &= (kM\mathbf{u} + \mathbf{d}) \cdot r\mathbf{u} = kMr\mathbf{u} \cdot \mathbf{u} + r\mathbf{d} \cdot \mathbf{u} \\ &= kMr + rd \cos \theta && \boxed{\text{where } d = \|\mathbf{d}\|}\end{aligned}$$

$$\text{we have } c^2 = kMr + rd \cos \theta \text{ or } r = \frac{c^2}{kM + d \cos \theta} = \frac{c^2/kM}{1 + (d/kM) \cos \theta}.$$

(h) First note that $c > 0$ (otherwise there is no orbit) and $d > 0$ (since the orbit is not a circle). We recognize the equation in (g) to be that of a conic section with eccentricity $e = d/kM$. Since the orbit of the planet is closed it must be an ellipse.

(i) At perihelion $c = \|\mathbf{c}\| = \|\mathbf{r} \times \mathbf{v}\| = r_0 v_0 \sin(\pi/r) = r_0 v_0$. Since r is minimum at this point, we want the denominator in the equation $r_0 = [c^2/kM]/[1 + (d/kM) \cos \theta]$ to be maximum. This occurs when $\theta = 0$. In this case

$$r_0 = \frac{r_0^2 v_0^2 / kM}{1 + d/kM} \quad \text{and} \quad d = r_0 v_0^2 - kM.$$

EXERCISES 9.3

Curvature and Components of Acceleration

1. $\mathbf{r}'(t) = -t \sin t \mathbf{i} + t \cos t \mathbf{j} + 2t \mathbf{k}$; $|\mathbf{r}'(t)| = \sqrt{t^2 \sin^2 t + t^2 \cos^2 t + 4t^2} = \sqrt{5}t$;
 $\mathbf{T}(t) = -\frac{\sin t}{\sqrt{5}} \mathbf{i} + \frac{\cos t}{\sqrt{5}} \mathbf{j} + \frac{2}{\sqrt{5}} \mathbf{k}$
2. $\mathbf{r}'(t) = e^t(-\sin t + \cos t) \mathbf{i} + e^t(\cos t + \sin t) \mathbf{j} + \sqrt{2}e^t \mathbf{k}$;
 $|\mathbf{r}'(t)| = [e^{2t}(\sin^2 t - 2 \sin t \cos t + \cos^2 t) + e^{2t}(\cos^2 t + 2 \sin t \cos t + \sin^2 t) + 2e^{2t}]^{1/2} = \sqrt{4e^{2t}} = 2e^t$;
 $\mathbf{T}(t) = \frac{1}{2}(-\sin t + \cos t) \mathbf{i} + \frac{1}{2}(\cos t + \sin t) \mathbf{j} + \frac{\sqrt{2}}{2} \mathbf{k}$
3. We assume $a > 0$. $\mathbf{r}'(t) = -a \sin t \mathbf{i} + a \cos t \mathbf{j} + c \mathbf{k}$; $|\mathbf{r}'(t)| = \sqrt{a^2 \sin^2 t + a^2 \cos^2 t + c^2} = \sqrt{a^2 + c^2}$;
 $\mathbf{T}(t) = -\frac{a \sin t}{\sqrt{a^2 + c^2}} \mathbf{i} + \frac{a \cos t}{\sqrt{a^2 + c^2}} \mathbf{j} + \frac{c}{\sqrt{a^2 + c^2}} \mathbf{k}$; $\frac{d\mathbf{T}}{dt} = -\frac{a \cos t}{\sqrt{a^2 + c^2}} \mathbf{i} - \frac{a \sin t}{\sqrt{a^2 + c^2}} \mathbf{j}$;
 $\left| \frac{d\mathbf{T}}{dt} \right| = \sqrt{\frac{a^2 \cos^2 t}{a^2 + c^2} + \frac{a^2 \sin^2 t}{a^2 + c^2}} = \frac{a}{\sqrt{a^2 + c^2}}$; $\mathbf{N} = -\cos t \mathbf{i} - \sin t \mathbf{j}$;
 $\mathbf{B} = \mathbf{T} \times \mathbf{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\frac{a \sin t}{\sqrt{a^2 + c^2}} & \frac{a \cos t}{\sqrt{a^2 + c^2}} & \frac{c}{\sqrt{a^2 + c^2}} \\ -\cos t & -\sin t & 0 \end{vmatrix} = \frac{c \sin t}{\sqrt{a^2 + c^2}} \mathbf{i} - \frac{c \cos t}{\sqrt{a^2 + c^2}} \mathbf{j} + \frac{a}{\sqrt{a^2 + c^2}} \mathbf{k}$;
 $\kappa = \frac{|d\mathbf{T}/dt|}{|\mathbf{r}'(t)|} = \frac{a/\sqrt{a^2 + c^2}}{\sqrt{a^2 + c^2}} = \frac{a}{a^2 + c^2}$

9.3 Curvature and Components of Acceleration

4. $\mathbf{r}'(t) = \mathbf{i} + t\mathbf{j} + t^2\mathbf{k}$, $\mathbf{r}'(1) = \mathbf{i} + \mathbf{j} + \mathbf{k}$; $|\mathbf{r}'(t)| = \sqrt{1+t^2+t^4}$, $|\mathbf{r}'(1)| = \sqrt{3}$;

$$\mathbf{T}(t) = (1+t^2+t^4)^{-1/2}(\mathbf{i} + t\mathbf{j} + t^2\mathbf{k}), \quad \mathbf{T}(1) = \frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} + \mathbf{k});$$

$$\begin{aligned} \frac{d\mathbf{T}}{dt} &= -\frac{1}{2}(1+t^2+t^4)^{-3/2}(2t+4t^3)\mathbf{i} + [(1+t^2+t^4)^{-1/2} - \frac{t}{2}(1+t^2+t^4)^{-3/2}(2t+4t^3)]\mathbf{j} \\ &\quad + [2t(1+t^2+t^4)^{-1/2} - \frac{t^2}{2}(1+t^2+t^4)^{-3/2}(2t+4t^3)]\mathbf{k}; \end{aligned}$$

$$\frac{d}{dt}\mathbf{T}(1) = -\frac{1}{\sqrt{3}}\mathbf{i} + \frac{1}{\sqrt{3}}\mathbf{k}, \quad \left| \frac{d}{dt}\mathbf{T}(1) \right| = \sqrt{\frac{1}{3} + \frac{1}{3}} = \frac{\sqrt{2}}{\sqrt{3}}; \quad \mathbf{N}(1) = -\frac{1}{\sqrt{2}}(\mathbf{i} - \mathbf{k});$$

$$\mathbf{B}(1) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{vmatrix} = \frac{1}{\sqrt{6}}(\mathbf{i} - 2\mathbf{j} + \mathbf{k}); \quad \kappa = \left| \frac{d}{dt}\mathbf{T}(1) \right| / |\mathbf{r}'(1)| = \frac{\sqrt{2}/\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{2}}{3}$$

5. From Example 2 in the text, a normal to the osculating plane is $\mathbf{B}(\pi/4) = \frac{1}{\sqrt{26}}(3\mathbf{i} - 3\mathbf{j} + 2\sqrt{2}\mathbf{k})$. The point on the curve when $t = \pi/4$ is $(\sqrt{2}, \sqrt{2}, 3\pi/4)$. An equation of the plane is $3(x - \sqrt{2}) - 3(y - \sqrt{2}) + 2\sqrt{2}(z - 3\pi/4) = 0$, $3x - 3y + 2\sqrt{2}z = 3\sqrt{2}\pi/2$, or $3\sqrt{2}x - 3\sqrt{2}y + 4z = 3\pi$.

6. From Problem 4, a normal to the osculating plane is $\mathbf{B}(1) = \frac{1}{\sqrt{6}}(\mathbf{i} - 2\mathbf{j} + \mathbf{k})$. The point on the curve when $t = 1$ is $(1, 1/2, 1/3)$. An equation of the plane is $(x - 1) - 2(y - 1/2) + (z - 1/3) = 0$ or $x - 2y + z = 1/3$.

7. $\mathbf{v}(t) = \mathbf{j} + 2t\mathbf{k}$, $|\mathbf{v}(t)| = \sqrt{1+4t^2}$; $\mathbf{a}(t) = 2\mathbf{k}$; $\mathbf{v} \cdot \mathbf{a} = 4t$, $\mathbf{v} \times \mathbf{a} = 2\mathbf{i}$, $|\mathbf{v} \times \mathbf{a}| = 2$;

$$a_T = \frac{4t}{\sqrt{1+4t^2}}, \quad a_N = \frac{2}{\sqrt{1+4t^2}}$$

8. $\mathbf{v}(t) = -3\sin t\mathbf{i} + 2\cos t\mathbf{j} + \mathbf{k}$,

$$|\mathbf{v}(t)| = \sqrt{9\sin^2 t + 4\cos^2 t + 1} = \sqrt{5\sin^2 t + 4\sin^2 t + 4\cos^2 t + 1} = \sqrt{5}\sqrt{\sin^2 t + 1};$$

$$\mathbf{a}(t) = -3\cos t\mathbf{i} - 2\sin t\mathbf{j}; \quad \mathbf{v} \cdot \mathbf{a} = 9\sin t \cos t - 4\sin t \cos t = 5\sin t \cos t,$$

$$\mathbf{v} \times \mathbf{a} = 2\sin t\mathbf{i} - 3\cos t\mathbf{j} + 6\mathbf{k}, \quad |\mathbf{v} \times \mathbf{a}| = \sqrt{4\sin^2 t + 9\cos^2 t + 36} = \sqrt{5}\sqrt{\cos^2 t + 8};$$

$$a_T = \frac{\sqrt{5}\sin t \cos t}{\sqrt{\sin^2 t + 1}}, \quad a_N = \sqrt{\frac{\cos^2 t + 8}{\sin^2 t + 1}}$$

9. $\mathbf{v}(t) = 2t\mathbf{i} + 2t\mathbf{j} + 4t\mathbf{k}$, $|\mathbf{v}(t)| = 2\sqrt{6}t$, $t > 0$; $\mathbf{a}(t) = 2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$; $\mathbf{v} \cdot \mathbf{a} = 24t$, $\mathbf{v} \times \mathbf{a} = \mathbf{0}$;

$$a_T = \frac{24t}{2\sqrt{6}t} = 2\sqrt{6}, \quad a_N = 0, \quad t > 0$$

10. $\mathbf{v}(t) = 2t\mathbf{i} - 3t^2\mathbf{j} + 4t^3\mathbf{k}$, $|\mathbf{v}(t)| = t\sqrt{4+9t^2+16t^4}$, $t > 0$; $\mathbf{a}(t) = 2\mathbf{i} - 6t\mathbf{j} + 12t^2\mathbf{k}$;

$$\mathbf{v} \cdot \mathbf{a} = 4t + 18t^3 + 48t^5; \quad \mathbf{v} \times \mathbf{a} = -12t^4\mathbf{i} - 16t^3\mathbf{j} - 6t^2\mathbf{k}, \quad |\mathbf{v} \times \mathbf{a}| = 2t^2\sqrt{36t^4 + 64t^2 + 9};$$

$$a_T = \frac{4 + 18t^2 + 48t^4}{\sqrt{4 + 9t^2 + 16t^4}}, \quad a_N = \frac{2t\sqrt{36t^4 + 64t^2 + 9}}{\sqrt{4 + 9t^2 + 16t^4}}, \quad t > 0$$

11. $\mathbf{v}(t) = 2\mathbf{i} + 2t\mathbf{j}$, $|\mathbf{v}(t)| = 2\sqrt{1+t^2}$; $\mathbf{a}(t) = 2\mathbf{j}$; $\mathbf{v} \cdot \mathbf{a} = 4t$; $\mathbf{v} \times \mathbf{a} = 4\mathbf{k}$, $|\mathbf{v} \times \mathbf{a}| = 4$;

$$a_T = \frac{2t}{\sqrt{1+t^2}}, \quad a_N = \frac{2}{\sqrt{1+t^2}}$$

12. $\mathbf{v}(t) = \frac{1}{1+t^2}\mathbf{i} + \frac{t}{1+t^2}\mathbf{j}$, $|\mathbf{v}(t)| = \frac{\sqrt{1+t^2}}{1+t^2}$; $\mathbf{a}(t) = -\frac{2t}{(1+t^2)^2}\mathbf{i} + \frac{1-t^2}{(1+t^2)^2}\mathbf{j}$;

$$\mathbf{v} \cdot \mathbf{a} = -\frac{2t}{(1+t^2)^3} + \frac{t-t^3}{(1+t^2)^3} = -\frac{t}{(1+t^2)^2}; \quad \mathbf{v} \times \mathbf{a} = \frac{1}{(1+t^2)^2}\mathbf{k}, \quad |\mathbf{v} \times \mathbf{a}| = \frac{1}{(1+t^2)^2};$$

$$a_T = -\frac{t/(1+t^2)^2}{\sqrt{1+t^2}/(1+t^2)} = -\frac{t}{(1+t^2)^{3/2}}, \quad a_N = \frac{1/(1+t^2)^2}{\sqrt{1+t^2}/(1+t^2)} = \frac{1}{(1+t^2)^{3/2}}$$

13. $\mathbf{v}(t) = -5 \sin t \mathbf{i} + 5 \cos t \mathbf{j}$, $|\mathbf{v}(t)| = 5$; $\mathbf{a}(t) = -5 \cos t \mathbf{i} - 5 \sin t \mathbf{j}$; $\mathbf{v} \cdot \mathbf{a} = 0$, $\mathbf{v} \times \mathbf{a} = 25 \mathbf{k}$, $|\mathbf{v} \times \mathbf{a}| = 25$;
 $a_T = 0$, $a_N = 5$
14. $\mathbf{v}(t) = \sinh t \mathbf{i} + \cosh t \mathbf{j}$, $|\mathbf{v}(t)| = \sqrt{\sinh^2 t + \cosh^2 t}$; $\mathbf{a}(t) = \cosh t \mathbf{i} + \sinh t \mathbf{j}$; $\mathbf{v} \cdot \mathbf{a} = 2 \sinh t \cosh t$;
 $\mathbf{v} \times \mathbf{a} = (\sinh^2 t - \cosh^2 t) \mathbf{k} = -\mathbf{k}$, $|\mathbf{v} \times \mathbf{a}| = 1$; $a_T = \frac{2 \sinh t \cosh t}{\sqrt{\sinh^2 t + \cosh^2 t}}$, $a_N = \frac{1}{\sqrt{\sinh^2 t + \cosh^2 t}}$
15. $\mathbf{v}(t) = -e^{-t}(\mathbf{i} + \mathbf{j} + \mathbf{k})$, $|\mathbf{v}(t)| = \sqrt{3} e^{-t}$; $\mathbf{a}(t) = e^{-t}(\mathbf{i} + \mathbf{j} + \mathbf{k})$; $\mathbf{v} \cdot \mathbf{a} = -3e^{-2t}$; $\mathbf{v} \times \mathbf{a} = \mathbf{0}$, $|\mathbf{v} \times \mathbf{a}| = 0$;
 $a_T = -\sqrt{3} e^{-t}$, $a_N = 0$
16. $\mathbf{v}(t) = \mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$, $|\mathbf{v}(t)| = \sqrt{21}$; $\mathbf{a}(t) = \mathbf{0}$; $\mathbf{v} \cdot \mathbf{a} = 0$, $\mathbf{v} \times \mathbf{a} = \mathbf{0}$, $|\mathbf{v} \times \mathbf{a}| = 0$; $a_T = 0$, $a_N = 0$
17. $\mathbf{v}(t) = -a \sin t \mathbf{i} + b \cos t \mathbf{j} + c \mathbf{k}$, $|\mathbf{v}(t)| = \sqrt{a^2 \sin^2 t + b^2 \cos^2 t + c^2}$; $\mathbf{a}(t) = -a \cos t \mathbf{i} - b \sin t \mathbf{j}$;
 $\mathbf{v} \times \mathbf{a} = bc \sin t \mathbf{i} - ac \cos t \mathbf{j} + ab \mathbf{k}$, $|\mathbf{v} \times \mathbf{a}| = \sqrt{b^2 c^2 \sin^2 t + a^2 c^2 \cos^2 t + a^2 b^2}$
 $\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3} = \frac{\sqrt{b^2 c^2 \sin^2 t + a^2 c^2 \cos^2 t + a^2 b^2}}{(a^2 \sin^2 t + b^2 \cos^2 t + c^2)^{3/2}}$;
18. (a) $\mathbf{v}(t) = -a \sin t \mathbf{i} + b \cos t \mathbf{j}$, $|\mathbf{v}(t)| = \sqrt{a^2 \sin^2 t + b^2 \cos^2 t}$; $\mathbf{a}(t) = -a \cos t \mathbf{i} - b \sin t \mathbf{j}$;
 $\mathbf{v} \times \mathbf{a} = ab \mathbf{k}$; $|\mathbf{v} \times \mathbf{a}| = ab$; $\kappa = \frac{ab}{(a^2 \sin^2 t + b^2 \cos^2 t)^{3/2}}$
 (b) When $a = b$, $|\mathbf{v}(t)| = a$, $|\mathbf{v} \times \mathbf{a}| = a^2$, and $\kappa = a^2/a^3 = 1/a$.
19. The equation of a line is $\mathbf{r}(t) = \mathbf{b} + t\mathbf{c}$, where $\mathbf{b}$ and $\mathbf{c}$ are constant vectors.
 $\mathbf{v}(t) = \mathbf{c}$, $|\mathbf{v}(t)| = |\mathbf{c}|$; $\mathbf{a}(t) = \mathbf{0}$; $\mathbf{v} \times \mathbf{a} = \mathbf{0}$, $|\mathbf{v} \times \mathbf{a}| = 0$; $\kappa = |\mathbf{v} \times \mathbf{a}|/|\mathbf{v}|^3 = 0$
20. $\mathbf{v}(t) = a(1 - \cos t)\mathbf{i} + a \sin t \mathbf{j}$; $\mathbf{v}(\pi) = 2a\mathbf{i}$, $|\mathbf{v}(\pi)| = 2a$; $\mathbf{a}(t) = a \sin t \mathbf{i} + a \cos t \mathbf{j}$, $\mathbf{a}(\pi) = -a\mathbf{j}$;
 $|\mathbf{v} \times \mathbf{a}| = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2a & 0 & 0 \\ 0 & -a & 0 \end{vmatrix} = -2a^2 \mathbf{k}$; $|\mathbf{v} \times \mathbf{a}| = 2a^2$; $\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3} = \frac{2a^2}{8a^3} = \frac{1}{4a}$
21. $\mathbf{v}(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j}$, $|\mathbf{v}(t)| = \sqrt{[f'(t)]^2 + [g'(t)]^2}$; $\mathbf{a}(t) = f''(t)\mathbf{i} + g''(t)\mathbf{j}$;
 $\mathbf{v} \times \mathbf{a} = [f'(t)g''(t) - g'(t)f''(t)]\mathbf{k}$, $|\mathbf{v} \times \mathbf{a}| = |f'(t)g''(t) - g'(t)f''(t)|$;
 $\kappa = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3} = \frac{|f'(t)g''(t) - g'(t)f''(t)|}{([f'(t)]^2 + [g'(t)]^2)^{3/2}}$
22. For $y = F(x)$, $\mathbf{r}(x) = x\mathbf{i} + F(x)\mathbf{j}$. We identify $f(x) = x$ and $g(x) = F(x)$ in Problem 21. Then $f'(x) = 1$,
 $f''(x) = 0$, $g'(x) = F'(x)$, $g''(x) = F''(x)$, and $\kappa = |F''(x)|/(1 + [F'(x)]^2)^{3/2}$.
23. $F(x) = x^2$, $F(0) = 0$, $F(1) = 1$; $F'(x) = 2x$, $F'(0) = 0$, $F'(1) = 2$; $F''(x) = 2$, $F''(0) = 2$, $F''(1) = 2$;
 $\kappa(0) = \frac{2}{(1 + 0^2)^{3/2}} = 2$; $\rho(0) = \frac{1}{2}$; $\kappa(1) = \frac{2}{(1 + 2^2)^{3/2}} = \frac{2}{5\sqrt{5}} \approx 0.18$;
 $\rho(1) = \frac{5\sqrt{5}}{2} \approx 5.59$; Since $2 > 2/5\sqrt{5}$, the curve is “sharper” at $(0, 0)$.
24. $F(x) = x^3$, $F(-1) = -1$, $F(1/2) = 1/8$; $F'(x) = 3x^2$, $F'(-1) = 3$, $F'(1/2) = 3/4$; $F''(x) = 6x$,
 $F''(-1) = -6$, $F''(1/2) = 3$; $\kappa(-1) = \frac{|-6|}{(1 + 3^2)^{3/2}} = \frac{6}{10\sqrt{10}} = \frac{3}{5\sqrt{10}} \approx 0.19$;
 $\rho(-1) = \frac{5\sqrt{10}}{3} \approx 5.27$; $\kappa(\frac{1}{2}) = \frac{3}{[1 + (3/4)^2]^{3/2}} = \frac{3}{125/64} = \frac{192}{125} \approx 1.54$; $\rho(\frac{1}{2}) = \frac{125}{192} \approx 0.65$
 Since $1.54 > 0.19$, the curve is “sharper” at $(1/2, 1/8)$.
25. At a point of inflection $(x_0, F(x_0))$, if $F''(x_0)$ exists then $F''(x_0) = 0$. Thus, assuming that $\lim_{x \rightarrow x_0} F''(x)$ exists, $F''(x)$ and hence κ is near 0 for x near x_0 .

9.3 Curvature and Components of Acceleration

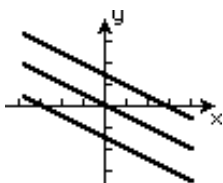
26. We use the fact that $\mathbf{T} \cdot \mathbf{N} = 0$ and $\mathbf{T} \cdot \mathbf{T} = \mathbf{N} \cdot \mathbf{N} = 1$. Then

$$|\mathbf{a}(t)|^2 = \mathbf{a} \cdot \mathbf{a} = (a_N \mathbf{N} + a_T \mathbf{T}) \cdot (a_N \mathbf{N} + a_T \mathbf{T}) = a_N^2 \mathbf{N} \cdot \mathbf{N} + 2a_N a_T \mathbf{N} \cdot \mathbf{T} + a_T^2 \mathbf{T} \cdot \mathbf{T} = a_N^2 + a_T^2.$$

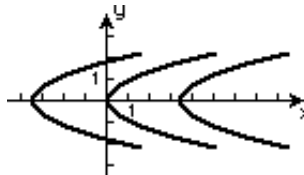
EXERCISES 9.4

Partial Derivatives

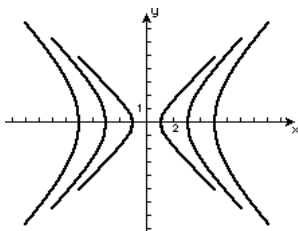
1. $y = -\frac{1}{2}x + C$



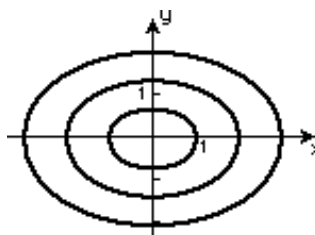
2. $x = y^2 - c$



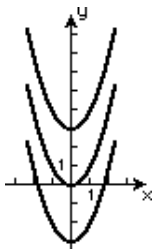
3. $x^2 - y^2 = 1 + c^2$



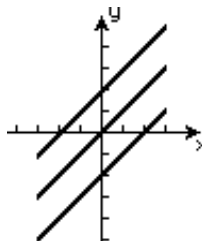
4. $4x^2 + 9y^2 = 36 - c^2, -6 \leq c \leq 6$



5. $y = x^2 + \ln c, c > 0$



6. $y = x + \tan c, -\pi/2 < c < \pi/2$



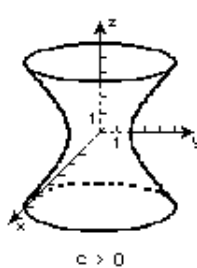
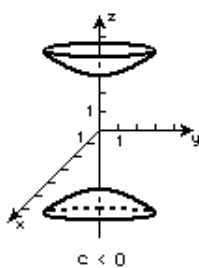
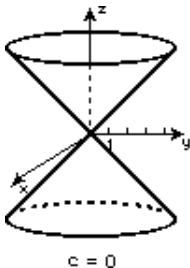
7. $x^2/9 + z^2/4 = c$; elliptical cylinder

8. $x^2 + y^2 + z^2 = c$; sphere

9. $x^2 + 3y^2 + 6z^2 = c$; ellipsoid

10. $4y - 2z + 1 = c$; plane

11.



12. Setting $x = -4$, $y = 2$, and $z = -3$ in $x^2/16 + y^2/4 + z^2/9 = c$ we obtain $c = 3$. The equation of the surface is $x^2/16 + y^2/4 + z^2/9 = 3$. Setting $y = z = 0$ we find the x -intercepts are $\pm 4\sqrt{3}$. Similarly, the y -intercepts are $\pm 2\sqrt{3}$ and the z -intercepts are $\pm 3\sqrt{3}$.
13. $z_x = 2x - y^2$; $z_y = -2xy + 20y^4$
14. $z_x = -3x^2 + 12xy^3$; $z_y = 18x^2y^2 + 10y$
15. $z_x = 20x^3y^3 - 2xy^6 + 30x^4$; $z_y = 15x^4y^2 - 6x^2y^5 - 4$
16. $z_x = 3x^2y^2 \sec^2(x^3y^2)$; $z_y = 2x^3y \sec^2(x^3y^2)$
17. $z_x = \frac{2}{\sqrt{x}(3y^2 + 1)}$; $z_y = -\frac{24y\sqrt{x}}{(3y^2 + 1)^2}$
18. $z_x = 12x^2 - 10x + 8$; $z_y = 0$
19. $z_x = -(x^3 - y^2)^{-2}(3x^2) = -3x^2(x^3 - y^2)^{-2}$; $z_y = -(x^3 - y^2)^{-2}(-2y) = 2y(x^3 - y^2)^{-2}$
20. $z_x = 6(-x^4 + 7y^2 + 3y)^5(-4x^3) = -24x^3(-x^4 + 7y^2 + 3y)^5$; $z_y = 6(-x^4 + 7y^2 + 3y)^5(14y + 3)$
21. $z_x = 2(\cos 5x)(-\sin 5x)(5) = -10 \sin 5x \cos 5x$; $z_y = 2(\sin 5y)(\cos 5y)(5) = 10 \sin 5y \cos 5y$
22. $z_x = (2x \tan^{-1} y^2)e^{x^2 \tan^{-1} y^2}$; $z_y = \frac{2x^2y}{1 + y^4} e^{x^2 \tan^{-1} y^2}$
23. $f_x = x(3x^2ye^{x^3y} + e^{x^3y}) = (3x^3y + 1)e^{x^3y}$; $f_y = x^4e^{x^3y}$
24. $f_\theta = \phi^2 \left(\cos \frac{\theta}{\phi} \right) \left(\frac{1}{\phi} \right) = \phi \cos \frac{\theta}{\phi}$; $f_\phi = \phi^2 \left(\cos \frac{\theta}{\phi} \right) \left(-\frac{\theta}{\phi^2} \right) + 2\phi \sin \frac{\theta}{\phi} = -\theta \cos \frac{\theta}{\phi} + 2\phi \sin \frac{\theta}{\phi}$
25. $f_x = \frac{(x + 2y)3 - (3x - y)}{(x + 2y)^2} = \frac{7y}{(x + 2y)^2}$; $f_y = \frac{(x + 2y)(-1) - (3x - y)(2)}{(x + 2y)^2} = \frac{-7x}{(x + 2y)^2}$
26. $f_x = \frac{(x^2 - y^2)^2y - xy[2(x^2 - y^2)2x]}{(x^2 - y^2)^4} = \frac{-3x^2y - y^3}{(x^2 - y^2)^3}$;
 $f_y = \frac{(x^2 - y^2)^2x - xy[2(x^2 - y^2)(-2y)]}{(x^2 - y^2)^4} = \frac{3xy^2 + x^3}{(x^2 - y^2)^3}$
27. $g_u = \frac{8u}{4u^2 + 5v^3}$; $g_v = \frac{15v^2}{4u^2 + 5v^3}$
28. $h_r = \frac{1}{2s\sqrt{r}} + \frac{\sqrt{s}}{r^2}$; $h_s = -\frac{\sqrt{r}}{s^2} - \frac{1}{2r\sqrt{s}}$
29. $w_x = \frac{y}{\sqrt{x}}$; $w_y = 2\sqrt{x} - y \left(\frac{1}{z} e^{y/z} \right) - e^{y/z} = 2\sqrt{x} - \left(\frac{y}{z} + 1 \right) e^{y/z}$; $w_z = -ye^{y/z} \left(-\frac{y}{z^2} \right) = \frac{y^2}{z^2} e^{y/z}$
30. $w_x = xy \left(\frac{1}{x} \right) + (\ln xz)y = y + y \ln xz$; $w_y = x \ln xz$; $w_z = \frac{xy}{z}$
31. $F_u = 2uw^2 - v^3 - vwt^2 \sin(ut^2)$; $F_v = -3uv^2 + w \cos(ut^2)$;
 $F_x = 4(2x^2t)^3(4xt) = 16xt(2x^2t)^3 = 128x^7t^4$; $F_t = -2uvw t \sin(ut^2) + 64x^8t^3$
32. $G_p = r^4s^5(p^2q^3)^{r^4s^5-1}(2pq^3) = 2pq^3r^4s^5(p^2q^3)^{r^4s^5-1}$;
 $G_q = r^4s^5(p^2q^3)^{r^4s^5-1}(3p^2q^2) = 3p^2q^2r^4s^5(p^2q^3)^{r^4s^5-1}$; $G_r = (p^2q^3)^{r^4s^5}(4r^3s^5) \ln(p^2q^3)$;
 $G_s = (p^2q^3)^{r^4s^5}(5r^4s^4) \ln(p^2q^3)$
33. $\frac{\partial z}{\partial x} = \frac{2x}{x^2 + y^2}$, $\frac{\partial^2 z}{\partial x^2} = \frac{(x^2 + y^2)2 - 2x(2x)}{(x^2 + y^2)^2} = \frac{2y^2 - 2x^2}{(x^2 + y^2)^2}$; $\frac{\partial z}{\partial y} = \frac{2y}{x^2 + y^2}$,
 $\frac{\partial^2 z}{\partial y^2} = \frac{(x^2 + y^2)2 - 2y(2y)}{(x^2 + y^2)^2} = \frac{2x^2 - 2y^2}{(x^2 + y^2)^2}$; $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{2y^2 - 2x^2 + 2x^2 - 2y^2}{(x^2 + y^2)^2} = 0$

9.4 Partial Derivatives

$$\begin{aligned}
 34. \quad \frac{\partial z}{\partial x} &= e^{x^2-y^2}(-2y \sin 2xy) + 2xe^{x^2-y^2} \cos 2xy \\
 \frac{\partial^2 z}{\partial x^2} &= e^{x^2-y^2}(-4y^2 \cos 2xy - 8xy \sin 2xy + 4x^2 \cos 2xy + 2 \cos 2xy) \\
 \frac{\partial z}{\partial y} &= e^{x^2-y^2}(-2x \sin 2xy) - 2ye^{x^2-y^2} \cos 2xy \\
 \frac{\partial^2 z}{\partial y^2} &= e^{x^2-y^2}(-4x^2 \cos 2xy + 8xy \sin 2xy + 4y^2 \cos 2xy - 2 \cos 2xy)
 \end{aligned}$$

Adding the second partial derivatives gives

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = [-4(y^2 + x^2) \cos 2xy + 4(x^2 + y^2) \cos 2xy] = 0.$$

$$\begin{aligned}
 35. \quad \frac{\partial u}{\partial x} &= \cos at \cos x, \quad \frac{\partial^2 u}{\partial x^2} = -\cos at \sin x; \quad \frac{\partial u}{\partial t} = -a \sin at \sin x, \quad \frac{\partial^2 u}{\partial t^2} = -a^2 \cos at \sin x; \\
 a^2 \frac{\partial^2 u}{\partial x^2} &= a^2(-\cos at \sin x) = \frac{\partial^2 u}{\partial t^2}
 \end{aligned}$$

$$\begin{aligned}
 36. \quad \frac{\partial u}{\partial x} &= -\sin(x+at) + \cos(x-at), \quad \frac{\partial^2 u}{\partial x^2} = -\cos(x+at) - \sin(x-at); \\
 \frac{\partial u}{\partial t} &= -a \sin(x+at) - a \cos(x-at), \quad \frac{\partial^2 u}{\partial t^2} = -a^2 \cos(x+at) - a^2 \sin(x-at); \\
 a^2 \frac{\partial^2 u}{\partial x^2} &= -a^2 \cos(x+at) - a^2 \sin(x-at) = \frac{\partial^2 u}{\partial t^2}
 \end{aligned}$$

$$\begin{aligned}
 37. \quad \frac{\partial C}{\partial x} &= -\frac{2x}{kt} t^{-1/2} e^{-x^2/kt}, \quad \frac{\partial^2 C}{\partial x^2} = \frac{4x^2}{k^2 t^2} t^{-1/2} e^{-x^2/kt} - \frac{2}{kt} t^{-1/2} e^{-x^2/kt}; \\
 \frac{\partial C}{\partial t} &= t^{-1/2} \frac{x^2}{kt^2} e^{-x^2/kt} - \frac{t^{-3/2}}{2} e^{-x^2/kt}, \quad \frac{k}{4} \frac{\partial^2 C}{\partial x^2} = \frac{x^2}{kt^2} t^{-1/2} e^{-x^2/kt} - \frac{t^{-1/2}}{2t} e^{-x^2/kt} = \frac{\partial C}{\partial t}
 \end{aligned}$$

$$38. \quad (\text{a}) \quad P_v = -k(T/V^2)$$

$$(\text{b}) \quad PV = kt, \quad PV_T = k, \quad V_T = k/P$$

$$(\text{c}) \quad PV = kT, \quad V = kT_p, \quad T_p = V/k$$

$$39. \quad z_x = v^2 e^{uv^2} (3x^2) + 2uve^{uv^2} (1) = 3x^2 v^2 e^{uv^2} + 2uve^{uv^2}; \quad z_y = v^2 e^{uv^2} (0) + 2uve^{uv^2} (-2y) = -4yuve^{uv^2}$$

$$\begin{aligned}
 40. \quad z_x &= (2u \cos 4v)(2xy^3) - (4u^2 \sin 4v)(3x^2) = 4xy^3 u \cos 4v - 12x^2 u^2 \sin 4v \\
 z_y &= (2u \cos 4v)(3x^2 y^2) - (4v^2 \sin 4v)(3y^2) = 6x^2 y^2 u \cos 4v - 12y^2 u^2 \sin 4v
 \end{aligned}$$

$$\begin{aligned}
 41. \quad z_u &= 4(4u^3) - 10y[2(2u-v)(2)] = 16u^3 - 40(2u-v)y \\
 z_v &= 4(-24v^2) - 10y[2(2u-v)(-1)] = -96v^2 + 20(2u-v)y
 \end{aligned}$$

$$\begin{aligned}
 42. \quad z_u &= \frac{2y}{(x+y)^2} \left(\frac{1}{v} \right) + \frac{-2x}{(x+y)^2} \left(-\frac{v^2}{u^2} \right) = \frac{2y}{v(x+y)^2} + \frac{2xv^2}{u^2(x+y)^2} \\
 z_v &= \frac{2y}{(x+y)^2} \left(-\frac{u}{v^2} \right) + \frac{-2x}{(x+y)^2} \left(\frac{2v}{u} \right) = -\frac{2yu}{v^2(x+y)^2} - \frac{4xv}{u(x+y)^2}
 \end{aligned}$$

$$\begin{aligned}
 43. \quad w_t &= \frac{3}{2}(u^2 + v^2)^{1/2}(2u)(-e^{-t} \sin \theta) + \frac{3}{2}(u^2 + v^2)^{1/2}(2v)(-e^{-t} \cos \theta) \\
 &= -3u(u^2 + v^2)^{1/2} e^{-t} \sin \theta - 3v(u^2 + v^2)^{1/2} e^{-t} \cos \theta \\
 w_\theta &= \frac{3}{2}(u^2 + v^2)^{1/2}(2u)e^{-t} \cos \theta + \frac{3}{2}(u^2 + v^2)^{1/2}(2v)(-e^{-t} \sin \theta) \\
 &= 3u(u^2 + v^2)^{1/2} e^{-t} \cos \theta - 3v(u^2 + v^2)^{1/2} e^{-t} \sin \theta
 \end{aligned}$$

44. $w_r = \frac{v/2\sqrt{uv}}{1+uv}(2r) + \frac{u/2\sqrt{uv}}{1+uv}(2rs^2) = \frac{rv}{\sqrt{uv}(1+uv)} + \frac{rs^2u}{\sqrt{uv}(1+uv)}$
 $w_s = \frac{v/2\sqrt{uv}}{1+uv}(-2s) + \frac{u/2\sqrt{uv}}{1+uv}(2r^2s) = \frac{-sv}{\sqrt{uv}(1+uv)} + \frac{r^2su}{\sqrt{uv}(1+uv)}$
45. $R_u = s^2t^4(e^{v^2}) + 2rst^4(-2uve^{-u^2}) + 4rs^2t^3(2uv^2e^{u^2v^2}) = s^2t^4e^{v^2} - 4uvrst^4e^{-u^2} + 8uv^2rs^2t^3e^{u^2v^2}$
 $R_v = s^2t^4(2uve^{v^2}) + 2rst^4(e^{-u^2}) + 4rs^2t^3(2u^2ve^{u^2v^2}) = 2s^2t^4uve^{v^2} + 2rst^4e^{-u^2} + 8rs^2t^3u^2ve^{u^2v^2}$
46. $Q_x = \frac{1}{P} \left(\frac{t^2}{\sqrt{1-x^2}} \right) + \frac{1}{q} \left(\frac{1}{t^2} \right) + \frac{1}{r} \left(\frac{1/t}{1+(x/t)^2} \right) = \frac{t^2}{p\sqrt{1-x^2}} + \frac{1}{qt^2} + \frac{t}{r(t^2+x^2)}$
 $Q_t = \frac{1}{p}(2t \sin^{-1} x) + \frac{1}{q} \left(-\frac{2x}{t^3} \right) + \frac{1}{r} \left(\frac{-x/t^2}{1+(x/t)^2} \right) = \frac{2t \sin^{-1} x}{p} - \frac{2x}{qt^3} - \frac{x}{r(t^2+x^2)}$
47. $w_t = \frac{2x}{2\sqrt{x^2+y^2}} \frac{u}{rs+tu} + \frac{2y}{2\sqrt{x^2+y^2}} \frac{\cosh rs}{u} = \frac{xu}{\sqrt{x^2+y^2}(rs+tu)} + \frac{y \cosh rs}{u\sqrt{x^2+y^2}}$
 $w_r = \frac{2x}{2\sqrt{x^2+y^2}} \frac{s}{rs+tu} + \frac{2y}{2\sqrt{x^2+y^2}} \frac{st \sinh rs}{u} = \frac{xs}{\sqrt{x^2+y^2}(rs+tu)} + \frac{yst \sinh rs}{u\sqrt{x^2+y^2}}$
 $w_u = \frac{2x}{2\sqrt{x^2+y^2}} \frac{t}{rs+tu} + \frac{2y}{2\sqrt{x^2+y^2}} \frac{-t \cosh rs}{u^2} = \frac{xt}{\sqrt{x^2+y^2}(rs+tu)} - \frac{yt \cosh rs}{u^2\sqrt{x^2+y^2}}$
48. $s_\phi = 2pe^{3\theta} + 2q[-\sin(\phi+\theta)] - 2r\theta^2 + 4(2) = 2pe^{3\theta} - 2q\sin(\phi+\theta) - 2r\theta^2 + 8$
 $s_\theta = 2p(3\phi e^{3\theta}) + 2q[-\sin(\phi+\theta)] - 2r(2\phi\theta) + 4(8) = 6p\phi e^{3\theta} - 2q\sin(\phi+\theta) - 4r\phi\theta + 32$
49. $\frac{dz}{dt} = \frac{2u}{u^2+v^2}(2t) + \frac{2v}{u^2+v^2}(-2t^{-3}) = \frac{4ut - 4vt^{-3}}{u^2+v^2}$
50. $\frac{dz}{dt} = (3u^2v - v^4)(-5e^{-5t}) + (u^3 - 4uv^3)(5 \sec 5t \tan 5t) = -5(3u^2v - v^4)e^{-5t} + 5(u^3 - 4uv^3) \sec 5t \tan 5t$
51. $\frac{dw}{dt} = -3\sin(3u+4v)(2) - 4\sin(3u+4v)(-1); \quad u(\pi) = 5\pi/2, \quad v(\pi) = -5\pi/4$
 $\left. \frac{dw}{dt} \right|_\pi = -6\sin\left(\frac{15\pi}{2} - 5\pi\right) + 4\sin\left(\frac{15\pi}{2} - 5\pi\right) = -2\sin\frac{5\pi}{2} = -2$
52. $\frac{dw}{dt} = ye^{xy} \left[\frac{-8}{(2t+1)^2} \right] + xe^{xy}(3); \quad x(0) = 4, \quad y(0) = 5; \quad \left. \frac{dw}{dt} \right|_0 = 5e^{20}(-8) + 4e^{20}(3) = -28e^{20}$
53. With $x = r \cos \theta$ and $y = r \sin \theta$

$$\begin{aligned} \frac{\partial u}{\partial r} &= \frac{\partial u}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial r} = \frac{\partial u}{\partial x} \cos \theta + \frac{\partial u}{\partial y} \sin \theta \\ \frac{\partial^2 u}{\partial r^2} &= \frac{\partial^2 u}{\partial x^2} \frac{\partial x}{\partial r} \cos \theta + \frac{\partial^2 u}{\partial y^2} \frac{\partial y}{\partial r} \sin \theta = \frac{\partial^2 u}{\partial x^2} \cos^2 \theta + \frac{\partial^2 u}{\partial y^2} \sin^2 \theta \\ \frac{\partial u}{\partial \theta} &= \frac{\partial u}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial \theta} = \frac{\partial u}{\partial x}(-r \sin \theta) + \frac{\partial u}{\partial y}(r \cos \theta) \\ \frac{\partial^2 u}{\partial \theta^2} &= \frac{\partial u}{\partial x}(-r \cos \theta) + \frac{\partial^2 u}{\partial x^2} \frac{\partial x}{\partial \theta}(-r \sin \theta) + \frac{\partial u}{\partial y}(-r \sin \theta) + \frac{\partial^2 u}{\partial y^2} \frac{\partial y}{\partial \theta}(r \cos \theta) \\ &= -r \frac{\partial u}{\partial x} \cos \theta + r^2 \frac{\partial^2 u}{\partial x^2} \sin^2 \theta - r \frac{\partial u}{\partial y} \sin \theta + r^2 \frac{\partial^2 u}{\partial y^2} \cos^2 \theta. \end{aligned}$$

9.4 Partial Derivatives

Using $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, we have

$$\begin{aligned}\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} &= \frac{\partial^2 u}{\partial x^2} \cos^2 \theta + \frac{\partial^2 u}{\partial y^2} \sin^2 \theta + \frac{1}{r} \left(\frac{\partial u}{\partial x} \cos \theta + \frac{\partial u}{\partial y} \sin \theta \right) \\ &\quad + \frac{1}{r^2} \left(-r \frac{\partial u}{\partial x} \cos \theta + r^2 \frac{\partial^2 u}{\partial x^2} \sin^2 \theta - r \frac{\partial u}{\partial y} \sin \theta + r^2 \frac{\partial^2 u}{\partial y^2} \cos^2 \theta \right) \\ &= \frac{\partial^2 u}{\partial x^2} (\cos^2 \theta + \sin^2 \theta) + \frac{\partial^2 u}{\partial y^2} (\sin^2 \theta + \cos^2 \theta) + \frac{\partial u}{\partial x} \left(\frac{1}{r} \cos \theta - \frac{1}{r} \cos \theta \right) \\ &\quad + \frac{\partial u}{\partial y} \left(\frac{1}{r} \sin \theta - \frac{1}{r} \sin \theta \right) \\ &= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.\end{aligned}$$

$$\begin{aligned}54. \quad \frac{dP}{dt} &= \frac{(V - 0.0427)(0.08)dT/dt}{(V - 0.0427)^2} - \frac{0.08T(dV/dt)}{(V - 0.0427)^2} + \frac{3.6}{V^3} \frac{dV}{dt} \\ &= \frac{0.08}{V - 0.0427} \frac{dT}{dt} + \left(\frac{3.6}{V^3} - \frac{0.08T}{(V - 0.0427)^2} \right) \frac{dV}{dt}\end{aligned}$$

55. Since $dT/dT = 1$ and $\partial P/\partial T = 0$,

$$0 = F_T = \frac{\partial F}{\partial P} \frac{\partial P}{\partial T} + \frac{\partial F}{\partial V} \frac{\partial V}{\partial T} + \frac{\partial F}{\partial T} \frac{dT}{dT} \implies \frac{\partial V}{\partial T} = -\frac{\partial F/\partial T}{\partial F/\partial V} = -\frac{1}{\partial T/\partial V}.$$

56. We are given $dE/dt = 2$ and $dR/dt = -1$. Then $\frac{dI}{dt} = \frac{\partial I}{\partial E} \frac{dE}{dt} + \frac{\partial I}{\partial R} \frac{dR}{dt} = \frac{1}{R}(2) - \frac{E}{R^2}(-1)$, and

$$\text{when } E = 60 \text{ and } R = 50, \quad \frac{dI}{dt} = \frac{2}{50} + \frac{60}{50^2} = \frac{1}{25} + \frac{3/5}{25} = \frac{8}{125} \text{ amp/min.}$$

57. Since the height of the triangle is $x \sin \theta$, the area is given by $A = \frac{1}{2}xy \sin \theta$. Then

$$\frac{dA}{dt} \frac{\partial A}{\partial x} \frac{dx}{dt} + \frac{\partial A}{\partial y} \frac{dy}{dt} + \frac{\partial A}{\partial \theta} \frac{d\theta}{dt} = \frac{1}{2}y \sin \theta \frac{dx}{dt} + \frac{1}{2}x \sin \theta \frac{dy}{dt} + \frac{1}{2}xy \cos \theta \frac{d\theta}{dt}.$$

When $x = 10$, $y = 8$, $\theta = \pi/6$, $dx/dt = 0.3$, $dy/dt = 0.5$, and $d\theta/dt = 0.1$,

$$\begin{aligned}\frac{dA}{dt} &= \frac{1}{2}(8) \left(\frac{1}{2} \right) (0.3) + \frac{1}{2}(10) \left(\frac{1}{2} \right) (0.5) + \frac{1}{2}(10)(8) \left(\frac{\sqrt{3}}{2} \right) (0.1) \\ &= 0.6 + 1.25 + 2\sqrt{3} = 1.85 + 2\sqrt{3} \approx 5.31 \text{ cm}^2/\text{s}.\end{aligned}$$

$$\begin{aligned}58. \quad \frac{dw}{dt} &= \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt} = \frac{x dx/dt + y dy/dt + z dz/dt}{\sqrt{x^2 + y^2 + z^2}} = \frac{-4x \sin t + 4y \cos t + 5z}{\sqrt{16 \cos^2 t + 16 \sin^2 t + 25t^2}} \\ &= \frac{-16 \sin t \cos t + 16 \sin t \cos t + 25t}{\sqrt{16 + 25t^2}} = \frac{25t}{\sqrt{16 + 25t^2}}\end{aligned}$$

$$\left. \frac{dw}{dt} \right|_{t=5\pi/2} = \frac{125\pi/2}{\sqrt{16 + 625\pi^2/4}} = \frac{125\pi}{\sqrt{64 + 625\pi^2}} \approx 4.9743$$

EXERCISES 9.5

Directional Derivative

1. $\nabla f = (2x - 3x^2y^2)\mathbf{i} + (4y^3 - 2x^3y)\mathbf{j}$
2. $\nabla f = 4xye^{-2x^2y}\mathbf{i} + (1 + 2x^2e^{-2x^2y})\mathbf{j}$
3. $\nabla F = \frac{y^2}{z^3}\mathbf{i} + \frac{2xy}{z^3}\mathbf{j} - \frac{3xy^2}{z^4}\mathbf{k}$
4. $\nabla F = y \cos yz \mathbf{i} + (x \cos yz - xyz \sin yz)\mathbf{j} - xy^2 \sin yz \mathbf{k}$
5. $\nabla f = 2x\mathbf{i} - 8y\mathbf{j}$; $\nabla f(2, 4) = 4\mathbf{i} - 32\mathbf{j}$
6. $\nabla f = \frac{3x^2}{2\sqrt{x^3y - y^4}}\mathbf{i} + \frac{x^3 - 4y^3}{2\sqrt{x^3y - y^4}}\mathbf{j}$; $\nabla f(3, 2) = \frac{27}{\sqrt{38}}\mathbf{i} - \frac{5}{2\sqrt{38}}\mathbf{j}$
7. $\nabla F = 2xz^2 \sin 4y\mathbf{i} + 4x^2z^2 \cos 4y\mathbf{j} + 2x^2z \sin 4y\mathbf{k}$
 $\nabla F(-2, \pi/3, 1) = -4 \sin \frac{4\pi}{3}\mathbf{i} + 16 \cos \frac{4\pi}{3}\mathbf{j} + 8 \sin \frac{4\pi}{3}\mathbf{k} = 2\sqrt{3}\mathbf{i} - 8\mathbf{j} - 4\sqrt{3}\mathbf{k}$
8. $\nabla F = \frac{2x}{x^2 + y^2 + z^2}\mathbf{i} + \frac{2y}{x^2 + y^2 + z^2}\mathbf{j} + \frac{2z}{x^2 + y^2 + z^2}\mathbf{k}$; $\nabla F(-4, 3, 5) = -\frac{4}{25}\mathbf{i} + \frac{3}{25}\mathbf{j} + \frac{1}{5}\mathbf{k}$
9. $D_{\mathbf{u}}f(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h\sqrt{3}/2, y + h/2) - f(x, y)}{h} = \lim_{h \rightarrow 0} \frac{(x + h\sqrt{3}/2)^2 + (y + h/2)^2 - x^2 - y^2}{h}$
 $= \lim_{h \rightarrow 0} \frac{h\sqrt{3}x + 3h^2/4 + hy + h^2/4}{h} = \lim_{h \rightarrow 0} (\sqrt{3}x + 3h/4 + y + h/4) = \sqrt{3}x + y$
10. $D_{\mathbf{u}}f(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h\sqrt{2}/2, y + h\sqrt{2}/2) - f(x, y)}{h} = \lim_{h \rightarrow 0} \frac{3x + 3h\sqrt{2}/2 - (y + h\sqrt{2}/2)^2 - 3x + y^2}{h}$
 $= \lim_{h \rightarrow 0} \frac{3h\sqrt{2}/2 - h\sqrt{2}y - h^2/2}{h} = \lim_{h \rightarrow 0} (3\sqrt{2}/2 - \sqrt{2}y - h/2) = 3\sqrt{2}/2 - \sqrt{2}y$
11. $\mathbf{u} = \frac{\sqrt{3}}{2}\mathbf{i} + \frac{1}{2}\mathbf{j}$; $\nabla f = 15x^2y^6\mathbf{i} + 30x^3y^5\mathbf{j}$; $\nabla f(-1, 1) = 15\mathbf{i} - 30\mathbf{j}$
 $D_{\mathbf{u}}f(-1, 1) = \frac{15\sqrt{3}}{2} - 15 = \frac{15}{2}(\sqrt{3} - 2)$
12. $\mathbf{u} = \frac{\sqrt{2}}{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j}$; $\nabla f = (4 + y^2)\mathbf{i} + (2xy - 5)\mathbf{j}$; $\nabla f(3, -1) = 5\mathbf{i} - 11\mathbf{j}$
 $D_{\mathbf{u}}f(3, -1) = \frac{5\sqrt{2}}{2} - \frac{11\sqrt{2}}{2} = -3\sqrt{2}$
13. $\mathbf{u} = \frac{\sqrt{10}}{10}\mathbf{i} - \frac{3\sqrt{10}}{10}\mathbf{j}$; $\nabla f = \frac{-y}{x^2 + y^2}\mathbf{i} + \frac{x}{x^2 + y^2}\mathbf{j}$; $\nabla f(2, -2) = \frac{1}{4}\mathbf{i} + \frac{1}{4}\mathbf{j}$
 $D_{\mathbf{u}}f(2, -2) = \frac{\sqrt{10}}{40} - \frac{3\sqrt{10}}{40} = -\frac{\sqrt{10}}{20}$
14. $\mathbf{u} = \frac{6}{10}\mathbf{i} + \frac{8}{10}\mathbf{j} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$; $\nabla f = \frac{y^2}{(x+y)^2}\mathbf{i} + \frac{x^2}{(x+y)^2}\mathbf{j}$; $\nabla f(2, -1) = \mathbf{i} + 4\mathbf{j}$
 $D_{\mathbf{u}}f(2, -1) = \frac{3}{5} + \frac{16}{5} = \frac{19}{5}$

9.5 Directional Derivative

15. $\mathbf{u} = (2\mathbf{i} + \mathbf{j})/\sqrt{5}$; $\nabla f = 2y(xy + 1)\mathbf{i} + 2x(xy + 1)\mathbf{j}$; $\nabla f(3, 2) = 28\mathbf{i} + 42\mathbf{j}$

$$D_{\mathbf{u}}f(3, 2) = \frac{2(28)}{\sqrt{5}} + \frac{42}{\sqrt{5}} = \frac{98}{\sqrt{5}}$$

16. $\mathbf{u} = -\mathbf{i}$; $\nabla f = 2x \tan y \mathbf{i} + x^2 \sec^2 y \mathbf{j}$; $\nabla f(1/2, \pi/3) = \sqrt{3}\mathbf{i} + \mathbf{j}$; $D_{\mathbf{u}}f(1/2, \pi/3) = -\sqrt{3}$

17. $\mathbf{u} = \frac{1}{\sqrt{2}}\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}$; $\nabla F = 2xy^2(2z + 1)^2\mathbf{i} + 2x^2y(2z + 1)^2\mathbf{j} + 4x^2y^2(2z + 1)\mathbf{k}$

$$\nabla F(1, -1, 1) = 18\mathbf{i} - 18\mathbf{j} + 12\mathbf{k}; \quad D_{\mathbf{u}}F(1, -1, 1) = -\frac{18}{\sqrt{2}} + \frac{12}{\sqrt{2}} = -\frac{6}{\sqrt{2}} = -3\sqrt{2}$$

18. $\mathbf{u} = \frac{1}{\sqrt{6}}\mathbf{i} - \frac{2}{\sqrt{6}}\mathbf{j} + \frac{1}{\sqrt{6}}\mathbf{k}$; $\nabla F = \frac{2x}{z^2}\mathbf{i} - \frac{2y}{z^2}\mathbf{j} + \frac{2y^2 - 2x^2}{z^3}\mathbf{k}$; $\nabla F(2, 4, -1) = 4\mathbf{i} - 8\mathbf{j} - 24\mathbf{k}$

$$D_{\mathbf{u}}F(2, 4, -1) = \frac{4}{\sqrt{6}} - \frac{16}{\sqrt{6}} - \frac{24}{\sqrt{6}} = -6\sqrt{6}$$

19. $\mathbf{u} = -\mathbf{k}$; $\nabla F = \frac{xy}{\sqrt{x^2y + 2y^2z}}\mathbf{i} + \frac{x^2 + 4z}{2\sqrt{x^2y + 2y^2z}}\mathbf{j} + \frac{y^2}{\sqrt{x^2y + 2y^2z}}\mathbf{k}$

$$\nabla F(-2, 2, 1) = -\mathbf{i} + \mathbf{j} + \mathbf{k}; \quad D_{\mathbf{u}}F(-2, 2, 1) = -1$$

20. $\mathbf{u} = -(4\mathbf{i} - 4\mathbf{j} + 2\mathbf{k})/\sqrt{36} = -\frac{2}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} - \frac{1}{3}\mathbf{k}$; $\nabla F = 2\mathbf{i} - 2y\mathbf{j} + 2z\mathbf{k}$; $\nabla F(4, -4, 2) = 2\mathbf{i} + 8\mathbf{j} + 4\mathbf{k}$

$$D_{\mathbf{u}}F(4, -4, 2) = -\frac{4}{3} + \frac{16}{3} - \frac{4}{3} = \frac{8}{3}$$

21. $\mathbf{u} = (-4\mathbf{i} - \mathbf{j})/\sqrt{17}$; $\nabla f = 2(x - y)\mathbf{i} - 2(x - y)\mathbf{j}$; $\nabla f(4, 2) = 4\mathbf{i} - 4\mathbf{j}$; $D_{\mathbf{u}}f(4, 2) = -\frac{16}{\sqrt{17}} \frac{4}{\sqrt{17}} = -\frac{12}{\sqrt{17}}$

22. $\mathbf{u} = (-2\mathbf{i} + 5\mathbf{j})/\sqrt{29}$; $\nabla f = (3x^2 - 5y)\mathbf{i} - (5x - 2y)\mathbf{j}$; $\nabla f(1, 1) = -2\mathbf{i} - 3\mathbf{j}$;

$$D_{\mathbf{u}}f(1, 1) = \frac{4}{\sqrt{29}} - \frac{15}{\sqrt{29}} = -\frac{11}{\sqrt{29}}$$

23. $\nabla f = 2e^{2x} \sin y \mathbf{i} + e^{2x} \cos y \mathbf{j}$; $\nabla f(0, \pi/4) = \sqrt{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j}$

The maximum $D_{\mathbf{u}}$ is $[(\sqrt{2})^2 + (\sqrt{2}/2)^2]^{1/2} = \sqrt{5/2}$ in the direction $\sqrt{2}\mathbf{i} + (\sqrt{2}/2)\mathbf{j}$.

24. $\nabla f = (xye^{x-y} + ye^{x-y})\mathbf{i} + (-xye^{x-y} + xe^{x-y})\mathbf{j}$; $\nabla f(5, 5) = 30\mathbf{i} - 20\mathbf{j}$

The maximum $D_{\mathbf{u}}$ is $[30^2 + (-20)^2]^{1/2} = 10\sqrt{13}$ in the direction $30\mathbf{i} - 20\mathbf{j}$.

25. $\nabla F = (2x + 4z)\mathbf{i} + 2z^2\mathbf{j} + (4x + 4yz)\mathbf{k}$; $\nabla F(1, 2, -1) = -2\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$

The maximum $D_{\mathbf{u}}$ is $[(-2)^2 + 2^2 + (-4)^2]^{1/2} = 2\sqrt{6}$ in the direction $-2\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$.

26. $\nabla F = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$; $\nabla F(3, 1, -5) = -5\mathbf{i} - 15\mathbf{j} + 3\mathbf{k}$

The maximum $D_{\mathbf{u}}$ is $[(-5)^2 + (-15)^2 + 3^2]^{1/2} = \sqrt{259}$ in the direction $-5\mathbf{i} - 15\mathbf{j} + 3\mathbf{k}$.

27. $\nabla f = 2x \sec^2(x^2 + y^2)\mathbf{i} + 2y \sec^2(x^2 + y^2)\mathbf{j}$;

$$\nabla f(\sqrt{\pi/6}, \sqrt{\pi/6}) = 2\sqrt{\pi/6} \sec^2(\pi/3)(\mathbf{i} + \mathbf{j}) = 8\sqrt{\pi/6}(\mathbf{i} + \mathbf{j})$$

The minimum $D_{\mathbf{u}}$ is $-8\sqrt{\pi/6}(1^2 + 1^2)^{1/2} = -8\sqrt{\pi/3}$ in the direction $-(\mathbf{i} + \mathbf{j})$.

28. $\nabla f = 3x^2\mathbf{i} - 3y^2\mathbf{j}$; $\nabla f(2, -2) = 12\mathbf{i} - 12\mathbf{j} = 12(\mathbf{i} - \mathbf{j})$

The minimum $D_{\mathbf{u}}$ is $-12[1^2 + (-1)^2]^{1/2} = -12\sqrt{2}$ in the direction $-(\mathbf{i} - \mathbf{j}) = -\mathbf{i} + \mathbf{j}$.

29. $\nabla F = \frac{\sqrt{z} e^y}{2\sqrt{x}}\mathbf{i} + \sqrt{xz} e^y \mathbf{j} + \frac{\sqrt{x}}{2\sqrt{z}}\mathbf{k}$; $\nabla F(16, 0, 9) = \frac{3}{8}\mathbf{i} + 12\mathbf{j} + \frac{2}{3}\mathbf{k}$. The minimum $D_{\mathbf{u}}$ is

$$-[(3/8)^2 + 12^2 + (2/3)^2]^{1/2} = -\sqrt{83,281}/24 \text{ in the direction } -\frac{3}{8}\mathbf{i} - 12\mathbf{j} - \frac{2}{3}\mathbf{k}.$$

30. $\nabla F = \frac{1}{x}\mathbf{i} + \frac{1}{y}\mathbf{j} - \frac{1}{z}\mathbf{k}$; $\nabla F(1/2, 1/6, 1/3) = 2\mathbf{i} + 6\mathbf{j} - 3\mathbf{k}$

The minimum $D_{\mathbf{u}}$ is $-[2^2 + 6^2 + (-3)^2]^{1/2} = -7$ in the direction $-2\mathbf{i} - 6\mathbf{j} + 3\mathbf{k}$.

31. Using implicit differentiation on $2x^2 + y^2 = 9$ we find $y' = -2x/y$. At $(2, 1)$ the slope of the tangent line is $-2(2)/1 = -4$. Thus, $\mathbf{u} = \pm(\mathbf{i} - 4\mathbf{j})/\sqrt{17}$. Now, $\nabla f = \mathbf{i} + 2y\mathbf{j}$ and $\nabla f(3, 4) = \mathbf{i} + 8\mathbf{j}$. Thus, $D_{\mathbf{u}} = \pm(1/\sqrt{17} - 32\sqrt{17}) = \pm 31/\sqrt{17}$.

32. $\nabla f = (2x + y - 1)\mathbf{i} + (x + 2y)\mathbf{j}$; $D_{\mathbf{u}}f(x, y) = \frac{2x + y - 1}{\sqrt{2}} + \frac{x + 2y}{\sqrt{2}} = \frac{3x + 3y - 1}{\sqrt{2}}$

Solving $(3x + 3y - 1)/\sqrt{2} = 0$ we see that $D_{\mathbf{u}}$ is 0 for all points on the line $3x + 3y = 1$.

33. (a) Vectors perpendicular to $4\mathbf{i} + 3\mathbf{j}$ are $\pm(3\mathbf{i} - 4\mathbf{j})$. Take $\mathbf{u} = \pm\left(\frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}\right)$.

(b) $\mathbf{u} = (4\mathbf{i} + 3\mathbf{j})/\sqrt{16 + 9} = \frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j}$

(c) $\mathbf{u} = -\frac{4}{5}\mathbf{i} - \frac{3}{5}\mathbf{j}$

34. $D_{-\mathbf{u}}f(a, b) = \nabla f(a, b) \cdot (-\mathbf{u}) = -\nabla f(a, b) \cdot \mathbf{u} = -D_{\mathbf{u}}f(a, b) = -6$

35. (a) $\nabla f = (3x^2 - 6xy^2)\mathbf{i} + (-6x^2y + 3y^2)\mathbf{j}$

$$D_{\mathbf{u}}f(x, y) = \frac{3(3x^2 - 6xy^2)}{\sqrt{10}} + \frac{-6x^2y + 3y^2}{\sqrt{10}} = \frac{9x^2 - 18xy^2 - 6x^2y + 3y^2}{\sqrt{10}}$$

(b) $F(x, y) = \frac{3}{\sqrt{10}}(3x^2 - 6xy^2 - 2x^2y + y^2)$; $\nabla F = \frac{3}{\sqrt{10}}[(6x - 6y^2 - 4xy)\mathbf{i} + (-12xy - 2x^2 + 2y)\mathbf{j}]$

$$\begin{aligned} D_{\mathbf{u}}F(x, y) &= \left(\frac{3}{\sqrt{10}}\right)\left(\frac{3}{\sqrt{10}}\right)(6x - 6y^2 - 4xy) + \left(\frac{1}{\sqrt{10}}\right)\left(\frac{3}{\sqrt{10}}\right)(-12xy - 2x^2 + 2y) \\ &= \frac{9}{5}(3x - 3y^2 - 2xy) + \frac{3}{5}(-6xy - x^2 + y) = \frac{1}{5}(27x - 27y^2 - 36xy - 3x^2 + 3y) \end{aligned}$$

36. $\nabla U = \frac{Gmx}{(x^2 + y^2)^{3/2}}\mathbf{i} + \frac{Gmy}{(x^2 + y^2)^{3/2}}\mathbf{j} = \frac{Gm}{(x^2 + y^2)^{3/2}}(x\mathbf{i} + y\mathbf{j})$

The maximum and minimum values of $D_{\mathbf{u}}U(x, y)$ are obtained when $\mathbf{u}$ is in the directions ∇U and $-\nabla U$, respectively. Thus, at a point (x, y) , not $(0, 0)$, the directions of maximum and minimum increase in U are $x\mathbf{i} + y\mathbf{j}$ and $-x\mathbf{i} - y\mathbf{j}$, respectively. A vector at (x, y) in the direction $\pm(x\mathbf{i} + y\mathbf{j})$ lies on a line through the origin.

37. $\nabla f = (3x^2 - 12)\mathbf{i} + (2y - 10)\mathbf{j}$. Setting $|\nabla f| = [(3x^2 - 12)^2 + (2y - 10)^2]^{1/2} = 0$, we obtain $3x^2 - 12 = 0$ and $2y - 10 = 0$. The points where $|\nabla f| = 0$ are $(2, 5)$ and $(-2, 5)$.

38. Let $\nabla f(a, b) = \alpha\mathbf{i} + \beta\mathbf{j}$. Then

$$D_{\mathbf{u}}f(a, b) = \nabla f(a, b) \cdot \mathbf{u} = \frac{5}{13}\alpha - \frac{12}{13}\beta = 7 \quad \text{and} \quad D_{\mathbf{v}}f(a, b) = \nabla f(a, b) \cdot \mathbf{v} = \frac{5}{13}\alpha - \frac{12}{13}\beta = 3.$$

Solving for α and β , we obtain $\alpha = 13$ and $\beta = -13/6$. Thus, $\nabla f(a, b) = 13\mathbf{i} - (13/6)\mathbf{j}$.

39. $\nabla T = 4x\mathbf{i} + 2y\mathbf{j}$; $\nabla T(4, 2) = 16\mathbf{i} + 4\mathbf{j}$. The minimum change in temperature (that is, the maximum decrease in temperature) is in the direction $-\nabla T(4, 3) = -16\mathbf{i} - 4\mathbf{j}$.

40. Let $x(t)\mathbf{i} + y(t)\mathbf{j}$ be the vector equation of the path. At (x, y) on this curve, the direction of a tangent vector is $x'(t)\mathbf{i} + y'(t)\mathbf{j}$. Since we want the direction of motion to be $-\nabla T(x, y)$, we have $x'(t)\mathbf{i} + y'(t)\mathbf{j} = -\nabla T(x, y) = 4x\mathbf{i} + 2y\mathbf{j}$. Separating variables in $dx/dt = 4x$, we obtain $dx/x = 4 dt$, $\ln x = 4t + c_1$, and $x = C_1e^{4t}$. Separating variables in $dy/dt = 2y$, we obtain $dy/y = 2 dt$, $\ln y = 2t + c_2$, and $y = C_2e^{2t}$. Since $x(0) = 4$ and $y(0) = 2$, we

9.5 Directional Derivative

have $x = 4e^{4t}$ and $y = 2e^{2t}$. The equation of the path is $4e^{4t}\mathbf{i} + 2e^{2t}\mathbf{j}$ for $t \geq 0$, or eliminating the parameter, $x = y^2$, $y \geq 0$.

41. Let $x(t)\mathbf{i} + y(t)\mathbf{j}$ be the vector equation of the path. At (x, y) on this curve, the direction of a tangent vector is $x'(t)\mathbf{i} + y'(t)\mathbf{j}$. Since we want the direction of motion to be $\nabla T(x, y)$, we have $x'(t)\mathbf{i} + y'(t)\mathbf{j} = \nabla T(x, y) = -4x\mathbf{i} - 2y\mathbf{j}$. Separating variables in $dx/dt = -4x$ we obtain $dx/x = -4dt$, $\ln x = -4t + c_1$ and $x = C_1e^{-4t}$. Separating variables in $dy/dt = -2y$ we obtain $dy/y = -2dt$, $\ln y = -2t + c_2$ and $y = C_2e^{-2t}$. Since $x(0) = 3$ and $y(0) = 4$, we have $x = 3e^{-4t}$ and $y = 4e^{-2t}$. The equation of the path is $3e^{-4t}\mathbf{i} + 4e^{-2t}\mathbf{j}$, or eliminating the parameter, $16x = 3y^2$, $y \geq 0$.

42. Substituting $x = 0$, $y = 0$, $z = 1$, and $T = 500$ into $T = k/(x^2 + y^2 + z^2)$ we see that $k = 500$ and $T(x, y, z) = 500/(x^2 + y^2 + z^2)$. To find the rate of change of T at $\langle 2, 3, 3 \rangle$ in the direction of $\langle 3, 1, 1 \rangle$ we first compute $\langle 3, 1, 1 \rangle - \langle 2, 3, 3 \rangle = \langle 1, -2, -2 \rangle$. Then $\mathbf{u} = \frac{1}{3}\langle 1, -2, -2 \rangle = \frac{1}{3}\mathbf{i} - \frac{2}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$. Now

$$\nabla T = -\frac{1000x}{(x^2 + y^2 + z^2)^2}\mathbf{i} - \frac{1000y}{(x^2 + y^2 + z^2)^2}\mathbf{j} - \frac{1000z}{(x^2 + y^2 + z^2)^2}\mathbf{k} \quad \text{and} \quad \nabla T(2, 3, 3) = -\frac{500}{121}\mathbf{i} - \frac{750}{121}\mathbf{j} - \frac{750}{121}\mathbf{k},$$

so

$$D_{\mathbf{u}}T(2, 3, 3) = \frac{1}{3}\left(-\frac{500}{121}\right) - \frac{2}{3}\left(-\frac{750}{121}\right) - \frac{2}{3}\left(-\frac{750}{121}\right) = \frac{2500}{363}.$$

The direction of maximum increase is $\nabla T(2, 3, 3) = -\frac{500}{121}\mathbf{i} - \frac{750}{121}\mathbf{j} - \frac{750}{121}\mathbf{k} = \frac{250}{121}(-2\mathbf{i} - 3\mathbf{j} - 3\mathbf{k})$, and the maximum rate of change of T is $|\nabla T(2, 3, 3)| = \frac{250}{121}\sqrt{4 + 9 + 9} = \frac{250}{121}\sqrt{22}$.

43. Since $\nabla f = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}$, we have $\partial f/\partial x = 3x^2 + y^3 + ye^{xy}$. Integrating, we obtain $f(x, y) = x^3 + xy^3 + e^{xy} + g(y)$. Then $f_y = 3xy^2 + xe^{xy} + g'(y) = -2y^2 + 3xy^2 + xe^{xy}$. Thus, $g'(y) = -2y^2$, $g(y) = -\frac{2}{3}y^3 + c$, and $f(x, y) = x^3 + xy^3 + e^{xy} - \frac{2}{3}y^3 + C$.

44. Let $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j}$. $D_{\mathbf{v}}f = (f_x\mathbf{i} + f_y\mathbf{j}) \cdot \mathbf{v} = v_1f_x + v_2f_y$

$$\begin{aligned} D_{\mathbf{u}}D_{\mathbf{v}}f &= \left[\frac{\partial}{\partial x}(v_1f_x + v_2f_y)\mathbf{i} + \frac{\partial}{\partial y}(v_1f_x + v_2f_y)\mathbf{j} \right] \cdot \mathbf{u} = [(v_1f_{xx} + v_2f_{yx})\mathbf{i} + (v_1f_{xy} + v_2f_{yy})\mathbf{j}] \cdot \mathbf{u} \\ &= u_1v_1f_{xx} + u_1v_2f_{yx} + u_2v_1f_{xy} + u_2v_2f_{yy} \end{aligned}$$

$$D_{\mathbf{u}}f = (f_x\mathbf{i} + f_y\mathbf{j}) \cdot \mathbf{u} = u_1f_x + u_2f_y$$

$$\begin{aligned} D_{\mathbf{v}}D_{\mathbf{u}}f &= \left[\frac{\partial}{\partial x}(u_1f_x + u_2f_y)\mathbf{i} + \frac{\partial}{\partial y}(u_1f_x + u_2f_y)\mathbf{j} \right] \cdot \mathbf{v} = [(u_1f_{xx} + u_2f_{yx})\mathbf{i} + (u_1f_{xy} + u_2f_{yy})\mathbf{j}] \cdot \mathbf{v} \\ &= u_1v_1f_{xx} + u_2v_1f_{yx} + u_1v_2f_{xy} + u_2v_2f_{yy} \end{aligned}$$

Since the second partial derivatives are continuous, $f_{xy} = f_{yx}$ and $D_{\mathbf{u}}D_{\mathbf{v}}f = D_{\mathbf{v}}D_{\mathbf{u}}f$. [Note that this result is a generalization of $f_{xy} = f_{yx}$ since $D_{\mathbf{i}}D_{\mathbf{j}}f = f_{yx}$ and $D_{\mathbf{j}}D_{\mathbf{i}}f = f_{xy}$.]

45. $\nabla(cf) = \frac{\partial}{\partial x}(cf)\mathbf{i} + \frac{\partial}{\partial y}(cf)\mathbf{j} = cf_x\mathbf{i} + cf_y\mathbf{j} = c(f_x\mathbf{i} + f_y\mathbf{j}) = c\nabla f$

46. $\nabla(f + g) = (f_x + g_x)\mathbf{i} + (f_y + g_y)\mathbf{j} = (f_x\mathbf{i} + f_y\mathbf{j}) + (g_x\mathbf{i} + g_y\mathbf{j}) = \nabla f + \nabla g$

47. $\nabla(fg) = (fg_x + f_xg)\mathbf{i} + (fg_y + f_yg)\mathbf{j} = f(g_x\mathbf{i} + g_y\mathbf{j}) + g(f_x\mathbf{i} + f_y\mathbf{j}) = f\nabla g + g\nabla f$

48. $\begin{aligned} \nabla(f/g) &= [(gf_x - f_gx)/g^2]\mathbf{i} + [(gf_y - fg_y)/g^2]\mathbf{j} = g(f_x\mathbf{i} + f_y\mathbf{j})/g^2 - f(g_x\mathbf{i} + g_y\mathbf{j})/g^2 \\ &= g\nabla f/g^2 - f\nabla g/g^2 = (g\nabla f - f\nabla g)/g^2 \end{aligned}$

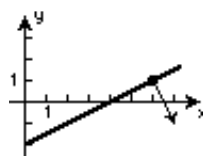
49.
$$\nabla \times F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f_1 & f_2 & f_3 \end{vmatrix} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) \mathbf{k}$$

EXERCISES 9.6

Tangent Planes and Normal Lines

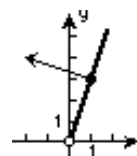
1. Since $f(6, 1) = 4$, the level curve is $x - 2y = 4$.

$$\nabla f = \mathbf{i} - 2\mathbf{j}; \quad \nabla f(6, 1) = \mathbf{i} - 2\mathbf{j}$$



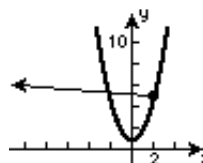
2. Since $f(1, 3) = 5$, the level curve is $y + 2x = 5$ or $y = 5 - 2x$, $x \neq 0$.

$$\nabla f = -\frac{y}{x^2}\mathbf{i} + \frac{1}{x}\mathbf{j}; \quad \nabla f(1, 3) = -3\mathbf{i} + \mathbf{j}$$



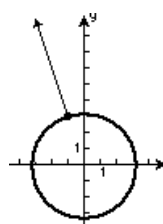
3. Since $f(2, 5) = 1$, the level curve is $y = x^2 + 1$.

$$\nabla f = -2x\mathbf{i} + \mathbf{j}; \quad \nabla f(2, 5) = -10\mathbf{i} + \mathbf{j}$$



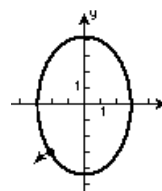
4. Since $f(-1, 3) = 10$, the level curve is $x^2 + y^2 = 10$.

$$\nabla f = 2x\mathbf{i} + 2y\mathbf{j}; \quad \nabla f(-1, 3) = -2\mathbf{i} + 6\mathbf{j}$$



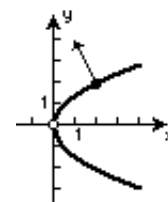
5. Since $f(-2, -3) = 2$, the level curve is $x^2/4 + y^2/9 = 2$

or $x^2/8 + y^2/18 = 1$. $\nabla f = \frac{x}{2}\mathbf{i} + \frac{2y}{9}\mathbf{j}; \quad \nabla f(-2, -3) = -\mathbf{i} - \frac{2}{3}\mathbf{j}$



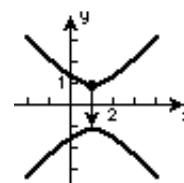
6. Since $f(2, 2) = 2$, the level curve is $y^2 = 2x$, $x \neq 0$. $\nabla f = -\frac{y^2}{x^2}\mathbf{i} + \frac{2y}{x}\mathbf{j};$

$$\nabla f(2, 2) = -\mathbf{i} + 2\mathbf{j}$$



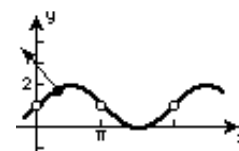
7. Since $f(1, 1) = -1$, the level curve is $(x - 1)^2 - y^2 = -1$ or $y^2 - (x - 1)^2 = 1$.

$$\nabla f = 2(x - 1)\mathbf{i} - 2y\mathbf{j}; \quad \nabla f(1, 1) = -2\mathbf{j}$$



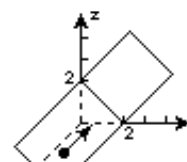
8. Since $f(\pi/6, 3/2) = 1$, the level curve is $y - 1 = \sin x$ or $y = 1 + \sin x$, $\sin x \neq 0$.

$$\nabla f = \frac{-(y - 1)\cos x}{\sin^2 x}\mathbf{i} + \frac{1}{\sin x}\mathbf{j}; \quad \nabla f(\pi/6, 3/2) = -\sqrt{3}\mathbf{i} + 2\mathbf{j}$$



9. Since $F(3, 1, 1) = 2$, the level surface is $y + z = 2$. $\nabla F = \mathbf{j} + \mathbf{k};$

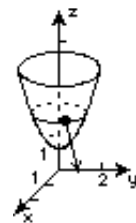
$$\nabla F(3, 1, 1) = \mathbf{j} + \mathbf{k}$$



9.6 Tangent Planes and Normal Lines

10. Since $F(1, 1, 3) = -1$, the level surface is $x^2 + y^2 - z = -1$ or $z = 1 + x^2 + y^2$.

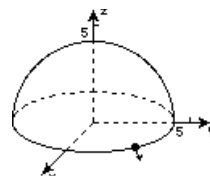
$$\nabla F = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}; \quad \nabla F(1, 1, 3) = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$$



11. Since $F(3, 4, 0) = 5$, the level surface is $x^2 + y^2 + z^2 = 25$.

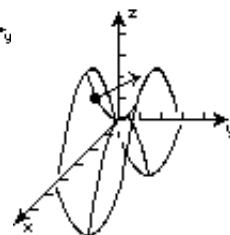
$$\nabla F = \frac{x}{\sqrt{x^2 + y^2 + z^2}}\mathbf{i} + \frac{y}{\sqrt{x^2 + y^2 + z^2}}\mathbf{j} + \frac{z}{\sqrt{x^2 + y^2 + z^2}}\mathbf{k};$$

$$\nabla F(3, 4, 0) = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$$



12. Since $F(0, -1, 1) = 0$, the level surface is $x^2 - y^2 + z = 0$ or $z = y^2 - x^2$.

$$\nabla F = 2x\mathbf{i} - 2y\mathbf{j} + \mathbf{k}; \quad \nabla F(0, -1, 1) = 2\mathbf{j} + \mathbf{k}$$



13. $F(x, y, z) = x^2 + y^2 - z$; $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}$. We want $\nabla F = c(4\mathbf{i} + \mathbf{j} + \frac{1}{2}\mathbf{k})$ or $2x = 4c$, $2y = c$, $-1 = c/2$. From the third equation $c = -2$. Thus, $x = -4$ and $y = -1$. Since $z = x^2 + y^2 = 16 + 1 = 17$, the point on the surface is $(-4, -1, -17)$.

14. $F(x, y, z) = x^3 + y^2 + z$; $\nabla F = 3x^2\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$. We want $\nabla F = c(27\mathbf{i} + 8\mathbf{j} + \mathbf{k})$ or $3x^2 = 27c$, $2y = 8c$, $1 = c$. From $c = 1$ we obtain $x = \pm 3$ and $y = 4$. Since $z = 15 - x^3 - y^2 = 15 - (\pm 3)^3 - 16 = -1 \mp 27$, the points on the surface are $(3, 4, -28)$ and $(-3, 4, 26)$.

15. $F(x, y, z) = x^2 + y^2 + z^2$; $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$. $\nabla F(-2, 2, 1) = -4\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$. The equation of the tangent plane is $-4(x + 2) + 4(y - 2) + 2(z - 1) = 0$ or $-2x + 2y + z = 9$.

16. $F(x, y, z) = 5x^2 - y^2 + 4z^2$; $\nabla F = 10x\mathbf{i} - 2y\mathbf{j} + 8z\mathbf{k}$; $\nabla F(2, 4, 1) = 20\mathbf{i} - 8\mathbf{j} + 8\mathbf{k}$. The equation of the tangent plane is $20(x - 2) - 8(y - 4) + 8(z - 1) = 0$ or $5x - 2y + 2z = 4$.

17. $F(x, y, z) = x^2 - y^2 - 3z^2$; $\nabla F = 2x\mathbf{i} - 2y\mathbf{j} - 6z\mathbf{k}$; $\nabla F(6, 2, 3) = 12\mathbf{i} - 4\mathbf{j} - 18\mathbf{k}$. The equation of the tangent plane is $12(x - 6) - 4(y - 2) - 18(z - 3) = 0$ or $6x - 2y - 9z = 5$.

18. $F(x, y, z) = xy + yz + zx$; $\nabla F = (y + z)\mathbf{i} + (x + z)\mathbf{j} + (y + x)\mathbf{k}$; $\nabla F(1, -3, -5) = -8\mathbf{i} - 4\mathbf{j} - 2\mathbf{k}$. The equation of the tangent plane is $-8(x - 1) - 4(y + 3) - 2(z + 5) = 0$ or $4x + 2y + z = -7$.

19. $F(x, y, z) = x^2 + y^2 + z$; $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$; $\nabla F(3, -4, 0) = 6\mathbf{i} - 8\mathbf{j} + \mathbf{k}$. The equation of the tangent plane is $6(x - 3) - 8(y + 4) + z = 0$ or $6x - 8y + z = 50$.

20. $F(x, y, z) = xz$; $\nabla F = z\mathbf{i} + x\mathbf{k}$; $\nabla F(2, 0, 3) = 3\mathbf{i} + 2\mathbf{k}$. The equation of the tangent plane is $3(x - 2) + 2(z - 3) = 0$ or $3x + 2z = 12$.

21. $F(x, y, z) = \cos(2x + y) - z$; $\nabla F = -2\sin(2x + y)\mathbf{i} - \sin(2x + y)\mathbf{j} - \mathbf{k}$; $\nabla F(\pi/2, \pi/4, -1/\sqrt{2}) = \sqrt{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j} - \mathbf{k}$.

$$\text{The equation of the tangent plane is } \sqrt{2}\left(x - \frac{\pi}{2}\right) + \frac{\sqrt{2}}{2}\left(y - \frac{\pi}{4}\right) - \left(z + \frac{1}{\sqrt{2}}\right) = 0,$$

$$2\left(x - \frac{\pi}{2}\right) + \left(y - \frac{\pi}{4}\right) - \sqrt{2}\left(z + \frac{1}{\sqrt{2}}\right) = 0, \text{ or } 2x + y - \sqrt{2}z = \frac{5\pi}{4} + 1.$$

22. $F(x, y, z) = x^2y^3 + 6z$; $\nabla F = 2xy^3\mathbf{i} + 3x^2y^2\mathbf{j} + 6\mathbf{k}$; $\nabla F(2, 1, 1) = 4\mathbf{i} + 12\mathbf{j} + 6\mathbf{k}$. The equation of the tangent plane is $4(x - 2) + 12(y - 1) + 6(z - 1) = 0$ or $2x + 6y + 3z = 13$.

23. $F(x, y, z) = \ln(x^2 + y^2) - z$; $\nabla F = \frac{2x}{x^2 + y^2} \mathbf{i} + \frac{2y}{x^2 + y^2} \mathbf{j} - \mathbf{k}$; $\nabla F(1/\sqrt{2}, 1/\sqrt{2}, 0) = \sqrt{2} \mathbf{i} + \sqrt{2} \mathbf{j} - \mathbf{k}$.

The equation of the tangent plane is $\sqrt{2} \left(x - \frac{1}{\sqrt{2}} \right) + \sqrt{2} \left(y - \frac{1}{\sqrt{2}} \right) - (z - 0) = 0$,

$$2 \left(x - \frac{1}{\sqrt{2}} \right) + 2 \left(y - \frac{1}{\sqrt{2}} \right) - \sqrt{2} z = 0, \text{ or } 2x + 2y - \sqrt{2} z = 2\sqrt{2}.$$

24. $F(x, y, z) = 8e^{-2y} \sin 4x - z$; $\nabla F = 32e^{-2y} \cos 4x \mathbf{i} - 16e^{-2y} \sin 4x \mathbf{j} - \mathbf{k}$; $\nabla F(\pi/24, 0, 4) = 16\sqrt{3} \mathbf{i} - 8\mathbf{j} - \mathbf{k}$.

The equation of the tangent plane is

$$16\sqrt{3}(x - \pi/24) - 8(y - 0) - (z - 4) = 0 \quad \text{or} \quad 16\sqrt{3}x - 8y - z = \frac{2\sqrt{3}\pi}{3} - 4.$$

25. The gradient of $F(x, y, z) = x^2 + y^2 + z^2$ is $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$, so the normal vector to the surface at (x_0, y_0, z_0) is $2x_0\mathbf{i} + 2y_0\mathbf{j} + 2z_0\mathbf{k}$. A normal vector to the plane $2x + 4y + 6z = 1$ is $2\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$. Since we want the tangent plane to be parallel to the given plane, we find c so that $2x_0 = 2c$, $2y_0 = 4c$, $2z_0 = 6c$ or $x_0 = c$, $y_0 = 2c$, $z_0 = 3c$. Now, (x_0, y_0, z_0) is on the surface, so $c^2 + (2c)^2 + (3c)^2 = 14c^2 = 7$ and $c = \pm 1/\sqrt{2}$. Thus, the points on the surface are $(\sqrt{2}/2, \sqrt{2}, 3\sqrt{2}/2)$ and $(-\sqrt{2}/2, -\sqrt{2}, -3\sqrt{2}/2)$.

26. The gradient of $F(x, y, z) = x^2 - 2y^2 - 3z^2$ is $\nabla F(x, y, z) = 2x\mathbf{i} - 4y\mathbf{j} - 6z\mathbf{k}$, so a normal vector to the surface at (x_0, y_0, z_0) is $\nabla F(x_0, y_0, z_0) = 2x_0\mathbf{i} - 4y_0\mathbf{j} - 6z_0\mathbf{k}$. A normal vector to the plane $8x + 4y + 6z = 5$ is $8\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$. Since we want the tangent plane to be parallel to the given plane, we find c so that $2x_0 = 8c$, $-4y_0 = 4c$, $-6z_0 = 6c$ or $x_0 = 4c$, $y_0 = -c$, $z_0 = -c$. Now, (x_0, y_0, z_0) is on the surface, so $(4c)^2 - 2(-c)^2 - 3(-c)^2 = 11c^2 = 33$ and $c = \pm\sqrt{3}$. Thus, the points on the surface are $(4\sqrt{3}, -\sqrt{3}, -\sqrt{3})$ and $(-4\sqrt{3}, \sqrt{3}, \sqrt{3})$.

27. The gradient of $F(x, y, z) = x^2 + 4x + y^2 + z^2 - 2z$ is $\nabla F = (2x + 4)\mathbf{i} + 2y\mathbf{j} + (2z - 2)\mathbf{k}$, so a normal to the surface at (x_0, y_0, z_0) is $(2x_0 + 4)\mathbf{i} + 2y_0\mathbf{j} + (2z_0 - 2)\mathbf{k}$. A horizontal plane has normal $c\mathbf{k}$ for $c \neq 0$. Thus, we want $2x_0 + 4 = 0$, $2y_0 = 0$, $2z_0 - 2 = c$ or $x_0 = -2$, $y_0 = 0$, $z_0 = c + 1$. Since (x_0, y_0, z_0) is on the surface, $(-2)^2 + 4(-2) + (c + 1)^2 - 2(c + 1) = c^2 - 5 = 11$ and $c = \pm 4$. The points on the surface are $(-2, 0, 5)$ and $(-2, 0, -3)$.

28. The gradient of $F(x, y, z) = x^2 + 3y^2 + 4z^2 - 2xy$ is $\nabla F = (2x - 2y)\mathbf{i} + (6y - 2x)\mathbf{j} + 8z\mathbf{k}$, so a normal to the surface at (x_0, y_0, z_0) is $2(x_0 - y_0)\mathbf{i} + 2(3y_0 - x_0)\mathbf{j} + 8z_0\mathbf{k}$.

(a) A normal to the xz plane is $c\mathbf{j}$ for $c \neq 0$. Thus, we want $2(x_0 - y_0) = 0$, $2(3y_0 - x_0) = c$, $8z_0 = 0$ or $x_0 = y_0$, $3y_0 - x_0 = c/2$, $z_0 = 0$. Solving the first two equations, we obtain $x_0 = y_0 = c/4$. Since (x_0, y_0, z_0) is on the surface, $(c/4)^2 + 3(c/4)^2 + 4(0)^2 - 2(c/4)(c/4) = 2c^2/16 = 16$ and $c = \pm 16/\sqrt{2}$. Thus, the points on the surface are $(4/\sqrt{2}, 4/\sqrt{2}, 0)$ and $(-4/\sqrt{2}, -4/\sqrt{2}, 0)$.

(b) A normal to the yz -plane is $c\mathbf{i}$ for $c \neq 0$. Thus, we want $2(x_0 - y_0) = c$, $2(3y_0 - x_0) = 0$, $8z_0 = 0$ or $x_0 - y_0 = c/2$, $x_0 = 3y_0$, $z_0 = 0$. Solving the first two equations, we obtain $x_0 = 3c/4$ and $y_0 = c/4$. Since (x_0, y_0, z_0) is on the surface, $(3c/4)^2 + 3(c/4)^2 + 4(0)^2 - 2(3c/4)(c/4) = 6c^2/16 = 16$ and $c = \pm 16\sqrt{6}$. Thus, the points on the surface are $(12/\sqrt{6}, 4/\sqrt{6}, 0)$ on the surface are $(12/\sqrt{6}, 4/\sqrt{6}, 0)$ and $(-12/\sqrt{6}, -4/\sqrt{6}, 0)$.

(c) A normal to the xy -plane is $c\mathbf{k}$ for $c \neq 0$. Thus, we want $2(x_0 - y_0) = 0$, $2(3y_0 - x_0) = 0$, $8z_0 = c$ or $x_0 = y_0$, $3y_0 - x_0 = 0$, $z_0 = c/8$. Solving the first two equations, we obtain $x_0 = y_0 = 0$. Since (x_0, y_0, z_0) is on the surface, $0^2 + 3(0)^2 + 4(c/8)^2 - 2(0)(0) = c^2/16 = 16$ and $c = \pm 16$. Thus, the points on the surface are $(0, 0, 2)$ and $(0, 0, -2)$.

29. If (x_0, y_0, z_0) is on $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$, then $x_0^2/a^2 + y_0^2/b^2 + z_0^2/c^2 = 1$ and (x_0, y_0, z_0) is on the plane $xx_0/a^2 + yy_0/b^2 + zz_0/c^2 = 1$. A normal to the surface at (x_0, y_0, z_0) is

$$\nabla F(x_0, y_0, z_0) = (2x_0/a^2)\mathbf{i} + (2y_0/b^2)\mathbf{j} + (2z_0/c^2)\mathbf{k}.$$

9.6 Tangent Planes and Normal Lines

A normal to the plane is $(x_0/a^2)\mathbf{i} + (y_0/b^2)\mathbf{j} + (z_0/c^2)\mathbf{k}$. Since the normal to the surface is a multiple of the normal to the plane, the normal vectors are parallel and the plane is tangent to the surface.

30. If (x_0, y_0, z_0) is on $x^2/a^2 - y^2/b^2 + z^2/c^2 = 1$, then $x_0^2/a^2 - y_0^2/b^2 + z_0^2/c^2 = 1$ and (x_0, y_0, z_0) is on the plane $xx_0/a^2 - yy_0/b^2 + zz_0/c^2 = 1$. A normal to the surface at (x_0, y_0, z_0) is

$$\nabla F(x_0, y_0, z_0) = (2x_0/a^2)\mathbf{i} - (2y_0/b^2)\mathbf{j} + (2z_0/c^2)\mathbf{k}.$$

A normal to the plane is $(x_0/a^2)\mathbf{i} - (y_0/b^2)\mathbf{j} + (z_0/c^2)\mathbf{k}$. Since the normal to the surface is a multiple of the normal to the plane, the normal vectors are parallel, and the plane is tangent to the surface.

31. Let $F(x, y, z) = x^2 + y^2 - z^2$. Then $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} - 2z\mathbf{k}$ and a normal to the surface at (x_0, y_0, z_0) is $x_0\mathbf{i} + y_0\mathbf{j} - z_0\mathbf{k}$. An equation of the tangent plane at (x_0, y_0, z_0) is $x_0(x - x_0) + y_0(y - y_0) - z_0(z - z_0) = 0$ or $x_0x + y_0y - z_0z = x_0^2 + y_0^2 - z_0^2$. Since (x_0, y_0, z_0) is on the surface, $z_0^2 = x_0^2 + y_0^2$ and $x_0^2 + y_0^2 - z_0^2 = 0$. Thus, the equation of the tangent plane is $x_0x + y_0y - z_0z = 0$, which passes through the origin.

32. Let $F(x, y, z) = \sqrt{x} + \sqrt{y} + \sqrt{z}$. Then $\nabla F = \frac{1}{2\sqrt{x}}\mathbf{i} + \frac{1}{2\sqrt{y}}\mathbf{j} + \frac{1}{2\sqrt{z}}\mathbf{k}$ and a normal to the surface at (x_0, y_0, z_0) is $\frac{1}{2\sqrt{x_0}}\mathbf{i} + \frac{1}{2\sqrt{y_0}}\mathbf{j} + \frac{1}{2\sqrt{z_0}}\mathbf{k}$. An equation of the tangent plane at (x_0, y_0, z_0) is

$$\frac{1}{2\sqrt{x_0}}(x - x_0) + \frac{1}{2\sqrt{y_0}}(y - y_0) + \frac{1}{2\sqrt{z_0}}(z - z_0) = 0$$

or

$$\frac{1}{\sqrt{x_0}}x + \frac{1}{\sqrt{y_0}}y + \frac{1}{\sqrt{z_0}}z = \sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0} = \sqrt{a}.$$

The sum of the intercepts is $\sqrt{x_0}\sqrt{a} + \sqrt{y_0}\sqrt{a} + \sqrt{z_0}\sqrt{a} = (\sqrt{x_0} + \sqrt{y_0} + \sqrt{z_0})\sqrt{a} = \sqrt{a} \cdot \sqrt{a} = a$.

33. $F(x, y, z) = x^2 + 2y^2 + z^2$; $\nabla F = 2x\mathbf{i} + 4y\mathbf{j} + 2z\mathbf{k}$; $\nabla F(1, -1, 1) = 2\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$. Parametric equations of the line are $x = 1 + 2t$, $y = -1 - 4t$, $z = 1 + 2t$.
34. $F(x, y, z) = 2x^2 - 4y^2 - z$; $\nabla F = 4x\mathbf{i} - 8y\mathbf{j} - \mathbf{k}$; $\nabla F(3, -2, 2) = 12\mathbf{i} + 16\mathbf{j} - \mathbf{k}$. Parametric equations of the line are $x = 3 + 12t$, $y = -2 + 16t$, $z = 2 - t$.
35. $F(x, y, z) = 4x^2 + 9y^2 - z$; $\nabla F = 8x\mathbf{i} + 18y\mathbf{j} - \mathbf{k}$; $\nabla F(1/2, 1/3, 3) = 4\mathbf{i} + 6\mathbf{j} - \mathbf{k}$. Symmetric equations of the line are $\frac{x - 1/2}{4} = \frac{y - 1/3}{6} = \frac{z - 3}{-1}$.
36. $F(x, y, z) = x^2 + y^2 - z^2$; $\nabla F = 2x\mathbf{i} + 2y\mathbf{j} - 2z\mathbf{k}$; $\nabla F(3, 4, 5) = 6\mathbf{i} + 8\mathbf{j} - 10\mathbf{k}$. Symmetric equations of the line are $\frac{x - 3}{6} = \frac{y - 4}{8} = \frac{z - 5}{-10}$.
37. A normal to the surface at (x_0, y_0, z_0) is $\nabla F(x_0, y_0, z_0) = 2x_0\mathbf{i} + 2y_0\mathbf{j} + 2z_0\mathbf{k}$. Parametric equations of the normal line are $x = x_0 + 2x_0t$, $y = y_0 + 2y_0t$, $z = z_0 + 2z_0t$. Letting $t = -1/2$, we see that the normal line passes through the origin.
38. The normal lines to $F(x, y, z) = 0$ and $G(x, y, z) = 0$ are $F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k}$ and $G_x\mathbf{i} + G_y\mathbf{j} + G_z\mathbf{k}$, respectively. These vectors are orthogonal if and only if their dot product is 0. Thus, the surfaces are orthogonal at P if and only if $F_xG_x + F_yG_y + F_zG_z = 0$.
39. Let $F(x, y, z) = x^2 + y^2 + z^2 - 25$ and $G(x, y, z) = -x^2 + y^2 + z^2$. Then

$$F_xG_x + F_yG_y + F_zG_z = (2x)(-2x) + (2y)(2y) + (2z)(2z) = 4(-x^2 + y^2 + z^2).$$

For (x, y, z) on both surfaces, $F(x, y, z) = G(x, y, z) = 0$. Thus, $F_xG_x + F_yG_y + F_zG_z = 4(0) = 0$ and the surfaces are orthogonal at points of intersection.

40. Let $F(x, y, z) = x^2 - y^2 + z^2 - 4$ and $G(x, y, z) = 1/xy^2 - z$. Then

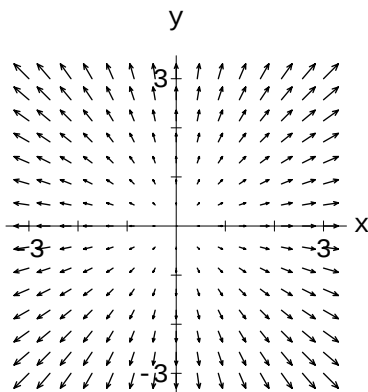
$$\begin{aligned} F_x G_x + F_y G_y + F_z G_z &= (2x)(-1/x^2 y^2) + (-2y)(-2/xy^3) + (2z)(-1) \\ &= -2/xy^2 + 4/xy^2 - 2z = 2(1/xy^2 - z). \end{aligned}$$

For (x, y, z) on both surfaces, $F(x, y, z) = G(x, y, z) = 0$. Thus, $F_x G_x + F_y G_y + F_z G_z = 2(0)$ and the surfaces are orthogonal at points of intersection.

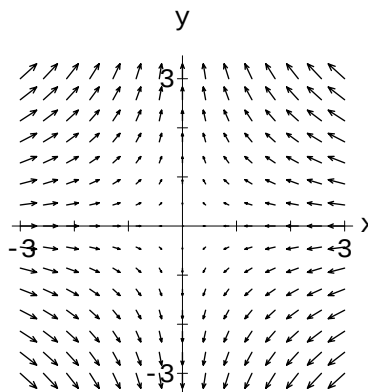
EXERCISES 9.7

Divergence and Curl

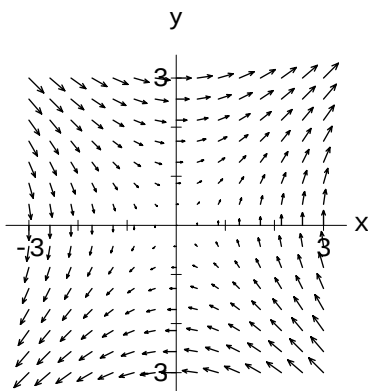
1.



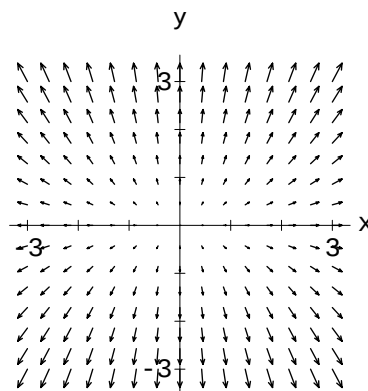
2.



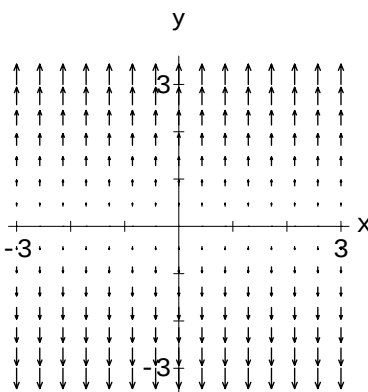
3.



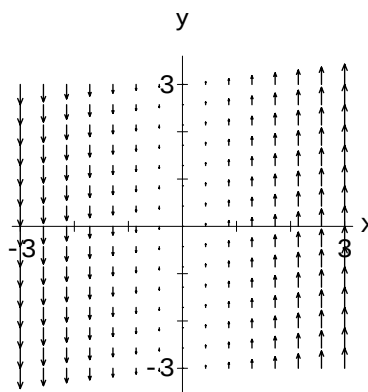
4.



5.



6.



9.7 Divergence and Curl

7. $\text{curl } \mathbf{F} = (x - y)\mathbf{i} + (x - y)\mathbf{j}$; $\text{div } \mathbf{F} = 2z$
8. $\text{curl } \mathbf{F} = -2x^2\mathbf{i} + (10y - 18x^2)\mathbf{j} + (4xz - 10z)\mathbf{k}$; $\text{div } \mathbf{F} = 0$
9. $\text{curl } \mathbf{F} = \mathbf{0}$; $\text{div } \mathbf{F} = 4y + 8z$
10. $\text{curl } \mathbf{F} = (xe^{2y} + ye^{-yz} + 2xye^{2y})\mathbf{i} - ye^{2y}\mathbf{j} + 3(x - y)^2\mathbf{k}$; $\text{div } \mathbf{F} = 3(x - y)^2 - ze^{-yz}$
11. $\text{curl } \mathbf{F} = (4y^3 - 6xz^2)\mathbf{i} + (2z^3 - 3x^2)\mathbf{k}$; $\text{div } \mathbf{F} = 6xy$
12. $\text{curl } \mathbf{F} = -x^3z\mathbf{i} + (3x^2yz - z)\mathbf{j} + (\frac{3}{2}x^2y^2 - y - 15y^2)\mathbf{k}$; $\text{div } \mathbf{F} = (x^3y - x) - (x^3y - x) = 0$
13. $\text{curl } \mathbf{F} = (3e^{-z} - 8yz)\mathbf{i} - xe^{-z}\mathbf{j}$; $\text{div } \mathbf{F} = e^{-z} + 4z^2 - 3ye^{-z}$
14. $\text{curl } \mathbf{F} = (2xyz^3 + 3y)\mathbf{i} + (y \ln x - y^2z^3)\mathbf{j} + (2 - z \ln x)\mathbf{k}$; $\text{div } \mathbf{F} = \frac{yz}{x} - 3z + 3xy^2z^2$
15. $\text{curl } \mathbf{F} = (xy^2e^y + 2xye^y + x^3ye^z + x^3yze^z)\mathbf{i} - y^2e^y\mathbf{j} + (-3x^2yze^z - xe^x)\mathbf{k}$; $\text{div } \mathbf{F} = xye^x + ye^x - x^3ze^z$
16. $\text{curl } \mathbf{F} = (5xye^{5xy} + e^{5xy} + 3xz^3 \sin xz^3 - \cos xz^3)\mathbf{i} + (x^2y \cos yz - 5y^2e^{5xy})\mathbf{j}$
 $+ (-z^4 \sin xz^3 - x^2z \cos yz)\mathbf{k}$; $\text{div } \mathbf{F} = 2x \sin yz$
17. $\text{div } \mathbf{r} = 1 + 1 + 1 = 3$
18. $\text{curl } \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix} = 0\mathbf{i} - 0\mathbf{j} + 0\mathbf{k} = \mathbf{0}$
19. $\mathbf{a} \times \nabla = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \end{vmatrix} = \left(a_2 \frac{\partial}{\partial z} - a_3 \frac{\partial}{\partial y}\right)\mathbf{i} + \left(a_3 \frac{\partial}{\partial x} - a_1 \frac{\partial}{\partial z}\right)\mathbf{j} + \left(a_1 \frac{\partial}{\partial y} - a_2 \frac{\partial}{\partial x}\right)\mathbf{k}$
 $(\mathbf{a} \times \nabla) \times \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_2 \frac{\partial}{\partial z} - a_3 & a_3 \frac{\partial}{\partial x} - a_1 \frac{\partial}{\partial z} & a_1 \frac{\partial}{\partial y} - a_2 \frac{\partial}{\partial x} \\ x & y & z \end{vmatrix}$
 $= (-a_1 - a_1)\mathbf{i} - (a_2 + a_2)\mathbf{j} + (-a_3 - a_3)\mathbf{k} = -2\mathbf{a}$
20. $\nabla \times (\mathbf{a} \times \mathbf{r}) = (\nabla \cdot \mathbf{r})\mathbf{a} - (\nabla \cdot \mathbf{a})\mathbf{r} = (1 + 1 + 1)\mathbf{a} - \left(a_1 \frac{\partial}{\partial x} + a_2 \frac{\partial}{\partial y} + a_3 \frac{\partial}{\partial z}\right)\mathbf{r} = 3\mathbf{a} - (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) = 2\mathbf{a}$
21. $\nabla \cdot (\mathbf{a} \times \mathbf{r}) = \begin{vmatrix} \partial/\partial x & \partial/\partial y & \partial/\partial z \\ a_1 & a_2 & a_3 \\ x & y & z \end{vmatrix} = \frac{\partial}{\partial x}(a_2z - a_3y) - \frac{\partial}{\partial y}(a_1z - a_3x) + \frac{\partial}{\partial z}(a_1y - a_2x) = 0$
22. $\nabla \times \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix} = \mathbf{0}$; $\mathbf{a} \times (\nabla \times \mathbf{r}) = \mathbf{a} \times \mathbf{0} = \mathbf{0}$
23. $\mathbf{r} \times \mathbf{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ a_1 & a_2 & a_3 \end{vmatrix} = (a_3y - a_2z)\mathbf{i} - (a_3x - a_1z)\mathbf{j} + (a_2x - a_1y)\mathbf{k}$; $\mathbf{r} \cdot \mathbf{r} = x^2 + y^2 + z^2$
 $\nabla \times [(\mathbf{r} \cdot \mathbf{r})\mathbf{a}] = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (\mathbf{r} \cdot \mathbf{r})a_1 & (\mathbf{r} \cdot \mathbf{r})a_2 & (\mathbf{r} \cdot \mathbf{r})a_3 \end{vmatrix}$
 $= (2ya_3 - 2za_2)\mathbf{i} - (2xa_3 - 2za_1)\mathbf{j} + (2xa_2 - 2ya_1)\mathbf{k} = 2(\mathbf{r} \times \mathbf{a})$
24. $\mathbf{r} \cdot \mathbf{a} = a_1x + a_2y + a_3z$; $\mathbf{r} \cdot \mathbf{r} = x^2 + y^2 + z^2$; $\nabla \cdot [(\mathbf{r} \cdot \mathbf{r})\mathbf{a}] = 2xa_1 + 2ya_2 + 2za_3 = 2(\mathbf{r} \cdot \mathbf{a})$

25. Let $\mathbf{F} = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$ and $\mathbf{G} = S(x, y, z)\mathbf{i} + T(x, y, z)\mathbf{j} + U(x, y, z)\mathbf{k}$.

$$\begin{aligned}\nabla \cdot (\mathbf{F} + \mathbf{G}) &= \nabla \cdot [(P + S)\mathbf{i} + (Q + T)\mathbf{j} + (R + U)\mathbf{k}] = P_x + S_x + Q_y + T_y + R_z + U_z \\ &= (P_x + Q_y + R_z) + (S_x + T_y + U_z) = \nabla \cdot \mathbf{F} + \nabla \cdot \mathbf{G}\end{aligned}$$

26. Let $\mathbf{F} = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$ and $\mathbf{G} = S(x, y, z)\mathbf{i} + T(x, y, z)\mathbf{j} + U(x, y, z)\mathbf{k}$.

$$\begin{aligned}\nabla \times (\mathbf{F} + \mathbf{G}) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P + S & Q + T & R + U \end{vmatrix} \\ &= (R_y + U_y - Q_z - T_z)\mathbf{i} - (R_x + U_x - P_z - S_z)\mathbf{j} + (Q_x + T_x - P_y - S_y)\mathbf{k} \\ &= (R_y - Q_z)\mathbf{i} - (R_x - P_z)\mathbf{j} + (Q_x - P_y)\mathbf{k} + (U_y - T_z)\mathbf{i} - (U_x - S_z)\mathbf{j} + (T_x - S_y)\mathbf{k} \\ &= \nabla \times \mathbf{F} + \nabla \times \mathbf{G}\end{aligned}$$

27. $\nabla \cdot (f\mathbf{F}) = \nabla \cdot (fP\mathbf{i} + fQ\mathbf{j} + fR\mathbf{k}) = fP_x + Pf_x + fQ_y + Qf_y + fR_z + Rf_z$

$$= f(P_x + Q_y + R_z) + (Pf_x + Qf_y + Rf_z) = f(\nabla \cdot \mathbf{F}) + \mathbf{F} \cdot (\nabla f)$$

28. $\nabla \times (f\mathbf{F}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ fP & fQ & fR \end{vmatrix}$

$$\begin{aligned}&= (fR_y + Rf_y - fQ_z - Qf_z)\mathbf{i} - (fR_x + Rf_x - fP_z - Pf_z)\mathbf{j} + (fQ_x + Qf_x - fP_y - Pf_y)\mathbf{k} \\ &= (fR_y - fQ_z)\mathbf{i} - (fR_x - fP_z)\mathbf{j} + (fQ_x - fP_y)\mathbf{k} + (Rf_y - Qf_z)\mathbf{i} \\ &\quad - (Rf_x - Pf_z)\mathbf{j} + (Qf_x - Pf_y)\mathbf{k}\end{aligned}$$

$$= f[(R_y - Q_z)\mathbf{i} - (R_x - P_z)\mathbf{j} + (Q_x - P_y)\mathbf{k}] + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ f_x & f_y & f_z \\ P & Q & R \end{vmatrix} = f(\nabla \times \mathbf{F}) + (\nabla f) \times \mathbf{F}$$

29. Assuming continuous second partial derivatives,

$$\begin{aligned}\text{curl}(\text{grad } f) &= \nabla \times (f_x\mathbf{i} + f_y\mathbf{j} + f_z\mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f_x & f_y & f_z \end{vmatrix} \\ &= (f_{zy} - f_{yz})\mathbf{i} - (f_{zx} - f_{xz})\mathbf{j} + (f_{yx} - f_{xy})\mathbf{k} = \mathbf{0}.\end{aligned}$$

30. Assuming continuous second partial derivatives,

$$\begin{aligned}\text{div}(\text{curl } \mathbf{F}) &= \nabla \cdot [(R_y - Q_z)\mathbf{i} - (R_x - P_z)\mathbf{j} + (Q_x - P_y)\mathbf{k}] \\ &= (R_{yx} - Q_{zx} - (R_{xy} - P_{zy}) + (Q_{xz} - P_{yz})) = 0.\end{aligned}$$

31. Let $\mathbf{F} = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$ and $\mathbf{G} = S(x, y, z)\mathbf{i} + T(x, y, z)\mathbf{j} + U(x, y, z)\mathbf{k}$.

$$\mathbf{F} \times \mathbf{G} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ P & Q & R \\ S & T & U \end{vmatrix} = (QU - RT)\mathbf{i} - (PU - RS)\mathbf{j} + (PT - QS)\mathbf{k}$$

$$\begin{aligned}\text{div}(\mathbf{F} \times \mathbf{G}) &= (QU_x + Q_xU - RT_x - R_xT) - (PU_y + P_yU - RS_y - R_yS) + (PT_z + P_zT - QS_z - Q_zS) \\ &= S(R_y - Q_z) + T(P_z - R_x) + U(Q_x - P_y) - P(U_y - T_z) - Q(S_z - U_x) - R(T_x - S_y) \\ &= \mathbf{G} \cdot (\text{curl } \mathbf{F}) - \mathbf{F} \cdot (\text{curl } \mathbf{G})\end{aligned}$$

32. Using Problems 26 and 29,

$$\begin{aligned}\text{curl}(\text{curl } \mathbf{F} + \text{grad } f) &= \nabla \times (\text{curl } \mathbf{F} + \text{grad } f) = \nabla \times (\text{curl } \mathbf{F}) + \nabla \times (\text{grad } f) \\ &= \text{curl}(\text{curl } \mathbf{F}) + \text{curl}(\text{grad } f) = \text{curl}(\text{curl } \mathbf{F}) + \mathbf{0} = \text{curl}(\text{curl } \mathbf{F}).\end{aligned}$$

9.7 Divergence and Curl

33. $\nabla \cdot \nabla f = \nabla \cdot (f_x \mathbf{i} + f_y \mathbf{j} + f_z \mathbf{k}) = f_{xx} + f_{yy} + f_{zz}$

34. Using Problem 27, $\nabla \cdot (f \nabla f) = f(\nabla \cdot \nabla f) + \nabla f \cdot \nabla f = f(\nabla^2 f) + |\nabla f|^2$.

35. $\text{curl } \mathbf{F} = -8yz\mathbf{i} - 2z\mathbf{j} - x\mathbf{k}$; $\text{curl}(\text{curl } \mathbf{F}) = 2\mathbf{i} - (8y - 1)\mathbf{j} + 8z\mathbf{k}$

36. (a) For $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$,

$$\begin{aligned}\text{curl}(\text{curl } \mathbf{F}) &= (Q_{xy} - P_{yy} - P_{zz} + R_{xz})\mathbf{i} + (R_{yz} - Q_{zz} - Q_{xx} + P_{yx})\mathbf{j} \\ &\quad + (P_{zx} - R_{xx} - R_{yy} + Q_{zy})\mathbf{k}\end{aligned}$$

and

$$\begin{aligned}-\nabla^2 \mathbf{F} + \text{grad}(\text{div } \mathbf{F}) &= -(P_{xx} + P_{yy} + P_{zz})\mathbf{i} - (Q_{xx} + Q_{yy} + Q_{zz})\mathbf{j} - (R_{xx} + R_{yy} + R_{zz})\mathbf{k} \\ &\quad + \text{grad}(P_x + Q_y + R_z) \\ &= -P_{xx}\mathbf{i} - Q_{yy}\mathbf{j} - R_{zz}\mathbf{k} + (-P_{yy} - P_{zz})\mathbf{i} + (-Q_{xx} - Q_{zz})\mathbf{j} \\ &\quad + (-R_{xx} - R_{yy})\mathbf{k} + (P_{xx} + Q_{yx} + R_{zx})\mathbf{i} + (P_{xy} + Q_{yy} + R_{zy})\mathbf{j} \\ &\quad + (P_{xz} + Q_{yz} + R_{zz})\mathbf{k} \\ &= (-P_{yy} - P_{zz} + Q_{yx} + R_{zx})\mathbf{i} + (-Q_{xx} - Q_{zz} + P_{xy} + R_{zy})\mathbf{j} \\ &\quad + (-R_{xx} - R_{yy} + P_{xz} + Q_{yz})\mathbf{k}.\end{aligned}$$

Thus, $\text{curl}(\text{curl } \mathbf{F}) = -\nabla^2 \mathbf{F} + \text{grad}(\text{div } \mathbf{F})$.

(b) For $\mathbf{F} = xy\mathbf{i} + 4yz^2\mathbf{j} + 2xz\mathbf{k}$, $\nabla^2 \mathbf{F} = 0\mathbf{i} + 8y\mathbf{j} + 0\mathbf{k}$, $\text{div } \mathbf{F} = y + 4z^2 + 2x$, and $\text{grad}(\text{div } \mathbf{F}) = 2\mathbf{i} + \mathbf{j} + 8z\mathbf{k}$.

Then $\text{curl}(\text{curl } \mathbf{F}) = -8y\mathbf{j} + 2\mathbf{i} + \mathbf{j} + 8z\mathbf{k} = 2\mathbf{i} + (1 - 8y)\mathbf{j} + 8z\mathbf{k}$.

37. $\frac{\partial f}{\partial x} = -x(x^2 + y^2 + z^2)^{-3/2}$
 $\frac{\partial f}{\partial y} = -y(x^2 + y^2 + z^2)^{-3/2}$
 $\frac{\partial f}{\partial z} = -z(x^2 + y^2 + z^2)^{-3/2}$
 $\frac{\partial^2 f}{\partial x^2} = 3x^2(x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2}$
 $\frac{\partial^2 f}{\partial y^2} = 3y^2(x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2}$
 $\frac{\partial^2 f}{\partial z^2} = 3z^2(x^2 + y^2 + z^2)^{-5/2} - (x^2 + y^2 + z^2)^{-3/2}$

Adding the second partial derivatives gives

$$\begin{aligned}\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} &= \frac{3(x^2 + y^2 + z^2)}{(x^2 + y^2 + z^2)^{5/2}} - 3(x^2 + y^2 + z^2)^{-3/2} \\ &= 3(x^2 + y^2 + z^2)^{-3/2} - 3(x^2 + y^2 + z^2)^{-3/2} = 0\end{aligned}$$

except when $x = y = z = 0$.

38. $f_x = \frac{1}{1 + \frac{4y^2}{(x^2 + y^2 - 1)^2}} \left(-\frac{4xy}{(x^2 + y^2 - 1)^2} \right) = -\frac{4xy}{(x^2 + y^2 - 1)^2 + 4y^2}$

$$f_{xx} = -\frac{[(x^2 + y^2 - 1)^2 + 4y^2]4y - 4xy[4x(x^2 + y^2 - 1)]}{[(x^2 + y^2 - 1)^2 + 4y^2]^2} = \frac{12x^4y - 4y^5 + 8x^2y^3 - 8x^2y - 8y^3 - 4y}{[(x^2 + y^2 - 1)^2 + 4y^2]^2}$$

$$f_y = \frac{1}{1 + \frac{4y^2}{(x^2 + y^2 - 1)^2}} \left[\frac{2(x^2 + y^2 - 1) - 4y^2}{(x^2 + y^2 - 1)^2} \right] = \frac{2(x^2 - y^2 - 1)}{(x^2 + y^2 - 1)^2 + 4y^2}$$

$$\begin{aligned}
 f_{yy} &= \frac{[(x^2 + y^2 - 1)^2 + 4y^2](-4y) - 2(x^2 - y^2 - 1)[4y(x^2 + y^2 - 1) + 8y]}{[(x^2 + y^2 - 1)^2 + 4y^2]^2} \\
 &= \frac{-12x^4y + 4y^5 - 8x^2y^3 + 8x^2y + 8y^3 + 4y}{[(x^2 + y^2 - 1)^2 + 4y^2]^2}
 \end{aligned}$$

$$\nabla^2 f = f_{xx} + f_{yy} = 0$$

$$\begin{aligned}
 39. \quad \text{curl } \mathbf{F} &= -Gm_1m_2 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x/|\mathbf{r}|^3 & y/|\mathbf{r}|^3 & z/|\mathbf{r}|^3 \end{vmatrix} \\
 &= -Gm_1m_2[(-3yz/|\mathbf{r}|^5 + 3yz/|\mathbf{r}|^5)\mathbf{i} - (-3xz/|\mathbf{r}|^5 + 3xz/|\mathbf{r}|^5)\mathbf{j} + (-3xy/|\mathbf{r}|^5 + 3xy/|\mathbf{r}|^5)\mathbf{k}] \\
 &= \mathbf{0}
 \end{aligned}$$

$$\text{div } \mathbf{F} = -Gm_1m_2 \left[\frac{-2x^2 + y^2 + z^2}{|\mathbf{r}|^{5/2}} + \frac{x^2 - 2y^2 + z^2}{|\mathbf{r}|^{5/2}} + \frac{x^2 + y^2 - 2z^2}{|\mathbf{r}|^{5/2}} \right] = 0$$

$$\begin{aligned}
 40. \quad \frac{1}{2} \text{curl } \mathbf{v} &= \frac{1}{2} \text{curl } (\boldsymbol{\omega} \times \mathbf{r}) = \frac{1}{2} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \omega_2z - \omega_3y & \omega_3x - \omega_1z & \omega_1y - \omega_2x \end{vmatrix} \\
 &= \frac{1}{2}[(\omega_1 + \omega_1)\mathbf{i} - (\omega_2 - \omega_2)\mathbf{j} + (\omega_3 + \omega_3)\mathbf{k}] = \omega_1\mathbf{i} + \omega_2\mathbf{j} + \omega_3\mathbf{k} = \boldsymbol{\omega}
 \end{aligned}$$

41. Using Problems 31 and 29,

$$\nabla \cdot \mathbf{F} = \text{div } (\nabla f \times \nabla g) = \nabla g \cdot (\text{curl } \nabla f) - \nabla f \cdot (\text{curl } \nabla g) = \nabla g \cdot \mathbf{0} - \nabla f \cdot \mathbf{0} = 0.$$

42. Recall that $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 0$. Then, using Problems 31, 29, and 28,

$$\begin{aligned}
 \nabla \cdot \mathbf{F} &= \text{div } (\nabla f \times f \nabla g) = f \nabla g \cdot (\text{curl } \nabla f) - \nabla f \cdot (\text{curl } f \nabla g) = f \nabla g \cdot \mathbf{0} - \nabla f \cdot (\nabla \times f \nabla g) \\
 &= -\nabla f \cdot [f(\nabla \times \nabla g) + (\nabla f \times \nabla g)] = -\nabla f \cdot [f \text{curl } \nabla g + (\nabla f \times \nabla g)] \\
 &= -\nabla f \cdot [f \mathbf{0} + (\nabla f \times \nabla g)] = -\nabla f \cdot (\nabla f \times \nabla g) = 0.
 \end{aligned}$$

43. (a) Expressing the vertical component of $\mathbf{V}$ in polar coordinates, we have

$$\frac{2xy}{(x^2 + y^2)^2} = \frac{2r^2 \sin \theta \cos \theta}{r^4} = \frac{\sin 2\theta}{r^2}.$$

Similarly,

$$\frac{x^2 - y^2}{(x^2 + y^2)^2} = \frac{r^2(\cos^2 \theta - \sin^2 \theta)}{r^4} = \frac{\cos 2\theta}{r^2}.$$

Since $\lim_{r \rightarrow \infty} (\sin 2\theta)/r^2 = \lim_{r \rightarrow \infty} (\cos 2\theta)/r^2 = 0$, $\mathbf{V} \approx A\mathbf{i}$ for r large or (x, y) far from the origin.

(b) Identifying $P(x, y) = A \left[1 - \frac{x^2 - y^2}{(x^2 + y^2)^2} \right]$, $Q(x, y) = -\frac{2Axy}{(x^2 + y^2)^2}$, and $R(x, y) = 0$, we have

$$P_y = \frac{2Ay(3x^2 - y^2)}{(x^2 + y^2)^3}, \quad Q_x = \frac{2Ay(3x^2 - y^2)}{(x^2 + y^2)^3}, \quad \text{and} \quad P_z = Q_z = R_x = R_y = 0.$$

Thus, $\text{curl } \mathbf{V} = (R_y - Q_z)\mathbf{i} + (P_z - R_x)\mathbf{j} + (Q_x - P_y)\mathbf{k} = \mathbf{0}$ and $\mathbf{V}$ is irrotational.

(c) Since $P_x = \frac{2Ax(x^2 - 3y^2)}{(x^2 + y^2)^3}$, $Q_y = \frac{2Ax(3y^2 - x^2)}{(x^2 + y^2)^3}$, and $R_z = 0$, $\nabla \cdot \mathbf{F} = P_x + Q_y + R_z = 0$ and $\mathbf{V}$ is incompressible.

9.7 Divergence and Curl

44. We first note that $\text{curl } (\partial \mathbf{H} / \partial t) = \partial(\text{curl } \mathbf{H}) / \partial t$ and $\text{curl } (\partial \mathbf{E} / \partial t) = \partial(\text{curl } \mathbf{E}) / \partial t$. Then, from Problem 36,

$$\begin{aligned} -\nabla^2 \mathbf{E} &= -\nabla^2 \mathbf{E} + \mathbf{0} = -\nabla^2 \mathbf{E} + \text{grad } 0 = -\nabla^2 \mathbf{E} + \text{grad } (\text{div } \mathbf{E}) = \text{curl } (\text{curl } \mathbf{E}) \\ &= \text{curl } \left(-\frac{1}{c} \frac{\partial \mathbf{H}}{\partial t} \right) = -\frac{1}{c} \frac{\partial}{\partial t} \text{curl } \mathbf{H} = -\frac{1}{c} \frac{\partial}{\partial t} \left(\frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} \right) = -\frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} \end{aligned}$$

and $\nabla^2 \mathbf{E} = \frac{1}{c^2} \partial^2 \mathbf{E} / \partial t^2$. Similarly,

$$\begin{aligned} -\nabla^2 \mathbf{H} &= -\nabla^2 \mathbf{H} + \text{grad } (\text{div } \mathbf{H}) = \text{curl } (\text{curl } \mathbf{H}) = \text{curl } \left(\frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} \right) = \frac{1}{c} \frac{\partial}{\partial t} \text{curl } \mathbf{E} \\ &= \frac{1}{c} \frac{\partial}{\partial t} \left(-\frac{1}{c} \frac{\partial \mathbf{H}}{\partial t} \right) = -\frac{1}{c^2} \frac{\partial^2 \mathbf{H}}{\partial t^2} \end{aligned}$$

and $\nabla^2 \mathbf{H} = \frac{1}{c^2} \partial^2 \mathbf{H} / \partial t^2$.

45. We note that $\text{div } \mathbf{F} = 2xyz - 2xyz + 1 = 1 \neq 0$. If $\mathbf{F} = \text{curl } \mathbf{G}$, then $\text{div}(\text{curl } \mathbf{G}) = \text{div } \mathbf{F} = 1$. But, by Problem 30, for any vector field $\mathbf{G}$, $\text{div}(\text{curl } \mathbf{G}) = 0$. Thus, $\mathbf{F}$ cannot be the curl of $\mathbf{G}$.

EXERCISES 9.8

Line Integrals

- $$\begin{aligned} \int_C 2xy \, dx &= \int_0^{\pi/4} 2(5 \cos t)(5 \sin t)(-5 \sin t) \, dt = -250 \int_0^{\pi/4} \sin^2 t \cos t \, dt = -250 \left(\frac{1}{3} \sin^3 t \right) \Big|_0^{\pi/4} = -\frac{125\sqrt{2}}{6} \\ \int_C 2xy \, dy &= \int_0^{\pi/4} 2(5 \cos t)(5 \sin t)(5 \cos t) \, dt = 250 \int_0^{\pi/4} \cos^2 t \sin t \, dt = 250 \left(-\frac{1}{3} \cos^3 t \right) \Big|_0^{\pi/4} \\ &= \frac{250}{3} \left(1 - \frac{\sqrt{2}}{4} \right) = \frac{125}{6} (4 - \sqrt{2}) \\ \int_C 2xy \, ds &= \int_0^{\pi/4} 2(5 \cos t)(5 \sin t) \sqrt{25 \sin^2 t + 25 \cos^2 t} \, dt = 250 \int_0^{\pi/4} \sin t \cos t \, dt \\ &= 250 \left(\frac{1}{2} \sin^2 t \right) \Big|_0^{\pi/4} = \frac{125}{2} \end{aligned}$$
- $$\begin{aligned} \int_C (x^3 + 2xy^2 + 2x) \, dx &= \int_0^1 [8t^3 + 2(2t)(t^4) + 2(2t)] 2 \, dt = 2 \int_0^1 (8t^3 + 4t^5 + 4t) \, dt \\ &= 2 \left(2t^4 + \frac{2}{3}t^6 + 2t^2 \right) \Big|_0^1 = \frac{28}{3} \\ \int_C (x^3 + 2xy^2 + 2x) \, dy &= \int_0^1 [8t^3 + 2(2t)(t^4) + 2(2t)] 2t \, dt = 2 \int_0^1 (8t^4 + 4t^6 + 4t^2) \, dt \\ &= 2 \left(\frac{8}{5}t^5 + \frac{4}{7}t^7 + \frac{4}{3}t^3 \right) \Big|_0^1 = \frac{736}{105} \\ \int_C (x^3 + 2xy^2 + 2x) \, ds &= \int_0^1 [8t^3 + 2(2t)(t^4) + 2(2t)] \sqrt{4 + 4t^2} \, dt = 8 \int_0^1 t(1 + t^2)^{5/2} \, dt \\ &= 8 \left(\frac{1}{7}(1 + t^2)^{7/2} \right) \Big|_0^1 = \frac{8}{7}(2^{7/2} - 1) \end{aligned}$$

$$\begin{aligned} 3. \quad \int_C (3x^2 + 6y^2) dx &= \int_{-1}^0 [3x^2 + 6(2x+1)^2] dx = \int_{-1}^0 (27x^2 + 24x + 6) dx = (9x^3 + 12x^2 + 6x) \Big|_{-1}^0 \\ &= -(-9 + 12 - 6) = 3 \end{aligned}$$

$$\int_C (3x^2 + 6y^2) dy = \int_{-1}^0 [3x^2 + 6(2x+1)^2] 2 dx = 6$$

$$\int_C (3x^2 + 6y^2) ds = \int_{-1}^0 [3x^2 + 6(2x+1)^2] \sqrt{1+4} dx = 3\sqrt{5}$$

$$4. \quad \int_C \frac{x^2}{y^3} dx = \int_1^8 \frac{x^2}{27x^2/8} dx = \frac{8}{27} \int_1^8 dx = \frac{56}{27}$$

$$\int_C \frac{x^2}{y^3} dy = \int_1^8 \frac{x^2}{27x^2/8} x^{-1/3} dx = \frac{8}{27} \int_1^8 x^{-1/3} dx = \frac{4}{9} x^{2/3} \Big|_1^8 = \frac{4}{3}$$

$$\int_C \frac{x^2}{y^3} ds = \int_1^8 \frac{x^2}{27x^2/8} \sqrt{1+x^{-2/3}} dx = \frac{8}{27} \int_1^8 x^{-1/3} \sqrt{1+x^{2/3}} dx = \frac{8}{27} (1+x^{2/3})^{3/2} \Big|_1^8 = \frac{8}{27} (5^{3/2} - 2^{3/2})$$

$$\begin{aligned} 5. \quad \int_C z dx &= \int_0^{\pi/2} t(-\sin t) dt \quad \boxed{\text{Integration by parts}} \\ &= (t \cos t - \sin t) \Big|_0^{\pi/2} = -1 \end{aligned}$$

$$\begin{aligned} \int_C z dy &= \int_0^{\pi/2} t \cos t dt \quad \boxed{\text{Integration by parts}} \\ &= (t \sin t + \cos t) \Big|_0^{\pi/2} = \frac{\pi}{2} - 1 \end{aligned}$$

$$\int_C z dz = \int_0^{\pi/2} t dt = \frac{1}{2} t^2 \Big|_0^{\pi/2} = \frac{\pi^2}{8}$$

$$\int_C z ds = \int_0^{\pi/2} t \sqrt{\sin^2 t + \cos^2 t + 1} dt = \sqrt{2} \int_0^{\pi/2} t dt = \frac{\pi^2 \sqrt{2}}{8}$$

$$6. \quad \int_C 4xyz dx = \int_0^1 4 \left(\frac{1}{3} t^3 \right) (t^2)(2t) t^2 dt = \frac{8}{3} \int_0^1 t^8 dt = \frac{8}{27} t^9 \Big|_0^1 = \frac{8}{27}$$

$$\int_C 4xyz dy = \int_0^1 4 \left(\frac{1}{3} t^3 \right) (t^2)(2t) 2t dt = \frac{16}{3} \int_0^1 t^7 dt = \frac{2}{3} t^8 \Big|_0^1 = \frac{2}{3}$$

$$\int_C 4xyz dz = \int_0^1 4 \left(\frac{1}{3} t^3 \right) (t^2)(2t) 2 dt = \frac{16}{3} \int_0^1 t^6 dt = \frac{16}{21} t^7 \Big|_0^1 = \frac{16}{21}$$

$$\int_C 4xyz ds = \int_0^1 4 \left(\frac{1}{3} t^3 \right) (t^2)(2t) \sqrt{t^4 + 4t^2 + 4} dt = \frac{8}{3} \int_0^1 t^6 (t^2 + 2) dt = \frac{8}{3} \left(\frac{1}{9} t^9 + \frac{2}{7} t^7 \right) \Big|_0^1 = \frac{200}{189}$$

7. Using x as the parameter, $dy = dx$ and

$$\begin{aligned} \int_C (2x + y) dx + xy dy &= \int_{-1}^2 (2x + x + 3 + x^2 + 3x) dx = \int_{-1}^2 (x^2 + 6x + 3) dx \\ &= \left(\frac{1}{3} x^3 + 3x^2 + 3x \right) \Big|_{-1}^2 = 21. \end{aligned}$$

9.8 Line Integrals

8. Using x as the parameter, $dy = 2x dx$ and

$$\begin{aligned}\int_C (2x + y) dx + xy dy &= \int_{-1}^2 (2x + x^2 + 1) dx + \int_{-1}^2 x(x^2 + 1) 2x dx = \int_{-1}^2 (2x^4 + 3x^2 + 2x + 1) dx \\ &= \left(\frac{2}{5}x^5 + x^3 + x^2 + x \right) \Big|_{-1}^2 = \frac{141}{5}.\end{aligned}$$

9. From $(-1, 2)$ to $(2, 2)$ we use x as a parameter with $y = 2$ and $dy = 0$. From $(2, 2)$ to $(2, 5)$ we use y as a parameter with $x = 2$ and $dx = 0$.

$$\int_C (2x + y) dx + xy dy = \int_{-1}^2 (2x + 2) dx + \int_2^5 2y dy = (x^2 + 2x) \Big|_{-1}^2 + y^2 \Big|_2^5 = 9 + 21 = 30$$

10. From $(-1, 2)$ to $(-1, 0)$ we use y as a parameter with $x = -1$ and $dx = 0$. From $(-1, 0)$ to $(2, 0)$ we use x as a parameter with $y = 0$ and $dy = 0$. From $(2, 0)$ to $(2, 5)$ we use y as a parameter with $x = 2$ and $dx = 0$.

$$\begin{aligned}\int_C (2x + y) dx + xy dy &= \int_2^0 (-1) y dy + \int_{-1}^2 2x dx + \int_0^5 2y dy = -\frac{1}{2}y^2 \Big|_2^0 + x^2 \Big|_{-1}^2 + y^2 \Big|_0^5 \\ &= 2 + 3 + 25 = 30\end{aligned}$$

11. Using x as the parameter, $dy = 2x dx$.

$$\int_C y dx + x dy = \int_0^1 x^2 dx + \int_0^1 x(2x) dx = \int_0^1 3x^2 dx = x^3 \Big|_0^1 = 1$$

12. Using x as the parameter, $dy = dx$.

$$\int_C y dx + x dy = \int_0^1 x dx + \int_0^1 x dx = \int_0^1 2x dx = x^2 \Big|_0^1 = 1$$

13. From $(0, 0)$ to $(0, 1)$ we use y as a parameter with $x = 0$ and $dx = 0$. From $(0, 1)$ to $(1, 1)$ we use x as a parameter with $y = 1$ and $dy = 0$.

$$\int_C y dx + x dy = 0 + \int_0^1 1 dx = 1$$

14. From $(0, 0)$ to $(1, 0)$ we use x as a parameter with $y = 0$ and $dy = 0$. From $(1, 0)$ to $(1, 1)$ we use y as a parameter with $x = 1$ and $dx = 0$.

$$\int_C y dx + x dy = 0 + \int_0^1 1 dy = 1$$

$$\begin{aligned}15. \quad \int_C (6x^2 + 2y^2) dx + 4xy dy &= \int_4^9 (6t + 2t^2) \frac{1}{2} t^{-1/2} dt + \int_4^9 4\sqrt{t} t dt = \int_4^9 (3t^{1/2} + 5t^{3/2}) dt \\ &= (2t^{3/2} + 2t^{5/2}) \Big|_4^9 = 460\end{aligned}$$

$$16. \quad \int_C (-y^2) dx + xy dy = \int_0^2 (-t^6) 2 dt + \int_0^2 (2t)(t^3) 3t^2 dt = \int_0^2 4t^6 dt = \frac{4}{7}t^7 \Big|_0^2 = \frac{512}{7}$$

$$\begin{aligned}17. \quad \int_C 2x^3 y dx + (3x + y) dy &= \int_{-1}^1 2(y^6) y 2y dy + \int_{-1}^1 (3y^2 + y) dy = \int_{-1}^1 (4y^8 + 3y^2 + y) dy \\ &= \left(\frac{4}{9}y^9 + y^3 + \frac{1}{2}y^2 \right) \Big|_{-1}^1 = \frac{26}{9}\end{aligned}$$

$$\begin{aligned}
 18. \quad \int_C 4x \, dx + 2y \, dy &= \int_{-1}^2 4(y^3 + 1)3y^2 \, dy + \int_{-1}^2 2y \, dy = \int_{-1}^2 (12y^5 + 12y^2 + 2y) \, dy \\
 &= (2y^6 + 4y^3 + y^2) \Big|_{-1}^2 = 165
 \end{aligned}$$

19. From $(-2, 0)$ to $(2, 0)$ we use x as a parameter with $y = dy = 0$. From $(2, 0)$ to $(-2, 0)$ we parameterize the semicircle as $x = 2 \cos \theta$ and $y = 2 \sin \theta$ for $0 \leq \theta \leq \pi$.

$$\begin{aligned}
 \oint_C (x^2 + y^2) \, dx - 2xy \, dy &= \int_{-2}^2 x^2 \, dx + \int_0^\pi 4(-2 \sin \theta \, d\theta) - \int_0^\pi 8 \cos \theta \sin \theta (2 \cos \theta \, d\theta) \\
 &= \frac{1}{3} x^3 \Big|_{-2}^2 - 8 \int_0^\pi (\sin \theta + 2 \cos^2 \theta \sin \theta) \, d\theta \\
 &= \frac{16}{3} - 8 \left(-\cos \theta - \frac{2}{3} \cos^3 \theta \right) \Big|_0^\pi = \frac{16}{3} - \frac{80}{3} = -\frac{64}{3}
 \end{aligned}$$

20. We start at $(0, 0)$ and use x as a parameter.

$$\begin{aligned}
 \oint_C (x^2 + y^2) \, dx - 2xy \, dy &= \int_0^1 (x^2 + x^4) \, dx - 2 \int_0^1 x x^2 (2x \, dx) + \int_1^0 (x^2 + x) \, dx \\
 &\quad - 2 \int_1^0 x \sqrt{x} \left(\frac{1}{2} x^{-1/2} \right) \, dx \\
 &= \int_0^1 (x^2 - 3x^4) \, dx + \int_1^0 x^2 \, dx = \int_0^1 (-3x^4) \, dx = -\frac{3}{5} x^5 \Big|_0^1 = -\frac{3}{5}
 \end{aligned}$$

21. From $(1, 1)$ to $(-1, 1)$ and $(-1, -1)$ to $(1, -1)$ we use x as a parameter with $y = 1$ and $y = -1$, respectively, and $dy = 0$. From $(-1, 1)$ to $(-1, -1)$ and $(1, -1)$ to $(1, 1)$ we use y as a parameter with $x = -1$ and $x = 1$, respectively, and $dx = 0$.

$$\begin{aligned}
 \oint_C x^2 y^3 \, dx - xy^2 \, dy &= \int_1^{-1} x^2(1) \, dx + \int_1^{-1} -(-1)y^2 \, dy + \int_{-1}^1 x^2(-1)^3 \, dx + \int_{-1}^1 -(1)y^2 \, dy \\
 &= \frac{1}{3} x^3 \Big|_1^{-1} + \frac{1}{3} y^3 \Big|_1^{-1} - \frac{1}{3} x^3 \Big|_{-1}^1 - \frac{1}{3} y^3 \Big|_{-1}^1 = -\frac{8}{3}
 \end{aligned}$$

22. From $(2, 4)$ to $(0, 4)$ we use x as a parameter with $y = 4$ and $dy = 0$. From $(0, 4)$ to $(0, 0)$ we use y as a parameter with $x = dx = 0$. From $(0, 0)$ to $(2, 4)$ we use $y = 2x$ and $dy = 2 \, dx$.

$$\begin{aligned}
 \oint_C x^2 y^3 \, dx - xy^2 \, dy &= \int_2^0 x^2(64) \, dx - \int_4^0 0 \, dy + \int_0^2 x^2(8x^3) \, dx - \int_0^2 x(4x^2)2 \, dx \\
 &= \frac{64}{3} x^3 \Big|_2^0 + \frac{4}{3} x^6 \Big|_0^2 - 2x^4 \Big|_0^2 = -\frac{512}{3} + \frac{256}{3} - 32 = -\frac{352}{3}
 \end{aligned}$$

$$\begin{aligned}
 23. \quad \oint_C (x^2 - y^2) \, ds &= \int_0^{2\pi} (25 \cos^2 \theta - 25 \sin^2 \theta) \sqrt{25 \sin^2 \theta + 25 \cos^2 \theta} \, d\theta = 125 \int_0^{2\pi} (\cos^2 \theta - \sin^2 \theta) \, d\theta \\
 &= 125 \int_0^{2\pi} \cos 2\theta \, d\theta = \frac{125}{2} \sin 2\theta \Big|_0^{2\pi} = 0
 \end{aligned}$$

$$24. \quad \oint_C y \, dx - x \, dy = \int_0^\pi 3 \sin t (-2 \sin t) \, dt - \int_0^\pi 2 \cos t (3 \cos t) \, dt = -6 \int_0^\pi (\sin^2 t + \cos^2 t) \, dt = -6 \int_0^\pi dt = -6\pi$$

Thus, $\int_{-C} y \, dx - x \, dy = 6\pi$.

9.8 Line Integrals

25. We parameterize the line segment from $(0, 0, 0)$ to $(2, 3, 4)$ by $x = 2t$, $y = 3t$, $z = 4t$ for $0 \leq t \leq 1$. We parameterize the line segment from $(2, 3, 4)$ to $(6, 8, 5)$ by $x = 2 + 4t$, $y = 3 + 5t$, $z = 4 + t$, $0 \leq t \leq 1$.

$$\begin{aligned}\oint_C y \, dx + z \, dy + x \, dz &= \int_0^1 3t(2 \, dt) + \int_0^1 4t(3 \, dt) + \int_0^1 2t(4 \, dt) + \int_0^1 (3 + 5t)(4 \, dt) \\ &\quad + \int_0^1 (4 + t)(5 \, dt) + \int_0^1 (2 + 4t) \, dt \\ &= \int_0^1 (55t + 34) \, dt = \left(\frac{55}{2}t^2 + 34t \right) \Big|_0^1 = \frac{123}{2}\end{aligned}$$

26.
$$\begin{aligned}\int_C y \, dx + z \, dy + x \, dz &= \int_0^2 t^3(3 \, dt) + \int_0^2 \left(\frac{5}{4}t^2 \right) (3t^2 \, dt) + \int_0^2 (3t) \left(\frac{5}{2}t \, dt \right) \\ &= \int_0^2 \left(3t^3 + \frac{15}{4}t^4 + \frac{15}{2}t^2 \right) dt = \left(\frac{3}{4}t^4 + \frac{3}{4}t^5 + \frac{5}{2}t^3 \right) \Big|_0^2 = 56\end{aligned}$$

27. From $(0, 0, 0)$ to $(6, 0, 0)$ we use x as a parameter with $y = dy = 0$ and $z = dz = 0$. From $(6, 0, 0)$ to $(6, 0, 5)$ we use z as a parameter with $x = 6$ and $dx = 0$ and $y = dy = 0$. From $(6, 0, 5)$ to $(6, 8, 5)$ we use y as a parameter with $x = 6$ and $dx = 0$ and $z = 5$ and $dz = 0$.

$$\int_C y \, dx + z \, dy + x \, dz = \int_0^6 0 + \int_0^5 6 \, dz + \int_0^8 5 \, dy = 70$$

28. We parametrize the line segment from $(0, 0, 0)$ to $(6, 8, 0)$ by $x = 6t$, $y = 8t$, $z = 0$ for $0 \leq t \leq 1$. From $(6, 8, 0)$ to $(6, 8, 5)$ we use z as a parameter with $x = 6$, $dx = 0$, and $y = 8$, $dy = 0$.

$$\int_C y \, dx + z \, dy + x \, dz = \int_0^1 8t(6 \, dt) + \int_0^5 6 \, dz = 24t^2 \Big|_0^1 + 30 = 54$$

29. $\mathbf{F} = e^{3t}\mathbf{i} - (e^{-4t})e^t\mathbf{j} = e^{3t}\mathbf{i} - e^{-3t}\mathbf{j}$; $d\mathbf{r} = (-2e^{-2t}\mathbf{i} + e^t\mathbf{j}) \, dt$; $\mathbf{F} \cdot d\mathbf{r} = (-2e^t - e^{-2t}) \, dt$;

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{\ln 2} (-2e^t - e^{-2t}) \, dt = \left(-2e^t + \frac{1}{2}e^{-2t} \right) \Big|_0^{\ln 2} = -\frac{31}{8} - \left(-\frac{3}{2} \right) = -\frac{19}{8}$$

30. $\mathbf{F} = e^t\mathbf{i} + te^{t^3}\mathbf{j} + t^3e^{t^6}\mathbf{k}$; $d\mathbf{r} = (\mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k}) \, dt$;

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 (e^t + 2t^2e^{t^3} + 3t^5e^{t^6}) \, dt = \left(e^t + \frac{2}{3}e^{t^3} + \frac{1}{2}e^{t^6} \right) \Big|_0^1 = \frac{13}{6}(e - 1)$$

31. Using x as a parameter, $\mathbf{r}(x) = x\mathbf{i} + \ln x\mathbf{j}$. Then $\mathbf{F} = \ln x\mathbf{i} + x\mathbf{j}$, $d\mathbf{r} = (\mathbf{i} + \frac{1}{x}\mathbf{j}) \, dx$, and

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_1^e (\ln x + 1) \, dx = (x \ln x) \Big|_1^e = e.$$

32. Let $\mathbf{r}_1 = (-2 + 2t)\mathbf{i} + (2 - 2t)\mathbf{j}$ and $\mathbf{r}_2 = 2t\mathbf{i} + 3t\mathbf{j}$ for $0 \leq t \leq 1$. Then

$$d\mathbf{r}_1 = 2\mathbf{i} - 2\mathbf{j}, \quad d\mathbf{r}_2 = 2\mathbf{i} + 3\mathbf{j},$$

$$\mathbf{F}_1 = 2(-2 + 2t)(2 - 2t)\mathbf{i} + 4(2 - 2t)^2\mathbf{j} = (-8t^2 + 16t - 8)\mathbf{i} + (16t^2 - 32t + 16)\mathbf{j},$$

$$\mathbf{F}_2 = 2(2t)(3t)\mathbf{i} + 4(3t)^2\mathbf{j} = 12t^2\mathbf{i} + 36t^2\mathbf{j},$$

and

$$\begin{aligned}W &= \int_{C_1} \mathbf{F}_1 \cdot d\mathbf{r}_1 + \int_{C_2} \mathbf{F}_2 \cdot d\mathbf{r}_2 = \int_0^1 (-16t^2 + 32t - 16 - 32t^2 + 64t - 32) \, dt + \int_0^1 (24t^2 + 108t^2) \, dt \\ &= \int_0^1 (84t^2 + 96t - 48) \, dt = (28t^3 + 48t^2 - 48t) \Big|_0^1 = 28.\end{aligned}$$

33. Let $\mathbf{r}_1 = (1 + 2t)\mathbf{i} + \mathbf{j}$, $\mathbf{r}_2 = 3\mathbf{i} + (1 + t)\mathbf{j}$, and $\mathbf{r}_3 = (3 - 2t)\mathbf{i} + (2 - t)\mathbf{j}$ for $0 \leq t \leq 1$.

Then

$$\begin{aligned} d\mathbf{r}_1 &= 2\mathbf{i}, & d\mathbf{r}_2 &= \mathbf{j}, & d\mathbf{r}_3 &= -2\mathbf{i} - \mathbf{j}, \\ \mathbf{F}_1 &= (1 + 2t + 2)\mathbf{i} + (6 - 2 - 4t)\mathbf{j} = (3 + 2t)\mathbf{i} + (4 - 4t)\mathbf{j}, \\ \mathbf{F}_2 &= (3 + 2 + 2t)\mathbf{i} + (6 + 6t - 6)\mathbf{j} = (5 + 2t)\mathbf{i} + 6t\mathbf{j}, \\ \mathbf{F}_3 &= (3 - 2t + 4 - 2t)\mathbf{i} + (12 - 6t - 6 + 4t)\mathbf{j} = (7 - 4t)\mathbf{i} + (6 - 2t)\mathbf{j}, \end{aligned}$$

and

$$\begin{aligned} W &= \int_{C_1} \mathbf{F}_1 \cdot d\mathbf{r}_1 + \int_{C_2} \mathbf{F}_2 \cdot d\mathbf{r}_2 + \int_{C_3} \mathbf{F}_3 \cdot d\mathbf{r}_3 \\ &= \int_0^1 (6 + 4t) dt + \int_0^1 6t dt + \int_0^1 (-14 + 8t - 6 + 2t) dt \\ &= \int_0^1 (-14 + 20t) dt = (-14t + 10t^2) \Big|_0^1 = -4. \end{aligned}$$

34. $\mathbf{F} = t^3\mathbf{i} + t^4\mathbf{j} + t^5\mathbf{k}$; $d\mathbf{r} = 3t^2\mathbf{i} + 2t\mathbf{j} + \mathbf{k}$; $W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_1^3 (3t^5 + 2t^5 + t^5) dt = \int_1^3 6t^5 dt = t^6 \Big|_1^3 = 728$

35. $\mathbf{r} = 3 \cos t \mathbf{i} + 3 \sin t \mathbf{j}$, $0 \leq t \leq 2\pi$; $d\mathbf{r} = -3 \sin t \mathbf{i} + 3 \cos t \mathbf{j}$; $\mathbf{F} = a\mathbf{i} + b\mathbf{j}$;

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (-3a \sin t + 3b \cos t) dt = (3a \cos t + 3b \sin t) \Big|_0^{2\pi} = 0$$

36. Let $\mathbf{r} = t\mathbf{i} + t\mathbf{j} + t\mathbf{k}$ for $1 \leq t \leq 3$. Then $d\mathbf{r} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, and

$$\begin{aligned} \mathbf{F} &= \frac{c}{|\mathbf{r}|^3}(t\mathbf{i} + t\mathbf{j} + t\mathbf{k}) = \frac{ct}{(\sqrt{3}t^2)^3}(\mathbf{i} + \mathbf{j} + \mathbf{k}) = \frac{c}{3\sqrt{3}t^2}(\mathbf{i} + \mathbf{j} + \mathbf{k}), \\ W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_1^3 \frac{c}{3\sqrt{3}t^2}(1 + 1 + 1) dt = \frac{c}{\sqrt{3}} \int_1^3 \frac{1}{t^2} dt = \frac{c}{\sqrt{3}} \left(-\frac{1}{t} \right) \Big|_1^3 = \frac{c}{\sqrt{3}} \left(-\frac{1}{3} + 1 \right) = \frac{2c}{3\sqrt{3}}. \end{aligned}$$

37. $\int_{C_1} y^2 dx + xy dy = \int_0^1 (4t + 2)^2 2 dt + \int_0^1 (2t + 1)(4t + 2) 4 dt = \int_0^1 (64t^2 + 64t + 16) dt$
 $= \left(\frac{64}{3}t^3 + 32t^2 + 16t \right) \Big|_0^1 = \frac{64}{3} + 32 + 16 = \frac{208}{3}$

$$\int_{C_2} y^2 dx + xy dy = \int_1^{\sqrt{3}} 4t^4(2t) dt + \int_1^{\sqrt{3}} 2t^4(4t) dt = \int_1^{\sqrt{3}} 16t^5 dt = \frac{8}{3}t^6 \Big|_1^{\sqrt{3}} = 72 - \frac{8}{3} = \frac{208}{3}$$

$$\int_{C_3} y^2 dx + xy dy = \int_e^{e^3} 4(\ln t)^2 \frac{1}{t} dt + \int_e^{e^3} 2(\ln t)^2 \frac{2}{t} dt = \int_e^{e^3} \frac{8}{t} (\ln t)^2 dt = \frac{8}{3} (\ln t)^3 \Big|_e^{e^3} = \frac{8}{3} (27 - 1) = \frac{208}{3}$$

38. $\int_{C_1} xy ds = \int_0^2 t(2t)\sqrt{1 + 4t^2} dt = 2\sqrt{5} \int_0^2 t^2 dt = 2\sqrt{5} \left(\frac{1}{3}t^3 \right) \Big|_0^2 = \frac{16\sqrt{5}}{3}$

$$\begin{aligned} \int_{C_2} xy ds &= \int_0^2 t(t^2)\sqrt{1 + 4t^2} dt = \int_0^2 t^3\sqrt{1 + 4t^2} dt \quad \boxed{u = 1 + 4t^2, \quad du = 8t dt, \quad t^2 = \frac{1}{4}(u - 1)} \\ &= \int_1^{17} \frac{1}{4}(u - 1)u^{1/2} \frac{1}{8} du = \frac{1}{32} \int_1^{17} (u^{3/2} - u^{1/2}) du = \frac{1}{32} \left(\frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2} \right) \Big|_1^{17} = \frac{391\sqrt{17} + 1}{120} \end{aligned}$$

$$\int_{C_3} xy ds = \int_2^3 (2t - 4)(4t - 8)\sqrt{4 + 16t^2} dt = 16\sqrt{5} \int_2^3 (t - 2)^2 dt = 16\sqrt{5} \left[\frac{1}{3}(t - 2)^3 \right] \Big|_2^3 = \frac{16\sqrt{5}}{3}$$

C_1 and C_3 are different parameterizations of the same curve, while C_1 and C_2 are different curves.

9.8 Line Integrals

39. Since $\mathbf{v} \cdot \mathbf{v} = v^2$, $\frac{d}{dt} v^2 = \frac{d}{dt}(\mathbf{v} \cdot \mathbf{v}) = \mathbf{v} \cdot \frac{d\mathbf{v}}{dt} + \frac{d\mathbf{v}}{dt} \cdot \mathbf{v} = 2 \frac{d\mathbf{v}}{dt} \cdot \mathbf{v}$. Then

$$\begin{aligned} W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b m\mathbf{a} \cdot \left(\frac{d\mathbf{r}}{dt} dt \right) = m \int_a^b \frac{d\mathbf{v}}{dt} \cdot \mathbf{v} dt = m \int_a^b \frac{1}{2} \left(\frac{d}{dt} v^2 \right) dt \\ &= \frac{1}{2} m(v^2) \Big|_a^b = \frac{1}{2} m[v(b)]^2 - \frac{1}{2} m[v(a)]^2. \end{aligned}$$

40. We are given $\rho = kx$. Then

$$\begin{aligned} m &= \int_C \rho ds = \int_0^\pi kx ds = k \int_0^\pi (1 + \cos t) \sqrt{\sin^2 t + \cos^2 t} dt = k \int_0^\pi (1 + \cos t) dt \\ &= k(t + \sin t) \Big|_0^\pi = k\pi. \end{aligned}$$

41. From Problem 40, $m = k\pi$ and $ds = dt$.

$$M_x = \int_C y\rho ds = \int_C kxy ds = k \int_0^\pi (1 + \cos t) \sin t dt = k \left(-\cos t + \frac{1}{2} \sin^2 t \right) \Big|_0^\pi = 2k$$

$$\begin{aligned} M_y &= \int_C x\rho ds = \int_C kx^2 ds = k \int_0^\pi (1 + \cos t)^2 dt = k \int_0^\pi (1 + 2\cos t + \cos^2 t) dt \\ &= k \left(t + 2\sin t + \frac{1}{2}t + \frac{1}{4}\sin 2t \right) \Big|_0^\pi = \frac{3}{2}k\pi \end{aligned}$$

$$\bar{x} = M_y/m = \frac{3k\pi/2}{k\pi} = \frac{3}{2}; \quad \bar{y} = M_x/m = \frac{2k}{k\pi} = \frac{2}{\pi}. \quad \text{The center of mass is } (3/2, 2/\pi).$$

42. On C_1 , $\mathbf{T} = \mathbf{i}$ and $\mathbf{F} \cdot \mathbf{T} = \text{comp}_{\mathbf{T}}\mathbf{F} \approx 1$. On C_2 , $\mathbf{T} = -\mathbf{j}$ and $\mathbf{F} \cdot \mathbf{T} = \text{comp}_{\mathbf{T}}\mathbf{F} \approx 2$. On C_3 , $\mathbf{T} = -\mathbf{i}$ and $\mathbf{F} \cdot \mathbf{T} = \text{comp}_{\mathbf{T}}\mathbf{F} \approx 1.5$. Using the fact that the lengths of C_1 , C_2 , and C_3 are 4, 5, and 5, respectively, we have

$$W = \int_C \mathbf{F} \cdot \mathbf{T} ds = \int_{C_1} \mathbf{F} \cdot \mathbf{T} ds + \int_{C_2} \mathbf{F} \cdot \mathbf{T} ds + \int_{C_3} \mathbf{F} \cdot \mathbf{T} ds \approx 1(4) + 2(5) + 1.5(5) = 21.5 \text{ ft-lb.}$$

EXERCISES 9.9

Independence of Path

1. (a) $P_y = 0 = Q_x$ and the integral is independent of path. $\phi_x = x^2$, $\phi = \frac{1}{3}x^3 + g(y)$,

$$\phi_y = g'(y) = y^2, \quad g(y) = \frac{1}{3}y^3, \quad \phi = \frac{1}{3}x^3 + \frac{1}{3}y^3, \quad \int_{(0,0)}^{(2,2)} x^2 dx + y^2 dy = \frac{1}{3}(x^3 + y^3) \Big|_{(0,0)}^{(2,2)} = \frac{16}{3}$$

(b) Use $y = x$ for $0 \leq x \leq 2$. $\int_{(0,0)}^{(2,2)} x^2 dx + y^2 dy = \int_0^2 (x^2 + x^2) dx = \frac{2}{3}x^3 \Big|_0^2 = \frac{16}{3}$

2. (a) $P_y = 2x = Q_x$ and the integral is independent of path. $\phi_x = 2xy$, $\phi = x^2y + g(y)$,

$$\phi_y = x^2 + g'(y) = x^2, \quad g(y) = 0, \quad \phi = x^2y, \quad \int_{(1,1)}^{(2,4)} 2xy dx + x^2 dy = x^2y \Big|_{(1,1)}^{(2,4)} = 16 - 1 = 15$$

- (b) Use $y = 3x - 2$ for $1 \leq x \leq 2$.

$$\int_{(1,1)}^{(2,4)} 2xy dx + x^2 dy = \int_1^2 [2x(3x - 2) + x^2(3)] dx = \int_1^2 (9x^2 - 4x) dx = (3x^3 - 2x^2) \Big|_1^2 = 15$$

3. (a) $P_y = 2 = Q_x$ and the integral is independent of path. $\phi_x = x + 2y$, $\phi = \frac{1}{2}x^2 + 2xy + g(y)$,

$$\phi_y = 2x + g'(y) = 2x - y, \quad g(y) = -\frac{1}{2}y^2, \quad \phi = \frac{1}{2}x^2 + 2xy - \frac{1}{2}y^2,$$

$$\int_{(1,0)}^{(3,2)} (x + 2y) dx + (2x - y) dy = \left(\frac{1}{2}x^2 + 2xy - \frac{1}{2}y^2 \right) \Big|_{(1,0)}^{(3,2)} = 14$$

- (b) Use $y = x - 1$ for $1 \leq x \leq 3$.

$$\begin{aligned} \int_{(1,0)}^{(3,2)} (x + 2y) dx + (2x - y) dy &= \int_1^3 [x + 2(x - 1) + 2x - (x - 1)] dx \\ &= \int_1^3 (4x - 1) dx = (2x^2 - x) \Big|_1^3 = 14 \end{aligned}$$

4. (a) $P_y = -\cos x \sin y = Q_x$ and the integral is independent of path. $\phi_x = \cos x \cos y$,
 $\phi = \sin x \cos y + g(y)$, $\phi_y = -\sin x \sin y + g'(y) = 1 - \sin x \sin y$, $g(y) = y$, $\phi = \sin x \cos y + y$,

$$\int_{(0,0)}^{(\pi/2,0)} \cos x \cos y dx + (1 - \sin x \sin y) dy = (\sin x \cos y + y) \Big|_{(0,0)}^{(\pi/2,0)} = 1$$

- (b) Use $y = 0$ for $0 \leq x \leq \pi/2$.

$$\int_{(0,0)}^{(\pi/2,0)} \cos x \cos y dx + (1 - \sin x \sin y) dy = \int_0^{\pi/2} \cos x dx = \sin x \Big|_0^{\pi/2} = 1$$

5. (a) $P_y = 1/y^2 = Q_x$ and the integral is independent of path. $\phi_x = -\frac{1}{y}$, $\phi = -\frac{x}{y} + g(y)$, $\phi_y = \frac{x}{y^2} + g'(y) = \frac{x}{y^2}$,

$$g(y) = 0, \quad \phi = -\frac{x}{y}, \quad \int_{(4,1)}^{(4,4)} -\frac{1}{y} dx + \frac{x}{y^2} dy = \left(-\frac{x}{y} \right) \Big|_{(4,1)}^{(4,4)} = 3$$

- (b) Use $x = 4$ for $1 \leq y \leq 4$.

$$\int_{(4,1)}^{(4,4)} -\frac{1}{y} dx + \frac{x}{y^2} dy = \int_1^4 \frac{4}{y^2} dy = -\frac{4}{y} \Big|_1^4 = 3$$

6. (a) $P_y = -xy(x^2 + y^2)^{-3/2} = Q_x$ and the integral is independent of path. $\phi_x = \frac{x}{\sqrt{x^2 + y^2}}$,

$$\phi = \sqrt{x^2 + y^2} + g(y), \quad \phi_y = \frac{y}{\sqrt{x^2 + y^2}} + g'(y) = \frac{y}{\sqrt{x^2 + y^2}}, \quad g(y) = 0, \quad \phi = \sqrt{x^2 + y^2},$$

$$\int_{(1,0)}^{(3,4)} \frac{x dx + y dy}{\sqrt{x^2 + y^2}} = \sqrt{x^2 + y^2} \Big|_{(1,0)}^{(3,4)} = 4$$

- (b) Use $y = 2x - 2$ for $1 \leq x \leq 3$.

$$\int_{(1,0)}^{(3,4)} \frac{x dx + y dy}{\sqrt{x^2 + y^2}} = \int_1^3 \frac{x + (2x - 2)2}{\sqrt{x^2 + (2x - 2)^2}} dx = \int_1^3 \frac{5x - 4}{\sqrt{5x^2 - 8x + 4}} = \sqrt{5x^2 - 8x + 4} \Big|_1^3 = 4$$

7. (a) $P_y = 4xy = Q_x$ and the integral is independent of path. $\phi_x = 2y^2x - 3$, $\phi = x^2y^2 - 3x + g(y)$,
 $\phi_y = 2x^2y + g'(y) = 2x^2y + 4$, $g(y) = 4y$, $\phi = x^2y^2 - 3x + 4y$,

$$\int_{(1,2)}^{(3,6)} (2y^2x - 3) dx + (2yx^2 + 4) dy = (x^2y^2 - 3x + 4y) \Big|_{(1,2)}^{(3,6)} = 330$$

9.9 Independence of Path

(b) Use $y = 2x$ for $1 \leq x \leq 3$.

$$\begin{aligned}\int_{(1,2)}^{(3,6)} (2y^2x - 3) dx + (2yx^2 + 4) dy &= \int_1^3 ([2(2x)^2x - 3] + [2(2x)x^2 + 4]2) dx \\ &= \int_1^3 (16x^3 + 5) dx = (4x^4 + 5x) \Big|_1^3 = 330\end{aligned}$$

8. (a) $P_y = 4 = Q_x$ and the integral is independent of path. $\phi_x = 5x + 4y$, $\phi = \frac{5}{2}x^2 + 4xy + g(y)$,
 $\phi_y = 4x + g'(y) = 4x - 8y^3$, $g(y) = -2y^4$, $\phi = \frac{5}{2}x^2 + 4xy - 2y^4$,

$$\int_{(-1,1)}^{(0,0)} (5x + 4y) dx + (4x - 8y^3) dy = \left(\frac{5}{2}x^2 + 4xy - 2y^4 \right) \Big|_{(-1,1)}^{(0,0)} = \frac{7}{2}$$

(b) Use $y = -x$ for $-1 \leq x \leq 0$.

$$\begin{aligned}\int_{(-1,1)}^{(0,0)} (5x + 4y) dx + (4x - 8y^3) dy &= \int_{-1}^0 [(5x - 4x) + (4x + 8x^3)(-1)] dx \\ &= \int_{-1}^0 (-3x - 8x^3) dx = \left(-\frac{3}{2}x^2 - 2x^4 \right) \Big|_{-1}^0 = \frac{7}{2}\end{aligned}$$

9. (a) $P_y = 3y^2 + 3x^2 = Q_x$ and the integral is independent of path. $\phi_x = y^3 + 3x^2y$,
 $\phi = xy^3 + x^3y + g(y)$, $\phi_y = 3xy^2 + x^3 + g'(y) = x^3 + 3y^2x + 1$, $g(y) = y$, $\phi = xy^3 + x^3y + y$,

$$\int_{(0,0)}^{(2,8)} (y^3 + 3x^2y) dx + (x^3 + 3y^2x + 1) dy = (xy^3 + x^3y + y) \Big|_{(0,0)}^{(2,8)} = 1096$$

(b) Use $y = 4x$ for $0 \leq x \leq 2$.

$$\begin{aligned}\int_{(0,0)}^{(2,8)} (y^3 + 3x^2y) dx + (x^3 + 3y^2x + 1) dy &= \int_0^2 [(64x^3 + 12x^3) + (x^3 + 48x^3 + 1)(4)] dx \\ &= \int_0^2 (272x^3 + 4) dx = (68x^4 + 4x) \Big|_0^2 = 1096\end{aligned}$$

10. (a) $P_y = -xy \cos xy - \sin xy - 20y^3 = Q_x$ and the integral is independent of path.

$$\phi_x = 2x - y \sin xy - 5y^4, \quad \phi = x^2 + \cos xy - 5xy^4 + g(y),$$

$$\phi_y = -x \sin xy - 20xy^3 + g'(y) = -20xy^3 - x \sin xy, \quad g(y) = 0, \quad \phi = x^2 + \cos xy - 5xy^4,$$

$$\int_{(-2,0)}^{(1,0)} (2x - y \sin xy - 5y^4) dx - (20xy^3 + x \sin xy) dy = (x^2 + \cos xy - 5xy^4) \Big|_{(-2,0)}^{(1,0)} = -3$$

(b) Use $y = 0$ for $-2 \leq x \leq 1$.

$$\int_{(-2,0)}^{(1,0)} (2x - y \sin xy - 5y^4) dx - (20xy^3 + x \sin xy) dy = \int_{-2}^1 2x dx = x^2 \Big|_{-2}^1 = -3$$

11. $P_y = 12x^3y^2 = Q_x$ and the vector field is a gradient field. $\phi_x = 4x^3y^3 + 3$, $\phi = x^4y^3 + 3x + g(y)$,
 $\phi_y = 3x^4y^2 + g'(y) = 3x^4y^2 + 1$, $g(y) = y$, $\phi = x^4y^3 + 3x + y$

12. $P_y = 6xy^2 = Q_x$ and the vector field is a gradient field. $\phi_x = 2xy^3$, $\phi = x^2y^3 + g(y)$,
 $\phi_y = 3x^2y^2 + g'(y) = 3x^2y^2 + 3y^2$, $g(y) = y^3$, $\phi = x^2y^3 + y^3$

13. $P_y = -2xy^3 \sin xy^2 + 2y \cos xy^2$, $Q_x = -2xy^3 \cos xy^2 - 2y \sin xy^2$ and the vector field is not a gradient field.

14. $P_y = -4xy(x^2 + y^2 + 1)^{-3} = Q_x$ and the vector field is a gradient field.

$$\phi_x = x(x^2 + y^2 + 1)^{-2}, \quad \phi = -\frac{1}{2}(x^2 + y^2 + 1)^{-1} + g(y), \quad \phi_y = y(x^2 + y^2 + 1)^{-2} + g'(y) = y(x^2 + y^2 + 1)^{-2},$$

$$g(y) = 0, \quad \phi = -\frac{1}{2}(x^2 + y^2 + 1)^{-1}$$

15. $P_y = 1 = Q_x$ and the vector field is a gradient field. $\phi_x = x^3 + y$, $\phi = \frac{1}{4}x^4 + xy + g(y)$, $\phi_y = x + g'(y) = x + y^3$,
 $g(y) = \frac{1}{4}y^4$, $\phi = \frac{1}{4}x^4 + xy + \frac{1}{4}y^4$

16. $P_y = 4e^{2y}$, $Q_x = e^{2y}$ and the vector field is not a gradient field.

17. Since $P_y = -e^{-y} = Q_x$, $\mathbf{F}$ is conservative and $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path. Thus, instead of the given curve we may use the simpler curve C_1 : $y = x$, $0 \leq x \leq 1$. Then

$$\begin{aligned} W &= \int_{C_1} (2x + e^{-y}) dx + (4y - xe^{-y}) dy \\ &= \int_0^1 (2x + e^{-x}) dx + \int_0^1 (4x - xe^{-x}) dx && \boxed{\text{Integration by parts}} \\ &= (x^2 - e^{-x}) \Big|_0^1 + (2x^2 + xe^{-x} + e^{-x}) \Big|_0^1 \\ &= [(1 - e^{-1}) - (-1)] + [(2 + e^{-1} + e^{-1}) - (1)] = 3 + e^{-1}. \end{aligned}$$

18. Since $P_y = -e^{-y} = Q_x$, $\mathbf{F}$ is conservative and $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of the path. Thus, instead of the given curve we may use the simpler curve C_1 : $y = 0$, $-2 \leq -x \leq 2$. Then $dy = 0$ and

$$W = \int_{C_1} (2x + e^{-y}) dx + (4y - xe^{-y}) dy = \int_2^{-2} (2x + 1) dx = (x^2 + x) \Big|_2^{-2} = (4 - 2) - (4 + 2) = -4.$$

19. $P_y = z = Q_x$, $Q_z = x = R_y$, $R_x = y = P_z$, and the integral is independent of path. Parameterize the line segment between the points by $x = 1 + t$, $y = 1 + 3t$, $z = 1 + 7t$, for $0 \leq t \leq 1$. Then $dx = dt$, $dy = 3 dt$, $dz = 7 dt$ and

$$\begin{aligned} \int_{(1,1,1)}^{(2,4,8)} yz dx + xz dy + xy dz &= \int_0^1 [(1 + 3t)(1 + 7t) + (1 + t)(1 + 7t)(3) + (1 + t)(1 + 3t)(7)] dt \\ &= \int_0^1 (11 + 62t + 63t^2) dt = (11t + 31t^2 + 21t^3) \Big|_0^1 = 63. \end{aligned}$$

20. $P_y = 0 = Q_x$, $Q_z = 0 = R_y$, $R_x = 0 = P_z$ and the integral is independent of path. Parameterize the line segment between the points by $x = t$, $y = t$, $z = t$, for $0 \leq t \leq 1$. Then $dx = dy = dz = dt$ and

$$\int_{(0,0,0)}^{(1,1,1)} 2x dx + 3y^2 dy + 4z^3 dz = \int_0^1 (2t + 3t^2 + 4t^3) dt = (t^2 + t^3 + t^4) \Big|_0^1 = 3.$$

21. $P_y = 2x \cos y = Q_x$, $Q_z = 0 = R_y$, $R_x = 3e^{3z} = P_z$, and the integral is independent of path. Integrating $\phi_x = 2x \sin y + e^{3z}$ we find $\phi = x^2 \sin y + xe^{3z} + g(y, z)$. Then $\phi_y = x^2 \cos y + g_y = Q = x^2 \cos y$, so $g_y = 0$, $g(y, z) = h(z)$, and $\phi = x^2 \sin y + xe^{3z} + h(z)$. Now $\phi_z = 3xe^{3z} + h'(z) = R = 3xe^{3z} + 5$, so $h'(z) = 5$ and $h(z) = 5z$. Thus $\phi = x^2 \sin y + xe^{3z} + 5z$ and

$$\begin{aligned} \int_{(1,0,0)}^{(2,\pi/2,1)} (2x \sin y + e^{3z}) dx + x^2 \cos y dy + (3xe^{3z} + 5) dz \\ = (x^2 \sin y + xe^{3z} + 5z) \Big|_{(1,0,0)}^{(2,\pi/2,1)} = [4(1) + 2e^3 + 5] - [0 + 1 + 0] = 8 + 2e^3. \end{aligned}$$

9.9 Independence of Path

22. $P_y = 0 = Q_x$, $Q_z = 0 = R_y$, $R_x = 0 = P_z$, and the integral is independent of path. Parameterize the line segment between the points by $x = 1 + 2t$, $y = 2 + 2t$, $z = 1$, for $0 \leq t \leq 1$. Then $dx = 2 dt$, $dz = 0$ and

$$\begin{aligned} \int_{(1,2,1)}^{(3,4,1)} (2x+1) dx + 3y^2 dy + \frac{1}{z} dz &= \int_0^1 [(2+4t+1)2 + 3(2+2t)^2] dt \\ &= \int_0^1 (24t^2 + 56t + 30) dt = (8t^3 + 28t^2 + 30t) \Big|_0^1 = 66. \end{aligned}$$

23. $P_y = 0 = Q_x$, $Q_z = 0 = R_y$, $R_x = 2e^{2z} = P_z$ and the integral is independent of path. Parameterize the line segment between the points by $x = 1 + t$, $y = 1 + t$, $z = \ln 3$, for $0 \leq t \leq 1$. Then $dx = dy = dt$, $dz = 0$ and

$$\int_{(1,1,\ln 3)}^{(2,2,\ln 3)} e^{2z} dx + 3y^2 dy + 2xe^{2z} dz = \int_0^1 [e^{2\ln 3} + 3(1+t)^2] dt = [9t + (1+t)^3] \Big|_0^1 = 16.$$

24. $P_y = 0 = Q_x$, $Q_z = 2y = R_y$, $R_x = 2x = P_z$ and the integral is independent of path. Parameterize the line segment between the points by $x = -2(1-t)$, $y = 3(1-t)$, $z = 1-t$, for $0 \leq t \leq 1$. Then $dx = 2 dt$, $dy = -3 dt$, $dz = -dt$, and

$$\begin{aligned} \int_{(-2,3,1)}^{(0,0,0)} 2xz dx + 2yz dy + (x^2 + y^2) dz \\ &= \int_0^1 [-4(1-t)^2(2) + 6(1-t)^2(-3) + 4(1-t)^2(-1) + 9(1-t)^2(-1)] dt \\ &= \int_0^1 -39(1-t)^2 dt = 13(1-t)^3 \Big|_0^1 = -13. \end{aligned}$$

25. $P_y = 1 - z \sin x = Q_x$, $Q_z = \cos x = R_y$, $R_x = -y \sin x = P_z$ and the integral is independent of path. Integrating $\phi_x = y - yz \sin x$ we find $\phi = xy + yz \cos x + g(y, z)$. Then $\phi_y = x + z \cos x + g_y(y, z) = Q = x + z \cos x$, so $g_y = 0$, $g(y, z) = h(z)$, and $\phi = xy + yz \cos x + h(z)$. Now $\phi_z = y \cos x + h'(z) = R = y \cos x$, so $h'(z) = 0$ and $\phi = xy + yz \cos x$. Since $\mathbf{r}(0) = 4\mathbf{j}$ and $\mathbf{r}(\pi/2) = \pi\mathbf{i} + \mathbf{j} + 4\mathbf{k}$,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = (xy + yz \cos x) \Big|_{(0,4,0)}^{(\pi,1,4)} = (\pi - 4) - (0 + 0) = \pi - 4.$$

26. $P_y = 0 = Q_x$, $Q_z = 0 = R_y$, $R_x = -e^z = P_z$ and the integral is independent of path. Integrating $\phi_x = 2 - e^z$ we find $\phi = 2x - xe^z + g(y, z)$. Then $\phi_y = g_y = 2y - 1$, so $g(y, z) = y^2 - y + h(z)$ and $\phi = 2x - xe^z + y^2 - y + h(z)$. Now $\phi_z = -xe^z + h'(z) = R = 2 - xe^z$, so $h'(z) = 2$, $h(z) = 2z$, and $\phi = 2x - xe^z + y^2 - y + 2z$. Thus

$$\int_C \mathbf{F} \cdot d\mathbf{r} = (2x - xe^z + y^2 - y + 2z) \Big|_{(-1,1,-1)}^{(2,4,8)} = (4 - 2e^4 + 16 - 4 + 16) - (-2 + e^{-1} + 1 - 1 - 2) = 36 - 2e^4 - e^{-1}.$$

27. Since $P_y = Gm_1m_2(2xy/|\mathbf{r}|^5) = Q_x$, $Q_z = Gm_1m_2(2yz/|\mathbf{r}|^5) = R_y$, and $R_x = Gm_1m_2(2xz/|\mathbf{r}|^5) = P_z$, the force field is conservative.

$$\phi_x = -Gm_1m_2 \frac{x}{(x^2 + y^2 + z^2)^{3/2}}, \quad \phi = Gm_1m_2(x^2 + y^2 + z^2)^{-1/2} + g(y, z),$$

$$\phi_y = -Gm_1m_2 \frac{y}{(x^2 + y^2 + z^2)^{3/2}} + g_y(y, z) = -Gm_1m_2 \frac{y}{(x^2 + y^2 + z^2)^{3/2}}, \quad g(y, z) = h(z),$$

$$\phi = Gm_1m_2(x^2 + y^2 + z^2)^{-1/2} + h(z),$$

$$\phi_z = -Gm_1m_2 \frac{z}{(x^2 + y^2 + z^2)^{3/2}} + h'(z) = -Gm_1m_2 \frac{z}{(x^2 + y^2 + z^2)^{3/2}},$$

$$h(z) = 0, \quad \phi = \frac{Gm_1m_2}{\sqrt{x^2 + y^2 + z^2}} = \frac{Gm_1m_2}{|\mathbf{r}|}$$

28. Since $P_y = 24xy^2z = Q_x$, $Q_z = 12x^2y^2 = R_y$, and $R_x = 8xy^3 = P_z$, $\mathbf{F}$ is conservative. Thus, the work done between two points is independent of the path. From $\phi_x = 8xy^3z$ we obtain $\phi = 4x^2y^3z$ which is a potential function for $\mathbf{F}$. Then

$$W = \int_{(2,0,0)}^{(1,\sqrt{3},\pi/3)} \mathbf{F} \cdot d\mathbf{r} = 4x^2y^3z \Big|_{(2,0,0)}^{(1,\sqrt{3},\pi/3)} = 4\sqrt{3}\pi \quad \text{and} \quad W = \int_{(2,0,0)}^{(0,2,\pi/2)} \mathbf{F} \cdot d\mathbf{r} = 0.$$

29. Since $\mathbf{F}$ is conservative, $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{-C_2} \mathbf{F} \cdot d\mathbf{r}$. Then, since the simply closed curve C is composed of C_1 and C_2 ,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} - \int_{-C_2} \mathbf{F} \cdot d\mathbf{r} = 0.$$



30. From $\mathbf{F} = (x^2 + y^2)^{n/2}(\mathbf{i} + y\mathbf{j})$ we obtain $P_y = nxy(x^2 + y^2)^{n/2-1} = Q_x$, so that $\mathbf{F}$ is conservative. From $\phi_x = x(x^2 + y^2)^{n/2}$ we obtain the potential function $\phi = (x^2 + y^2)^{(n+2)/2}/(n+2)$. Then

$$W = \int_{(x_1,y_1)}^{(x_2,y_2)} \mathbf{F} \cdot d\mathbf{r} = \left(\frac{(x^2 + y^2)^{(n+2)/2}}{n+2} \right) \Big|_{(x_1,y_1)}^{(x_2,y_2)} = \frac{1}{n+2} \left[(x_2^2 + y_2^2)^{(n+2)/2} - (x_1^2 + y_1^2)^{(n+2)/2} \right].$$

31. From the solution to Problem 39 in Exercises 9.8, $\frac{d\mathbf{v}}{dt} \cdot \frac{d\mathbf{r}}{dt} = \frac{d\mathbf{v}}{dt} \cdot \mathbf{v} = \frac{1}{2} \frac{d}{dt} v^2$. Then, using $\frac{dp}{dt} = \frac{\partial p}{\partial x} \frac{dx}{dt} + \frac{\partial p}{\partial y} \frac{dy}{dt} = \nabla p \cdot \frac{d\mathbf{r}}{dt}$, we have

$$\begin{aligned} \int m \frac{d\mathbf{v}}{dt} \cdot \frac{d\mathbf{r}}{dt} dt + \int \nabla p \cdot \frac{d\mathbf{r}}{dt} dt &= \int 0 dt \\ \frac{1}{2} m \int \frac{d}{dt} v^2 dt + \int \frac{dp}{dt} dt &= \text{constant} \\ \frac{1}{2} m v^2 + p &= \text{constant}. \end{aligned}$$

32. By Problem 31, the sum of kinetic and potential energies in a conservative force field is constant. That is, it is independent of points A and B , so $p(B) + K(B) = p(A) + K(A)$.

EXERCISES 9.10

Double Integrals

- $\int_{-1}^3 (6xy - 5e^y) dx = (3x^2y - 5xe^y) \Big|_{-1}^3 = (27y - 15e^y) - (3y + 5e^y) = 24y - 20e^y$
- $\int_1^2 \tan xy dy = \frac{1}{x} \ln |\sec xy| \Big|_1^2 = \frac{1}{x} \ln |\sec 2x - \sec x|$
- $\int_1^{3x} x^3 e^{xy} dy = x^2 e^{xy} \Big|_1^{3x} = x^2 (e^{3x^2} - e^x)$
- $\int_{\sqrt{y}}^{y^3} (8x^3y - 4xy^2) dx = (2x^4y - 2x^2y^2) \Big|_{\sqrt{y}}^{y^3} = (2y^{13} - 2y^8) - (2y^3 - 2y^3) = 2y^{13} - 2y^8$

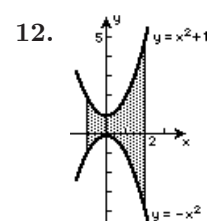
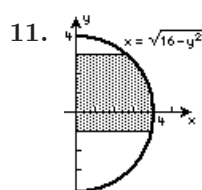
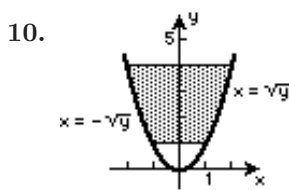
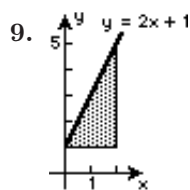
9.10 Double Integrals

$$5. \int_0^{2x} \frac{xy}{x^2 + y^2} dy = \frac{x}{2} \ln(x^2 + y^2) \Big|_0^{2x} = \frac{x}{2} [\ln(x^2 + 4x^2) - \ln x^2] = \frac{x}{2} \ln 5$$

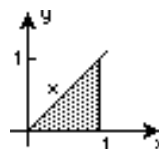
$$6. \int_{x^3}^x e^{2y/x} dy = \frac{x}{2} e^{2y/x} \Big|_{x^3}^x = \frac{x}{2} (e^{2x/x} - e^{2x^3/x}) = \frac{x}{2} (e^2 - e^{2x^2})$$

$$7. \int_{\tan y}^{\sec y} (2x + \cos y) dx = (x^2 + x \cos y) \Big|_{\tan y}^{\sec y} = \sec^2 y + \sec y \cos y - \tan^2 y - \tan y \cos y \\ = \sec^2 y + 1 - \tan^2 y - \sin y = 2 - \sin y$$

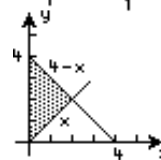
$$8. \int_{\sqrt{y}}^1 y \ln x dx \quad \boxed{\text{Integration by parts}} \\ = y(x \ln x - x) \Big|_{\sqrt{y}}^1 = y(0 - 1) - y(\sqrt{y} \ln \sqrt{y} - \sqrt{y}) = -y - y\sqrt{y} \left(\frac{1}{2} \ln y - 1 \right)$$



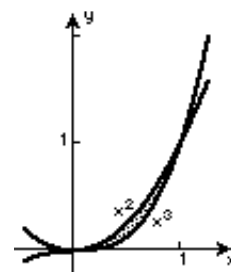
$$13. \iint_R x^3 y^2 dA = \int_0^1 \int_0^x x^3 y^2 dy dx = \int_0^1 \frac{1}{3} x^3 y^3 \Big|_0^x dx = \frac{1}{3} \int_0^1 x^6 dx = \frac{1}{21} x^7 \Big|_0^1 = \frac{1}{21}$$



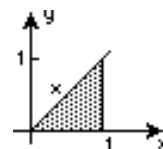
$$14. \iint_R (x + 1) dA = \int_0^2 \int_x^{4-x} (x + 1) dy dx = \int_0^2 (xy + y) \Big|_x^{4-x} dx \\ = \int_0^2 [(4x - x^2 + 4 - x) - (x^2 + x)] dx = \int_0^2 (2x - 2x^2 + 4) dx \\ = \left(x^2 - \frac{2}{3} x^3 + 4x \right) \Big|_0^2 = \frac{20}{3}$$



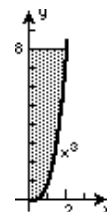
$$15. \iint_R (2x + 4y + 1) dA = \int_0^1 \int_{x^3}^{x^2} (2x + 4y + 1) dy dx = \int_0^1 (2xy + 2y^2 + y) \Big|_{x^3}^{x^2} dx \\ = \int_0^1 [(2x^3 + 2x^4 + x^2) - (2x^4 + 2x^6 + x^3)] dx \\ = \int_0^1 (x^3 + x^2 - 2x^6) dx = \left(\frac{1}{4} x^4 + \frac{1}{3} x^3 - \frac{2}{7} x^7 \right) \Big|_0^1 \\ = \frac{1}{4} + \frac{1}{3} - \frac{2}{7} = \frac{25}{84}$$



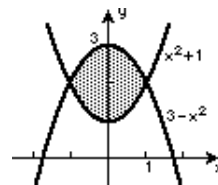
$$16. \iint_R x e^y dA = \int_0^1 \int_0^x x e^y dy dx = \int_0^1 x e^y \Big|_0^x dx = \int_0^1 (x e^x - x) dx \\ \boxed{\text{Integration by parts}} \\ = \left(x e^x - e^x - \frac{1}{2} x^2 \right) \Big|_0^1 = \left(e - e - \frac{1}{2} \right) - (-1) = \frac{1}{2}$$



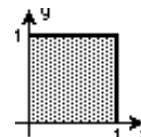
$$17. \iint_R 2xy dA = \int_0^2 \int_{x^3}^8 2xy dy dx = \int_0^2 xy^2 \Big|_{x^3}^8 dx = \int_0^2 (64x - x^7) dx = \left(32x^2 - \frac{1}{8} x^8 \right) \Big|_0^2 = 96$$



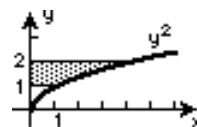
$$\begin{aligned}
 18. \iint_R \frac{x}{\sqrt{y}} dA &= \int_{-1}^1 \int_{x^2+1}^{3-x^2} xy^{-1/2} dy dx = \int_{-1}^1 2x\sqrt{y} \Big|_{x^2+1}^{3-x^2} dx \\
 &= 2 \int_{-1}^1 (x\sqrt{3-x^2} - x\sqrt{x^2+1}) dx \\
 &= 2 \left[-\frac{1}{3}(3-x^2)^{3/2} - \frac{1}{3}(x^2+1)^{3/2} \right]_{-1}^1 \\
 &= -\frac{2}{3}[(2^{3/2} + 2^{3/2}) - (2^{3/2} + 2^{3/2})] = 0
 \end{aligned}$$



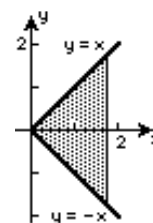
$$\begin{aligned}
 19. \iint_R \frac{y}{1+xy} dA &= \int_0^1 \int_0^1 \frac{y}{1+xy} dx dy = \int_0^1 \ln(1+xy) \Big|_0^1 dy = \int_0^1 \ln(1+y) dy \\
 &= [(1+y)\ln(1+y) - (1+y)] \Big|_0^1 = (2\ln 2 - 2) - (-1) = 2\ln 2 - 1
 \end{aligned}$$



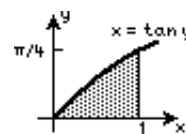
$$\begin{aligned}
 20. \iint_R \sin \frac{\pi x}{y} dA &= \int_1^2 \int_0^{y^2} \sin \frac{\pi x}{y} dx dy = \int_1^2 \left(-\frac{y}{\pi} \cos \frac{\pi x}{y} \right) \Big|_0^{y^2} dy \\
 &= \int_1^2 \left(-\frac{y}{\pi} \cos \pi y + \frac{y}{\pi} \right) dy \quad \boxed{\text{Integration by parts}} \\
 &= \left(-\frac{y}{\pi^2} \sin \pi y - \frac{1}{\pi^3} \cos \pi y + \frac{y^2}{2\pi} \right) \Big|_1^2 = \left(-\frac{1}{\pi^3} + \frac{2}{\pi} \right) - \left(\frac{1}{\pi^3} + \frac{1}{2\pi} \right) \\
 &= \frac{3\pi^2 - 4}{2\pi^3}
 \end{aligned}$$



$$\begin{aligned}
 21. \iint_R \sqrt{x^2+1} dA &= \int_0^{\sqrt{3}} \int_{-x}^x \sqrt{x^2+1} dy dx = \int_0^{\sqrt{3}} y\sqrt{x^2+1} \Big|_{-x}^x dx \\
 &= \int_0^{\sqrt{3}} (x\sqrt{x^2+1} + x\sqrt{x^2+1}) dx = \int_0^{\sqrt{3}} 2x\sqrt{x^2+1} dx \\
 &= \frac{2}{3}(x^2+1)^{3/2} \Big|_0^{\sqrt{3}} = \frac{2}{3}(4^{3/2} - 1^{3/2}) = \frac{14}{3}
 \end{aligned}$$



$$\begin{aligned}
 22. \iint_R x dA &= \int_0^{\pi/4} \int_{\tan y}^1 x dx dy = \int_0^{\pi/4} \frac{1}{2}x^2 \Big|_{\tan y}^1 dy = \frac{1}{2} \int_0^{\pi/4} (1 - \tan^2 y) dy \\
 &= \frac{1}{2} \int_0^{\pi/4} (2 - \sec^2 y) dy = \frac{1}{2}(2y - \tan y) \Big|_0^{\pi/4} = \frac{1}{2} \left(\frac{\pi}{2} - 1 \right) = \frac{\pi}{4} - \frac{1}{2}
 \end{aligned}$$



23. The correct integral is (c).

$$\begin{aligned}
 V &= 2 \int_{-2}^2 \int_0^{\sqrt{4-y^2}} (4-y) dx dy = 2 \int_{-2}^2 (4-y)x \Big|_0^{\sqrt{4-y^2}} dy = 2 \int_{-2}^2 (4-y)\sqrt{4-y^2} dy \\
 &= 2 \left[2y\sqrt{4-y^2} + 8 \sin^{-1} \frac{y}{2} + \frac{1}{3}(4-y^2)^{3/2} \right]_{-2}^2 = 2(4\pi - (-4\pi)) = 16\pi
 \end{aligned}$$

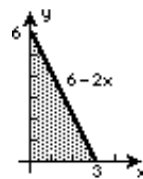
24. The correct integral is (b).

$$\begin{aligned}
 V &= 8 \int_0^2 \int_0^{\sqrt{4-y^2}} (4-y^2)^{1/2} dx dy = 8 \int_0^2 (4-y^2)^{1/2} x \Big|_0^{\sqrt{4-y^2}} dy = 8 \int_0^2 (4-y^2) dy \\
 &= 8 \left(4y - \frac{1}{3}y^3 \right) \Big|_0^2 = \frac{128}{3}
 \end{aligned}$$

9.10 Double Integrals

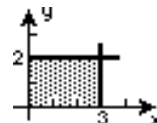
25. Setting $z = 0$ we have $y = 6 - 2x$.

$$\begin{aligned} V &= \int_0^3 \int_0^{6-2x} (6-2x-y) dy dx = \int_0^3 \left(6y - 2xy - \frac{1}{2}y^2 \right) \Big|_0^{6-2x} dx \\ &= \int_0^3 [6(6-2x) - 2x(6-2x) - \frac{1}{2}(6-2x)^2] dx = \int_0^3 (18 - 12x + 2x^2) dx \\ &= \left(18x - 6x^2 + \frac{2}{3}x^3 \right) \Big|_0^3 = 18 \end{aligned}$$



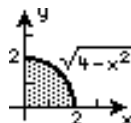
26. Setting $z = 0$ we have $y = \pm 2$.

$$V = \int_0^3 \int_0^2 (4-y^2) dy dx = \int_0^3 \left(4y - \frac{1}{3}y^3 \right) \Big|_0^2 dx = \int_0^3 \frac{16}{3} dx = 16$$



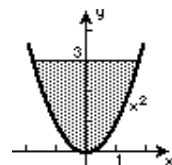
27. Solving for z , we have $x = 2 - \frac{1}{2}x + \frac{1}{2}y$. Setting $z = 0$, we see that this surface (plane) intersects the xy -plane in the line $y = x - 4$. since $z(0, 0) = 2 > 0$, the surface lies above the xy -plane over the quarter-circular region.

$$\begin{aligned} V &= \int_0^2 \int_0^{\sqrt{4-x^2}} \left(2 - \frac{1}{2}x + \frac{1}{2}y \right) dy dx = \int_0^2 \left(2y - \frac{1}{2}xy + \frac{1}{4}y^2 \right) \Big|_0^{\sqrt{4-x^2}} dx \\ &= \int_0^2 \left(2\sqrt{4-x^2} - \frac{1}{2}x\sqrt{4-x^2} + 1 - \frac{1}{4}x^2 \right) dx = \left[x\sqrt{4-x^2} + 4\sin^{-1} \frac{x}{2} + \frac{1}{6}(4-x^2)^{3/2} + x - \frac{1}{12}x^3 \right] \Big|_0^2 \\ &= \left(2\pi + 2 - \frac{2}{3} \right) - \frac{4}{3} = 2\pi \end{aligned}$$



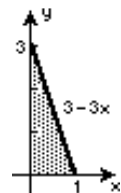
28. Setting $z = 0$ we have $y = 3$. Using symmetry,

$$\begin{aligned} V &= 2 \int_0^{\sqrt{3}} \int_{x^2}^3 (3-y) dy dx = 2 \int_0^{\sqrt{3}} \left(3y - \frac{1}{2}y^2 \right) \Big|_{x^2}^3 dx = 2 \int_0^{\sqrt{3}} \left(\frac{9}{2} - 3x^2 + \frac{1}{2}x^4 \right) dx \\ &= 2 \left(\frac{9}{2}x - x^3 + \frac{1}{10}x^5 \right) \Big|_0^{\sqrt{3}} = 2 \left(\frac{9}{2}\sqrt{3} - 3\sqrt{3} + \frac{9}{10}\sqrt{3} \right) = \frac{24\sqrt{3}}{5} \end{aligned}$$



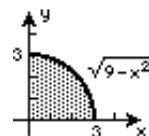
29. Note that $z = 1 + x^2 + y^2$ is always positive. Then

$$\begin{aligned} V &= \int_0^1 \int_0^{3-3x} (1+x^2+y^2) dy dx = \int_0^1 \left(y + x^2y + \frac{1}{3}y^3 \right) \Big|_0^{3-3x} dx \\ &= \int_0^1 [(3-3x) + x^2(3-3x) + 9(1-x)^3] dx = \int_0^1 (12-30x+30x^2-12x^3) dx \\ &= (12x-15x^2+10x^3-3x^4) \Big|_0^1 = 4. \end{aligned}$$



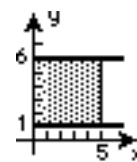
30. In the first octant, $z = x + y$ is nonnegative. Then

$$\begin{aligned} V &= \int_0^3 \int_0^{\sqrt{9-x^2}} (x+y) dy dx = \int_0^3 \left(xy + \frac{1}{2}y^2 \right) \Big|_0^{\sqrt{9-x^2}} dx \\ &= \int_0^3 \left(x\sqrt{9-x^2} + \frac{9}{2} - \frac{1}{2}x^2 \right) dx = \left[-\frac{1}{3}(9-x^2)^{3/2} + \frac{9}{2}x - \frac{1}{6}x^3 \right] \Big|_0^3 = \left(\frac{27}{2} - \frac{9}{2} \right) - (-9) = 18. \end{aligned}$$



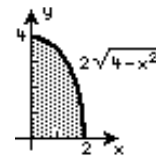
31. In the first octant $z = 6/y$ is positive. Then

$$V = \int_1^6 \int_0^5 \frac{6}{y} dx dy = \int_1^6 \frac{6x}{y} \bigg|_0^5 dy = 30 \int_1^6 \frac{dy}{y} = 30 \ln y \bigg|_1^6 = 30 \ln 6.$$



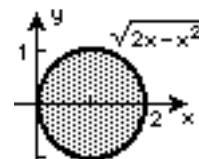
32. Setting $z = 0$, we have $x^2/4 + y^2/16 = 1$. Using symmetry,

$$\begin{aligned} V &= 4 \int_0^2 \int_0^{2\sqrt{4-x^2}} \left(4 - x^2 - \frac{1}{4}y^2 \right) dy dx = 4 \int_0^2 \left(4y - x^2y - \frac{1}{12}y^3 \right) \bigg|_0^{2\sqrt{4-x^2}} dx \\ &= 4 \int_0^2 \left[8\sqrt{4-x^2} - 2x^2\sqrt{4-x^2} - \frac{2}{3}(4-x^2)^{3/2} \right] dx \quad \boxed{\text{Trig substitution}} \\ &= 4 \left[4x\sqrt{4-x^2} + 16\sin^{-1} \frac{x}{2} - \frac{1}{4}x(2x^2-4)\sqrt{4-x^2} - 4\sin^{-1} \frac{x}{2} + \frac{1}{12}x(2x^2-20)\sqrt{4-x^2} - 4\sin \frac{x}{2} \right] \bigg|_0^2 \\ &= 4 \left(\frac{16\pi}{2} - \frac{4\pi}{2} - \frac{4\pi}{2} \right) - (0) = 16\pi. \end{aligned}$$



33. Note that $z = 4 - y^2$ is positive for $|y| \leq 1$. Using symmetry,

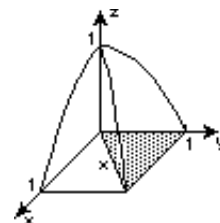
$$\begin{aligned} V &= 2 \int_0^2 \int_0^{\sqrt{2x-x^2}} (4 - y^2) dy dx = 2 \int_0^2 \left(4y - \frac{1}{3}y^3 \right) \bigg|_0^{\sqrt{2x-x^2}} dx \\ &= 2 \int_0^2 \left[4\sqrt{2x-x^2} - \frac{1}{3}(2x-x^2)\sqrt{2x-x^2} \right] dx \\ &= 2 \int_0^2 \left(4\sqrt{1-(x-1)^2} - \frac{1}{3}[1-(x-1)^2]\sqrt{1-(x-1)^2} \right) dx \quad \boxed{u = x-1, du = dx} \\ &= 2 \int_{-1}^1 \left[4\sqrt{1-u^2} - \frac{1}{3}(1-u^2)\sqrt{1-u^2} \right] du = 2 \int_{-1}^1 \left(\frac{11}{3}\sqrt{1-u^2} + \frac{1}{3}u^2\sqrt{1-u^2} \right) du \\ &\quad \boxed{\text{Trig substitution}} \\ &= 2 \left[\frac{11}{6}u\sqrt{1-u^2} + \frac{11}{6}\sin u + \frac{1}{24}x(2x^2-1)\sqrt{1-u^2} + \frac{1}{24}\sin^{-1} u \right] \bigg|_{-1}^1 \\ &= 2 \left[\left(\frac{11}{6} \frac{\pi}{2} + \frac{1}{24} \frac{\pi}{2} \right) - \left(-\frac{11}{6} \frac{\pi}{2} - \frac{1}{24} \frac{\pi}{2} \right) \right] = \frac{15\pi}{4}. \end{aligned}$$



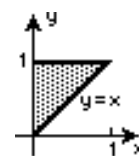
34. From $z = 1 - x^2$ and $z = 1 - y^2$ we have $1 - x^2 = 1 - y^2$ or $y = x$ (in the first octant).

Thus, the surfaces intersect in the plane $y = x$. Using symmetry,

$$\begin{aligned} V &= 2 \int_0^1 \int_x^1 (1 - y^2) dy dx = 2 \int_0^1 \left(y - \frac{1}{3}y^3 \right) \bigg|_x^1 dx = 2 \int_0^1 \left(\frac{2}{3} - x + \frac{1}{3}x^3 \right) dx \\ &= 2 \left(\frac{2}{3}x - \frac{1}{2}x^2 + \frac{1}{12}x^4 \right) \bigg|_0^1 = \frac{1}{2}. \end{aligned}$$

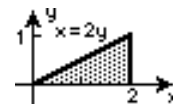


35.
$$\begin{aligned} \int_0^1 \int_x^1 x^2 \sqrt{1+y^4} dy dx &= \int_0^1 \int_0^y x^2 \sqrt{1+y^4} dx dy = \int_0^1 \frac{1}{3}x^3 \sqrt{1+y^4} \bigg|_0^y dy \\ &= \frac{1}{3} \int_0^1 y^3 \sqrt{1+y^4} dy = \frac{1}{3} \left[\frac{1}{6}(1+y^4)^{3/2} \right] \bigg|_0^1 = \frac{1}{18}(2\sqrt{2}-1) \end{aligned}$$

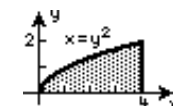


9.10 Double Integrals

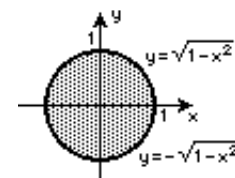
$$\begin{aligned}
 36. \quad \int_0^1 \int_{2y}^2 e^{-y/x} dx dy &= \int_0^2 \int_0^{x/2} e^{-y/x} dy dx = \int_0^2 -xe^{-y/x} \Big|_0^{x/2} = \int_0^2 (-xe^{-1/2} + x) dx \\
 &= \int_0^2 (1 - e^{-1/2})x dx = \frac{1}{2}(1 - e^{-1/2})x^2 \Big|_0^2 = 2(1 - e^{-1/2})
 \end{aligned}$$



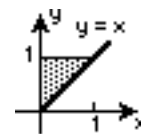
$$\begin{aligned}
 37. \quad \int_0^2 \int_{y^2}^4 \cos x^{3/2} dx dy &= \int_0^4 \int_0^{\sqrt{x}} \cos x^{3/2} dy dx = \int_0^4 y \cos x^{3/2} \Big|_0^{\sqrt{x}} dx \\
 &= \int_0^4 \sqrt{x} \cos x^{3/2} dx = \frac{2}{3} \sin x^{3/2} \Big|_0^4 = \frac{2}{3} \sin 8
 \end{aligned}$$



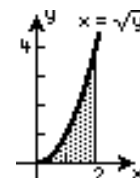
$$\begin{aligned}
 38. \quad \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} x \sqrt{1-x^2-y^2} dy dx &= \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} x \sqrt{1-x^2-y^2} dx dy \\
 &= \int_{-1}^1 \left[-\frac{1}{3}(1-x^2-y^2)^{3/2} \right] \Big|_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy = -\frac{1}{3} \int_{-1}^1 (0-0) dy = 0
 \end{aligned}$$



$$\begin{aligned}
 39. \quad \int_0^1 \int_x^1 \frac{1}{1+y^4} dy dx &= \int_0^1 \int_0^y \frac{1}{1+y^4} dx dy = \int_0^1 \frac{x}{1+y^4} \Big|_0^y dy = \int_0^1 \frac{y}{1+y^4} dy \\
 &= \frac{1}{2} \tan^{-1} y^2 \Big|_0^1 = \frac{\pi}{8}
 \end{aligned}$$



$$\begin{aligned}
 40. \quad \int_0^4 \int_{\sqrt{y}}^2 \sqrt{x^3+1} dx dy &= \int_0^2 \int_0^{x^2} \sqrt{x^3+1} dy dx = \int_0^2 y \sqrt{x^3+1} \Big|_0^{x^2} dx \\
 &= \int_0^2 x^2 \sqrt{x^3+1} dx = \frac{2}{9} (x^3+1)^{3/2} \Big|_0^2 = \frac{2}{9} (9^{3/2} - 1^{3/2}) = \frac{52}{9}
 \end{aligned}$$

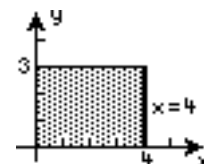


$$41. \quad m = \int_0^3 \int_0^4 xy dx dy = \int_0^3 \frac{1}{2} x^2 y \Big|_0^4 dy = \int_0^3 8y dy = 4y^2 \Big|_0^3 = 36$$

$$M_y = \int_0^3 \int_0^4 x^2 y dx dy = \int_0^3 \frac{1}{3} x^3 y \Big|_0^4 dy = \int_0^3 \frac{64}{3} y dy = \frac{32}{3} y^2 \Big|_0^3 = 96$$

$$M_x = \int_0^3 \int_0^4 xy^2 dx dy = \int_0^3 \frac{1}{2} x^2 y^2 \Big|_0^4 dy = \int_0^3 8y^2 dy = \frac{8}{3} y^3 \Big|_0^3 = 72$$

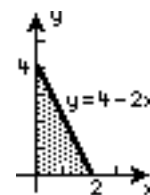
$\bar{x} = M_y/m = 96/36 = 8/3$; $\bar{y} = M_x/m = 72/36 = 2$. The center of mass is $(8/3, 2)$.



$$\begin{aligned}
 42. \quad m &= \int_0^2 \int_0^{4-2x} x^2 dy dx = \int_0^2 x^2 y \Big|_0^{4-2x} dx = \int_0^2 x^2 (4-2x) dx \\
 &= \int_0^2 (4x^2 - 2x^3) dx = \left(\frac{4}{3} x^3 - \frac{1}{2} x^4 \right) \Big|_0^2 = \frac{32}{3} - 8 = \frac{8}{3}
 \end{aligned}$$

$$\begin{aligned}
 M_y &= \int_0^2 \int_0^{4-2x} x^3 dy dx = \int_0^2 x^3 y \Big|_0^{4-2x} dx = \int_0^2 x^3 (4-2x) dx = \int_0^2 (4x^3 - 2x^4) dx \\
 &= \left(x^4 - \frac{2}{5} x^5 \right) \Big|_0^2 = 16 - \frac{64}{5} = \frac{16}{5}
 \end{aligned}$$

$$\begin{aligned}
 M_x &= \int_0^2 \int_0^{4-2x} x^2 y dy dx = \int_0^2 \frac{1}{2} x^2 y^2 \Big|_0^{4-2x} dx = \frac{1}{2} \int_0^2 x^2 (4-2x)^2 dx = \frac{1}{2} \int_0^2 (16x^2 - 16x^3 + 4x^4) dx \\
 &= 2 \int_0^2 (4x^2 - 4x^3 + x^4) dx = 2 \left(\frac{4}{3} x^3 - x^4 + \frac{1}{5} x^5 \right) \Big|_0^2 = 2 \left(\frac{32}{3} - 16 + \frac{32}{5} \right) = \frac{32}{15}
 \end{aligned}$$



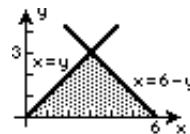
$\bar{x} = M_y/m = \frac{16/5}{8/3} = 6/5$; $\bar{y} = M_x/m = \frac{32/15}{8/3} = 4/5$. The center of mass is $(6/5, 4/5)$.

43. Since both the region and ρ are symmetric with respect to the line $x = 3$, $\bar{x} = 3$.

$$\begin{aligned} m &= \int_0^3 \int_y^{6-y} 2y \, dx \, dy = \int_0^3 2xy \Big|_y^{6-y} \, dy = \int_0^3 2y(6-y-y) \, dy = \int_0^3 (12y - 4y^2) \, dy \\ &= \left(6y^2 - \frac{4}{3}y^3 \right) \Big|_0^3 = 18 \end{aligned}$$

$$\begin{aligned} M_x &= \int_0^3 \int_y^{6-y} 2y^2 \, dx \, dy = \int_0^3 2xy^2 \Big|_y^{6-y} \, dy = \int_0^3 2y^2(6-y-y) \, dy = \int_0^3 (12y^2 - 4y^3) \, dy \\ &= (4y^3 - y^4) \Big|_0^3 = 27 \end{aligned}$$

$\bar{y} = M_x/m = 27/18 = 3/2$. The center of mass is $(3, 3/2)$.

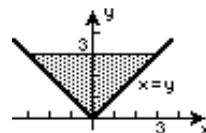


44. Since both the region and ρ are symmetric with respect to the y -axis, $\bar{x} = 0$. Using symmetry,

$$\begin{aligned} m &= \int_0^3 \int_0^y (x^2 + y^2) \, dx \, dy = \int_0^3 \left(\frac{1}{3}x^3 + xy^2 \right) \Big|_0^y \, dy = \int_0^3 \left(\frac{1}{3}y^3 + y^3 \right) \, dy \\ &= \frac{4}{3} \int_0^3 y^3 \, dy = \frac{1}{3}y^4 \Big|_0^3 = 27. \end{aligned}$$

$$\begin{aligned} M_x &= \int_0^3 \int_0^y (x^2y + y^3) \, dx \, dy = \int_0^3 \left(\frac{1}{3}x^3y + xy^3 \right) \Big|_0^y \, dy = \int_0^3 \left(\frac{1}{3}y^4 + y^4 \right) \, dy = \frac{4}{3} \int_0^3 y^4 \, dy \\ &= \frac{4}{15}y^5 \Big|_0^3 = \frac{324}{5} \end{aligned}$$

$\bar{y} = M_x/m = \frac{324/5}{27} = 12/5$. The center of mass is $(0, 12/5)$.

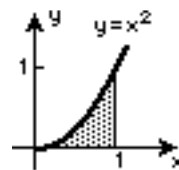


45.
$$\begin{aligned} m &= \int_0^1 \int_0^{x^2} (x+y) \, dy \, dx = \int_0^1 \left(xy + \frac{1}{2}y^2 \right) \Big|_0^{x^2} \, dx = \int_0^1 \left(x^3 + \frac{1}{2}x^4 \right) \, dx \\ &= \left(\frac{1}{4}x^4 + \frac{1}{10}x^5 \right) \Big|_0^1 = \frac{7}{20} \end{aligned}$$

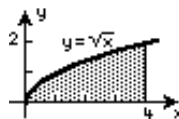
$$\begin{aligned} M_y &= \int_0^1 \int_0^{x^2} (x^2 + xy) \, dy \, dx = \int_0^1 \left(x^2y + \frac{1}{2}xy^2 \right) \Big|_0^{x^2} \, dx = \int_0^1 \left(x^4 + \frac{1}{2}x^5 \right) \, dx \\ &= \left(\frac{1}{5}x^5 + \frac{1}{12}x^6 \right) \Big|_0^1 = \frac{17}{60} \end{aligned}$$

$$M_x = \int_0^1 \int_0^{x^2} (xy + y^2) \, dy \, dx = \int_0^1 \left(\frac{1}{2}xy^2 + \frac{1}{3}y^3 \right) \Big|_0^{x^2} \, dx = \int_0^1 \left(\frac{1}{2}x^5 + \frac{1}{3}x^6 \right) \, dx = \left(\frac{1}{12}x^6 + \frac{1}{21}x^7 \right) \Big|_0^1 = \frac{11}{84}$$

$\bar{x} = M_y/m = \frac{17/60}{7/20} = 17/21$; $\bar{y} = M_x/m = \frac{11/84}{7/20} = 55/147$. The center of mass is $(17/21, 55/147)$.



46.
$$\begin{aligned} m &= \int_0^4 \int_0^{\sqrt{x}} (y+5) \, dy \, dx = \int_0^4 \left(\frac{1}{2}y^2 + 5y \right) \Big|_0^{\sqrt{x}} \, dx = \int_0^4 \left(\frac{1}{2}x + 5\sqrt{x} \right) \, dx \\ &= \left(\frac{1}{4}x^2 + \frac{10}{3}x^{3/2} \right) \Big|_0^4 = \frac{92}{3} \end{aligned}$$



9.10 Double Integrals

$$\begin{aligned} M_y &= \int_0^4 \int_0^{\sqrt{x}} (xy + 5x) dy dx = \int_0^4 \left(\frac{1}{2}xy^2 + 5xy \right) \Big|_0^{\sqrt{x}} dx = \int_0^4 \left(\frac{1}{2}x^2 + 5x^{3/2} \right) dx \\ &= \left(\frac{1}{6}x^3 + 2x^{5/2} \right) \Big|_0^4 = \frac{224}{3} \end{aligned}$$

$$\begin{aligned} M_x &= \int_0^4 \int_0^{\sqrt{x}} (y^2 + 5y) dy dx = \int_0^4 \left(\frac{1}{3}y^3 + \frac{5}{2}y^2 \right) \Big|_0^{\sqrt{x}} dx = \int_0^4 \left(\frac{1}{3}x^{3/2} + \frac{5}{2}x \right) dx \\ &= \left(\frac{2}{15}x^{5/2} + \frac{5}{4}x^2 \right) \Big|_0^4 = \frac{364}{15} \end{aligned}$$

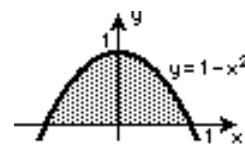
$$\bar{x} = M_y/m = \frac{224/3}{92/3} = 56/23; \quad \bar{y} = M_x/m = \frac{364/15}{92/3} = 91/115. \text{ The center of mass is } (56/23, 91/115).$$

47. The density is $\rho = ky$. Since both the region and ρ are symmetric with respect to the y -axis, $\bar{x} = 0$. Using symmetry,

$$\begin{aligned} m &= 2 \int_0^1 \int_0^{1-x^2} ky dy dx = 2k \int_0^1 \frac{1}{2}y^2 \Big|_0^{1-x^2} dx = k \int_0^1 (1-x^2)^2 dx \\ &= k \int_0^1 (1-2x^2+x^4) dx = k \left(x - \frac{2}{3}x^3 + \frac{1}{5}x^5 \right) \Big|_0^1 = k \left(1 - \frac{2}{3} + \frac{1}{5} \right) = \frac{8}{15}k \end{aligned}$$

$$\begin{aligned} M_x &= 2 \int_0^1 \int_0^{1-x^2} ky^2 dy dx = 2k \int_0^1 \frac{1}{3}y^3 \Big|_0^{1-x^2} dx = \frac{2}{3}k \int_0^1 (1-x^2)^3 dx = \frac{2}{3}k \int_0^1 (1-3x^2+3x^4-x^6) dx \\ &= \frac{2}{3}k \left(x - x^3 + \frac{3}{5}x^5 - \frac{1}{7}x^7 \right) \Big|_0^1 = \frac{2}{3}k \left(1 - 1 + \frac{3}{5} - \frac{1}{7} \right) = \frac{32}{105}k \end{aligned}$$

$$\bar{y} = M_x/m = \frac{32k/105}{8k/15} = 4/7. \text{ The center of mass is } (0, 4/7).$$



48. The density is $\rho = kx$.

$$m = \int_0^\pi \int_0^{\sin x} kx dy dx = \int_0^\pi kxy \Big|_0^{\sin x} dx = \int_0^\pi kx \sin x dx$$

Integration by parts

$$= k(\sin x - x \cos x) \Big|_0^\pi = k\pi$$

$$M_y = \int_0^\pi \int_0^{\sin x} kx^2 dy dx = \int_0^\pi kx^2 y \Big|_0^{\sin x} dx = \int_0^\pi kx^2 \sin x dx$$

Integration by parts

$$= k(-x^2 \cos x + 2 \cos x + 2x \sin x) \Big|_0^\pi = k[(\pi^2 - 2) - 2] = k(\pi^2 - 4)$$

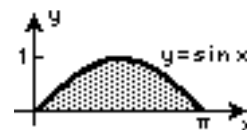
$$M_x = \int_0^\pi \int_0^{\sin x} kxy dy dx = \int_0^\pi \frac{1}{2}kxy^2 \Big|_0^{\sin x} dx = \int_0^\pi \frac{1}{2}kx \sin^2 x dx = \int_0^\pi \frac{1}{4}kx(1 - \cos 2x) dx$$

$$= \frac{1}{4}k \left[\int_0^\pi x dx - \int_0^\pi x \cos 2x dx \right]$$

Integration by parts

$$= \frac{1}{4}k \left[\frac{1}{2}x^2 \Big|_0^\pi - \frac{1}{4}(\cos 2x + 2x \sin 2x) \Big|_0^\pi \right] = \frac{1}{4}k \left(\frac{1}{2}\pi^2 \right) = \frac{1}{8}k\pi^2$$

$$\bar{x} = M_y/m = \frac{k(\pi^2 - 4)}{k\pi} = \pi - 4/\pi; \quad \bar{y} = M_x/m = \frac{k\pi^2/8}{k\pi} = \pi/8. \text{ The center of mass is } (\pi - 4/\pi, \pi/8).$$



$$49. m = \int_0^1 \int_0^{e^x} y^3 dy dx = \int_0^1 \frac{1}{4} y^4 \Big|_0^{e^x} dx = \int_0^1 \frac{1}{4} e^{4x} dx = \frac{1}{16} e^{4x} \Big|_0^1 = \frac{1}{16} (e^4 - 1)$$

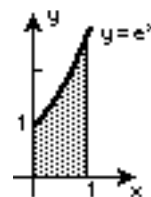
$$M_y = \int_0^1 \int_0^{e^x} xy^3 dy dx = \int_0^1 \frac{1}{4} xy^4 \Big|_0^{e^x} dx = \int_0^1 \frac{1}{4} x e^{4x} dx \quad \boxed{\text{Integration by parts}}$$

$$= \frac{1}{4} \left(\frac{1}{4} x e^{4x} - \frac{1}{16} e^{4x} \right) \Big|_0^1 = \frac{1}{4} \left(\frac{3}{16} e^4 + \frac{1}{16} \right) = \frac{1}{64} (3e^4 + 1)$$

$$M_x = \int_0^1 \int_0^{e^x} y^4 dy dx = \int_0^1 \frac{1}{5} y^5 \Big|_0^{e^x} dx = \int_0^1 \frac{1}{5} e^{5x} dx = \frac{1}{25} e^{5x} \Big|_0^1 = \frac{1}{25} (e^5 - 1)$$

$$\bar{x} = M_y/m = \frac{(3e^4 + 1)/64}{(e^4 - 1)/16} = \frac{3e^4 + 1}{4(e^4 - 1)}; \quad \bar{y} = M_x/m = \frac{(e^5 - 1)/25}{(e^4 - 1)/16} = \frac{16(e^5 - 1)}{25(e^4 - 1)}$$

The center of mass is $\left(\frac{3e^4 + 1}{4(e^4 - 1)}, \frac{16(e^5 - 1)}{25(e^4 - 1)} \right) \approx (0.77, 1.76)$.



50. Since both the region and ρ are symmetric with respect to the y -axis, $\bar{x} = 0$. Using symmetry,

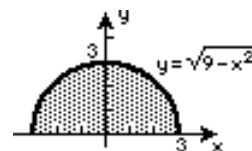
$$m = 2 \int_0^3 \int_0^{\sqrt{9-x^2}} x^2 dy dx = 2 \int_0^3 x^2 y \Big|_0^{\sqrt{9-x^2}} dx = 2 \int_0^3 x^2 \sqrt{9-x^2} dx$$

$\boxed{\text{Trig substitution}}$

$$= 2 \left[\frac{x}{8} (2x^2 - 9) \sqrt{9-x^2} + \frac{81}{8} \sin^{-1} \frac{x}{3} \right] \Big|_0^3 = \frac{81}{4} \frac{\pi}{2} = \frac{81\pi}{2}.$$

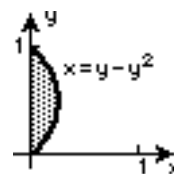
$$M_x = 2 \int_0^3 \int_0^{\sqrt{9-x^2}} x^2 y dy dx = 2 \int_0^3 \frac{1}{2} x^2 y^2 \Big|_0^{\sqrt{9-x^2}} dy dx = \int_0^3 x^2 (9-x^2) dx = \left(3x^2 - \frac{1}{5} x^5 \right) \Big|_0^3 = \frac{162}{5}$$

$$\bar{y} = M_x/m = \frac{162/5}{81\pi/8} = 16/5\pi. \text{ The center of mass is } (0, 16/5\pi).$$



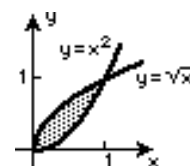
$$51. I_x = \int_0^1 \int_0^{y-y^2} 2xy^2 dx dy = \int_0^1 x^2 y^2 \Big|_0^{y-y^2} dy = \int_0^1 (y-y^2)^2 y^2 dy$$

$$= \int_0^1 (y^4 - 2y^5 + y^6) dy = \left(\frac{1}{5} y^5 - \frac{2}{6} y^6 + \frac{1}{7} y^7 \right) \Big|_0^1 = \frac{1}{105}$$



$$52. I_x = \int_0^1 \int_{x^2}^{\sqrt{x}} x^2 y^2 dy dx = \int_0^1 \frac{1}{3} x^2 y^3 \Big|_{x^2}^{\sqrt{x}} dx = \frac{1}{3} \int_0^1 (x^{7/2} - x^8) dx$$

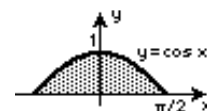
$$= \frac{1}{3} \left(\frac{2}{9} x^{9/2} - \frac{1}{9} x^9 \right) \Big|_0^1 = \frac{1}{27}$$



53. Using symmetry,

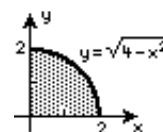
$$I_x = 2 \int_0^{\pi/2} \int_0^{\cos x} ky^2 dy dx = 2k \int_0^{\pi/2} \frac{1}{3} y^3 \Big|_0^{\cos x} dx = \frac{2}{3} k \int_0^{\pi/2} \cos^3 x dx$$

$$= \frac{2}{3} k \int_0^{\pi/2} \cos x (1 - \sin^2 x) dx = \frac{2}{3} k \left(\sin x - \frac{1}{3} \sin^3 x \right) \Big|_0^{\pi/2} = \frac{4}{9} k.$$



$$54. I_x = \int_0^2 \int_0^{\sqrt{4-x^2}} y^3 dy dx = \int_0^2 \frac{1}{4} y^4 \Big|_0^{\sqrt{4-x^2}} dx = \frac{1}{4} \int_0^2 (4-x^2)^2 dx$$

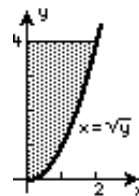
$$= \frac{1}{4} \int_0^2 (16 - 8x^2 + x^4) dx = \frac{1}{4} \left(16x - \frac{8}{3} x^3 + \frac{1}{5} x^5 \right) \Big|_0^2 = \frac{1}{4} \left(32 - \frac{64}{3} + \frac{32}{5} \right)$$



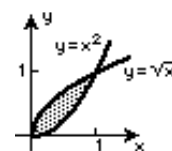
9.10 Double Integrals

$$= 8 \left(1 - \frac{2}{3} + \frac{1}{5} \right) = \frac{64}{15}$$

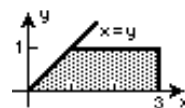
$$\begin{aligned} 55. \quad I_y &= \int_0^4 \int_0^{\sqrt{y}} x^2 y \, dx \, dy = \int_0^4 \frac{1}{3} x^3 y \Big|_0^{\sqrt{y}} dy = \frac{1}{3} \int_0^4 y^{3/2} y \, dy = \frac{1}{3} \int_0^4 y^{5/2} dy \\ &= \frac{1}{3} \left(\frac{2}{7} y^{7/2} \right) \Big|_0^4 = \frac{2}{21} (4^{7/2}) = \frac{256}{21} \end{aligned}$$



$$56. \quad I_y = \int_0^1 \int_{x^2}^{\sqrt{x}} x^4 \, dy \, dx = \int_0^1 x^4 y \Big|_{x^2}^{\sqrt{x}} dx = \int_0^1 (x^{9/2} - x^6) dx = \left(\frac{2}{11} x^{11/2} - \frac{1}{7} x^7 \right) \Big|_0^1 = \frac{3}{77}$$

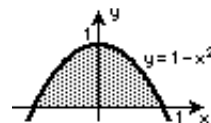


$$\begin{aligned} 57. \quad I_y &= \int_0^1 \int_y^3 (4x^3 + 3x^2 y) \, dx \, dy = \int_0^1 (x^4 + x^3 y) \Big|_y^3 dy = \int_0^1 (81 + 27y - 2y^4) dy \\ &= \left(81y + \frac{27}{2} y^2 - \frac{2}{5} y^5 \right) \Big|_0^1 = \frac{941}{10} \end{aligned}$$



58. The density is $\rho = ky$. Using symmetry,

$$\begin{aligned} I_y &= 2 \int_0^1 \int_0^{1-x^2} kx^2 y \, dy \, dx = 2 \int_0^1 \frac{1}{2} kx^2 y^2 \Big|_0^{1-x^2} dx = k \int_0^1 x^2 (1-x^2)^2 dx \\ &= k \int_0^1 (x^2 - 2x^4 + x^6) dx = k \left(\frac{1}{3} x^3 - \frac{2}{5} x^5 + \frac{1}{7} x^7 \right) \Big|_0^1 = \frac{8k}{105}. \end{aligned}$$

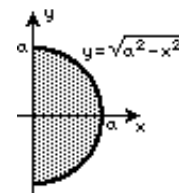


59. Using symmetry,

$$\begin{aligned} m &= 2 \int_0^a \int_0^{\sqrt{a^2-y^2}} x \, dx \, dy = 2 \int_0^a \frac{1}{2} x^2 \Big|_0^{\sqrt{a^2-y^2}} dy = \int_0^a (a^2 - y^2) dy \\ &= \left(a^2 y - \frac{1}{3} y^3 \right) \Big|_0^a = \frac{2}{3} a^3. \end{aligned}$$

$$\begin{aligned} I_y &= 2 \int_0^a \int_0^{\sqrt{a^2-y^2}} x^3 \, dx \, dy = 2 \int_0^a \frac{1}{4} x^4 \Big|_0^{\sqrt{a^2-y^2}} dy = \frac{1}{2} \int_0^a (a^2 - y^2)^2 dy \\ &= \frac{1}{2} \int_0^a (a^4 - 2a^2 y^2 + y^4) dy = \frac{1}{2} \left(a^4 y - \frac{2}{3} a^2 y^3 + \frac{1}{5} y^5 \right) \Big|_0^a = \frac{4}{15} a^5 \end{aligned}$$

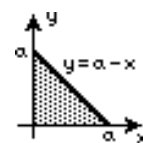
$$R_g = \sqrt{\frac{I_y}{m}} = \sqrt{\frac{4a^5/15}{2a^3/3}} = \sqrt{\frac{2}{5}} a$$



$$60. \quad m = \int_0^a \int_0^{a-x} k \, dy \, dx = \int_0^a ky \Big|_0^{a-x} dx = k \int_0^a (a-x) dx = k \left(ax - \frac{1}{2} x^2 \right) \Big|_0^a = \frac{1}{2} ka^2$$

$$\begin{aligned} I_x &= \int_0^a \int_0^{a-x} ky^2 \, dy \, dx = \int_0^a \frac{1}{3} ky^3 \Big|_0^{a-x} dx = \frac{1}{3} k \int_0^a (a-x)^3 dx \\ &= \frac{1}{3} k \int_0^a (a^3 - 3a^2 x + 3ax^2 - x^3) dx = \frac{1}{3} k \left(a^3 x - \frac{3}{2} a^2 x^2 + ax^3 - \frac{1}{4} x^4 \right) \Big|_0^a = \frac{1}{12} ka^4 \end{aligned}$$

$$R_g = \sqrt{\frac{I_x}{m}} = \sqrt{\frac{ka^4/12}{ka^2/2}} = \sqrt{\frac{1}{6}} a$$



61. (a) Using symmetry,

$$\begin{aligned}
 I_x &= 4 \int_0^a \int_0^{b\sqrt{a^2-x^2}/a} y^2 dy dx = \frac{4b^3}{3a^3} \int_0^a (a^2 - x^2)^{3/2} dx & \boxed{x = a \sin \theta, \quad dx = a \cos \theta d\theta} \\
 &= \frac{4}{3} ab^3 \int_0^{\pi/2} \cos^4 \theta d\theta = \frac{4}{3} ab^3 \int_0^{\pi/2} \frac{1}{4} (1 + \cos 2\theta)^2 d\theta = \frac{1}{3} ab^3 \int_0^{\pi/2} \left(1 + \cos 2\theta + \frac{1}{2} + \frac{1}{2} \cos 4\theta \right) d\theta \\
 &= \frac{1}{3} ab^3 \left(\frac{3}{2} \theta + \frac{1}{2} \sin 2\theta + \frac{1}{8} \sin 4\theta \right) \Big|_0^{\pi/2} = \frac{ab^3 \pi}{4}.
 \end{aligned}$$

(b) Using symmetry,

$$\begin{aligned}
 I_y &= 4 \int_0^a \int_0^{b\sqrt{a^2-x^2}/a} x^2 dy dx = \frac{4b}{a} \int_0^a x^2 \sqrt{a^2 - x^2} dx & \boxed{x = a \sin \theta, \quad dx = a \cos \theta d\theta} \\
 &= 4a^3 b \int_0^{\pi/2} \sin^2 \theta \cos^2 \theta d\theta = 4a^3 b \int_0^{\pi/2} \frac{1}{4} (1 - \cos^2 2\theta) d\theta \\
 &= a^3 b \int_0^{\pi/2} \left(1 - \frac{1}{2} - \frac{1}{2} \cos 4\theta \right) d\theta = a^3 b \left(\frac{1}{2} \theta - \frac{1}{8} \sin 4\theta \right) \Big|_0^{\pi/2} = \frac{a^3 b \pi}{4}.
 \end{aligned}$$

(c) Using $m = \pi ab$, $R_g = \sqrt{I_x/m} = \frac{1}{2} \sqrt{ab^3 \pi / \pi ab} = \frac{1}{2} b$.

(d) $R_g = \sqrt{I_y/m} = \frac{1}{2} \sqrt{a^3 b \pi / \pi ab} = \frac{1}{2} a$

62. The equation of the ellipse is $9x^2/a^2 + 4y^2/b^2 = 1$ and the equation of the parabola is $y = \pm(9bx^2/8a^2 - b/2)$. Letting I_e and I_p represent the moments of inertia of the ellipse and parabola, respectively, about the x -axis, we have

$$\begin{aligned}
 I_e &= 2 \int_{-a/3}^0 \int_0^{b\sqrt{a^2-9x^2}/2a} y^2 dy dx = \frac{b^3}{12a^3} \int_{-a/3}^0 (a^2 - 9x^2)^{3/2} dx & \boxed{x = \frac{a}{3} \sin \theta, \quad dx = \frac{a}{3} \cos \theta d\theta} \\
 &= \frac{b^3}{12a^3} \frac{a^4}{3} \int_{-\pi/3}^0 \cos^4 \theta d\theta = \frac{b^3 a}{36} \frac{3\pi}{16} = \frac{ab^3 \pi}{192}
 \end{aligned}$$

and

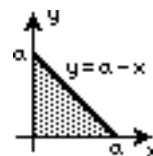
$$\begin{aligned}
 I_p &= 2 \int_0^{2a/3} \int_0^{b/2-9bx^2/8a^2} y^2 dy dx = \frac{2}{3} \int_0^{2a/3} \left(\frac{b}{2} - \frac{9b}{8a^2} x^2 \right)^3 dx = \frac{2}{3} \frac{b^3}{8} \int_0^{2a/3} \left(1 - \frac{9}{4a^2} x^2 \right)^3 dx \\
 &= \frac{b^3}{12} \int_0^{2a/3} \left(1 - \frac{27}{4a^2} x^2 + \frac{243}{16a^4} x^4 - \frac{729}{64a^6} x^6 \right) dx = \frac{b^3}{12} \left(x - \frac{9}{4a^2} x^3 + \frac{243}{80a^4} x^5 - \frac{729}{64a^6} x^7 \right) \Big|_0^{2a/3} \\
 &= \frac{b^3}{12} \frac{32a}{105} = \frac{8ab^3}{315}.
 \end{aligned}$$

Then $I_x = I_e + I_p = \frac{ab^3 \pi}{192} + \frac{8ab^3}{315}$.

63. From the solution to Problem 60, $m = \frac{1}{2}ka^2$ and $I_x = \frac{1}{12}ka^4$.

$$\begin{aligned}
 I_y &= \int_0^a \int_0^{a-x} kx^2 dy dx = \int_0^a kx^2 y \Big|_0^{a-x} dx = k \int_0^a x^2(a-x) dx \\
 &= k \left(\frac{1}{3} ax^3 - \frac{1}{4} x^4 \right) \Big|_0^a = \frac{1}{12} ka^4
 \end{aligned}$$

$$I_0 = I_x + I_y = \frac{1}{12} ka^4 + \frac{1}{12} ka^4 = \frac{1}{6} ka^4$$

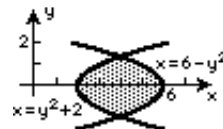


9.10 Double Integrals

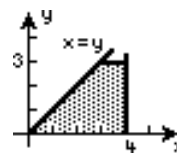
64. From the solution to Problem 52, $I_x = \frac{1}{27}$, and from the solution to Problem 56, $I_y = \frac{3}{77}$. Thus, $I_0 = I_x + I_y = \frac{1}{27} + \frac{3}{77} = \frac{158}{2079}$.

65. The density is $\rho = k/(x^2 + y^2)$. Using symmetry,

$$\begin{aligned} I_0 &= 2 \int_0^{\sqrt{2}} \int_{y^2+2}^{6-y^2} (x^2 + y^2) \frac{k}{x^2 + y^2} dx dy = 2 \int_0^{\sqrt{2}} kx \Big|_{y^2+2}^{6-y^2} dy \\ &= 2k \int_0^{\sqrt{2}} (6 - y^2 - y^2 - 2) dy = 2k \left(4y - \frac{2}{3}y^3 \right) \Big|_0^{\sqrt{2}} = 2k \left(\frac{8}{3}\sqrt{2} \right) = \frac{16\sqrt{2}}{3} k. \end{aligned}$$



66.
$$\begin{aligned} I_0 &= \int_0^3 \int_y^4 k(x^2 + y^2) dx dy = k \int_0^3 \left(\frac{1}{3}x^3 + xy^2 \right) \Big|_y^4 dy \\ &= k \int_0^3 \left(\frac{64}{3} + 4y^2 - \frac{1}{3}y^3 - y^3 \right) dy = k \left(\frac{64}{3}y + \frac{4}{3}y^3 - \frac{1}{3}y^4 \right) \Big|_0^3 = 73k \end{aligned}$$

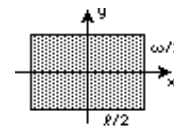


67. From the solution to Problem 60, $m = \frac{1}{2}ka^2$, and from the solution to Problem 63, $I_0 = \frac{1}{6}ka^4$. Then

$$R_g = \sqrt{I_0/m} = \sqrt{\frac{ka^4/6}{ka^2/2}} = \sqrt{\frac{1}{3}} a.$$

68. Since the plate is homogeneous, the density is $\rho = m/\ell\omega$. Using symmetry,

$$\begin{aligned} I_0 &= 4 \int_0^{\ell/2} \int_0^{\omega/2} \frac{m}{\ell\omega} (x^2 + y^2) dy dx = \frac{4m}{\ell\omega} \int_0^{\ell/2} \left(x^2y + \frac{1}{3}y^3 \right) \Big|_0^{\omega/2} dx \\ &= \frac{4m}{\ell\omega} \int_0^{\ell/2} \left(\frac{\omega}{2}x^2 + \frac{\omega^3}{24} \right) dx = \frac{4m}{\ell\omega} \left(\frac{\omega}{6}x^3 + \frac{\omega^3}{24}x \right) \Big|_0^{\ell/2} = \frac{4m}{\ell\omega} \left(\frac{\omega\ell^3}{48} + \frac{\ell\omega^3}{48} \right) = m \frac{\ell^2 + \omega^2}{12}. \end{aligned}$$

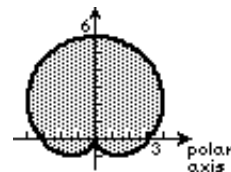


EXERCISES 9.11

Double Integrals in Polar Coordinates

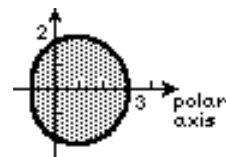
1. Using symmetry,

$$\begin{aligned} A &= 2 \int_{-\pi/2}^{\pi/2} \int_0^{3+3\sin\theta} r dr d\theta = 2 \int_{-\pi/2}^{\pi/2} \frac{1}{2}r^2 \Big|_0^{3+3\sin\theta} d\theta = \int_{-\pi/2}^{\pi/2} 9(1 + \sin\theta)^2 d\theta \\ &= 9 \int_{-\pi/2}^{\pi/2} (1 + 2\sin\theta + \sin^2\theta) d\theta = 9 \left(\theta - 2\cos\theta + \frac{1}{2}\theta - \frac{1}{4}\sin 2\theta \right) \Big|_{-\pi/2}^{\pi/2} \\ &= 9 \left[\frac{3}{2} \frac{\pi}{2} - \frac{3}{2} \left(-\frac{\pi}{2} \right) \right] = \frac{27\pi}{2} \end{aligned}$$



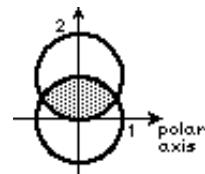
2. Using symmetry,

$$\begin{aligned} A &= 2 \int_0^\pi \int_0^{2+\cos\theta} r \, dr \, d\theta = 2 \int_0^\pi \frac{1}{2} r^2 \Big|_0^{2+\cos\theta} d\theta = \int_0^\pi (2+\cos\theta)^2 d\theta \\ &= \int_0^\pi (4 + 4\cos\theta + \cos^2\theta) d\theta = \left(4\theta + 4\sin\theta + \frac{1}{2}\theta + \frac{1}{4}\cos 2\theta \right) \Big|_0^\pi \\ &= \left(4\pi + \frac{\pi}{2} + \frac{1}{4} \right) - \left(\frac{1}{4} \right) = \frac{9\pi}{2}. \end{aligned}$$

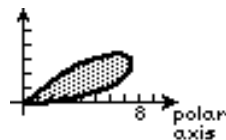


3. Solving $r = 2\sin\theta$ and $r = 1$, we obtain $\sin\theta = 1/2$ or $\theta = \pi/6$. Using symmetry,

$$\begin{aligned} A &= 2 \int_0^{\pi/6} \int_0^{2\sin\theta} r \, dr \, d\theta + 2 \int_{\pi/6}^{\pi/2} \int_0^1 r \, dr \, d\theta \\ &= 2 \int_0^{\pi/6} \frac{1}{2} r^2 \Big|_0^{2\sin\theta} d\theta + 2 \int_{\pi/6}^{\pi/2} \frac{1}{2} r^2 \Big|_0^1 d\theta = \int_0^{\pi/6} 4\sin^2\theta \, d\theta + \int_{\pi/6}^{\pi/2} d\theta \\ &= (2\theta - \sin 2\theta) \Big|_0^{\pi/6} + \left(\frac{\pi}{2} - \frac{\pi}{6} \right) = \frac{\pi}{3} - \frac{\sqrt{3}}{2} + \frac{\pi}{3} = \frac{4\pi - 3\sqrt{3}}{6} \end{aligned}$$



$$\begin{aligned} 4. \quad A &= \int_0^{\pi/4} \int_0^{8\sin 4\theta} r \, dr \, d\theta = \int_0^{\pi/4} \frac{1}{2} r^2 \Big|_0^{8\sin 4\theta} d\theta = \frac{1}{2} \int_0^{\pi/4} 64 \sin^2 4\theta \, d\theta \\ &= 32 \left(\frac{1}{2}\theta - \frac{1}{16} \sin 8\theta \right) \Big|_0^{\pi/4} = 4\pi \end{aligned}$$

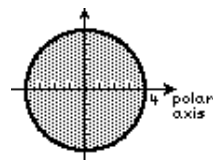


5. Using symmetry,

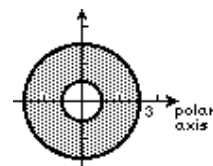
$$\begin{aligned} V &= 2 \int_0^{\pi/6} \int_0^{5\cos 3\theta} 4r \, dr \, d\theta = 4 \int_0^{\pi/6} r^2 \Big|_0^{5\cos 3\theta} d\theta = 4 \int_0^{\pi/6} 25 \cos^2 3\theta \, d\theta \\ &= 100 \left(\frac{1}{2}\theta + \frac{1}{12} \sin 6\theta \right) \Big|_0^{\pi/6} = \frac{25\pi}{3} \end{aligned}$$



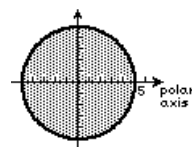
$$\begin{aligned} 6. \quad V &= \int_0^{2\pi} \int_0^2 \sqrt{9-r^2} \, r \, dr \, d\theta = \int_0^{2\pi} -\frac{1}{3} (9-r^2)^{3/2} \Big|_0^2 d\theta \\ &= -\frac{1}{3} \int_0^{2\pi} (5^{3/2} - 27) \, d\theta = \frac{1}{3} (27 - 5^{3/2}) 2\pi = \frac{2\pi(27 - 5\sqrt{5})}{3} \end{aligned}$$



$$\begin{aligned} 7. \quad V &= \int_0^{2\pi} \int_1^3 \sqrt{16-r^2} \, r \, dr \, d\theta = \int_0^{2\pi} -\frac{1}{3} (16-r^2)^{3/2} \Big|_1^3 d\theta \\ &= -\frac{1}{3} \int_0^{2\pi} (7^{3/2} - 15^{3/2}) \, d\theta = \frac{1}{3} (15^{3/2} - 7^{3/2}) 2\pi = \frac{2\pi(15\sqrt{15} - 7\sqrt{7})}{3} \end{aligned}$$

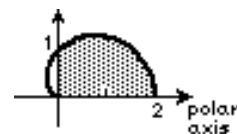


$$8. \quad V = \int_0^{2\pi} \int_0^5 \sqrt{r^2} \, r \, dr \, d\theta = \int_0^{2\pi} \frac{1}{3} r^3 \Big|_0^5 d\theta = \int_0^{2\pi} \frac{125}{3} \, d\theta = \frac{250\pi}{3}$$



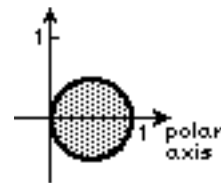
9.11 Double Integrals in Polar Coordinates

$$\begin{aligned}
 9. \quad V &= \int_0^{\pi/2} \int_0^{1+\cos\theta} (r \sin \theta) r \, dr \, d\theta = \int_0^{\pi/2} \left. \frac{1}{3} r^3 \sin \theta \right|_0^{1+\cos\theta} d\theta \\
 &= \frac{1}{3} \int_0^{\pi/2} (1 + \cos \theta)^3 \sin \theta \, d\theta = \frac{1}{3} \left[-\frac{1}{4} (1 + \cos \theta)^4 \right] \Big|_0^{\pi/2} = -\frac{1}{12} (1 - 2^4) = \frac{5}{4}
 \end{aligned}$$

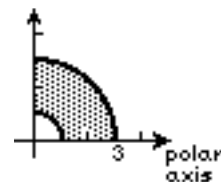


10. Using symmetry,

$$\begin{aligned}
 V &= 2 \int_0^{\pi/2} \int_0^{\cos\theta} (2 + r^2) r \, dr \, d\theta = 2 \int_0^{\pi/2} \left(r^2 + \frac{1}{4} r^4 \right) \Big|_0^{\cos\theta} d\theta \\
 &= 2 \int_0^{\pi/2} \left(\cos^2 \theta + \frac{1}{4} \cos^4 \theta \right) d\theta = 2 \int_0^{\pi/2} \left[\cos^2 \theta + \frac{1}{4} \left(\frac{1 + \cos 2\theta}{2} \right)^2 \right] d\theta \\
 &= \int_0^{\pi/2} \left(2 \cos^2 \theta + \frac{1}{8} + \frac{1}{4} \cos 2\theta + \frac{1}{8} \cos^2 2\theta \right) d\theta \\
 &= \left(\theta + \frac{1}{2} \sin 2\theta + \frac{1}{8} \theta + \frac{1}{8} \sin 2\theta + \frac{1}{16} \theta + \frac{1}{64} \sin 4\theta \right) \Big|_0^{\pi/2} = \frac{19\pi}{32}.
 \end{aligned}$$



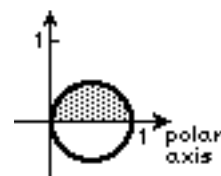
$$\begin{aligned}
 11. \quad m &= \int_0^{\pi/2} \int_1^3 k r \, dr \, d\theta = k \int_0^{\pi/2} \left. \frac{1}{2} r^2 \right|_1^3 d\theta = \frac{1}{2} k \int_0^{\pi/2} 8 \, d\theta = 2k\pi \\
 M_y &= \int_0^{\pi/2} \int_1^3 k x r \, dr \, d\theta = k \int_0^{\pi/2} \int_1^3 r^2 \cos \theta \, dr \, d\theta = k \int_0^{\pi/2} \left. \frac{1}{3} r^3 \cos \theta \right|_1^3 d\theta \\
 &= \frac{1}{3} k \int_0^{\pi/2} 26 \cos \theta \, d\theta = \frac{26}{3} k \sin \theta \Big|_0^{\pi/2} = \frac{26}{3} k \\
 \bar{x} &= M_y/m = \frac{26k/3}{2k\pi} = \frac{13}{3\pi}. \text{ Since the region and density function are symmetric about the ray } \theta = \pi/4, \\
 \bar{y} &= \bar{x} = 13/3\pi \text{ and the center of mass is } (13/3\pi, 13/3\pi).
 \end{aligned}$$



12. The interior of the upper-half circle is traced from $\theta = 0$ to $\pi/2$. The density is kr . Since both the region and the density are symmetric about the polar axis, $\bar{y} = 0$.

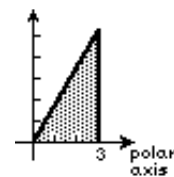
$$\begin{aligned}
 m &= \int_0^{\pi/2} \int_0^{\cos\theta} k r^2 \, dr \, d\theta = k \int_0^{\pi/2} \left. \frac{1}{3} r^3 \right|_0^{\cos\theta} d\theta = \frac{k}{3} \int_0^{\pi/2} \cos^3 \theta \, d\theta \\
 &= \frac{k}{3} \left(\frac{2}{3} + \frac{1}{3} \cos^2 \theta \right) \sin \theta \Big|_0^{\pi/2} = \frac{2k}{9} \\
 M_y &= k \int_0^{\pi/2} \int_0^{\cos\theta} (r \cos \theta)(r)(r \, dr \, d\theta) = k \int_0^{\pi/2} \int_0^{\cos\theta} r^3 \cos \theta \, dr \, d\theta = k \int_0^{\pi/2} \left. \frac{1}{4} r^4 \cos \theta \right|_0^{\cos\theta} d\theta \\
 &= \frac{k}{4} \int_0^{\pi/2} \cos^5 \theta \, d\theta = \frac{k}{4} \left(\sin \theta - \frac{2}{3} \sin^3 \theta + \frac{1}{5} \sin^5 \theta \right) \Big|_0^{\pi/2} = \frac{2k}{15}
 \end{aligned}$$

Thus, $\bar{x} = \frac{2k/15}{2k/9} = 3/5$ and the center of mass is $(3/5, 0)$.



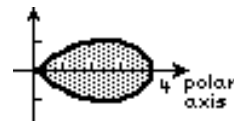
13. In polar coordinates the line $x = 3$ becomes $r \cos \theta = 3$ or $r = 3 \sec \theta$. The angle of inclination of the line $y = \sqrt{3}x$ is $\pi/3$.

$$\begin{aligned}
 m &= \int_0^{\pi/3} \int_0^{3 \sec \theta} r^2 r \, dr \, d\theta = \int_0^{\pi/3} \left. \frac{1}{4} r^4 \right|_0^{3 \sec \theta} d\theta = \frac{81}{4} \int_0^{\pi/3} \sec^4 \theta \, d\theta \\
 &= \frac{81}{4} \int_0^{\pi/3} (1 + \tan^2 \theta) \sec^2 \theta \, d\theta = \frac{81}{4} \left(\tan \theta + \frac{1}{3} \tan^3 \theta \right) \Big|_0^{\pi/3} = \frac{81}{4} (\sqrt{3} + \sqrt{3}) = \frac{81}{2} \sqrt{3}
 \end{aligned}$$



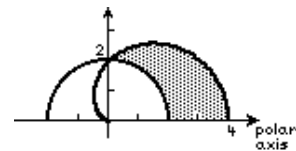
$$\begin{aligned}
M_y &= \int_0^{\pi/3} \int_0^{3 \sec \theta} x r^2 r \, dr \, d\theta = \int_0^{\pi/3} \int_0^{3 \sec \theta} r^4 \cos \theta \, dr \, d\theta = \int_0^{\pi/3} \frac{1}{5} r^5 \cos \theta \Big|_0^{3 \sec \theta} d\theta \\
&= \frac{243}{5} \int_0^{\pi/3} \sec^5 \theta \cos \theta \, d\theta = \frac{243}{5} \int_0^{\pi/3} \sec^4 \theta \, d\theta = \frac{243}{5} (2\sqrt{3}) = \frac{486}{5} \sqrt{3} \\
M_x &= \int_0^{\pi/3} \int_0^{3 \sec \theta} y r^2 r \, dr \, d\theta = \int_0^{\pi/3} \int_0^{3 \sec \theta} r^4 \sin \theta \, dr \, d\theta = \int_0^{\pi/3} \frac{1}{5} r^5 \sin \theta \Big|_0^{3 \sec \theta} d\theta \\
&= \frac{243}{5} \int_0^{\pi/3} \sec^5 \theta \sin \theta \, d\theta = \frac{243}{5} \int_0^{\pi/3} \tan \theta \sec^4 \theta \, d\theta = \frac{243}{5} \int_0^{\pi/3} \tan \theta (1 + \tan^2 \theta) \sec^2 \theta \, d\theta \\
&= \frac{243}{5} \int_0^{\pi/3} (\tan \theta + \tan^3 \theta) \sec^2 \theta \, d\theta = \frac{243}{5} \left(\frac{1}{2} \tan^2 \theta + \frac{1}{4} \tan^4 \theta \right) \Big|_0^{\pi/3} = \frac{243}{5} \left(\frac{3}{2} + \frac{9}{4} \right) = \frac{729}{4} \\
\bar{x} = \frac{M_y}{m} &= \frac{486\sqrt{3}/5}{81\sqrt{3}/2} = 12/5; \quad \bar{y} = \frac{M_x}{m} = \frac{729/4}{81\sqrt{3}/2} = 3\sqrt{3}/2. \text{ The center of mass is } (12/5, 3\sqrt{3}/2).
\end{aligned}$$

14. Since both the region and the density are symmetric about the x -axis, $\bar{y} = 0$. Using symmetry,



$$\begin{aligned}
m &= 2 \int_0^{\pi/4} \int_0^{4 \cos 2\theta} k r \, dr \, d\theta = 2k \int_0^{\pi/4} \frac{1}{2} r^2 \Big|_0^{4 \cos 2\theta} d\theta = 16k \int_0^{\pi/4} \cos^2 2\theta \, d\theta \\
&= 16k \left(\frac{1}{2} \theta + \frac{1}{8} \sin 4\theta \right) \Big|_0^{\pi/4} = 2k\pi \\
M_y &= 2 \int_0^{\pi/4} \int_0^{4 \cos 2\theta} k x r \, dr \, d\theta = 2k \int_0^{\pi/4} \int_0^{4 \cos 2\theta} r^2 \cos \theta \, dr \, d\theta = 2k \int_0^{\pi/4} \frac{1}{3} r^3 \cos \theta \Big|_0^{4 \cos 2\theta} d\theta \\
&= \frac{128}{3} k \int_0^{\pi/4} \cos^3 2\theta \cos \theta \, d\theta = \frac{128}{3} k \int_0^{\pi/4} (1 - 2 \sin^2 \theta)^3 \cos \theta \, d\theta \\
&= \frac{128}{3} k \int_0^{\pi/4} (1 - 6 \sin^2 \theta + 12 \sin^4 \theta - 8 \sin^6 \theta) \cos \theta \, d\theta = \frac{128}{3} k \left(\sin \theta - 2 \sin^3 \theta + \frac{12}{5} \sin^5 \theta - \frac{8}{7} \sin^7 \theta \right) \Big|_0^{\pi/4} \\
&= \frac{128}{3} k \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} + \frac{3\sqrt{2}}{10} - \frac{\sqrt{2}}{14} \right) = \frac{1024}{105} \sqrt{2} k \\
\bar{x} = M_y/m &= \frac{1024\sqrt{2}k/105}{2k\pi} = \frac{512\sqrt{2}}{105\pi}. \text{ The center of mass is } (512\sqrt{2}/105\pi, 0) \text{ or approximately } (2.20, 0).
\end{aligned}$$

15. The density is $\rho = k/r$.



$$\begin{aligned}
m &= \int_0^{\pi/2} \int_2^{2+2 \cos \theta} \frac{k}{r} r \, dr \, d\theta = k \int_0^{\pi/2} \int_2^{2+2 \cos \theta} dr \, d\theta \\
&= k \int_0^{\pi/2} 2 \cos \theta \, d\theta = 2k(\sin \theta) \Big|_0^{\pi/2} = 2k \\
M_y &= \int_0^{\pi/2} \int_2^{2+2 \cos \theta} x \frac{k}{r} r \, dr \, d\theta = k \int_0^{\pi/2} \int_2^{2+2 \cos \theta} r \cos \theta \, dr \, d\theta = k \int_0^{\pi/2} \frac{1}{2} r^2 \Big|_2^{2+2 \cos \theta} \cos \theta \, d\theta \\
&= \frac{1}{2} k \int_0^{\pi/2} (8 \cos \theta + 4 \cos^2 \theta) \cos \theta \, d\theta = 2k \int_0^{\pi/2} (2 \cos^2 \theta + \cos \theta - \sin^2 \theta \cos \theta) \, d\theta \\
&= 2k \left(\theta + \frac{1}{2} \sin 2\theta + \sin \theta - \frac{1}{3} \sin^3 \theta \right) \Big|_0^{\pi/2} = 2k \left(\frac{\pi}{2} + \frac{2}{3} \right) = \frac{3\pi + 4}{3} k
\end{aligned}$$

9.11 Double Integrals in Polar Coordinates

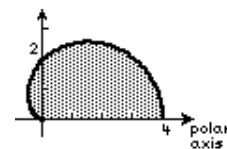
$$M_x = \int_0^{\pi/2} \int_2^{2+2\cos\theta} y \frac{k}{r} r dr d\theta = k \int_0^{\pi/2} \int_2^{2+2\cos\theta} r \sin\theta dr d\theta = k \int_0^{\pi/2} \frac{1}{2} r^2 \Big|_2^{2+2\cos\theta} \sin\theta d\theta$$

$$= \frac{1}{2} k \int_0^{\pi/2} (8\cos\theta + 4\cos^2\theta) \sin\theta d\theta = \frac{1}{2} k \left(-4\cos^2\theta - \frac{4}{3}\cos^3\theta \right) \Big|_0^{\pi/2} = \frac{1}{2} k \left[-\left(-4 - \frac{4}{3}\right) \right] = \frac{8}{3} k$$

$$\bar{x} = M_y/m = \frac{(3\pi+4)k/3}{2k} = \frac{3\pi+4}{6}; \quad \bar{y} = M_x/m = \frac{8k/3}{2k} = \frac{4}{3}. \text{ The center of mass is } ((3\pi+4)/6, 4/3).$$

$$16. \quad m = \int_0^\pi \int_0^{2+2\cos\theta} kr dr d\theta = k \int_0^\pi \frac{1}{2} r^2 \Big|_0^{2+2\cos\theta} d\theta = 2k \int_0^\pi (1+\cos\theta)^2 d\theta$$

$$= 2k \int_0^\pi (1+2\cos\theta+\cos^2\theta) d\theta = 2k \left(\theta + 2\sin\theta + \frac{1}{2}\theta + \frac{1}{4}\sin 2\theta \right) \Big|_0^\pi = 3\pi k$$



$$M_y = \int_0^\pi \int_0^{2+2\cos\theta} kxr dr d\theta = k \int_0^\pi \int_0^{2+2\cos\theta} r^2 \cos\theta dr d\theta = k \int_0^\pi \frac{1}{3} r^3 \Big|_0^{2+2\cos\theta} \cos\theta d\theta$$

$$= \frac{8}{3} k \int_0^\pi (1+\cos\theta)^3 \cos\theta d\theta = \frac{8}{3} k \int_0^\pi (\cos\theta + 3\cos^2\theta + 3\cos^3\theta + \cos^4\theta) d\theta$$

$$= \frac{8}{3} k \left[\sin\theta + \left(\frac{3}{2}\theta + \frac{3}{4}\sin 2\theta \right) + (3\sin\theta - \sin^3\theta) + \left(\frac{3}{8}\theta + \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta \right) \right] \Big|_0^\pi = \frac{8}{3} k \left(\frac{15}{8}\pi \right) = 5\pi k$$

$$M_x = \int_0^\pi \int_0^{2+2\cos\theta} kyr dr d\theta = k \int_0^\pi \int_0^{2+2\cos\theta} r^2 \sin\theta dr d\theta = k \int_0^\pi \frac{1}{3} r^3 \Big|_0^{2+2\cos\theta} \sin\theta d\theta$$

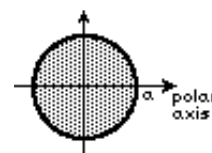
$$= \frac{8}{3} k \int_0^\pi (1+\cos\theta)^3 \sin\theta d\theta = \frac{8}{3} k \int_0^\pi (1+3\cos\theta+3\cos^2\theta+\cos^3\theta) \sin\theta d\theta$$

$$= \frac{8}{3} k \left(-\cos\theta - \frac{3}{2}\cos^2\theta - \cos^3\theta - \frac{1}{4}\cos^4\theta \right) \Big|_0^\pi = \frac{8}{3} k \left[\frac{1}{4} - \left(-\frac{15}{4}\right) \right] = \frac{32}{3} k$$

$$\bar{x} = M_y/m = \frac{5\pi k}{3\pi k} = 5/3; \quad \bar{y} = M_x/m = \frac{32k/3}{3\pi k} = 32/9\pi. \text{ The center of mass is } (5/3, 32/9\pi).$$

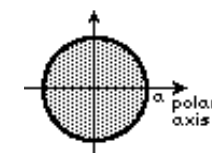
$$17. \quad I_x = \int_0^{2\pi} \int_0^a y^2 kr dr d\theta = k \int_0^{2\pi} \int_0^a r^3 \sin^2\theta dr d\theta = k \int_0^{2\pi} \frac{1}{4} r^4 \sin^2\theta \Big|_0^a d\theta$$

$$= \frac{ka^4}{4} \int_0^{2\pi} \sin^2\theta d\theta = \frac{ka^4}{4} \left(\frac{1}{2}\theta - \frac{1}{4}\sin 2\theta \right) \Big|_0^{2\pi} = \frac{k\pi a^4}{4}$$



$$18. \quad I_x = \int_0^{2\pi} \int_0^a y^2 \frac{1}{1+r^4} r dr d\theta = \int_0^{2\pi} \int_0^a \frac{r^3}{1+r^4} \sin^2\theta dr d\theta$$

$$= \int_0^{2\pi} \frac{1}{4} \ln(1+r^4) \Big|_0^a \sin^2\theta d\theta = \frac{1}{4} \ln(1+a^4) \left(\frac{1}{2}\theta - \frac{1}{4}\sin 2\theta \right) \Big|_0^{2\pi} = \frac{\pi}{4} \ln(1+a^4)$$

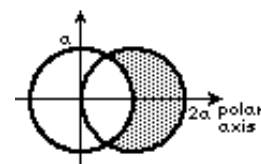


19. Solving $a = 2a \cos\theta$, $\cos\theta = 1/2$ or $\theta = \pi/3$. The density is k/r^3 . Using symmetry,

$$I_y = 2 \int_0^{\pi/3} \int_a^{2a\cos\theta} x^2 \frac{k}{r^3} r dr d\theta = 2k \int_0^{\pi/3} \int_a^{2a\cos\theta} \cos^2\theta dr d\theta$$

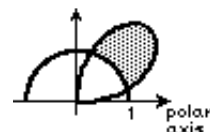
$$= 2k \int_0^{\pi/3} (2a\cos^3\theta - a\cos^2\theta) d\theta = 2ak \left(2\sin\theta - \frac{2}{3}\sin^3\theta - \frac{1}{2}\theta - \frac{1}{4}\sin 2\theta \right) \Big|_0^{\pi/3}$$

$$= 2ak \left(\sqrt{3} - \frac{\sqrt{3}}{4} - \frac{\pi}{6} - \frac{\sqrt{3}}{8} \right) = \frac{5ak\sqrt{3}}{4} - \frac{ak\pi}{3}$$

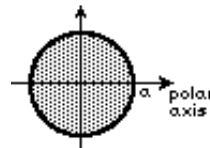


20. Solving $1 = 2 \sin 2\theta$, we obtain $\sin 2\theta = 1/2$ or $\theta = \pi/12$ and $\theta = 5\pi/12$.

$$\begin{aligned} I_y &= \int_{\pi/12}^{5\pi/12} \int_1^{2\sin 2\theta} x^2 \sec^2 \theta r dr d\theta = \int_{\pi/12}^{5\pi/12} \int_1^{2\sin 2\theta} r^3 dr d\theta \\ &= \int_{\pi/12}^{5\pi/12} \left. \frac{1}{4} r^4 \right|_1^{2\sin 2\theta} d\theta = 4 \int_{\pi/12}^{5\pi/12} \sin^4 2\theta d\theta = 2 \left(\frac{3}{4} \theta - \frac{1}{4} \sin 4\theta + \frac{1}{32} \sin 8\theta \right) \Big|_{\pi/12}^{5\pi/12} \\ &= 2 \left[\left(\frac{5\pi}{16} + \frac{\sqrt{3}}{8} - \frac{\sqrt{3}}{64} \right) - \left(\frac{\pi}{16} - \frac{\sqrt{3}}{8} + \frac{\sqrt{3}}{64} \right) \right] = \frac{8\pi + 7\sqrt{3}}{16} \end{aligned}$$

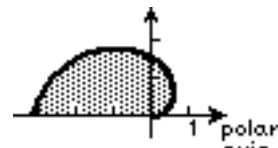


21. From the solution to Problem 17, $I_x = k\pi a^4/4$. By symmetry, $I_y = I_x$. Thus $I_0 = k\pi a^4/2$.



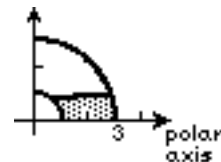
22. The density is $\rho = kr$.

$$\begin{aligned} I_0 &= \int_0^\pi \int_0^\theta r^2 (kr) r dr d\theta = k \int_0^\pi \int_0^\theta r^4 dr d\theta = k \int_0^\pi \left. \frac{1}{5} r^5 \right|_0^\theta d\theta \\ &= \frac{1}{5} k \int_0^\pi \theta^5 d\theta = \frac{1}{5} k \left(\frac{1}{6} \theta^6 \right) \Big|_0^\pi = \frac{k\pi^6}{30} \end{aligned}$$

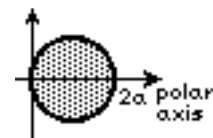


23. The density is $\rho = k/r$.

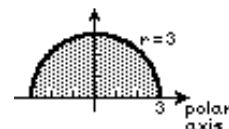
$$I_0 = \int_1^3 \int_0^{1/r} r^2 \frac{k}{r} r d\theta dr = k \int_1^3 \int_0^{1/r} r^2 d\theta dr = k \int_1^3 r^2 \left(\frac{1}{r} \right) dr = k \left(\frac{1}{2} r^2 \right) \Big|_1^3 = 4k$$



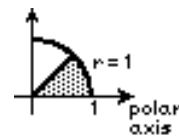
$$\begin{aligned} 24. \quad I_0 &= \int_0^\pi \int_0^{2a \cos \theta} r^2 k r dr d\theta = k \int_0^\pi \left. \frac{1}{4} r^4 \right|_0^{2a \cos \theta} d\theta = 4ka^4 \int_0^\pi \cos^4 \theta d\theta \\ &= 4ka^4 \left(\frac{3}{8} \theta + \frac{1}{4} \sin 2\theta + \frac{1}{32} \sin 4\theta \right) \Big|_0^\pi = 4ka^4 \left(\frac{3\pi}{8} \right) = \frac{3k\pi a^4}{2} \end{aligned}$$



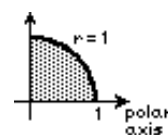
$$25. \quad \int_{-3}^3 \int_0^{\sqrt{9-x^2}} \sqrt{x^2+y^2} dy dx = \int_0^\pi \int_0^3 |r| r dr d\theta = \int_0^\pi \left. \frac{1}{3} r^3 \right|_0^3 d\theta = 9 \int_0^\pi d\theta = 9\pi$$



$$\begin{aligned} 26. \quad \int_0^{\sqrt{2}/2} \int_y^{\sqrt{1-y^2}} \frac{y^2}{\sqrt{x^2+y^2}} dx dy &= \int_0^{\pi/4} \int_0^1 \frac{r^2 \sin^2 \theta}{|r|} r dr d\theta \\ &= \int_0^{\pi/4} \int_0^1 r^2 \sin^2 \theta dr d\theta = \int_0^{\pi/4} \left. \frac{1}{3} r^3 \sin^2 \theta \right|_0^1 d\theta = \frac{1}{3} \int_0^{\pi/4} \sin^2 \theta d\theta \\ &= \frac{1}{3} \left(\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right) \Big|_0^{\pi/4} = \frac{\pi - 2}{24} \end{aligned}$$

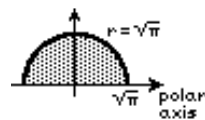


$$\begin{aligned} 27. \quad \int_0^1 \int_0^{\sqrt{1-y^2}} e^{x^2+y^2} dx dy &= \int_0^{\pi/2} \int_0^1 e^{r^2} r dr d\theta = \int_0^{\pi/2} \left. \frac{1}{2} e^{r^2} \right|_0^1 d\theta \\ &= \frac{1}{2} \int_0^{\pi/2} (e - 1) d\theta = \frac{\pi(e - 1)}{4} \end{aligned}$$

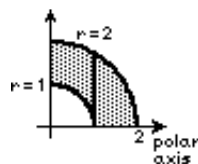


9.11 Double Integrals in Polar Coordinates

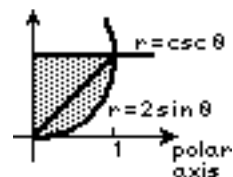
$$\begin{aligned}
 28. \quad \int_{-\sqrt{\pi}}^{\sqrt{\pi}} \int_0^{\sqrt{\pi-x^2}} \sin(x^2+y^2) \, dy \, dx &= \int_0^{\pi} \int_0^{\sqrt{\pi}} (\sin r^2) r \, dr \, d\theta = \int_0^{\pi} -\frac{1}{2} \cos r^2 \Big|_0^{\sqrt{\pi}} d\theta \\
 &= -\frac{1}{2} \int_0^{\pi} (-1-1) \, d\theta = \pi
 \end{aligned}$$



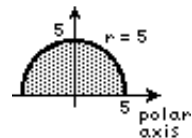
$$\begin{aligned}
 29. \quad \int_0^1 \int_{\sqrt{1-x^2}}^{\sqrt{4-x^2}} \frac{x^2}{x^2+y^2} \, dy \, dx + \int_1^2 \int_0^{\sqrt{4-x^2}} \frac{x^2}{x^2+y^2} \, dy \, dx \\
 = \int_0^{\pi/2} \int_1^2 \frac{r^2 \cos^2 \theta}{r^2} r \, dr \, d\theta = \int_0^{\pi/2} \int_1^2 r \cos^2 \theta \, dr \, d\theta \\
 = \int_0^{\pi/2} \frac{1}{2} r^2 \Big|_1^2 \cos^2 \theta \, d\theta = \frac{3}{2} \int_0^{\pi/2} \cos^2 \theta \, d\theta = \frac{3}{2} \left(\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right) \Big|_0^{\pi/2} = \frac{3\pi}{8}
 \end{aligned}$$



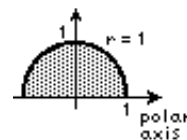
$$\begin{aligned}
 30. \quad \int_0^1 \int_0^{\sqrt{2y-y^2}} (1-x^2-y^2) \, dx \, dy \\
 = \int_0^{\pi/4} \int_0^{2\sin\theta} (1-r^2) r \, dr \, d\theta + \int_{\pi/4}^{\pi/2} \int_0^{\csc\theta} (1-r^2) r \, dr \, d\theta \\
 = \int_0^{\pi/4} \left(\frac{1}{2} r^2 - \frac{1}{4} r^4 \right) \Big|_0^{2\sin\theta} d\theta + \int_{\pi/4}^{\pi/2} \left(\frac{1}{2} r^2 - \frac{1}{4} r^4 \right) \Big|_0^{\csc\theta} d\theta \\
 = \int_0^{\pi/4} (2\sin^2\theta - 4\sin^4\theta) \, d\theta + \int_{\pi/4}^{\pi/2} \left(\frac{1}{2} \csc^2\theta - \frac{1}{4} \csc^4\theta \right) d\theta \\
 = \left[\theta - \frac{1}{2} \sin 2\theta - \left(\frac{3}{2} \theta - \sin 2\theta + \frac{1}{8} \sin 4\theta \right) \right] + \left[-\frac{1}{2} \cot \theta - \frac{1}{4} \left(-\cot \theta - \frac{1}{3} \cot^3 \theta \right) \right] \Big|_{\pi/4}^{\pi/2} \\
 = \left(-\frac{\pi}{8} + \frac{1}{2} \right) + \left[0 - \left(-\frac{1}{4} + \frac{1}{12} \right) \right] = \frac{16-3\pi}{24}
 \end{aligned}$$



$$\begin{aligned}
 31. \quad \int_{-5}^5 \int_0^{\sqrt{25-x^2}} (4x+3y) \, dy \, dx = \int_0^{\pi} \int_0^5 (4r \cos \theta + 3r \sin \theta) r \, dr \, d\theta \\
 = \int_0^{\pi} \int_0^5 (4r^2 \cos \theta + 3r^2 \sin \theta) \, dr \, d\theta = \int_0^{\pi} \left(\frac{4}{3} r^3 \cos \theta + r^3 \sin \theta \right) \Big|_0^5 d\theta \\
 = \int_0^{\pi} \left(\frac{500}{3} \cos \theta + 125 \sin \theta \right) d\theta = \left(\frac{500}{3} \sin \theta - 125 \cos \theta \right) \Big|_0^{\pi} = 250
 \end{aligned}$$



$$\begin{aligned}
 32. \quad \int_0^1 \int_0^{\sqrt{1-y^2}} \frac{1}{1+\sqrt{x^2+y^2}} \, dx \, dy = \int_0^{\pi/2} \int_0^1 \frac{1}{1+r} r \, dr \, d\theta \\
 = \int_0^{\pi/2} \int_0^1 \left(1 - \frac{1}{1+r} \right) dr \, d\theta = \int_0^{\pi/2} [r - \ln(1+r)] \Big|_0^1 d\theta \\
 = \int_0^{\pi/2} (1 - \ln 2) d\theta = \frac{\pi}{2} (1 - \ln 2)
 \end{aligned}$$



33. The volume of the cylindrical portion of the tank is $V_c = \pi(4.2)^2 19.3 \approx 1069.56 \text{ m}^3$. We take the equation of the ellipsoid to be

$$\frac{x^2}{(4.2)^2} + \frac{z^2}{(5.15)^2} = 1 \quad \text{or} \quad z = \pm \frac{5.15}{4.2} \sqrt{(4.2)^2 - x^2 - y^2}.$$

The volume of the ellipsoid is

$$\begin{aligned}
 V_e &= 2 \left(\frac{5.15}{4.2} \right) \iint_R \sqrt{(4.2)^2 - x^2 - y^2} \, dx \, dy = \frac{10.3}{4.2} \int_0^{2\pi} \int_0^{4.2} [(4.2)^2 - r^2]^{1/2} r \, dr \, d\theta \\
 &= \frac{10.3}{4.2} \int_0^{2\pi} \left[\left(-\frac{1}{2} \right) \frac{2}{3} [(4.2)^2 - r^2]^{3/2} \right]_0^{4.2} d\theta = \frac{10.3}{4.2} \frac{1}{3} \int_0^{2\pi} (4.2)^3 d\theta \\
 &= \frac{2\pi}{3} \frac{10.3}{4.2} (4.2)^3 \approx 380.53.
 \end{aligned}$$

The volume of the tank is approximately $1069.56 + 380.53 = 1450.09 \text{ m}^3$.

$$\begin{aligned}
 34. \quad \iint_R (x+y) \, dA &= \int_0^{\pi/2} \int_{2\sin\theta}^2 (r\cos\theta + r\sin\theta) r \, dr \, d\theta = \int_0^{\pi/2} \int_{2\sin\theta}^2 r^2 (\cos\theta + \sin\theta) \, dr \, d\theta \\
 &= \int_0^{\pi/2} \frac{1}{3} r^3 (\cos\theta + \sin\theta) \Big|_{2\sin\theta}^2 d\theta = \frac{8}{3} \int_0^{\pi/2} (\cos\theta + \sin\theta - \sin^3\theta \cos\theta - \sin^4\theta) \, d\theta \\
 &= \frac{8}{3} \left(\sin\theta - \cos\theta - \frac{1}{4} \sin^4\theta + \frac{1}{4} \sin^3\theta \cos\theta - \frac{3}{8}\theta + \frac{3}{16} \sin 2\theta \right) \Big|_0^{\pi/2} \\
 &= \frac{8}{3} \left[\left(1 - \frac{1}{4} - \frac{3\pi}{16} \right) - (-1) \right] = \frac{28 - 3\pi}{6}
 \end{aligned}$$

$$\begin{aligned}
 35. \quad I^2 &= \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} \, dx \, dy = \int_0^{\pi/2} \int_0^\infty e^{-r^2} r \, dr \, d\theta = \int_0^{\pi/2} \lim_{t \rightarrow \infty} \left. -\frac{1}{2} e^{-r^2} \right|_0^t d\theta \\
 &= \int_0^{\pi/2} \lim_{t \rightarrow \infty} \left(-\frac{1}{2} e^{-t^2} + \frac{1}{2} \right) d\theta = \int_0^{\pi/2} \frac{1}{2} d\theta = \frac{\pi}{4}; \quad I = \frac{\sqrt{\pi}}{2}
 \end{aligned}$$

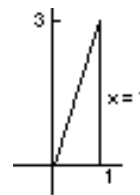
EXERCISES 9.12

Green's Theorem

1. The sides of the triangle are C_1 : $y = 0, 0 \leq x \leq 1$; C_2 : $x = 1, 0 \leq y \leq 3$; C_3 : $y = 3x, 0 \leq -x \leq 1$.

$$\begin{aligned}
 \oint_C (x-y) \, dx + xy \, dy &= \int_0^1 x \, dx + \int_0^3 y \, dy + \int_1^0 (x-3x) \, dx + \int_1^0 x(3x) \, 3 \, dx \\
 &= \left(\frac{1}{2} x^2 \right) \Big|_0^1 + \left(\frac{1}{2} y^2 \right) \Big|_0^3 + (-x^2) \Big|_0^1 + (3x^2) \Big|_1^0 = \frac{1}{2} + \frac{9}{2} + 1 - 3 = 3
 \end{aligned}$$

$$\begin{aligned}
 \iint_R (y+1) \, dA &= \int_0^1 \int_0^{3x} (y+1) \, dy \, dx = \int_0^1 \left(\frac{1}{2} y^2 + y \right) \Big|_0^{3x} dx = \int_0^1 \left(\frac{9}{2} x^2 + 3x \right) dx \\
 &= \left(\frac{3}{2} x^3 + \frac{3}{2} x^2 \right) \Big|_0^1 = 3
 \end{aligned}$$



2. The sides of the rectangle are C_1 : $y = 0, -1 \leq x \leq 1$; C_2 : $x = 1, 0 \leq y \leq 1$; C_3 : $y = 1, 1 \geq x \geq -1$; C_4 : $x = -1, 1 \geq y \geq 0$.

9.12 Green's Theorem

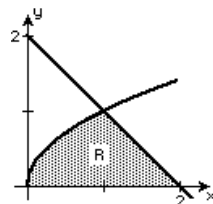
$$\begin{aligned}\oint_C 3x^2y \, dx + (x^2 - 5y) \, dy &= \int_{-1}^1 0 \, dx + \int_0^1 (1 - 5y) \, dy = \int_{-1}^1 0 \, dx + \int_0^1 (1 - 5y) \, dy \\ &= \int_1^{-1} 3x^2 \, dx + \int_1^0 (1 - 5y) \, dy = \left(y - \frac{5}{2}y^2\right) \Big|_0^1 + x^3 \Big|_1^{-1} + \left(y - \frac{5}{2}y^2\right) \Big|_1^0 = -2 \\ \iint_R (2x - 3x^2) \, dA &= \int_0^1 \int_{-1}^1 (2x - 3x^2) \, dx \, dy = \int_0^1 (x^2 - x^3) \Big|_{-1}^1 \, dy = \int_0^1 (-2) \, dy = -2\end{aligned}$$

$$\begin{aligned}3. \quad \oint_C -y^2 \, dx + x^2 \, dy &= \int_0^{2\pi} (-9\sin^2 t)(-3\sin t) \, dt + \int_0^{2\pi} 9\cos^2 t(3\cos t) \, dt \\ &= 27 \int_0^{2\pi} [(1 - \cos^2 t)\sin t + (1 - \sin^2 t)\cos t] \, dt \\ &= 27 \left(-\cos t + \frac{1}{3}\cos^3 t + \sin t - \frac{1}{3}\sin^3 t\right) \Big|_0^{2\pi} = 27(0) = 0 \\ \iint_R (2x + 2y) \, dA &= 2 \int_0^{2\pi} \int_0^3 (r\cos\theta + r\sin\theta)r \, dr \, d\theta = 2 \int_0^{2\pi} \int_0^3 r^2(\cos\theta + \sin\theta) \, dr \, d\theta \\ &= 2 \int_0^{2\pi} \left[\frac{1}{3}r^3(\cos\theta + \sin\theta)\right] \Big|_0^3 \, d\theta = 18 \int_0^{2\pi} (\cos\theta + \sin\theta) \, d\theta \\ &= 18(\sin\theta - \cos\theta) \Big|_0^{2\pi} = 18(0) = 0\end{aligned}$$

4. The sides of the region are C_1 : $y = 0$, $0 \leq x \leq 2$; C_2 : $y = -x + 2$, $2 \geq x \geq 1$; C_3 : $y = \sqrt{x}$, $1 \geq x \geq 0$.

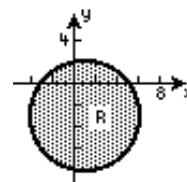
$$\begin{aligned}\oint_C -2y^2 \, dx + 4xy \, dy &= \int_0^2 0 \, dx + \int_2^1 -2(-x + 2)^2 \, dx + \int_2^1 4x(-x + 2)(-dx) \\ &\quad + \int_1^0 -2x \, dx + \int_1^0 4x\sqrt{x} \left(\frac{1}{2\sqrt{x}}\right) \, dx \\ &= 0 + \frac{2}{3} + \frac{8}{3} + 1 - 1 = \frac{10}{3}\end{aligned}$$

$$\iint_R 8y \, dA = \int_0^1 \int_{y^2}^{2-y} 8y \, dx \, dy = \int_0^1 8y(2 - y - y^2) \, dy = \left(8y^2 - \frac{8}{3}y^3 - 2y^4\right) \Big|_0^1 = \frac{10}{3}$$



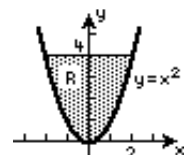
5. $P = 2y$, $P_y = 2$, $Q = 5x$, $Q_x = 5$

$$\oint_C 2y \, dx + 5x \, dy = \iint_R (5 - 2) \, dA = 3 \iint_R dA = 3(25\pi) = 75\pi$$



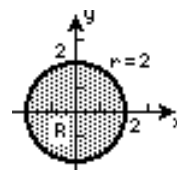
6. $P = x + y^2$, $P_y = 2y$, $Q = 2x^2 - y$, $Q_x = 4x$

$$\begin{aligned}\oint_C (x + y^2) \, dx + (2x^2 - y) \, dy &= \iint_R (4x - 2y) \, dA = \int_{-2}^2 \int_{x^2}^4 (4x - 2y) \, dy \, dx \\ &= \int_{-2}^2 (4xy - y^2) \Big|_{x^2}^4 \, dx = \int_{-2}^2 (16x - 16 - 4x^3 + x^4) \, dx \\ &= \left(8x^2 - 16x - x^4 + \frac{1}{5}x^5\right) \Big|_{-2}^2 = -\frac{96}{5}\end{aligned}$$



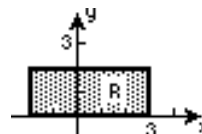
7. $P = x^4 - 2y^3$, $P_y = -6y^2$, $Q = 2x^3 - y^4$, $Q_x = 6x^2$. Using polar coordinates,

$$\begin{aligned}\oint_C (x^4 - 2y^3) dx + (2x^3 - y^4) dy &= \iint_R (6x^2 + 6y^2) dA = \int_0^{2\pi} \int_0^2 6r^2 r dr d\theta \\ &= \int_0^{2\pi} \left(\frac{3}{2} r^4 \right) \bigg|_0^2 d\theta = \int_0^{2\pi} 24 d\theta = 48\pi.\end{aligned}$$



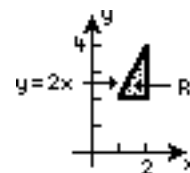
8. $P = x - 3y$, $P_y = -3$, $Q = 4x + y$, $Q_x = 4$

$$\oint_C (x - 3y) dx + 4(x + y) dy = \iint_R (4 + 3) dA = 7(10) = 70$$



9. $P = 2xy$, $P_y = 2x$, $Q = 3xy^2$, $Q_x = 3y^2$

$$\begin{aligned}\oint_C 2xy dx + 3xy^2 dy &= \iint_R (3y^2 - 2x) dA = \int_1^2 \int_2^{2x} (3y^2 - 2x) dy dx \\ &= \int_1^2 (y^3 - 2xy) \bigg|_2^{2x} dx = \int_1^2 (8x^3 - 4x^2 - 8 + 4x) dx \\ &= \left(2x^4 - \frac{4}{3}x^3 - 8x + 2x^2 \right) \bigg|_1^2 = \frac{40}{3} - \left(-\frac{16}{3} \right) = \frac{56}{3}\end{aligned}$$

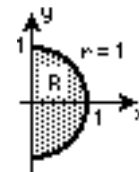


10. $P = e^{2x} \sin 2y$, $P_y = 2e^{2x} \cos 2y$, $Q = e^{2x} \cos 2y$, $Q_x = 2e^{2x} \cos 2y$

$$\oint_C = e^{2x} \sin 2y dx + e^{2x} \cos 2y dy = \iint_R 0 dA = 0$$

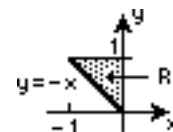
11. $P = xy$, $P_y = x$, $Q = x^2$, $Q_x = 2x$. Using polar coordinates,

$$\begin{aligned}\oint_C xy dx + x^2 dy &= \iint_R (2x - x) dA = \int_{-\pi/2}^{\pi/2} \int_0^1 r \cos \theta r dr d\theta \\ &= \int_{-\pi/2}^{\pi/2} \left(\frac{1}{3} r^3 \cos \theta \right) \bigg|_0^1 d\theta = \int_{-\pi/2}^{\pi/2} \frac{1}{3} \cos \theta d\theta = \frac{1}{3} \sin \theta \bigg|_{-\pi/2}^{\pi/2} = \frac{2}{3}\end{aligned}$$



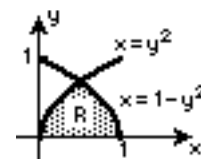
12. $P = e^{x^2}$, $P_y = 0$, $Q = 2 \tan^{-1} x$, $Q_x = \frac{2}{1+x^2}$

$$\begin{aligned}\oint_C e^{x^2} dx + 2 \tan^{-1} x dy &= \iint_R \frac{2}{1+x^2} dA = \int_{-1}^0 \int_{-x}^1 \frac{2}{1+x^2} dy dx \\ &= \int_{-1}^0 \left(\frac{2y}{1+x^2} \right) \bigg|_{-x}^1 dx = \int_{-1}^0 \left(\frac{2}{1+x^2} + \frac{2x}{1+x^2} \right) dx \\ &= [2 \tan^{-1} x + \ln(1+x^2)] \bigg|_{-1}^0 = 0 - \left(-\frac{\pi}{2} + \ln 2 \right) = \frac{\pi}{2} - \ln 2\end{aligned}$$



13. $P = \frac{1}{3}y^3$, $P_y = y^2$, $Q = xy + xy^2$, $Q_x = y + y^2$

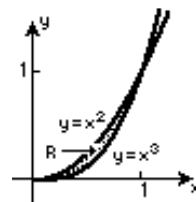
$$\begin{aligned}\oint_C \frac{1}{3}y^3 dx + (xy + xy^2) dy &= \iint_R y dA = \int_0^{1/\sqrt{2}} \int_{y^2}^{1-y^2} y dx dy \\ &= \int_0^{1/\sqrt{2}} (xy) \bigg|_{y^2}^{1-y^2} dy = \int_0^{1/\sqrt{2}} (y - y^3 - y^3) dy \\ &= \left(\frac{1}{2}y^2 - \frac{1}{2}y^4 \right) \bigg|_0^{1/\sqrt{2}} = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}\end{aligned}$$



9.12 Green's Theorem

14. $P = xy^2$, $P_y = 2xy$, $Q = 3 \cos y$, $Q_x = 0$

$$\begin{aligned}\oint_C xy^2 dx + 3 \cos y dy &= \iint_R (-2xy) dA = - \int_0^1 \int_{x^3}^{x^2} 2xy dy dx \\ &= - \int_0^1 (xy) \Big|_{x^3}^{x^2} dx = - \int_0^1 (x^3 - x^4) dx = \left(\frac{1}{4}x^4 - \frac{1}{5}x^5 \right) \Big|_0^1 = -\frac{1}{20}\end{aligned}$$



15. $P = ay$, $P_y = a$, $Q = bx$, $Q_x = b$. $\oint_C ay dx + bx dy = \iint_R (b - a) dA = (b - a) \times (\text{area bounded by } C)$

16. $P = P(x)$, $P_y = 0$, $Q = Q(y)$, $Q_x = 0$. $\oint_C P(x) dx + Q(y) dy = \iint_R 0 dA = 0$

17. For the first integral: $P = 0$, $P_y = 0$, $Q = x$, $Q_x = 1$; $\oint_C x dy = \iint_R 1 dA = \text{area of } R$.

For the second integral: $P = y$, $P_y = 1$, $Q = 0$, $Q_x = 0$; $-\oint_C y dx = -\iint_R -1 dA = \text{area of } R$.

Thus, $\oint_C x dy = -\oint_C y dx$.

18. $P = -y$, $P_y = -1$, $Q = x$, $Q_x = 1$. $\frac{1}{2} \oint_C -y dx + x dy = \frac{1}{2} \iint_R 2 dA = \iint_R dA = \text{area of } R$

19. $A = \iint_R dA = \oint_C x dy = \int_0^{2\pi} a \cos^3 t (3a \sin^2 t \cos t dt) = 3a^2 \int_0^{2\pi} \sin^2 t \cos^4 t dt$
 $= 3a^2 \left(\frac{1}{16}t - \frac{1}{64} \sin 4t + \frac{1}{48} \sin^3 2t \right) \Big|_0^{2\pi} = \frac{3}{8} \pi a^2$

20. $A = \iint_R dA = \oint_C x dy = \int_0^{2\pi} a \cos t (b \cos t dt) = ab \int_0^{2\pi} \cos^2 t dt = ab \left(\frac{1}{2}t + \frac{1}{4} \sin 2t \right) \Big|_0^{2\pi} = \pi ab$

21. (a) Parameterize C by $x = x_1 + (x_2 - x_1)t$ and $y = y_1 + (y_2 - y_1)t$ for $0 \leq t \leq 1$. Then

$$\begin{aligned}\int_C -y dx + x dy &= \int_0^1 -[y_1 + (y_2 - y_1)t](x_2 - x_1) dt + \int_0^1 [x_1 + (x_2 - x_1)t](y_2 - y_1) dt \\ &= -(x_2 - x_1) \left[y_1 t + \frac{1}{2}(y_2 - y_1)t^2 \right] \Big|_0^1 + (y_2 - y_1) \left[x_1 t + \frac{1}{2}(x_2 - x_1)t^2 \right] \Big|_0^1 \\ &= -(x_2 - x_1) \left[y_1 + \frac{1}{2}(y_2 - y_1) \right] + (y_2 - y_1) \left[x_1 + \frac{1}{2}(x_2 - x_1) \right] = x_1 y_2 - x_2 y_1.\end{aligned}$$

(b) Let C_i be the line segment from (x_i, y_i) to (x_{i+1}, y_{i+1}) for $i = 1, 2, \dots, n-1$, and C_2 the line segment from (x_n, y_n) to (x_1, y_1) . Then

$$\begin{aligned}A &= \frac{1}{2} \oint_C -y dx + x dy \quad \boxed{\text{Using Problem 18}} \\ &= \frac{1}{2} \left[\int_{C_1} -y dx + x dy + \int_{C_2} -y dx + x dy + \dots + \int_{C_{n-1}} -y dx + x dy + \int_{C_n} -y dx + x dy \right] \\ &= \frac{1}{2} (x_1 y_2 - x_2 y_1) + \frac{1}{2} (x_2 y_3 - x_3 y_2) + \frac{1}{2} (x_{n-1} y_n - x_n y_{n-1}) + \frac{1}{2} (x_n y_1 - x_1 y_n).\end{aligned}$$

22. From part (b) of Problem 21

$$\begin{aligned}A &= \frac{1}{2} [(-1)(1) - (1)(3)] + \frac{1}{2} [(1)(2) - (4)(1)] + \frac{1}{2} [(4)(5) - (3)(2)] + \frac{1}{2} [(3)(3) - (-1)(5)] \\ &= \frac{1}{2} (-4 - 2 + 14 + 14) = 11.\end{aligned}$$

23. $P = 4x^2 - y^3$, $P_y = -3y^2$; $Q = x^3 + y^2$, $Q_x = 3x^2$.

$$\begin{aligned}\oint_C (4x^2 - y^3) dx + (x^3 + y^2) dy &= \iint_R (3x^2 + 3y^2) dA = \int_0^{2\pi} \int_1^2 3r^2 (r dr d\theta) = \int_0^{2\pi} \left(\frac{3}{4} r^4 \right) \Big|_1^2 d\theta \\ &= \int_0^{2\pi} \frac{45}{4} d\theta = \frac{45\pi}{2}\end{aligned}$$

24. $P = \cos x^2 - y$, $P_y = -1$; $Q = \sqrt{y^3 + 1}$, $Q_x = 0$

$$\oint_C (\cos x^2 - y) dx + \sqrt{y^3 + 1} dy = \iint_R (0 + 1) dA = \iint_R dA = (6\sqrt{2})^2 - \pi(2)(4) = 72 - 8\pi$$

25. We first observe that $P_y = (y^4 - 3x^2y^2)/(x^2 + y^2)^3 = Q_x$. Letting C' be the circle $x^2 + y^2 = \frac{1}{4}$ we have

$$\begin{aligned}\oint_C \frac{-y^3 dx + xy^2 dy}{(x^2 + y^2)^2} &= \oint_{C'} \frac{-y^3 dx + xy^2 dy}{(x^2 + y^2)^2} \\ &\quad \boxed{x = \frac{1}{4} \cos t, \quad dx = -\frac{1}{4} \sin t dt, \quad y = \frac{1}{4} \sin t, \quad dy = \frac{1}{4} \cos t dt} \\ &= \int_0^{2\pi} \frac{-\frac{1}{64} \sin^3 t (-\frac{1}{4} \sin t dt) + \frac{1}{4} \cos t (\frac{1}{16} \sin^2 t) (\frac{1}{4} \cos t dt)}{1/256} \\ &= \int_0^{2\pi} (\sin^4 t + \sin^2 t \cos^2 t) dt = \int_0^{2\pi} (\sin^4 t + (\sin^2 t - \sin^4 t) dt) \\ &= \int_0^{2\pi} \sin^2 t dt = \left(\frac{1}{2} t - \frac{1}{4} \sin 2t \right) \Big|_0^{2\pi} = \pi\end{aligned}$$

26. We first observe that $P_y = [4y^2 - (x+1)^2]/[(x+1)^2 + 4y^2]^2 = Q_x$. Letting C' be the ellipse $(x+1)^2 + 4y^2 = 4$ we have

$$\begin{aligned}\oint_C \frac{-y}{(x+1)^2 + 4y^2} dx + \frac{x+1}{(x+1)^2 + 4y^2} dy &= \oint_{C'} \frac{-y}{(x+1)^2 + 4y^2} dx + \frac{x+1}{(x+1)^2 + 4y^2} dy \\ &\quad \boxed{x+1 = 2 \cos t, \quad dx = -2 \sin t dt, \quad y = \sin t, \quad dy = \cos t dt} \\ &= \int_0^{2\pi} \left[\frac{-\sin t}{4} (-2 \sin t) + \frac{2 \cos t}{4} \cos t \right] dt = \frac{1}{2} \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = \pi.\end{aligned}$$

27. Writing $\iint_R x^2 dA = \iint_R (Q_x - P_y) dA$ we identify $Q = 0$ and $P = -x^2y$. Then, with C : $x = 3 \cos t$, $y = 2 \sin t$, $0 \leq t \leq 2\pi$, we have

$$\begin{aligned}\iint_R x^2 dA &= \oint_C P dx + Q dy = \oint_C -x^2y dx = - \int_0^{2\pi} 9 \cos^2 t (2 \sin t) (-3 \sin t) dt \\ &= \frac{54}{4} \int_0^{2\pi} 4 \sin^2 t \cos^2 t dt = \frac{27}{2} \int_0^{2\pi} \sin^2 2t dt = \frac{27}{4} \int_0^{2\pi} (1 - \cos 4t) dt \\ &= \frac{27}{4} \left(t - \frac{1}{4} \sin 4t \right) \Big|_0^{2\pi} = \frac{27\pi}{2}.\end{aligned}$$

28. Writing $\iint_R [1 - 2(y-1)] dA = \iint_R (Q_x - P_y) dA$ we identify $Q = x$ and $P = (y-1)^2$. Then, with

C_1 : $x = \cos t$, $y - 1 = \sin t$, $-\pi/2 \leq t \leq \pi/2$, and C_2 : $x = 0$, $2 \geq y \geq 0$,

9.12 Green's Theorem

$$\begin{aligned}
 \iint_R [1 - 2(y - 1)] dA &= \int_{C_1} P dx + Q dy + \int_{C_2} P dx + Q dy = \int_{C_1} (y - 1)^2 dx + x dy + \int_{C_2} 0 dy \\
 &= \int_{-\pi/2}^{\pi/2} [\sin^2 t(-\sin t) + \cos t \cos t] dt = \int_{-\pi/2}^{\pi/2} [\cos^2 t - (1 - \cos^2 t) \sin t] dt \\
 &= \int_{-\pi/2}^{\pi/2} \left[\frac{1}{2}(1 + \cos 2t) - \sin t + \cos^2 t \sin t \right] dt \\
 &= \left(\frac{1}{2}t + \frac{1}{4} \sin 2t + \cos t - \frac{1}{3} \cos^3 t \right) \Big|_{-\pi/2}^{\pi/2} = \frac{\pi}{4} - \left(-\frac{\pi}{4} \right) = \frac{\pi}{2}.
 \end{aligned}$$

29. $P = x - y$, $P_y = -1$, $Q = x + y$, $Q_x = 1$; $W = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R 2 dA = 2 \times \text{area} = 2 \left(\frac{3\pi}{4} \right) = \frac{3}{2}\pi$

30. $P = -xy^2$, $P_y = -2xy$, $Q = x^2y$, $Q_x = 2xy$. Using polar coordinates,

$$\begin{aligned}
 W &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R 4xy dA = \int_0^{\pi/2} \int_1^2 4(r \cos \theta)(r \sin \theta)r dr d\theta = \int_0^{\pi/2} (r^4 \cos \theta \sin \theta) \Big|_1^2 d\theta \\
 &= 15 \int_0^{\pi/2} \sin \theta \cos \theta d\theta = \frac{15}{2} \sin^2 \theta \Big|_0^{\pi/2} = \frac{15}{2}.
 \end{aligned}$$

31. Since $\int_A^B P dx + Q dy$ is independent of path, $P_y = Q_x$ by Theorem 9.9. Then, by Green's Theorem

$$\oint_C P dx + Q dy = \iint_R (Q_x - P_y) dA = \iint_R 0 dA = 0.$$

32. Let $P = 0$ and $Q = x^2$. Then $Q_x - P_y = 2x$ and

$$\frac{1}{2A} \oint_C x^2 dy = \frac{1}{2A} \iint_R 2x dA = \frac{\iint_R x dA}{A} = \bar{x}.$$

Let $P = y^2$ and $Q = 0$. Then $Q_x - P_y = -2y$ and

$$-\frac{1}{2A} \oint_C y^2 dx = -\frac{1}{2A} \iint_R -2y dA = \frac{\iint_R y dA}{A} = \bar{y}.$$

33. Using Green's Theorem,

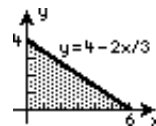
$$\begin{aligned}
 W &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C -y dx + x dy = \iint_R 2 dA = 2 \int_0^{2\pi} \int_0^{1+\cos \theta} r dr d\theta \\
 &= 2 \int_0^{2\pi} \left(\frac{1}{2} r^2 \right) \Big|_0^{1+\cos \theta} d\theta = \int_0^{2\pi} (1 + 2 \cos \theta + \cos^2 \theta) d\theta \\
 &= \left(\theta + 2 \sin \theta + \frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right) \Big|_0^{2\pi} = 3\pi.
 \end{aligned}$$

EXERCISES 9.13

Surface Integrals

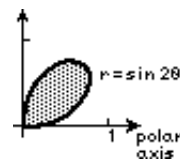
1. Letting $z = 0$, we have $2x + 3y = 12$. Using $f(x, y) = z = 3 - \frac{1}{2}x - \frac{3}{4}y$ we have $f_x = -\frac{1}{2}$, $f_y = -\frac{3}{4}$, $1 + f_x^2 + f_y^2 = \frac{29}{16}$. Then

$$\begin{aligned} A &= \int_0^6 \int_0^{4-2x/3} \sqrt{29/16} dy dx = \frac{\sqrt{29}}{4} \int_0^6 \left(4 - \frac{2}{3}x\right) dx = \frac{\sqrt{29}}{4} \left(4x - \frac{1}{3}x^2\right) \Big|_0^6 \\ &= \frac{\sqrt{29}}{4}(24 - 12) = 3\sqrt{29}. \end{aligned}$$



2. We see from the graph in Problem 1 that the plane is entirely above the region bounded by $r = \sin 2\theta$ in the first octant. Using $f(x, y) = z = 3 - \frac{1}{2}x - \frac{3}{4}y$ we have $f_x = -\frac{1}{2}$, $f_y = -\frac{3}{4}$, $1 + f_x^2 + f_y^2 = \frac{29}{16}$. Then

$$\begin{aligned} A &= \int_0^{\pi/2} \int_0^{\sin 2\theta} \sqrt{29/16} r dr d\theta = \frac{\sqrt{29}}{4} \int_0^{\pi/2} \frac{1}{2} r^2 \Big|_0^{\sin 2\theta} d\theta = \frac{\sqrt{29}}{8} \int_0^{\pi/2} \sin^2 2\theta d\theta \\ &= \frac{\sqrt{29}}{8} \left(\frac{1}{2}\theta - \frac{1}{8}\sin 4\theta \right) \Big|_0^{\pi/2} = \frac{\sqrt{29}\pi}{32}. \end{aligned}$$

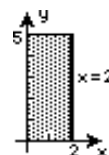


3. Using $f(x, y) = z = \sqrt{16 - x^2}$ we see that for $0 \leq x \leq 2$ and $0 \leq y \leq 5$, $z > 0$.

Thus, the surface is entirely above the region. Now $f_x = -\frac{x}{\sqrt{16 - x^2}}$, $f_y = 0$,

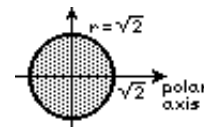
$$1 + f_x^2 + f_y^2 = 1 + \frac{x^2}{16 - x^2} = \frac{16}{16 - x^2} \text{ and}$$

$$A = \int_0^5 \int_0^2 \frac{4}{\sqrt{16 - x^2}} dx dy = 4 \int_0^5 \sin^{-1} \frac{x}{4} \Big|_0^2 dy = 4 \int_0^5 \frac{\pi}{6} dy = \frac{10\pi}{3}.$$



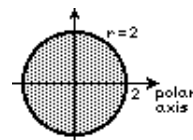
4. The region in the xy -plane beneath the surface is bounded by the graph of $x^2 + y^2 = 2$. Using $f(x, y) = z = x^2 + y^2$ we have $f_x = 2x$, $f_y = 2y$, $1 + f_x^2 + f_y^2 = 1 + 4(x^2 + y^2)$. Then,

$$A = \int_0^{2\pi} \int_0^{\sqrt{2}} \sqrt{1 + 4r^2} r dr d\theta = \int_0^{2\pi} \frac{1}{12} (1 + 4r^2)^{3/2} \Big|_0^{\sqrt{2}} d\theta = \frac{1}{12} \int_0^{2\pi} (27 - 1) d\theta = \frac{13\pi}{3}.$$



5. Letting $z = 0$ we have $x^2 + y^2 = 4$. Using $f(x, y) = z = 4 - (x^2 + y^2)$ we have $f_x = -2x$, $f_y = -2y$, $1 + f_x^2 + f_y^2 = 1 + 4(x^2 + y^2)$. Then

$$\begin{aligned} A &= \int_0^{2\pi} \int_0^2 \sqrt{1 + 4r^2} r dr d\theta = \int_0^{2\pi} \frac{1}{3} (1 + 4r^2)^{3/2} \Big|_0^2 d\theta \\ &= \frac{1}{12} \int_0^{2\pi} (17^{3/2} - 1) d\theta = \frac{\pi}{6} (17^{3/2} - 1). \end{aligned}$$

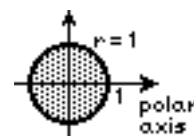


9.13 Surface Integrals

6. The surfaces $x^2 + y^2 + z^2 = 2$ and $z^2 = x^2 + y^2$ intersect on the cylinder $2x^2 + 2y^2 = 2$ or $x^2 + y^2 = 1$. There are portions of the sphere within the cone both above and below the xy -plane. Using $f(x, y) = \sqrt{2 - x^2 - y^2}$ we have $f_x = -\frac{x}{\sqrt{2 - x^2 - y^2}}$,

$$f_y = -\frac{y}{\sqrt{2 - x^2 - y^2}}, \quad 1 + f_x^2 + f_y^2 = \frac{2}{2 - x^2 - y^2}. \quad \text{Then}$$

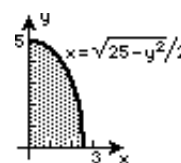
$$\begin{aligned} A &= 2 \left[\int_0^{2\pi} \int_0^1 \frac{\sqrt{2}}{\sqrt{2 - r^2}} r \, dr \, d\theta \right] = 2\sqrt{2} \int_0^{2\pi} -\sqrt{2 - r^2} \Big|_0^1 d\theta \\ &= 2\sqrt{2} \int_0^{2\pi} (\sqrt{2} - 1) d\theta = 4\pi\sqrt{2}(\sqrt{2} - 1). \end{aligned}$$



7. Using $f(x, y) = z = \sqrt{25 - x^2 - y^2}$ we have $f_x = -\frac{x}{\sqrt{25 - x^2 - y^2}}$,

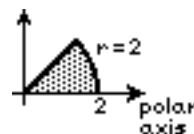
$$f_y = -\frac{y}{\sqrt{25 - x^2 - y^2}}, \quad 1 + f_x^2 + f_y^2 = \frac{25}{25 - x^2 - y^2}. \quad \text{Then}$$

$$\begin{aligned} A &= \int_0^5 \int_0^{\sqrt{25 - y^2}/2} \frac{5}{\sqrt{25 - x^2 - y^2}} \, dx \, dy = 5 \int_0^5 \sin^{-1} \frac{x}{\sqrt{25 - y^2}} \Big|_0^{\sqrt{25 - y^2}/2} dy \\ &= 5 \int_0^5 \frac{\pi}{6} \, dy = \frac{25\pi}{6}. \end{aligned}$$



8. In the first octant, the graph of $z = x^2 - y^2$ intersects the xy -plane in the line $y = x$. The surface is in the first octant for $x > y$. Using $f(x, y) = z = x^2 - y^2$ we have $f_x = 2x$, $f_y = -2y$, $1 + f_x^2 + f_y^2 = 1 + 4x^2 + 4y^2$. Then

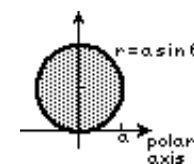
$$\begin{aligned} A &= \int_0^{\pi/4} \int_0^2 \sqrt{1 + 4r^2} \, r \, dr \, d\theta = \int_0^{\pi/4} \frac{1}{12} (1 + 4r^2)^{3/2} \Big|_0^2 d\theta \\ &= \frac{1}{12} \int_0^{\pi/4} (17^{3/2} - 1) d\theta = \frac{\pi}{48} (17^{3/2} - 1). \end{aligned}$$



9. There are portions of the sphere within the cylinder both above and below the xy -plane.

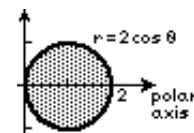
Using $f(x, y) = z = \sqrt{a^2 - x^2 - y^2}$ we have $f_x = -\frac{x}{\sqrt{a^2 - x^2 - y^2}}$, $f_y = -\frac{y}{\sqrt{a^2 - x^2 - y^2}}$, $1 + f_x^2 + f_y^2 = \frac{a^2}{a^2 - x^2 - y^2}$. Then, using symmetry,

$$\begin{aligned} A &= 2 \left[2 \int_0^{\pi/2} \int_0^{a \sin \theta} \frac{a}{\sqrt{a^2 - r^2}} r \, dr \, d\theta \right] = 4a \int_0^{\pi/2} -\sqrt{a^2 - r^2} \Big|_0^{a \sin \theta} d\theta \\ &= 4a \int_0^{\pi/2} (a - a\sqrt{1 - \sin^2 \theta}) d\theta = 4a^2 \int_0^{\pi/2} (1 - \cos \theta) d\theta \\ &= 4a^2 (\theta - \sin \theta) \Big|_0^{\pi/2} = 4a^2 \left(\frac{\pi}{2} - 1 \right) = 2a^2 (\pi - 2). \end{aligned}$$



10. There are portions of the cone within the cylinder both above and below the xy -plane. Using $f(x, y) = \frac{1}{2}\sqrt{x^2 + y^2}$, we have $f_x = \frac{x}{2\sqrt{x^2 + y^2}}$, $f_y = \frac{y}{2\sqrt{x^2 + y^2}}$, $1 + f_x^2 + f_y^2 = \frac{5}{4}$.

Then, using symmetry,



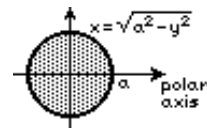
$$\begin{aligned}
 A &= 2 \left[2 \int_0^{\pi/2} \int_0^{2 \cos \theta} \sqrt{\frac{5}{4}} r \, dr \, d\theta \right] = 2\sqrt{5} \int_0^{\pi/2} \frac{1}{2} r^2 \Big|_0^{2 \cos \theta} d\theta \\
 &= 4\sqrt{5} \int_0^{\pi/2} \cos^2 \theta \, d\theta = 4\sqrt{5} \left(\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right) \Big|_0^{\pi/2} = \sqrt{5} \pi.
 \end{aligned}$$

11. There are portions of the surface in each octant with areas equal to the area of the portion

in the first octant. Using $f(x, y) = z = \sqrt{a^2 - y^2}$ we have $f_x = 0$, $f_y = \frac{y}{\sqrt{a^2 - y^2}}$,

$1 + f_x^2 + f_y^2 = \frac{a^2}{a^2 - y^2}$. Then

$$A = 8 \int_0^a \int_0^{\sqrt{a^2 - y^2}} \frac{a}{\sqrt{a^2 - y^2}} \, dx \, dy = 8a \int_0^a \frac{x}{\sqrt{a^2 - y^2}} \Big|_0^{\sqrt{a^2 - y^2}} dy = 8a \int_0^a dy = 8a^2.$$



12. From Example 1, the area of the portion of the hemisphere within $x^2 + y^2 = b^2$ is $2\pi a(a - \sqrt{a^2 - b^2})$. Thus,

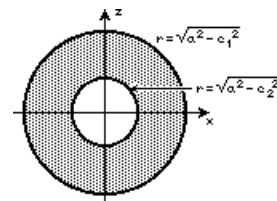
the area of the sphere is $A = 2 \lim_{b \rightarrow a} 2\pi a(a - \sqrt{a^2 - b^2}) = 2(2\pi a^2) = 4\pi a^2$.

13. The projection of the surface onto the xz -plane is shown in the graph. Using $f(x, z) =$

$y = \sqrt{a^2 - x^2 - z^2}$ we have $f_x = -\frac{x}{\sqrt{a^2 - x^2 - z^2}}$,

$f_z = -\frac{z}{\sqrt{a^2 - x^2 - z^2}}$, $1 + f_x^2 + f_z^2 = \frac{a^2}{a^2 - x^2 - z^2}$. Then

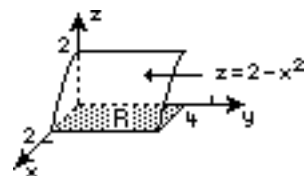
$$\begin{aligned}
 A &= \int_0^{2\pi} \int_{\sqrt{a^2 - c_2^2}}^{\sqrt{a^2 - c_1^2}} \frac{a}{\sqrt{a^2 - r^2}} r \, dr \, d\theta = a \int_0^{2\pi} -\sqrt{a^2 - r^2} \Big|_{\sqrt{a^2 - c_2^2}}^{\sqrt{a^2 - c_1^2}} d\theta \\
 &= a \int_0^{2\pi} (c_2 - c_1) \, d\theta = 2\pi a(c_2 - c_1).
 \end{aligned}$$



14. The surface area of the cylinder $x^2 + z^2 = a^2$ from $y = c_1$ to $y = c_2$ is the area of a cylinder of radius a and height $c_2 - c_1$. This is $2\pi a(c_2 - c_1)$.

15. $z_x = -2x$, $z_y = 0$; $dS = \sqrt{1 + 4x^2} \, dA$

$$\begin{aligned}
 \iint_S x \, dS &= \int_0^4 \int_0^{\sqrt{2}} x \sqrt{1 + 4x^2} \, dx \, dy = \int_0^4 \frac{1}{12} (1 + 4x^2)^{3/2} \Big|_0^{\sqrt{2}} dy \\
 &= \int_0^4 \frac{13}{6} \, dy = \frac{26}{3}
 \end{aligned}$$



16. See Problem 15.

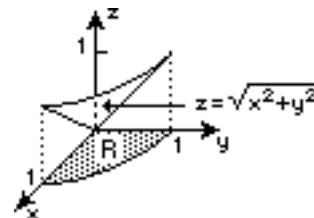
$$\begin{aligned}
 \iint_S xy(9 - 4z) \, dS &= \iint_S xy(1 + 4x^2) \, dS = \int_0^4 \int_0^{\sqrt{2}} xy(1 + 4x^2)^{3/2} \, dx \, dy \\
 &= \int_0^4 \frac{y}{20} (1 + 4x^2)^{5/2} \Big|_0^{\sqrt{2}} dy = \int_0^4 \frac{242}{20} y \, dy = \frac{121}{10} \int_0^4 y \, dy = \frac{121}{10} \left(\frac{1}{2} y^2 \right) \Big|_0^4 = \frac{484}{5}
 \end{aligned}$$

9.13 Surface Integrals

17. $z_x = \frac{x}{\sqrt{x^2 + y^2}}, z_y = \frac{y}{\sqrt{x^2 + y^2}}; dS = \sqrt{2} dA.$

Using polar coordinates,

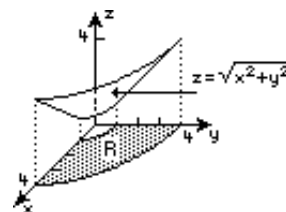
$$\begin{aligned}\iint_S xz^3 dS &= \iint_R x(x^2 + y^2)^{3/2} \sqrt{2} dA = \sqrt{2} \int_0^{2\pi} \int_0^1 (r \cos \theta) r^{3/2} r dr d\theta \\ &= \sqrt{2} \int_0^{2\pi} \int_0^1 r^{7/2} \cos \theta dr d\theta = \sqrt{2} \int_0^{2\pi} \frac{2}{9} r^{9/2} \cos \theta \Big|_0^1 d\theta \\ &= \sqrt{2} \int_0^{2\pi} \frac{2}{9} \cos \theta d\theta = \frac{2\sqrt{2}}{9} \sin \theta \Big|_0^{2\pi} = 0.\end{aligned}$$



18. $z_x = \frac{x}{\sqrt{x^2 + y^2}}, z_y = \frac{y}{\sqrt{x^2 + y^2}}; dS = \sqrt{2} dA.$

Using polar coordinates,

$$\begin{aligned}\iint_S (x + y + z) dS &= \iint_R (x + y + \sqrt{x^2 + y^2}) \sqrt{2} dA \\ &= \sqrt{2} \int_0^{2\pi} \int_1^4 (r \cos \theta + r \sin \theta + r) r dr d\theta \\ &= \sqrt{2} \int_0^{2\pi} \int_1^4 r^2 (1 + \cos \theta + \sin \theta) dr d\theta = \sqrt{2} \int_0^{2\pi} \frac{1}{3} r^3 (1 + \cos \theta + \sin \theta) \Big|_1^4 d\theta \\ &= \frac{63\sqrt{2}}{3} \int_0^{2\pi} (1 + \cos \theta + \sin \theta) d\theta = 21\sqrt{2} (\theta + \sin \theta - \cos \theta) \Big|_0^{2\pi} = 42\sqrt{2} \pi.\end{aligned}$$

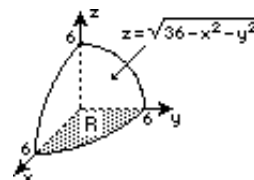


19. $z = \sqrt{36 - x^2 - y^2}, z_x = -\frac{x}{\sqrt{36 - x^2 - y^2}}, z_y = -\frac{y}{\sqrt{36 - x^2 - y^2}};$

$$dS = \sqrt{1 + \frac{x^2}{36 - x^2 - y^2} + \frac{y^2}{36 - x^2 - y^2}} dA = \frac{6}{\sqrt{36 - x^2 - y^2}} dA.$$

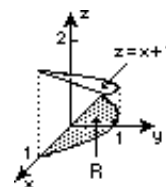
Using polar coordinates,

$$\begin{aligned}\iint_S (x^2 + y^2) z dS &= \iint_R (x^2 + y^2) \sqrt{36 - x^2 - y^2} \frac{6}{\sqrt{36 - x^2 - y^2}} dA \\ &= 6 \int_0^{2\pi} \int_0^6 r^2 r dr d\theta = 6 \int_0^{2\pi} \frac{1}{4} r^4 \Big|_0^6 d\theta = 6 \int_0^{2\pi} 324 d\theta = 972\pi.\end{aligned}$$



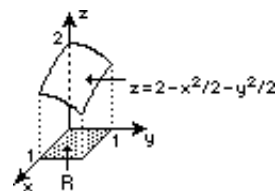
20. $z_x = 1, z_y = 0; dS = \sqrt{2} dA$

$$\begin{aligned}\iint_S z^2 dS &= \int_{-1}^1 \int_0^{1-x^2} (x+1)^2 \sqrt{2} dy dx = \sqrt{2} \int_{-1}^1 y(x+1)^2 \Big|_0^{1-x^2} dx \\ &= \sqrt{2} \int_{-1}^1 (1-x^2)(x+1)^2 dx = \sqrt{2} \int_{-1}^1 (1+2x-2x^3-x^4) dx \\ &= \sqrt{2} \left(x + x^2 - \frac{1}{2} x^4 - \frac{1}{5} x^5 \right) \Big|_{-1}^1 = \frac{8\sqrt{2}}{5}\end{aligned}$$



21. $z_x = -x$, $z_y = -y$; $dS = \sqrt{1+x^2+y^2} dA$

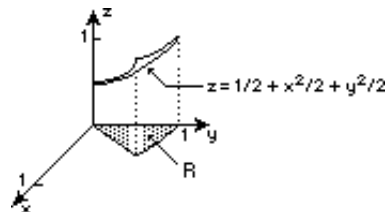
$$\begin{aligned}\iint_S xy dS &= \int_0^1 \int_0^1 xy \sqrt{1+x^2+y^2} dx dy = \int_0^1 \frac{1}{3} y(1+x^2+y^2)^{3/2} \Big|_0^1 dy \\ &= \int_0^1 \left[\frac{1}{3} y(2+y^2)^{3/2} - \frac{1}{3} y(1+y^2)^{3/2} \right] dy \\ &= \left[\frac{1}{15} (2+y^2)^{5/2} - \frac{1}{15} (1+y^2)^{5/2} \right] \Big|_0^1 = \frac{1}{15} (3^{5/2} - 2^{7/2} + 1)\end{aligned}$$



22. $z = \frac{1}{2} + \frac{1}{2}x^2 + \frac{1}{2}y^2$, $z_x = x$, $z_y = y$; $dS = \sqrt{1+x^2+y^2} dA$

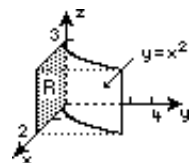
Using polar coordinates,

$$\begin{aligned}\iint_S 2z dS &= \iint_R (1+x^2+y^2) \sqrt{1+x^2+y^2} dA \\ &= \int_{\pi/3}^{\pi/2} \int_0^1 (1+r^2) \sqrt{1+r^2} r dr d\theta \\ &= \int_{\pi/3}^{\pi/2} \int_0^1 (1+r^2)^{3/2} r dr d\theta = \int_{\pi/3}^{\pi/2} \frac{1}{5} (1+r^2)^{5/2} \Big|_0^1 d\theta = \frac{1}{5} \int_{\pi/3}^{\pi/2} (2^{5/2} - 1) d\theta \\ &= \frac{4\sqrt{2}-1}{5} \left(\frac{\pi}{2} - \frac{\pi}{3} \right) = \frac{(4\sqrt{2}-1)\pi}{30}.\end{aligned}$$



23. $y_x = 2x$, $y_z = 0$; $dS = \sqrt{1+4x^2} dA$

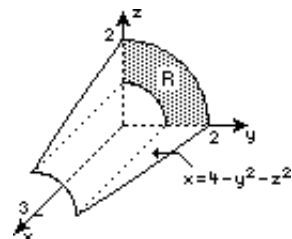
$$\begin{aligned}\iint_S 24\sqrt{y} z dS &= \int_0^3 \int_0^2 24xz \sqrt{1+4x^2} dx dz = \int_0^3 2z(1+4x^2)^{3/2} \Big|_0^2 dz \\ &= 2(17^{3/2} - 1) \int_0^3 z dz = 2(17^{3/2} - 1) \left(\frac{1}{2} z^2 \right) \Big|_0^3 = 9(17^{3/2} - 1)\end{aligned}$$



24. $x_y = -2y$, $x_z = -2z$; $dS = \sqrt{1+4y^2+4z^2} dA$

Using polar coordinates,

$$\begin{aligned}\iint_S (1+4y^2+4z^2)^{1/2} dS &= \int_0^{\pi/2} \int_1^2 (1+4r^2) r dr d\theta \\ &= \int_0^{\pi/2} \frac{1}{16} (1+4r^2)^2 \Big|_1^2 d\theta = \frac{1}{16} \int_0^{\pi/2} 12 d\theta = \frac{3\pi}{8}.\end{aligned}$$



25. Write the equation of the surface as $y = \frac{1}{2}(6-x-3z)$. Then $y_x = -\frac{1}{2}$, $y_z = -\frac{3}{2}$; $dS = \sqrt{1+1/4+9/4} = \frac{\sqrt{14}}{2}$.

$$\begin{aligned}\iint_S (3z^2 + 4yz) dS &= \int_0^2 \int_0^{6-3z} \left[3z^2 + 4z \frac{1}{2}(6-x-3z) \right] \frac{\sqrt{14}}{2} dx dz \\ &= \frac{\sqrt{14}}{2} \int_0^2 [3z^2 x - z(6-x-3z)^2] \Big|_0^{6-3z} dz \\ &= \frac{\sqrt{14}}{2} \int_0^2 ([3z^2(6-3z) - 0] - [0 - z(6-3z)^2]) dz \\ &= \frac{\sqrt{14}}{2} \int_0^2 (36z - 18z^2) dz = \frac{\sqrt{14}}{2} (18z^2 - 6z^3) \Big|_0^2 = \frac{\sqrt{14}}{2} (72 - 48) = 12\sqrt{14}\end{aligned}$$

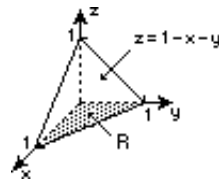
9.13 Surface Integrals

26. Write the equation of the surface as $x = 6 - 2y - 3z$. Then $x_y = -2$, $x_z = -3$; $dS = \sqrt{1+4+9} = \sqrt{14}$.

$$\begin{aligned}\iint_S (3z^2 + 4yz) dS &= \int_0^2 \int_0^{3-3z/2} (3z^2 + 4yz) \sqrt{14} dy dz = \sqrt{14} \int_0^2 (3yz + 2y^2 z) \Big|_0^{3-3z/2} dz \\ &= \sqrt{14} \int_0^2 \left[9z \left(1 - \frac{z}{2}\right) + 18z \left(1 - \frac{z}{2}\right)^2 \right] dz = \sqrt{14} \int_0^2 \left(27z - \frac{45}{2} z^2 + \frac{9}{2} z^3 \right) dz \\ &= \sqrt{14} \left(\frac{27}{2} z^2 - \frac{15}{2} z^3 + \frac{9}{8} z^4 \right) \Big|_0^2 = \sqrt{14} (54 - 60 + 18) = 2\sqrt{14}\end{aligned}$$

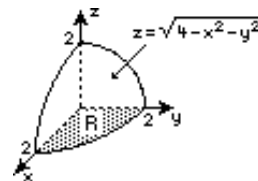
27. The density is $\rho = kx^2$. The surface is $z = 1 - x - y$. Then $z_x = -1$, $z_y = -1$; $dS = \sqrt{3} dA$.

$$\begin{aligned}m &= \iint_S kx^2 dS = k \int_0^1 \int_0^{1-x} x^2 \sqrt{3} dy dx = \sqrt{3} k \int_0^1 \frac{1}{3} x^3 \Big|_0^{1-x} dx \\ &= \frac{\sqrt{3}}{3} k \int_0^1 (1-x)^3 dx = \frac{\sqrt{3}}{3} k \left[-\frac{1}{4} (1-x)^4 \right] \Big|_0^1 = \frac{\sqrt{3}}{12} k\end{aligned}$$



28. $z_x = -\frac{x}{\sqrt{4-x^2-y^2}}$, $z_y = -\frac{y}{\sqrt{4-x^2-y^2}}$;

$$dS = \sqrt{1 + \frac{x^2}{4-x^2-y^2} + \frac{y^2}{4-x^2-y^2}} dA = \frac{2}{\sqrt{4-x^2-y^2}} dA.$$



Using symmetry and polar coordinates,

$$\begin{aligned}m &= 4 \iint_S |xy| dS = 4 \int_0^{\pi/2} \int_0^2 (r^2 \cos \theta \sin \theta) \frac{2}{\sqrt{4-r^2}} r dr d\theta \\ &= 4 \int_0^{\pi/2} \int_0^2 r^2 (4-r^2)^{-1/2} \sin 2\theta (r dr) d\theta \quad \boxed{u = 4 - r^2, du = -2r dr, r^2 = 4 - u} \\ &= 4 \int_0^{\pi/2} \int_4^0 (4-u) u^{-1/2} \sin 2\theta \left(-\frac{1}{2} du \right) d\theta = -2 \int_0^{\pi/2} \int_4^0 (4u^{-1/2} - u^{1/2}) \sin 2\theta du d\theta \\ &= -2 \int_0^{\pi/2} \left(8u^{1/2} - \frac{2}{3} u^{3/2} \right) \Big|_4^0 \sin 2\theta d\theta = -2 \int_0^{\pi/2} \left(-\frac{32}{3} \sin 2\theta \right) d\theta = \frac{64}{3} \left(-\frac{1}{2} \cos 2\theta \right) \Big|_0^{\pi/2} = \frac{64}{3}.\end{aligned}$$

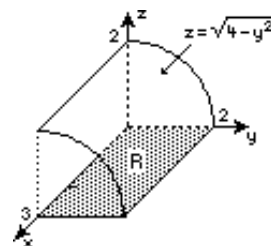
29. The surface is $g(x, y, z) = y^2 + z^2 - 4 = 0$. $\nabla g = 2y\mathbf{j} + 2z\mathbf{k}$,

$$|\nabla g| = 2\sqrt{y^2 + z^2}; \quad \mathbf{n} = \frac{y\mathbf{j} + z\mathbf{k}}{\sqrt{y^2 + z^2}};$$

$$\mathbf{F} \cdot \mathbf{n} = \frac{2yz}{\sqrt{y^2 + z^2}} + \frac{yz}{\sqrt{y^2 + z^2}} = \frac{3yz}{\sqrt{y^2 + z^2}}; \quad z = \sqrt{4 - y^2}, \quad z_x = 0,$$

$$z_y = -\frac{y}{\sqrt{4 - y^2}}; \quad dS = \sqrt{1 + \frac{y^2}{4 - y^2}} dA = \frac{2}{\sqrt{4 - y^2}} dA$$

$$\begin{aligned}\text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_R \frac{3yz}{\sqrt{y^2 + z^2}} \frac{2}{\sqrt{4 - y^2}} dA = \iint_R \frac{3y\sqrt{4 - y^2}}{\sqrt{y^2 + 4 - y^2}} \frac{2}{\sqrt{4 - y^2}} dA \\ &= \int_0^3 \int_0^2 3y dy dx = \int_0^3 \frac{3}{2} y^2 \Big|_0^2 dx = \int_0^3 6 dx = 18\end{aligned}$$

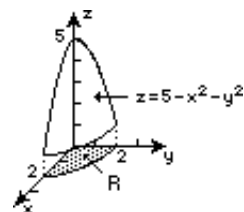


30. The surface is $g(x, y, z) = x^2 + y^2 + z - 5 = 0$. $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$,

$$|\nabla g| = \sqrt{1 + 4x^2 + 4y^2}; \quad \mathbf{n} = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{1 + 4x^2 + 4y^2}}; \quad \mathbf{F} \cdot \mathbf{n} = \frac{z}{\sqrt{1 + 4x^2 + 4y^2}};$$

$$z_x = -2x, \quad z_y = -2y, \quad dS = \sqrt{1 + 4x^2 + 4y^2} \, dA. \text{ Using polar coordinates,}$$

$$\begin{aligned} \text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R \frac{z}{\sqrt{1 + 4x^2 + 4y^2}} \sqrt{1 + 4x^2 + 4y^2} \, dA = \iint_R (5 - x^2 - y^2) \, dA \\ &= \int_0^{2\pi} \int_0^2 (5 - r^2)r \, dr \, d\theta = \int_0^{2\pi} \left(\frac{5}{2}r^2 - \frac{1}{4}r^4 \right) \Big|_0^2 d\theta = \int_0^{2\pi} 6 \, d\theta = 12\pi. \end{aligned}$$



31. From Problem 30, $\mathbf{n} = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{1 + 4x^2 + 4y^2}}$. Then $\mathbf{F} \cdot \mathbf{n} = \frac{2x^2 + 2y^2 + z}{\sqrt{1 + 4x^2 + 4y^2}}$. Also, from Problem 30,

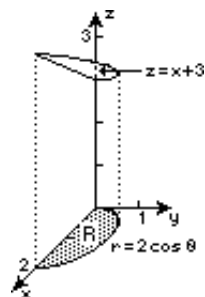
$$dS = \sqrt{1 + 4x^2 + 4y^2} \, dA. \text{ Using polar coordinates,}$$

$$\begin{aligned} \text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R \frac{2x^2 + 2y^2 + z}{\sqrt{1 + 4x^2 + 4y^2}} \sqrt{1 + 4x^2 + 4y^2} \, dA = \iint_R (2x^2 + 2y^2 + 5 - x^2 - y^2) \, dA \\ &= \int_0^{2\pi} \int_0^2 (r^2 + 5)r \, dr \, d\theta = \int_0^{2\pi} \left(\frac{1}{4}r^4 + \frac{5}{2}r^2 \right) \Big|_0^2 d\theta = \int_0^{2\pi} 14 \, d\theta = 28\pi. \end{aligned}$$

32. The surface is $g(x, y, z) = z - x - 3 = 0$. $\nabla g = -\mathbf{i} + \mathbf{k}$, $|\nabla g| = \sqrt{2}$; $\mathbf{n} = \frac{-\mathbf{i} + \mathbf{k}}{\sqrt{2}}$;

$$\mathbf{F} \cdot \mathbf{n} = \frac{1}{\sqrt{2}} x^3 y + \frac{1}{\sqrt{2}} x y^3; \quad z_x = 1, \quad z_y = 0, \quad dS = \sqrt{2} \, dA. \text{ Using polar coordinates,}$$

$$\begin{aligned} \text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R \frac{1}{\sqrt{2}} (x^3 y + x y^3) \sqrt{2} \, dA = \iint_R x y (x^2 + y^2) \, dA \\ &= \int_0^{\pi/2} \int_0^{2 \cos \theta} (r^2 \cos \theta \sin \theta) r^2 r \, dr \, d\theta = \int_0^{\pi/2} \int_0^{2 \cos \theta} r^5 \cos \theta \sin \theta \, dr \, d\theta \\ &= \int_0^{\pi/2} \frac{1}{6} r^6 \cos \theta \sin \theta \Big|_0^{2 \cos \theta} d\theta = \frac{1}{6} \int_0^{\pi/2} 64 \cos^7 \theta \sin \theta \, d\theta = \frac{32}{3} \left(-\frac{1}{8} \cos^8 \theta \right) \Big|_0^{\pi/2} = \frac{4}{3}. \end{aligned}$$

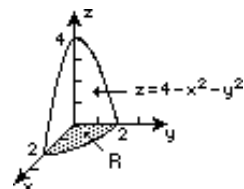


33. The surface is $g(x, y, z) = x^2 + y^2 + z - 4 = 0$. $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$,

$$|\nabla g| = \sqrt{4x^2 + 4y^2 + 1}; \quad \mathbf{n} = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}; \quad \mathbf{F} \cdot \mathbf{n} = \frac{x^3 + y^3 + z}{\sqrt{4x^2 + 4y^2 + 1}};$$

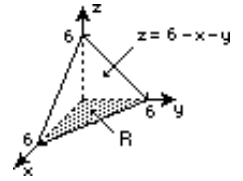
$$z_x = -2x, \quad z_y = -2y, \quad dS = \sqrt{1 + 4x^2 + 4y^2} \, dA. \text{ Using polar coordinates,}$$

$$\begin{aligned} \text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R (x^3 + y^3 + z) \, dA = \iint_R (4 - x^2 - y^2 + x^3 + y^3) \, dA \\ &= \int_0^{2\pi} \int_0^2 (4 - r^2 + r^3 \cos^3 \theta + r^3 \sin^3 \theta) r \, dr \, d\theta \\ &= \int_0^{2\pi} \left(2r^2 - \frac{1}{4}r^4 + \frac{1}{5}r^5 \cos^3 \theta + \frac{1}{5}r^5 \sin^3 \theta \right) \Big|_0^2 d\theta \\ &= \int_0^{2\pi} \left(4 + \frac{32}{5} \cos^3 \theta + \frac{32}{5} \sin^3 \theta \right) d\theta = 4\theta \Big|_0^{2\pi} + 0 + 0 = 8\pi. \end{aligned}$$



9.13 Surface Integrals

34. The surface is $g(x, y, z) = x + y + z - 6$. $\nabla g = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $|\nabla g| = \sqrt{3}$; $\mathbf{n} = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$;
 $\mathbf{F} \cdot \mathbf{n} = (e^y + e^x + 18y)/\sqrt{3}$; $z_x = -1$, $z_y = -1$, $dS = \sqrt{1+1+1} dA = \sqrt{3} dA$.



$$\begin{aligned} \text{Flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_R (e^y + e^x + 18y) dA = \int_0^6 \int_0^{6-x} (e^y + e^x + 18y) dy dx \\ &= \int_0^6 (e^y + ye^x + 9y^2) \Big|_0^{6-x} dx = \int_0^6 [e^{6-x} + (6-x)e^x + 9(6-x)^2 - 1] dx \\ &= [-e^{6-x} + 6e^x - xe^x + e^x - 3(6-x)^3 - x] \Big|_0^6 \\ &= (-1 + 6e^6 - 6e^6 + e^6 - 6) - (-e^6 + 6 + 1 - 648) = 2e^6 + 634 \approx 1440.86 \end{aligned}$$

35. For S_1 : $g(x, y, z) = x^2 + y^2 - z$, $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}$, $|\nabla g| = \sqrt{4x^2 + 4y^2 + 1}$; $\mathbf{n}_1 = \frac{2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}$;

$$\mathbf{F} \cdot \mathbf{n}_1 = \frac{2xy^2 + 2x^2y - 5z}{\sqrt{4x^2 + 4y^2 + 1}}; \quad z_x = 2x, \quad z_y = 2y, \quad dS_1 = \sqrt{1 + 4x^2 + 4y^2} dA. \quad \text{For } S_2: g(x, y, z) = z - 1,$$

$\nabla g = \mathbf{k}$, $|\nabla g| = 1$; $\mathbf{n}_2 = \mathbf{k}$; $\mathbf{F} \cdot \mathbf{n}_2 = 5z$; $z_x = 0$, $z_y = 0$, $dS_2 = dA$. Using polar coordinates and $R: x^2 + y^2 \leq 1$ we have

$$\begin{aligned} \text{Flux} &= \iint_{S_1} \mathbf{F} \cdot \mathbf{n}_1 dS_1 + \iint_{S_2} \mathbf{F} \cdot \mathbf{n}_2 dS_2 = \iint_R (2xy^2 + 2x^2y - 5z) dA + \iint_R 5z dA \\ &= \iint_R [2xy^2 + 2x^2y - 5(x^2 + y^2) + 5(1)] dA \\ &= \int_0^{2\pi} \int_0^1 (2r^3 \cos \theta \sin^2 \theta + 2r^3 \cos^2 \theta \sin \theta - 5r^2 + 5)r dr d\theta \\ &= \int_0^{2\pi} \left(\frac{2}{5} r^5 \cos \theta \sin^2 \theta + \frac{2}{5} r^5 \cos^2 \theta \sin \theta - \frac{5}{4} r^4 + \frac{5}{2} r^2 \right) \Big|_0^1 d\theta \\ &= \int_0^{2\pi} \left[\frac{2}{5} (\cos \theta \sin^2 \theta + \cos^2 \theta \sin \theta) + \frac{5}{4} \right] d\theta = \frac{2}{5} \left(\frac{1}{3} \sin^3 \theta - \frac{1}{3} \cos^3 \theta \right) \Big|_0^{2\pi} + \frac{5}{4} \theta \Big|_0^{2\pi} \\ &= \frac{2}{5} \left[-\frac{1}{3} - \left(-\frac{1}{3} \right) \right] + \frac{5}{2} \pi = \frac{5}{2} \pi. \end{aligned}$$

36. For S_1 : $g(x, y, z) = x^2 + y^2 + z - 4$, $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$, $|\nabla g| = \sqrt{4x^2 + 4y^2 + 1}$;

$$\mathbf{n}_1 = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}; \quad \mathbf{F} \cdot \mathbf{n}_1 = 6z^2/\sqrt{4x^2 + 4y^2 + 1}; \quad z_x = -2x, \quad z_y = -2y,$$

$$dS_1 = \sqrt{1 + 4x^2 + 4y^2} dA. \quad \text{For } S_2: g(x, y, z) = x^2 + y^2 - z, \quad \nabla g = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k},$$

$$|\nabla g| = \sqrt{4x^2 + 4y^2 + 1}; \quad \mathbf{n}_2 = \frac{2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}; \quad \mathbf{F} \cdot \mathbf{n}_2 = -6z^2/\sqrt{4x^2 + 4y^2 + 1}; \quad z_x = 2x, \quad z_y = 2y,$$

$dS_2 = \sqrt{1 + 4x^2 + 4y^2} dA$. Using polar coordinates and $R: x^2 + y^2 \leq 2$ we have

$$\begin{aligned} \text{Flux} &= \iint_{S_1} \mathbf{F} \cdot \mathbf{n}_1 dS_1 + \iint_{S_2} \mathbf{F} \cdot \mathbf{n}_2 dS_2 = \iint_R 6z^2 dA + \iint_R -6z^2 dA \\ &= \iint_R [6(4 - x^2 - y^2)^2 - 6(x^2 + y^2)^2] dA = 6 \int_0^{2\pi} \int_0^{\sqrt{2}} [(4 - r^2)^2 - r^4] r dr d\theta \\ &= 6 \int_0^{2\pi} \left[-\frac{1}{6} (4 - r^2)^3 - \frac{1}{6} r^6 \right] \Big|_0^{\sqrt{2}} d\theta = - \int_0^{2\pi} [(2^3 - 4^3) + (\sqrt{2})^6] d\theta = \int_0^{2\pi} 48 d\theta = 96\pi. \end{aligned}$$

37. The surface is $g(x, y, z) = x^2 + y^2 + z^2 - a^2 = 0$. $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$,

$$|\nabla g| = 2\sqrt{x^2 + y^2 + z^2}; \quad \mathbf{n} = \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{\sqrt{x^2 + y^2 + z^2}};$$

$$\mathbf{F} \cdot \mathbf{n} = -(2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}) \cdot \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{\sqrt{x^2 + y^2 + z^2}} = -\frac{2x^2 + 2y^2 + 2z^2}{\sqrt{x^2 + y^2 + z^2}} = -2\sqrt{x^2 + y^2 + z^2} = -2a.$$

$$\text{Flux} = \iint_S -2a \, dS = -2a \times \text{area} = -2a(4\pi a^2) = -8\pi a^3$$

38. $\mathbf{n}_1 = \mathbf{k}$, $\mathbf{n}_2 = -\mathbf{i}$, $\mathbf{n}_3 = \mathbf{j}$, $\mathbf{n}_4 = -\mathbf{k}$, $\mathbf{n}_5 = \mathbf{i}$, $\mathbf{n}_6 = -\mathbf{j}$; $\mathbf{F} \cdot \mathbf{n}_1 = z = 1$, $\mathbf{F} \cdot \mathbf{n}_2 = -x = 0$, $\mathbf{F} \cdot \mathbf{n}_3 = y = 1$,

$$\mathbf{F} \cdot \mathbf{n}_4 = -z = 0, \mathbf{F} \cdot \mathbf{n}_5 = x = 1, \mathbf{F} \cdot \mathbf{n}_6 = -y = 0; \quad \text{Flux} = \iint_{S_1} 1 \, dS + \iint_{S_3} 1 \, dS + \iint_{S_5} 1 \, dS = 3$$

39. Referring to the solution to Problem 37, we find $\mathbf{n} = \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{\sqrt{x^2 + y^2 + z^2}}$ and $dS = \frac{a}{\sqrt{a^2 - x^2 - y^2}} \, dA$.

Now

$$\mathbf{F} \cdot \mathbf{n} = kq \frac{\mathbf{r}}{|\mathbf{r}|^3} \cdot \frac{\mathbf{r}}{|\mathbf{r}|} = \frac{kq}{|\mathbf{r}|^4} |\mathbf{r}|^2 = \frac{kq}{|\mathbf{r}|^2} = \frac{kq}{x^2 + y^2 + z^2} = \frac{kq}{a^2}$$

and

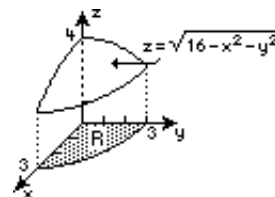
$$\text{Flux} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_S \frac{kq}{a^2} \, dS = \frac{kq}{a^2} \times \text{area} = \frac{kq}{a^2} (4\pi a^2) = 4\pi kq.$$

40. We are given $\sigma = kz$. Now $z_x = \frac{x}{\sqrt{16 - x^2 - y^2}}$, $z_y = -\frac{y}{\sqrt{16 - x^2 - y^2}}$;

$$dS = \sqrt{1 + \frac{x^2}{16 - x^2 - y^2} + \frac{y^2}{16 - x^2 - y^2}} \, dA = \frac{4}{\sqrt{16 - x^2 - y^2}} \, dA$$

Using polar coordinates,

$$\begin{aligned} Q &= \iint_S kz \, dS = k \iint_R \sqrt{16 - x^2 - y^2} \frac{4}{\sqrt{16 - x^2 - y^2}} \, dA = 4k \int_0^{2\pi} \int_0^3 r \, dr \, d\theta \\ &= 4k \int_0^{2\pi} \frac{1}{2} r^2 \Big|_0^3 \, d\theta = 4k \int_0^{2\pi} \frac{9}{2} \, d\theta = 36\pi k. \end{aligned}$$



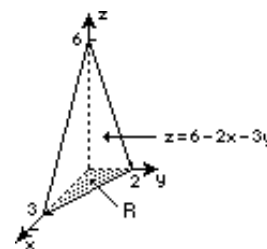
41. The surface is $z = 6 - 2x - 3y$. Then $z_x = -2$, $z_y = -3$, $dS = \sqrt{1 + 4 + 9} = \sqrt{14} \, dA$.

The area of the surface is

$$\begin{aligned} A(s) &= \iint_S dS = \int_0^3 \int_0^{2-2x/3} \sqrt{14} \, dy \, dx = \sqrt{14} \int_0^3 \left(2 - \frac{2}{3}x\right) \, dx \\ &= \sqrt{14} \left(2x - \frac{1}{3}x^2\right) \Big|_0^3 = 3\sqrt{14}. \end{aligned}$$

$$\begin{aligned} \bar{x} &= \frac{1}{3\sqrt{14}} \iint_S x \, dS = \frac{1}{3\sqrt{14}} \int_0^3 \int_0^{2-2x/3} \sqrt{14} x \, dy \, dx = \frac{1}{3} \int_0^3 xy \Big|_0^{2-2x/3} \, dx \\ &= \frac{1}{3} \int_0^3 \left(2x - \frac{2}{3}x^2\right) \, dx = \frac{1}{3} \left(x^2 - \frac{2}{9}x^3\right) \Big|_0^3 = 1 \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{1}{3\sqrt{14}} \iint_S y \, dS = \frac{1}{3\sqrt{14}} \int_0^3 \int_0^{2-2x/3} \sqrt{14} y \, dy \, dx = \frac{1}{3} \int_0^3 \frac{1}{2} y^2 \Big|_0^{2-2x/3} \, dx \\ &= \frac{1}{6} \int_0^3 \left(2 - \frac{2}{3}x\right)^2 \, dx = \frac{1}{6} \left[-\frac{1}{2} \left(2 - \frac{2}{3}x\right)^3\right] \Big|_0^3 = \frac{2}{3} \end{aligned}$$



9.13 Surface Integrals

$$\begin{aligned}\bar{z} &= \frac{1}{3\sqrt{14}} \iint_S z \, dS = \frac{1}{3\sqrt{14}} \int_0^3 \int_0^{2-2x/3} (6-2x-3y)\sqrt{14} \, dy \, dx \\ &= \frac{1}{3} \int_0^3 \left(6y - 2xy - \frac{3}{2}y^2 \right) \Big|_0^{2-2x/3} dx = \frac{1}{3} \int_0^3 \left(6 - 4x + \frac{2}{3}x^2 \right) dx = \frac{1}{3} \left(6x - 2x^2 + \frac{2}{9}x^3 \right) \Big|_0^3 = 2\end{aligned}$$

The centroid is $(1, 2/3, 2)$.

42. The area of the hemisphere is $A(s) = 2\pi a^2$. By symmetry, $\bar{x} = \bar{y} = 0$.

$$z_x = -\frac{x}{\sqrt{a^2 - x^2 - y^2}}, \quad z_y = -\frac{y}{\sqrt{a^2 - x^2 - y^2}};$$

$$dS = \sqrt{1 + \frac{x^2}{a^2 - x^2 - y^2} + \frac{y^2}{a^2 - x^2 - y^2}} \, dA = \frac{a}{\sqrt{a^2 - x^2 - y^2}} \, dA$$

Using polar coordinates,

$$\begin{aligned}z &= \iint_S \frac{z \, dS}{2\pi a^2} = \frac{1}{2\pi a^2} \iint_R \sqrt{a^2 - x^2 - y^2} \frac{a}{\sqrt{a^2 - x^2 - y^2}} \, dA = \frac{1}{2\pi a} \int_0^{2\pi} \int_0^a r \, dr \, d\theta \\ &= \frac{1}{2\pi a} \int_0^{2\pi} \frac{1}{2} r^2 \Big|_0^a d\theta = \frac{1}{2\pi a} \int_0^{2\pi} \frac{1}{2} a^2 d\theta = \frac{a}{2}.\end{aligned}$$

The centroid is $(0, 0, a/2)$.

43. The surface is $g(x, y, z) = z - f(x, y) = 0$. $\nabla g = -f_x \mathbf{i} - f_y \mathbf{j} + \mathbf{k}$, $|\nabla g| = \sqrt{f_x^2 + f_y^2 + 1}$;

$$\mathbf{n} = \frac{-f_x \mathbf{i} - f_y \mathbf{j} + \mathbf{k}}{\sqrt{1 + f_x^2 + f_y^2}}; \quad \mathbf{F} \cdot \mathbf{n} = \frac{-Pf_x - Qf_y + R}{\sqrt{1 + f_x^2 + f_y^2}}; \quad dS = \sqrt{1 + f_x^2 + f_y^2} \, dA$$

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_R \frac{-Pf_x - Qf_y + R}{\sqrt{1 + f_x^2 + f_y^2}} \sqrt{1 + f_x^2 + f_y^2} \, dA = \iint_R (-Pf_x - Qf_y + R) \, dA$$

EXERCISES 9.14

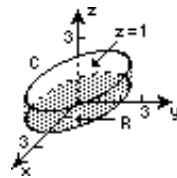
Stokes' Theorem

1. **Surface Integral:** $\text{curl } \mathbf{F} = -10\mathbf{k}$. Letting $g(x, y, z) = z - 1$, we have $\nabla g = \mathbf{k}$ and $\mathbf{n} = \mathbf{k}$. Then

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_S (-10) \, dS = -10 \times (\text{area of } S) = -10(4\pi) = -40\pi.$$

Line Integral: Parameterize the curve C by $x = 2 \cos t$, $y = 2 \sin t$, $z = 1$, for $0 \leq t \leq 2\pi$. Then

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C 5y \, dx - 5x \, dy + 3 \, dz = \int_0^{2\pi} [10 \sin t(-2 \sin t) - 10 \cos t(2 \cos t)] \, dt \\ &= \int_0^{2\pi} (-20 \sin^2 t - 20 \cos^2 t) \, dt = \int_0^{2\pi} -20 \, dt = -40\pi.\end{aligned}$$

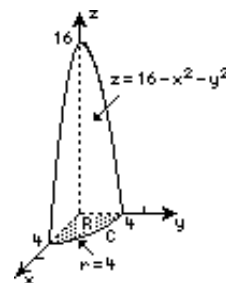


- 2. Surface Integral:** $\text{curl } \mathbf{F} = 4\mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$. Letting $g(x, y, z) = x^2 + y^2 + z - 16$, $\nabla g = 2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}$, and $\mathbf{n} = (2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k})/\sqrt{4x^2 + 4y^2 + 1}$. Thus,

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_S \frac{8x - 4y - 3}{\sqrt{4x^2 + 4y^2 + 1}} \, dS.$$

Letting the surface be $z = 16 - x^2 - y^2$, we have $z_x = -2x$, $z_y = -2y$, and $dS = \sqrt{1 + 4x^2 + 4y^2} \, dA$. Then, using polar coordinates,

$$\begin{aligned} \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS &= \iint_R (8x - 4y - 3) \, dA = \int_0^{2\pi} \int_0^4 (8r \cos \theta - 4r \sin \theta - 3) r \, dr \, d\theta \\ &= \int_0^{2\pi} \left(\frac{8}{3} r^3 \cos \theta - \frac{4}{3} r^3 \sin \theta - \frac{3}{2} r^2 \right) \Big|_0^4 \, d\theta = \int_0^{2\pi} \left(\frac{512}{3} \cos \theta - \frac{256}{3} \sin \theta - 24 \right) \, d\theta \\ &= \left(\frac{512}{3} \sin \theta + \frac{256}{3} \cos \theta - 24\theta \right) \Big|_0^{2\pi} = -48\pi. \end{aligned}$$

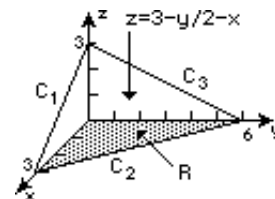


Line Integral: Parameterize the curve C by $x = 4 \cos t$, $y = 4 \sin t$, $z = 0$, for $0 \leq t \leq 2\pi$. Then,

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \oint_C 2z \, dx - 3x \, dy + 4y \, dz = \int_0^{2\pi} [-12 \cos t (4 \cos t)] \, dt \\ &= \int_0^{2\pi} -48 \cos^2 t \, dt = (-24t - 12 \sin 2t) \Big|_0^{2\pi} = -48\pi. \end{aligned}$$

- 3. Surface Integral:** $\text{curl } \mathbf{F} = \mathbf{i} + \mathbf{j} + \mathbf{k}$. Letting $g(x, y, z) = 2x + y + 2z - 6$, we have $\nabla g = 2\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{n} = (2\mathbf{i} + \mathbf{j} + 2\mathbf{k})/3$. Then $\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_S \frac{5}{3} \, dS$. Letting the surface be $z = 3 - \frac{1}{2}y - x$ we have $z_x = -1$, $z_y = -\frac{1}{2}$, and $dS = \sqrt{1 + (-1)^2 + (-\frac{1}{2})^2} \, dA = \frac{3}{2} \, dA$. Then

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = \iint_R \frac{5}{3} \left(\frac{3}{2} \right) \, dA = \frac{5}{2} \times (\text{area of } R) = \frac{5}{2}(9) = \frac{45}{2}.$$



Line Integral: C_1 : $z = 3 - x$, $0 \leq x \leq 3$, $y = 0$; C_2 : $y = 6 - 2x$, $3 \geq x \geq 0$, $z = 0$; C_3 : $z = 3 - y/2$, $6 \geq y \geq 0$, $x = 0$.

$$\begin{aligned} \oint_C z \, dx + x \, dy + y \, dz &= \iint_{C_1} z \, dx + \int_{C_2} x \, dy + \int_{C_3} y \, dz \\ &= \int_0^3 (3 - x) \, dx + \int_3^0 x(-2 \, dx) + \int_6^0 y(-dy/2) \\ &= \left(3x - \frac{1}{2}x^2 \right) \Big|_0^3 - x^2 \Big|_3^0 - \frac{1}{4}y^2 \Big|_6^0 = \frac{9}{2} - (0 - 9) - \frac{1}{4}(0 - 36) = \frac{45}{2} \end{aligned}$$

- 4. Surface Integral:** $\text{curl } \mathbf{F} = \mathbf{0}$ and $\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dS = 0$.

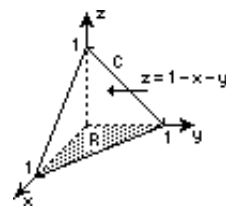
Line Integral: the curve is $x = \cos t$, $y = \sin t$, $z = 0$, $0 \leq t \leq 2\pi$.

$$\oint_C x \, dx + y \, dy + z \, dz = \int_0^{2\pi} [\cos t(-\sin t) + \sin t(\cos t)] \, dt = 0.$$

9.14 Stokes' Theorem

5. $\text{curl } \mathbf{F} = 2\mathbf{i} + \mathbf{j}$. A unit vector normal to the plane is $\mathbf{n} = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$. Taking the equation of the plane to be $z = 1 - x - y$, we have $z_x = z_y = -1$. Thus, $dS = \sqrt{1+1+1} dA = \sqrt{3} dA$ and

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_S \sqrt{3} dS = \sqrt{3} \iint_R \sqrt{3} dA \\ &= 3 \times (\text{area of } R) = 3(1/2) = 3/2.\end{aligned}$$



6. $\text{curl } \mathbf{F} = -2xz\mathbf{i} + z^2\mathbf{k}$. A unit vector normal to the plane is $\mathbf{n} = (\mathbf{j} + \mathbf{k})/\sqrt{2}$. From $z = 1 - y$, we have $z_x = 0$ and $z_y = -1$. Thus, $dS = \sqrt{1+1} dA = \sqrt{2} dA$ and

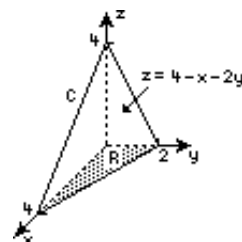
$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R \frac{1}{\sqrt{2}} z^2 \sqrt{2} dA = \iint_R (1-y)^2 dA \\ &= \int_0^2 \int_0^1 (1-y)^2 dy dx = \int_0^2 \left[-\frac{1}{3}(1-y)^3 \right]_0^1 dx = \int_0^2 \frac{1}{3} dx = \frac{2}{3}.\end{aligned}$$

7. $\text{curl } \mathbf{F} = -2y\mathbf{i} - z\mathbf{j} - x\mathbf{k}$. A unit vector normal to the plane is $\mathbf{n} = (\mathbf{j} + \mathbf{k})/\sqrt{2}$. From $z = 1 - y$ we have $z_x = 0$ and $z_y = -1$. Then $dS = \sqrt{1+1} dA = \sqrt{2} dA$ and

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R \left[-\frac{1}{\sqrt{2}}(z+x) \right] \sqrt{2} dA = \iint_R (y-x-1) dA \\ &= \int_0^2 \int_0^1 (y-x-1) dy dx = \int_0^2 \left(\frac{1}{2}y^2 - xy - y \right) \Big|_0^1 dx = \int_0^2 \left(-x - \frac{1}{2} \right) dx \\ &= \left(-\frac{1}{2}x^2 - \frac{1}{2}x \right) \Big|_0^2 = -3.\end{aligned}$$

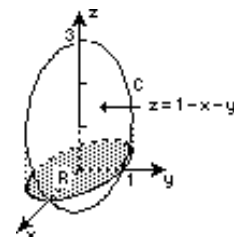
8. $\text{curl } \mathbf{F} = 2\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. Letting $g(x, y, z) = x + 2y + z - 4$, we have $\nabla g = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$ and $\mathbf{n} = (\mathbf{i} + 2\mathbf{j} + \mathbf{k})/\sqrt{6}$. From $z = 4 - x - 2y$ we have $z_x = -1$ and $z_y = -2$. Then $dS = \sqrt{6} dA$ and

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R \frac{1}{\sqrt{6}} (9) \sqrt{6} dA = \iint_R 9 dA = 9(4) = 36.$$



9. $\text{curl } \mathbf{F} = (-3x^2 - 3y^2)\mathbf{k}$. A unit vector normal to the plane is $\mathbf{n} = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$. From $z = 1 - x - y$, we have $z_x = z_y = -1$ and $dS = \sqrt{3} dA$. Then, using polar coordinates,

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R (-\sqrt{3}x^2 - \sqrt{3}y^2) \sqrt{3} dA \\ &= 3 \iint_R (-x^2 - y^2) dA = 3 \int_0^{2\pi} \int_0^1 (-r^2) r dr d\theta \\ &= 3 \int_0^{2\pi} \left[-\frac{1}{4}r^4 \right]_0^1 d\theta = 3 \int_0^{2\pi} -\frac{1}{4} d\theta = -\frac{3\pi}{2}.\end{aligned}$$



10. $\text{curl } \mathbf{F} = 2xyz\mathbf{i} - y^2z\mathbf{j} + (1-x^2)\mathbf{k}$. A unit vector normal to the surface is $\mathbf{n} = \frac{2y\mathbf{j} + \mathbf{k}}{\sqrt{4y^2 + 1}}$. From $z = 9 - y^2$ we have $z_x = 0$, $z_y = -2y$ and $dS = \sqrt{1+4y^2} dA$. Then

$$\begin{aligned}
\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} dS = \iint_R (-2y^3z + 1 - x^2) dA = \int_0^3 \int_0^{y/2} [-2y^3(9 - y^2) + 1 - x^2] dx dy \\
&= \int_0^3 \left(-18y^3x + 2y^5x + x - \frac{1}{3}x^3 \right) \Big|_0^{y/2} dy = \int_0^3 \left(-9y^4 + y^6 + \frac{1}{2}y - \frac{1}{24}y^3 \right) dy \\
&= \left(-\frac{9}{5}y^5 + \frac{1}{7}y^7 + \frac{1}{4}y^2 - \frac{1}{96}y^4 \right) \Big|_0^3 \approx 123.57.
\end{aligned}$$

11. $\operatorname{curl} \mathbf{F} = 3x^2y^2\mathbf{k}$. A unit vector normal to the surface is

$$\mathbf{n} = \frac{8x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}}{\sqrt{64x^2 + 4y^2 + 4z^2}} = \frac{4x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{\sqrt{16x^2 + y^2 + z^2}}.$$

From $z_x = -\frac{4x}{\sqrt{4-4x^2-y^2}}$, $z_y = -\frac{y}{\sqrt{4-4x^2-y^2}}$ we obtain $dS = 2\sqrt{\frac{1+3x^2}{4-4x^2-y^2}} dA$. Then

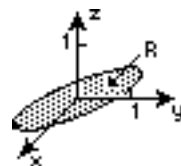
$$\begin{aligned}
\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} dS = \iint_R \frac{3x^2y^2z}{\sqrt{16x^2 + y^2 + z^2}} \left(2\sqrt{\frac{1+3x^2}{4-4x^2-y^2}} \right) dA \\
&= \iint_R 3x^2y^2 dA \quad \boxed{\text{Using symmetry}} \\
&= 12 \int_0^1 \int_0^{2\sqrt{1-x^2}} x^2y^2 dy dx = 12 \int_0^1 \left(\frac{1}{3}x^2y^3 \right) \Big|_0^{2\sqrt{1-x^2}} dx \\
&= 32 \int_0^1 x^2(1-x^2)^{3/2} dx \quad \boxed{x = \sin t, \quad dx = \cos t dt} \\
&= 32 \int_0^{\pi/2} \sin^2 t \cos^4 t dt = \pi.
\end{aligned}$$

12. $\operatorname{curl} \mathbf{F} = \mathbf{i} + \mathbf{j} + \mathbf{k}$. Taking the surface S bounded by C to be the portion of the plane $x + y + z = 0$ inside C , we have $\mathbf{n} = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$ and $dS = \sqrt{3} dA$.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} dS = \iint_S \sqrt{3} dS = \sqrt{3} \iint_R \sqrt{3} dA = 3 \times (\text{area of } R)$$

The region R is obtained by eliminating z from the equations of the plane and the sphere. This gives $x^2 + xy + y^2 = \frac{1}{2}$. Rotating axes, we see that R is enclosed by the ellipse $X^2/(1/3) + Y^2/1 = 1$ in a rotated coordinate system. Thus,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 3 \times (\text{area of } R) = 3 \left(\pi \frac{1}{\sqrt{3}} 1 \right) = \sqrt{3}\pi.$$



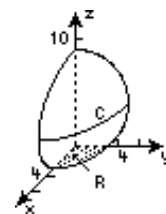
13. Parameterize C by $x = 4 \cos t$, $y = 2 \sin t$, $z = 4$, for $0 \leq t \leq 2\pi$. Then

$$\begin{aligned}
\iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n} dS &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C 6yz dx + 5x dy + yze^{x^2} dz \\
&= \int_0^{2\pi} [6(2 \sin t)(4)(-4 \sin t) + 5(4 \cos t)(2 \cos t) + 0] dt \\
&= 8 \int_0^{2\pi} (-24 \sin^2 t + 5 \cos^2 t) dt = 8 \int_0^{2\pi} (5 - 29 \sin^2 t) dt = -152\pi.
\end{aligned}$$

9.14 Stokes' Theorem

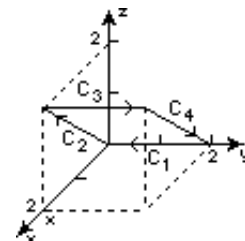
14. Parameterize C by $x = 5 \cos t$, $y = 5 \sin t$, $z = 4$, for $0 \leq t \leq 2\pi$. Then,

$$\begin{aligned} \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS &= \oint_C \mathbf{F} \cdot \mathbf{r} = \oint_C y dx + (y - x) dy + z^2 dz \\ &= \int_0^{2\pi} [(5 \sin t)(-5 \sin t) + (5 \sin t - 5 \cos t)(5 \cos t)] dt \\ &= \int_0^{2\pi} (25 \sin t \cos t - 25) dt = \left(\frac{25}{2} \sin^2 t - 25t \right) \Big|_0^{2\pi} = -50\pi. \end{aligned}$$



15. Parameterize C by C_1 : $x = 0$, $z = 0$, $2 \geq y \geq 0$; C_2 : $z = x$, $y = 0$, $0 \leq x \leq 2$; C_3 : $x = 2$, $z = 2$, $0 \leq y \leq 2$; C_4 : $z = x$, $y = 2$, $2 \geq x \geq 0$. Then

$$\begin{aligned} \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS &= \oint_C \mathbf{F} \cdot \mathbf{r} = \oint_C 3x^2 dx + 8x^3 y dy + 3x^2 y dz \\ &= \int_{C_1} 0 dx + 0 dy + 0 dz + \int_{C_2} 3x^2 dx + \int_{C_3} 64 dy + \int_{C_4} 3x^2 dx + 6x^2 dx \\ &= \int_0^2 3x^2 dx + \int_0^2 64 dy + \int_2^0 9x^2 dx = x^3 \Big|_0^2 + 64y \Big|_0^2 + 3x^3 \Big|_2^0 = 112. \end{aligned}$$



16. Parameterize C by $x = \cos t$, $y = \sin t$, $z = \sin t$, $0 \leq t \leq 2\pi$. Then

$$\begin{aligned} \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS &= \oint_C \mathbf{F} \cdot \mathbf{r} = \oint_C 2xy^2 z dx + 2x^2 yz dy + (x^2 y^2 - 6x) dz \\ &= \int_0^{2\pi} [2 \cos t \sin^2 t \sin t (-\sin t) + 2 \cos^2 t \sin t \sin t \cos t \\ &\quad + (\cos^2 t \sin^2 t - 6 \cos t) \cos t] dt \\ &= \int_0^{2\pi} (-2 \cos t \sin^4 t + 3 \cos^3 t \sin^2 t - 6 \cos^2 t) dt = -6\pi. \end{aligned}$$

17. We take the surface to be $z = 0$. Then $\mathbf{n} = \mathbf{k}$ and $dS = dA$. Since $\text{curl } \mathbf{F} = \frac{1}{1+y^2} \mathbf{i} + 2ze^{x^2} \mathbf{j} + y^2 \mathbf{k}$,

$$\begin{aligned} \oint_C z^2 e^{x^2} dx + xy dy + \tan^{-1} y dz &= \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_S y^2 dS = \iint_R y^2 dA \\ &= \int_0^{2\pi} \int_0^3 r^2 \sin^2 \theta r dr d\theta = \int_0^{2\pi} \frac{1}{4} r^4 \sin^2 \theta \Big|_0^3 d\theta \\ &= \frac{81}{4} \int_0^{2\pi} \sin^2 \theta d\theta = \frac{81\pi}{4}. \end{aligned}$$

18. (a) $\text{curl } \mathbf{F} = xz\mathbf{i} - yz\mathbf{j}$. A unit vector normal to the surface is $\mathbf{n} = \frac{2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}$ and

$dS = \sqrt{1 + 4x^2 + 4y^2} dA$. Then, using $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$, we have

$$\begin{aligned} \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS &= \iint_R (2x^2 z - 2y^2 z) dA = \iint_R (2x^2 - 2y^2)(1 - x^2 - y^2) dA \\ &= \iint_R (2x^2 - 2y^2 - 2x^4 + 2y^4) dA \\ &= \int_0^{2\pi} \int_0^1 (2r^2 \cos^2 \theta - 2r^2 \sin^2 \theta - 2r^4 \cos^4 \theta + 2r^4 \sin^4 \theta) r dr d\theta \\ &= 2 \int_0^{2\pi} \int_0^1 [r^3 \cos 2\theta - r^5 (\cos^2 \theta - \sin^2 \theta)(\cos^2 \theta + \sin^2 \theta)] dr d\theta \end{aligned}$$

$$\begin{aligned}
&= 2 \int_0^{2\pi} \int_0^1 (r^3 \cos 2\theta - r^5 \cos 2\theta) dr d\theta = 2 \int_0^{2\pi} \cos 2\theta \left(\frac{1}{4}r^4 - \frac{1}{6}r^6 \right) \bigg|_0^1 d\theta \\
&= \frac{1}{6} \int_0^{2\pi} \cos 2\theta d\theta = 0.
\end{aligned}$$

(b) We take the surface to be $z = 0$. Then $\mathbf{n} = \mathbf{k}$, $\text{curl } \mathbf{F} \cdot \mathbf{n} = \text{curl } \mathbf{F} \cdot \mathbf{k} = 0$ and $\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = 0$.

(c) By Stokes' Theorem, using $z = 0$, we have

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C xyz dz = \oint_C xy(0) dz = 0.$$

EXERCISES 9.15

Triple Integrals

- $$\begin{aligned}
1. \quad \int_2^4 \int_{-2}^2 \int_{-1}^1 (x + y + z) dx dy dz &= \int_2^4 \int_{-2}^2 \left(\frac{1}{2}x^2 + xy + xz \right) \bigg|_{-1}^1 dy dz \\
&= \int_2^4 \int_{-2}^2 (2y + 2z) dy dz = \int_2^4 (y^2 + 2yz) \bigg|_{-2}^2 dz = \int_2^4 8z dz = 4z^2 \bigg|_2^4 = 48
\end{aligned}$$
- $$\begin{aligned}
2. \quad \int_1^3 \int_1^x \int_2^{xy} 24xyz dz dy dx &= \int_1^3 \int_1^x 24xyz \bigg|_2^{xy} dy dx = \int_1^3 \int_1^x (24x^2y^2 - 48xy) dy dx \\
&= \int_1^3 (8x^2y^3 - 24xy^2) \bigg|_1^x dx = \int_1^3 (8x^5 - 24x^3 - 8x^2 + 24x) dx \\
&= \left(\frac{4}{3}x^6 - 6x^4 - \frac{8}{3}x^3 + 12x^2 \right) \bigg|_1^3 = 522 - \frac{14}{3} = \frac{1552}{3}
\end{aligned}$$
- $$\begin{aligned}
3. \quad \int_0^6 \int_0^{6-x} \int_0^{6-x-z} dy dz dx &= \int_0^6 \int_0^{6-x} (6 - x - z) dz dx = \int_0^6 \left(6z - xz - \frac{1}{2}z^2 \right) \bigg|_0^{6-x} dx \\
&= \int_0^6 \left[6(6-x) - x(6-x) - \frac{1}{2}(6-x)^2 \right] dx = \int_0^6 \left(18 - 6x + \frac{1}{2}x^2 \right) dx \\
&= \left(18x - 3x^2 + \frac{1}{6}x^3 \right) \bigg|_0^6 = 36
\end{aligned}$$
- $$\begin{aligned}
4. \quad \int_0^1 \int_0^{1-x} \int_0^{\sqrt{y}} 4x^2z^3 dz dy dx &= \int_0^1 \int_0^{1-x} x^2z^4 \bigg|_0^{\sqrt{y}} dy dx = \int_0^1 \int_0^{1-x} x^2y^2 dy dx \\
&= \int_0^1 \frac{1}{3}x^2y^3 \bigg|_0^{1-x} dx = \frac{1}{3} \int_0^1 x^2(1-x)^3 dx = \frac{1}{3} \int_0^1 (x^2 - 3x^3 + 3x^4 - x^5) dx \\
&= \frac{1}{3} \left(\frac{1}{3}x^3 - \frac{3}{4}x^4 + \frac{3}{5}x^5 - \frac{1}{6}x^6 \right) \bigg|_0^1 = \frac{1}{180}
\end{aligned}$$

9.15 Triple Integrals

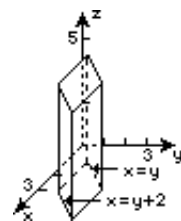
$$\begin{aligned}
 5. \quad \int_0^{\pi/2} \int_0^{y^2} \int_0^y \cos \frac{x}{y} dz dx dy &= \int_0^{\pi/2} \int_0^{y^2} y \cos \frac{x}{y} dx dy = \int_0^{\pi/2} y^2 \sin \frac{x}{y} \Big|_0^{y^2} dy \\
 &= \int_0^{\pi/2} y^2 \sin y dy \quad \boxed{\text{Integration by parts}} \\
 &= (-y^2 \cos y + 2 \cos y + 2y \sin y) \Big|_0^{\pi/2} = \pi - 2
 \end{aligned}$$

$$\begin{aligned}
 6. \quad \int_0^{\sqrt{2}} \int_{\sqrt{y}}^2 \int_0^{e^{x^2}} x dz dx dy &= \int_0^{\sqrt{2}} \int_{\sqrt{y}}^2 x e^{x^2} dx dy = \int_0^{\sqrt{2}} \frac{1}{2} e^{x^2} \Big|_{\sqrt{y}}^2 dy = \frac{1}{2} \int_0^{\sqrt{2}} (e^4 - e^y) dy \\
 &= \frac{1}{2} (ye^4 - e^y) \Big|_0^{\sqrt{2}} = \frac{1}{2} [(e^4 \sqrt{2} - e^{\sqrt{2}}) - (-1)] = \frac{1}{2} (1 + e^4 \sqrt{2} - e^{\sqrt{2}})
 \end{aligned}$$

$$\begin{aligned}
 7. \quad \int_0^1 \int_0^1 \int_0^{2-x^2-y^2} xy e^z dz dx dy &= \int_0^1 \int_0^1 xy e^z \Big|_0^{2-x^2-y^2} dx dy = \int_0^1 \int_0^1 (xy e^{2-x^2-y^2} - xy) dx dy \\
 &= \int_0^1 \left(-\frac{1}{2} y e^{2-x^2-y^2} - \frac{1}{2} x^2 y \right) \Big|_0^1 dy = \int_0^1 \left(-\frac{1}{2} y e^{1-y^2} - \frac{1}{2} y + \frac{1}{2} y e^{2-y^2} \right) dy \\
 &= \left(\frac{1}{4} e^{1-y^2} - \frac{1}{4} y^2 - \frac{1}{4} e^{2-y^2} \right) \Big|_0^1 = \left(\frac{1}{4} - \frac{1}{4} - \frac{1}{4} e \right) - \left(\frac{1}{4} e - \frac{1}{4} e^2 \right) = \frac{1}{4} e^2 - \frac{1}{2} e
 \end{aligned}$$

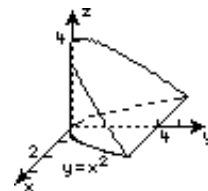
$$\begin{aligned}
 8. \quad \int_0^4 \int_0^{1/2} \int_0^{x^2} \frac{1}{\sqrt{x^2 - y^2}} dy dx dz &= \int_0^4 \int_0^{1/2} \sin^{-1} \frac{y}{x} \Big|_0^{x^2} dx dz = \int_0^4 \int_0^{1/2} \sin^{-1} x dx dz \\
 &\quad \boxed{\text{Integration by parts}} \\
 &= \int_0^4 (x \sin^{-1} x + \sqrt{1-x^2}) \Big|_0^{1/2} dz = \int_0^4 \left(\frac{1}{2} \frac{\pi}{6} + \frac{\sqrt{3}}{2} - 1 \right) dz = \frac{\pi}{3} + 2\sqrt{3} - 4
 \end{aligned}$$

$$\begin{aligned}
 9. \quad \iiint_D z dV &= \int_0^5 \int_1^3 \int_y^{y+2} z dx dy dz = \int_0^5 \int_1^3 xz \Big|_y^{y+2} dy dz = \int_0^5 \int_1^3 2z dy dz \\
 &= \int_0^5 2yz \Big|_1^3 dz = \int_0^5 4z dz = 2z^2 \Big|_0^5 = 50
 \end{aligned}$$

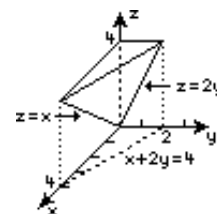


10. Using symmetry,

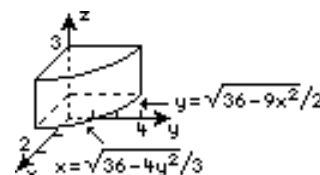
$$\begin{aligned}
 \iiint_D (x^2 + y^2) dV &= 2 \int_0^2 \int_{x^2}^4 \int_0^{4-y} (x^2 + y^2) dz dy dx = 2 \int_0^2 \int_{x^2}^4 (x^2 + y^2) z \Big|_0^{4-y} dy dx \\
 &= 2 \int_0^2 \int_{x^2}^4 (4x^2 - x^2 y + 4y^2 - y^3) dy dx \\
 &= 2 \int_0^2 \left(4x^2 y - \frac{1}{2} x^2 y^2 + \frac{4}{3} y^3 - \frac{1}{4} y^4 \right) \Big|_{x^2}^4 dx \\
 &= 2 \int_0^2 \left[\left(8x^2 + \frac{64}{3} \right) - \left(4x^4 + \frac{5}{6} x^6 - \frac{1}{4} x^8 \right) \right] dx \\
 &= 2 \left(\frac{8}{3} x^3 + \frac{64}{3} x - \frac{4}{5} x^5 - \frac{5}{42} x^7 + \frac{1}{36} x^9 \right) \Big|_0^2 = \frac{23,552}{315}.
 \end{aligned}$$



11. The other five integrals are $\int_0^4 \int_0^{2-x/2} \int_{x+2y}^4 F(x, y, z) dz dy dx$,
 $\int_0^4 \int_0^z \int_0^{(z-x)/2} F(x, y, z) dy dx dz$, $\int_0^4 \int_x^4 \int_0^{(z-x)/2} F(x, y, z) dy dz dx$,
 $\int_0^4 \int_0^{z/2} \int_0^{z-2y} F(x, y, z) dx dy dz$, $\int_0^2 \int_{2y}^4 \int_0^{z-2y} F(x, y, z) dx dz dy$.



12. The other five integrals are $\int_0^3 \int_0^{\sqrt{36-4y^2}/3} \int_1^3 F(x, y, z) dz dx dy$,
 $\int_1^3 \int_0^2 \int_0^{\sqrt{36-9x^2}/2} F(x, y, z) dy dx dz$, $\int_1^3 \int_0^3 \int_0^{\sqrt{36-4y^2}/3} F(x, y, z) dx dy dz$,
 $\int_0^3 \int_1^3 \int_0^{\sqrt{36-4y^2}/3} F(x, y, z) dx dz dy$, $\int_0^2 \int_1^3 \int_0^{\sqrt{36-9x^2}/2} F(x, y, z) dy dz dx$.

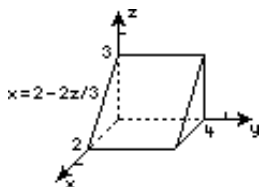


13. (a) $V = \int_0^2 \int_{x^3}^8 \int_0^4 dz dy dx$ (b) $V = \int_0^8 \int_0^4 \int_0^{y^{1/3}} dx dz dy$ (c) $V = \int_0^4 \int_0^2 \int_{x^2}^8 dy dx dz$

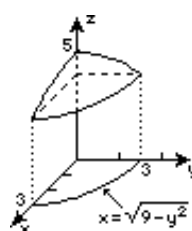
14. Solving $z = \sqrt{x}$ and $x + z = 2$, we obtain $x = 1$, $z = 1$. (a) $V = \int_0^3 \int_0^1 \int_{z^2}^{2-z} dx dz dy$

(b) $V = \int_0^1 \int_{z^2}^{2-z} \int_0^3 dy dx dz$ (c) $V = \int_0^3 \int_0^1 \int_0^{\sqrt{x}} dz dx dy + \int_0^3 \int_1^2 \int_0^{2-x} dz dx dy$

15.

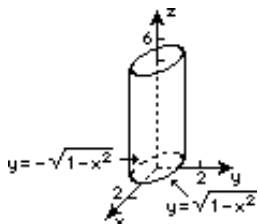


16.

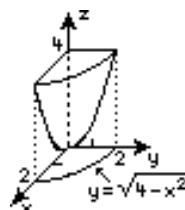


The region in the first octant is shown.

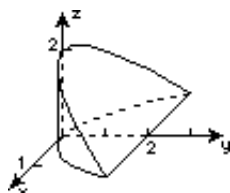
17.



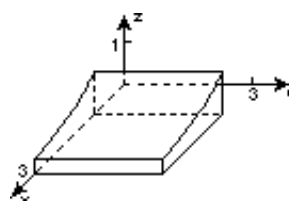
18.



19.



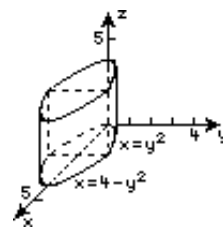
20.



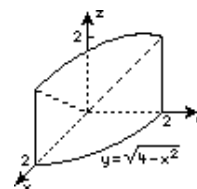
9.15 Triple Integrals

21. Solving $x = y^2$ and $4 - x = y^2$, we obtain $x = 2$, $y = \pm\sqrt{2}$. Using symmetry,

$$\begin{aligned} V &= 2 \int_0^3 \int_0^{\sqrt{2}} \int_{y^2}^{4-y^2} dx \, dy \, dz = 2 \int_0^3 \int_0^{\sqrt{2}} (4 - 2y^2) dy \, dz \\ &= 2 \int_0^3 \left(4y - \frac{2}{3}y^3 \right) \Big|_0^{\sqrt{2}} dz = 2 \int_0^3 \frac{8\sqrt{2}}{3} dz = 16\sqrt{2}. \end{aligned}$$

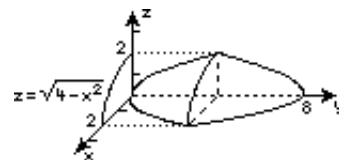


22.
$$\begin{aligned} V &= \int_0^2 \int_0^{\sqrt{4-x^2}} \int_0^{x+y} dz \, dy \, dx = \int_0^2 \int_0^{\sqrt{4-x^2}} z \Big|_0^{x+y} dy \, dx \\ &= \int_0^2 \int_0^{\sqrt{4-x^2}} (x+y) dy \, dx = \int_0^2 \left(xy + \frac{1}{2}y^2 \right) \Big|_0^{\sqrt{4-x^2}} dx \\ &= \int_0^2 \left[x\sqrt{4-x^2} + \frac{1}{2}(4-x^2) \right] dx = \left[-\frac{1}{3}(4-x^2)^{3/2} + 2x - \frac{1}{6}x^3 \right] \Big|_0^2 \\ &= \left(4 - \frac{4}{3} \right) - \left(-\frac{8}{3} \right) = \frac{16}{3} \end{aligned}$$



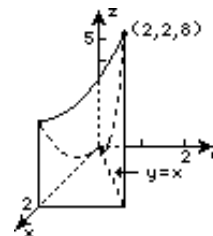
23. Adding the two equations, we obtain $2y = 8$. Thus, the paraboloids intersect in the plane $y = 4$. Their intersection is a circle of radius 2. Using symmetry,

$$\begin{aligned} V &= 4 \int_0^2 \int_0^{\sqrt{4-x^2}} \int_{x^2+z^2}^{8-x^2-z^2} dy \, dz \, dx = 4 \int_0^2 \int_0^{\sqrt{4-x^2}} (8 - 2x^2 - 2z^2) dz \, dx \\ &= 4 \int_0^2 \left[2(4-x^2)z - \frac{2}{3}z^3 \right] \Big|_0^{\sqrt{4-x^2}} dx = 4 \int_0^2 \frac{4}{3}(4-x^2)^{3/2} dx \quad \boxed{\text{Trig substitution}} \\ &= \frac{16}{3} \left[-\frac{x}{8}(2x^2 - 20)\sqrt{4-x^2} + 6 \sin^{-1} \frac{x}{2} \right] \Big|_0^2 = 16\pi. \end{aligned}$$



24. Solving $x = 2$, $y = x$, and $z = x^2 + y^2$, we obtain the point $(2, 2, 8)$.

$$\begin{aligned} V &= \int_0^2 \int_0^x \int_0^{x^2+y^2} dz \, dy \, dx = \int_0^2 \int_0^x (x^2 + y^2) dy \, dx = \int_0^2 \left(x^2y + \frac{1}{3}y^3 \right) \Big|_0^x dx \\ &= \int_0^2 \frac{4}{3}x^3 dx = \frac{1}{3}x^4 \Big|_0^2 = \frac{16}{3}. \end{aligned}$$



25. We are given $\rho(x, y, z) = kz$.

$$\begin{aligned} m &= \int_0^8 \int_0^4 \int_0^{y^{1/3}} kz \, dx \, dz \, dy = k \int_0^8 \int_0^4 xz \Big|_0^{y^{1/3}} dz \, dy = k \int_0^8 \int_0^4 y^{1/3} z \, dz \, dy \\ &= k \int_0^8 \frac{1}{2}y^{1/3}z^2 \Big|_0^4 dy = 8k \int_0^8 y^{1/3} dy = 8k \left(\frac{3}{4}y^{4/3} \right) \Big|_0^8 = 96k \\ M_{xy} &= \int_0^8 \int_0^4 \int_0^{y^{1/3}} kz^2 \, dx \, dz \, dy = k \int_0^8 \int_0^4 xz^2 \Big|_0^{y^{1/3}} dz \, dy = k \int_0^8 \int_0^4 y^{1/3} z^2 \, dz \, dy \\ &= k \int_0^8 \frac{1}{3}y^{1/3}z^3 \Big|_0^4 dy = \frac{64}{3}k \int_0^8 y^{1/3} dy = \frac{64}{3}k \left(\frac{3}{4}y^{4/3} \right) \Big|_0^8 = 256k \end{aligned}$$

$$\begin{aligned}
 M_{xz} &= \int_0^8 \int_0^4 \int_0^{y^{1/3}} kyz \, dx \, dz \, dy = k \int_0^8 \int_0^4 xyz \Big|_0^{y^{1/3}} dz \, dy = k \int_0^8 \int_0^4 y^{4/3} z \, dz \, dy \\
 &= k \int_0^8 \frac{1}{2} y^{4/3} z^2 \Big|_0^4 dy = 8k \int_0^8 y^{4/3} dy = 8k \left(\frac{3}{7} y^{7/3} \right) \Big|_0^8 = \frac{3072}{7} k
 \end{aligned}$$

$$\begin{aligned}
 M_{yz} &= \int_0^8 \int_0^4 \int_0^{y^{1/3}} kxz \, dx \, dz \, dy = k \int_0^8 \int_0^4 \frac{1}{2} x^2 z \Big|_0^{y^{1/3}} dz \, dy = \frac{1}{2} k \int_0^8 \int_0^4 y^{2/3} z \, dz \, dy \\
 &= \frac{1}{2} k \int_0^8 \frac{1}{2} y^{2/3} z^2 \Big|_0^4 dy = 4k \int_0^8 y^{2/3} dy = 4k \left(\frac{3}{5} y^{5/3} \right) \Big|_0^8 = \frac{384}{5} k
 \end{aligned}$$

$$\bar{x} = M_{yz}/m = \frac{384k/5}{96k} = 4/5; \quad \bar{y} = M_{xz}/m = \frac{3072k/7}{96k} = 32/7; \quad \bar{z} = M_{xy}/m = \frac{256k}{96k} = 8/3$$

The center of mass is $(4/5, 32/7, 8/3)$.

26. We use the form of the integral in Problem 14(b) of this section. Without loss of generality, we take $\rho = 1$.

$$m = \int_0^1 \int_{z^2}^{2-z} \int_0^3 dy \, dx \, dz = \int_0^1 \int_{z^2}^{2-z} 3 \, dx \, dz = 3 \int_0^1 (2 - z - z^2) \, dz = 3 \left(2z - \frac{1}{2} z^2 - \frac{1}{3} z^3 \right) \Big|_0^1 = \frac{7}{2}$$

$$\begin{aligned}
 M_{xy} &= \int_0^1 \int_{z^2}^{2-z} \int_0^3 z \, dy \, dx \, dz = \int_0^1 \int_{z^2}^{2-z} yz \Big|_0^3 dx \, dz = \int_0^1 \int_{z^2}^{2-z} 3z \, dx \, dz \\
 &= 3 \int_0^1 xz \Big|_{z^2}^{2-z} dz = 3 \int_0^1 (2z - z^2 - z^3) \, dz = 3 \left(z^2 - \frac{1}{3} z^3 - \frac{1}{4} z^4 \right) \Big|_0^1 = \frac{5}{4}
 \end{aligned}$$

$$\begin{aligned}
 M_{xz} &= \int_0^1 \int_{z^2}^{2-z} \int_0^3 y \, dy \, dx \, dz = \int_0^1 \int_{z^2}^{2-z} \frac{1}{2} y^2 \Big|_0^3 dx \, dz = \frac{9}{2} \int_0^1 \int_{z^2}^{2-z} dx \, dz \\
 &= \frac{9}{2} \int_0^1 (2 - z - z^2) \, dz = \frac{9}{2} \left(2z - \frac{1}{2} z^2 - \frac{1}{3} z^3 \right) \Big|_0^1 = \frac{21}{4}
 \end{aligned}$$

$$\begin{aligned}
 M_{yz} &= \int_0^1 \int_{z^2}^{2-z} \int_0^3 x \, dy \, dx \, dz = \int_0^1 \int_{z^2}^{2-z} xy \Big|_0^3 dx \, dz = \int_0^1 \int_{z^2}^{2-z} 3x \, dx \, dz \\
 &= 3 \int_0^1 \frac{1}{2} x^2 \Big|_{z^2}^{2-z} dz = \frac{3}{2} \int_0^1 (4 - 4z + z^2 - z^4) \, dz = \frac{3}{2} \left(4z - 2z^2 + \frac{1}{3} z^3 - \frac{1}{5} z^5 \right) \Big|_0^1 = \frac{16}{5}
 \end{aligned}$$

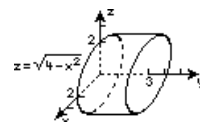
$$\bar{x} = M_{yz}/m = \frac{16/5}{7/2} = 32/35, \quad \bar{y} = M_{xz}/m = \frac{21/4}{7/2} = 3/2, \quad \bar{z} = M_{xy}/m = \frac{5/4}{7/2} = 5/14.$$

The centroid is $(32/35, 3/2, 5/14)$.

27. The density is $\rho(x, y, z) = ky$. Since both the region and the density function are symmetric with respect to the xy - and yz -planes, $\bar{x} = \bar{z} = 0$. Using symmetry,

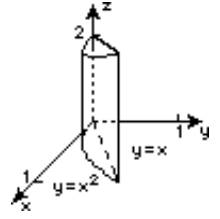
$$\begin{aligned}
 m &= 4 \int_0^3 \int_0^2 \int_0^{\sqrt{4-x^2}} ky \, dz \, dx \, dy = 4k \int_0^3 \int_0^2 yz \Big|_0^{\sqrt{4-x^2}} dx \, dy = 4k \int_0^3 \int_0^2 y\sqrt{4-x^2} \, dx \, dy \\
 &= 4k \int_0^3 y \left(\frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1} \frac{x}{2} \right) \Big|_0^2 dy = 4k \int_0^3 \pi y \, dy = 4\pi k \left(\frac{1}{2} y^2 \right) \Big|_0^3 = 18\pi k \\
 M_{xz} &= 4 \int_0^3 \int_0^2 \int_0^{\sqrt{4-x^2}} ky^2 \, dz \, dx \, dy = 4k \int_0^3 \int_0^2 y^2 z \Big|_0^{\sqrt{4-x^2}} dx \, dy = 4k \int_0^3 \int_0^2 y^2 \sqrt{4-x^2} \, dx \, dy \\
 &= 4k \int_0^3 y^2 \left(\frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1} \frac{x}{2} \right) \Big|_0^2 dy = 4k \int_0^3 \pi y^2 \, dy = 4\pi k \left(\frac{1}{3} y^3 \right) \Big|_0^3 = 36\pi k.
 \end{aligned}$$

$$\bar{y} = M_{xz}/m = \frac{36\pi k}{18\pi k} = 2. \quad \text{The center of mass is } (0, 2, 0).$$



9.15 Triple Integrals

28. The density is $\rho(x, y, z) = kz$.



$$\begin{aligned} m &= \int_0^1 \int_{x^2}^x \int_0^{y+2} kz \, dz \, dy \, dx = k \int_0^1 \int_{x^2}^x \frac{1}{2} z^2 \Big|_0^{y+2} dy \, dx \\ &= \frac{1}{2} k \int_0^1 \int_{x^2}^x (y+2)^2 dy \, dx = \frac{1}{2} k \int_0^1 \frac{1}{3} (y+2)^3 \Big|_{x^2}^x dx \\ &= \frac{1}{6} k \int_0^1 [(x+2)^3 - (x^2+2)^3] dx = \frac{1}{6} k \int_0^1 [(x+2)^3 - (x^6 + 6x^4 + 12x^2 + 8)] dx \\ &= \frac{1}{6} k \left[\frac{1}{4} (x+2)^4 - \frac{1}{7} x^7 - \frac{6}{5} x^5 - 4x^3 - 8x \right] \Big|_0^1 = \frac{407}{840} k \end{aligned}$$

$$\begin{aligned} M_{xy} &= \int_0^1 \int_{x^2}^x \int_0^{y+2} kz^2 \, dz \, dy \, dx = k \int_0^1 \int_{x^2}^x \frac{1}{3} z^3 \Big|_0^{y+2} dy \, dx = \frac{1}{3} k \int_0^1 \int_{x^2}^x (y+2)^3 dy \, dx \\ &= \frac{1}{3} k \int_0^1 \frac{1}{4} (y+2)^4 \Big|_{x^2}^x dx = \frac{1}{12} k \int_0^1 [(x+2)^4 - (x^2+2)^4] dx \\ &= \frac{1}{12} k \int_0^1 [(x+2)^4 - (x^8 + 8x^6 + 24x^4 + 32x^2 + 16)] dx \\ &= \frac{1}{12} k \left[\frac{1}{5} (x+2)^5 - \frac{1}{9} x^9 - \frac{8}{7} x^7 - \frac{24}{5} x^5 - \frac{32}{3} x^3 - 16x \right] \Big|_0^1 = \frac{1493}{1890} k \end{aligned}$$

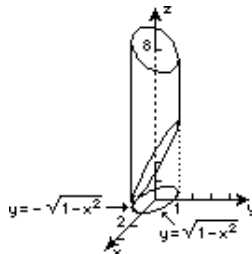
$$\begin{aligned} M_{xz} &= \int_0^1 \int_{x^2}^x \int_0^{y+2} kyz \, dz \, dy \, dx = k \int_0^1 \int_{x^2}^x \frac{1}{2} yz^2 \Big|_0^{y+2} dy \, dx = \frac{1}{2} k \int_0^1 \int_{x^2}^x y(y+2)^2 dy \, dx \\ &= \frac{1}{2} k \int_0^1 \int_{x^2}^x (y^3 + 4y^2 + 4y) dy \, dx = \frac{1}{2} k \int_0^1 \left(\frac{1}{4} y^4 + \frac{4}{3} y^3 + 2y^2 \right) \Big|_{x^2}^x dx \\ &= \frac{1}{2} k \int_0^1 \left(-\frac{1}{4} x^8 - \frac{4}{3} x^6 - 74x^4 + \frac{4}{3} x^3 + 2x^2 \right) dx \\ &= \frac{1}{2} k \left(-\frac{1}{36} x^9 - \frac{4}{21} x^7 - \frac{7}{20} x^5 + \frac{1}{3} x^4 + \frac{2}{3} x^3 \right) \Big|_0^1 = \frac{68}{315} k \end{aligned}$$

$$\begin{aligned} M_{yz} &= \int_0^1 \int_{x^2}^x \int_0^{y+2} kxz \, dz \, dy \, dx = k \int_0^1 \int_{x^2}^x \frac{1}{2} xz^2 \Big|_0^{y+2} dy \, dx = \frac{1}{2} k \int_0^1 \int_{x^2}^x x(y+2)^2 dy \, dx \\ &= \frac{1}{2} k \int_0^1 \frac{1}{3} x(y+2)^3 \Big|_{x^2}^x dx = \frac{1}{6} k \int_0^1 [x(x+2)^3 - x(x^2+2)^3] dx \\ &= \frac{1}{6} k \int_0^1 [x^4 + 6x^3 + 12x^2 + 8x - x(x^2+2)^3] dx \\ &= \frac{1}{6} k \left[\frac{1}{5} x^5 + \frac{3}{2} x^4 + 4x^3 + 4x^2 - \frac{1}{8} (x^2+2)^4 \right] \Big|_0^1 = \frac{21}{80} k \end{aligned}$$

$$\bar{x} = M_{yz}/m = \frac{21k/80}{407k/840} = 441/814, \quad \bar{y} = M_{xz}/m = \frac{68k/315}{407k/840} = 544/1221,$$

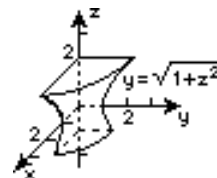
$$\bar{z} = M_{xy}/m = \frac{1493k/1890}{407k/840} = 5972/3663. \text{ The center of mass is } (441/814, 544/1221, 5972/3663).$$

29. $m = \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{2+2y}^{8-y} (x+y+4) dz dy dx$



30. Both the region and the density function are symmetric with respect to the xz - and

yz -planes. Thus, $m = 4 \int_{-1}^2 \int_0^{\sqrt{1+z^2}} \int_0^{\sqrt{1+z^2-y^2}} z^2 dx dy dz$.



31. We are given $\rho(x, y, z) = kz$.

$$\begin{aligned} I_y &= \int_0^8 \int_0^4 \int_0^{y^{1/3}} kz(x^2 + z^2) dx dz dy = k \int_0^8 \int_0^4 \left(\frac{1}{3}x^3 z + xz^3 \right) \Big|_0^{y^{1/3}} dz dy \\ &= k \int_0^8 \int_0^4 \left(\frac{1}{3}yz + y^{1/3}z^3 \right) dz dy = k \int_0^8 \left(\frac{1}{6}yz^2 + \frac{1}{4}y^{1/3}z^4 \right) \Big|_0^4 dy \\ &= k \int_0^8 \left(\frac{8}{3}y + 64y^{1/3} \right) dy = k \left(\frac{4}{3}y^2 + 48y^{4/3} \right) \Big|_0^8 = \frac{2560}{3}k \end{aligned}$$

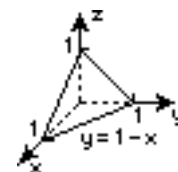
From Problem 25, $m = 96k$. Thus, $R_g = \sqrt{I_y/m} = \sqrt{\frac{2560k/3}{96k}} = \frac{4\sqrt{5}}{3}$.

32. We are given $\rho(x, y, z) = k$.

$$\begin{aligned} I_x &= \int_0^1 \int_{z^2}^{2-z} \int_0^3 k(y^2 + z^2) dy dx dz = k \int_0^1 \int_{z^2}^{2-z} \left(\frac{1}{3}y^3 + yz^2 \right) \Big|_0^3 dx dz = k \int_0^1 \int_{z^2}^{2-z} (9 + 3z^2) dx dz \\ &= k \int_0^1 (9x + 3xz^2) \Big|_{z^2}^{2-z} dz = k \int_0^1 (18 - 9z - 3z^2 - 3z^3 - 3z^4) dz \\ &= k \left(18z - \frac{9}{2}z^2 - z^3 - \frac{3}{4}z^4 - \frac{3}{5}z^5 \right) \Big|_0^1 = \frac{223}{20}k \\ m &= \int_0^1 \int_{z^2}^{2-z} \int_0^3 k dy dx dz = k \int_0^1 \int_{z^2}^{2-z} 3 dx dz = 3k \int_0^1 (2 - z - z^2) dz = 3k \left(2z - \frac{1}{2}z^2 - \frac{1}{3}z^3 \right) \Big|_0^1 = \frac{7}{2}k \\ R_g &= \sqrt{\frac{I_x}{m}} = \sqrt{\frac{223k/20}{7k/2}} = \sqrt{\frac{223}{70}} \end{aligned}$$

33. $I_z = k \int_0^1 \int_0^{1-x} \int_0^{1-x-y} (x^2 + y^2) dz dy dx = k \int_0^1 \int_0^{1-x} (x^2 + y^2)(1-x-y) dy dx$

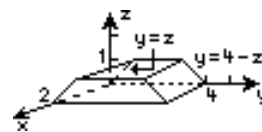
$$\begin{aligned} &= k \int_0^1 \int_0^{1-x} (x^2 - x^3 - x^2y + y^2 - xy^2 - y^3) dy dx \\ &= k \int_0^1 \left[(x^2 - x^3)y - \frac{1}{2}x^2y^2 + \frac{1}{3}(1-x)y^3 - \frac{1}{4}y^4 \right] \Big|_0^{1-x} dx \\ &= k \int_0^1 \left[\frac{1}{2}x^2 - x^3 + \frac{1}{2}x^4 + \frac{1}{12}(1-x)^4 \right] dx = k \left[\frac{1}{6}x^6 - \frac{1}{4}x^4 + \frac{1}{10}x^5 - \frac{1}{60}(1-x)^5 \right] \Big|_0^1 = \frac{k}{30} \end{aligned}$$



9.15 Triple Integrals

34. We are given $\rho(x, y, z) = kx$.

$$\begin{aligned} I_y &= \int_0^1 \int_0^2 \int_z^{4-z} kx(x^2 + z^2) dy dx dz = k \int_0^1 \int_0^2 (x^3 + xz^2) y \Big|_z^{4-z} dx dz \\ &= k \int_0^1 \int_0^2 (x^3 + xz^2)(4 - 2z) dx dz = k \int_0^1 \left(\frac{1}{4}x^4 + \frac{1}{2}x^2 z^2 \right) (4 - 2z) \Big|_0^2 dz \\ &= k \int_0^1 (4 + 2z^2)(4 - 2z) dz = 4k \int_0^1 (4 - 2z + 2z^2 - z^3) dz = 4k \left(4z - z^2 + \frac{2}{3}z^3 - \frac{1}{4}z^4 \right) \Big|_0^1 = \frac{41}{3}k \end{aligned}$$



35. $x = 10 \cos 3\pi/4 = -5\sqrt{2}$; $y = 10 \sin 3\pi/4 = 5\sqrt{2}$; $(-5\sqrt{2}, 5\sqrt{2}, 5)$

36. $x = 2 \cos 5\pi/6 = -\sqrt{3}$; $y = 2 \sin 5\pi/6 = 1$; $(-\sqrt{3}, 1, -3)$

37. $x = \sqrt{3} \cos \pi/3 = \sqrt{3}/2$; $y = \sqrt{3} \sin \pi/3 = 3/2$; $(\sqrt{3}/2, 3/2, -4)$

38. $x = 4 \cos 7\pi/4 = 2\sqrt{2}$; $y = 4 \sin 7\pi/4 = -2\sqrt{2}$; $(2\sqrt{2}, -2\sqrt{2}, 0)$

39. With $x = 1$ and $y = -1$ we have $r^2 = 2$ and $\tan \theta = -1$. The point is $(\sqrt{2}, -\pi/4, -9)$.

40. With $x = 2\sqrt{3}$ and $y = 2$ we have $r^2 = 16$ and $\tan \theta = 1/\sqrt{3}$. The point is $(4, \pi/6, 17)$.

41. With $x = -\sqrt{2}$ and $y = \sqrt{6}$ we have $r^2 = 8$ and $\tan \theta = -\sqrt{3}$. The point is $(2\sqrt{2}, 2\pi/3, 2)$.

42. With $x = 1$ and $y = 2$ we have $r^2 = 5$ and $\tan \theta = 2$. The point is $(\sqrt{5}, \tan^{-1} 2, 7)$.

43. $r^2 + z^2 = 25$

44. $r \cos \theta + r \sin \theta - z = 1$

45. $r^2 - z^2 = 1$

46. $r^2 \cos^2 \theta + z^2 = 16$

47. $z = x^2 + y^2$

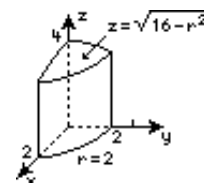
48. $z = 2y$

49. $r \cos \theta = 5$, $x = 5$

50. $\tan \theta = 1/\sqrt{3}$, $y/x = 1/\sqrt{3}$, $x = \sqrt{3}y$, $x > 0$

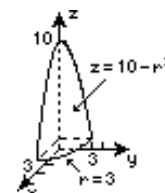
51. The equations are $r^2 = 4$, $r^2 + z^2 = 16$, and $z = 0$.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^2 \int_0^{\sqrt{16-r^2}} r dz dr d\theta = \int_0^{2\pi} \int_0^2 r \sqrt{16-r^2} dr d\theta \\ &= \int_0^{2\pi} -\frac{1}{3}(16-r^2)^{3/2} \Big|_0^2 d\theta = \int_0^{2\pi} \frac{1}{3}(64-24\sqrt{3}) d\theta = \frac{2\pi}{3}(64-24\sqrt{3}) \end{aligned}$$



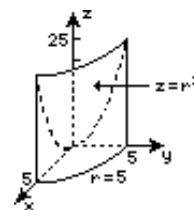
52. The equation is $z = 10 - r^2$.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^3 \int_1^{10-r^2} r dz dr d\theta = \int_0^{2\pi} \int_0^3 r(9-r^2) dr d\theta = \int_0^{2\pi} \left(\frac{9}{2}r^2 - \frac{1}{4}r^4 \right) \Big|_0^3 d\theta \\ &= \int_0^{2\pi} \frac{81}{4} d\theta = \frac{81\pi}{2}. \end{aligned}$$



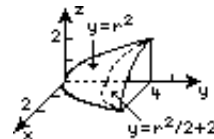
53. The equations are $z = r^2$, $r = 5$, and $z = 0$.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^5 \int_0^{r^2} r dz dr d\theta = \int_0^{2\pi} \int_0^5 r^3 dr d\theta = \int_0^{2\pi} \frac{1}{4}r^4 \Big|_0^5 d\theta \\ &= \int_0^{2\pi} \frac{625}{4} d\theta = \frac{625\pi}{2} \end{aligned}$$



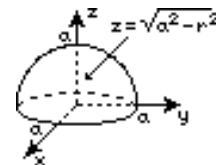
54. Substituting the first equation into the second, we see that the surfaces intersect in the plane $y = 4$. Using polar coordinates in the xz -plane, the equations of the surfaces become $y = r^2$ and $y = \frac{1}{2}r^2 + 2$.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^2 \int_{r^2}^{r^2/2+2} r \, dy \, dr \, d\theta = \int_0^{2\pi} \int_0^2 r \left(\frac{r^2}{2} + 2 - r^2 \right) dr \, d\theta \\ &= \int_0^{2\pi} \int_0^2 \left(2r - \frac{1}{2}r^3 \right) dr \, d\theta = \int_0^{2\pi} \left(r^2 - \frac{1}{8}r^4 \right) \Big|_0^2 d\theta = \int_0^{2\pi} 2 \, d\theta = 4\pi \end{aligned}$$



55. The equation is $z = \sqrt{a^2 - r^2}$. By symmetry, $\bar{x} = \bar{y} = 0$.

$$\begin{aligned} m &= \int_0^{2\pi} \int_0^a \int_0^{\sqrt{a^2-r^2}} r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^a r \sqrt{a^2 - r^2} \, dr \, d\theta \\ &= \int_0^{2\pi} -\frac{1}{3}(a^2 - r^2)^{3/2} \Big|_0^a d\theta = \int_0^{2\pi} \frac{1}{3}a^3 \, d\theta = \frac{2}{3}\pi a^3 \end{aligned}$$

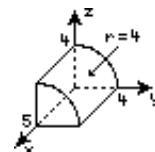


$$\begin{aligned} M_{xy} &= \int_0^{2\pi} \int_0^a \int_0^{\sqrt{a^2-r^2}} zr \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^a \frac{1}{2}rz^2 \Big|_0^{\sqrt{a^2-r^2}} dr \, d\theta = \frac{1}{2} \int_0^{2\pi} \int_0^a r(a^2 - r^2) \, dr \, d\theta \\ &= \frac{1}{2} \int_0^{2\pi} \left(\frac{1}{2}a^2r^2 - \frac{1}{4}r^4 \right) \Big|_0^a d\theta = \frac{1}{2} \int_0^{2\pi} \frac{1}{4}a^4 \, d\theta = \frac{1}{4}\pi a^4 \end{aligned}$$

$$\bar{z} = M_{xy}/m = \frac{\pi a^4/4}{2\pi a^3/3} = 3a/8. \text{ The centroid is } (0, 0, 3a/8).$$

56. We use polar coordinates in the yz -plane. The density is $\rho(x, y, z) = kz$. By symmetry, $\bar{y} = \bar{z} = 0$.

$$\begin{aligned} m &= \int_0^{2\pi} \int_0^4 \int_0^5 kxr \, dx \, dr \, d\theta = k \int_0^{2\pi} \int_0^4 \frac{1}{2}rz^2 \Big|_0^5 dr \, d\theta = \frac{k}{2} \int_0^{2\pi} \int_0^4 25r \, dr \, d\theta \\ &= \frac{25k}{2} \int_0^{2\pi} \frac{1}{2}r^2 \Big|_0^4 d\theta = \frac{25k}{2} \int_0^{2\pi} 8 \, d\theta = 200k\pi \end{aligned}$$

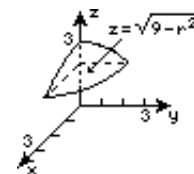


$$\begin{aligned} M_{yz} &= \int_0^{2\pi} \int_0^4 \int_0^5 kx^2r \, dx \, dr \, d\theta = k \int_0^{2\pi} \int_0^4 \frac{1}{3}rx^3 \Big|_0^5 dr \, d\theta = \frac{1}{3}k \int_0^{2\pi} \int_0^4 125r \, dr \, d\theta \\ &= \frac{1}{3}k \int_0^{2\pi} \frac{125}{2}r^2 \Big|_0^4 d\theta = \frac{1}{3}k \int_0^{2\pi} 1000 \, d\theta = \frac{2000}{3}k\pi \end{aligned}$$

$$\bar{x} = M_{yz}/m = \frac{2000k\pi/3}{200k\pi} = 10/3. \text{ The center of mass of the given solid is } (10/3, 0, 0).$$

57. The equation is $z = \sqrt{9 - r^2}$ and the density is $\rho = k/r^2$. When $z = 2$, $r = \sqrt{5}$.

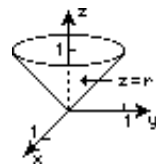
$$\begin{aligned} I_z &= \int_0^{2\pi} \int_0^{\sqrt{5}} \int_2^{\sqrt{9-r^2}} r^2(k/r^2)r \, dz \, dr \, d\theta = k \int_0^{2\pi} \int_0^{\sqrt{5}} rz \Big|_2^{\sqrt{9-r^2}} dr \, d\theta \\ &= k \int_0^{2\pi} \int_0^{\sqrt{5}} (r\sqrt{9-r^2} - 2r) \, dr \, d\theta = k \int_0^{2\pi} \left[-\frac{1}{3}(9-r^2)^{3/2} - r^2 \right] \Big|_0^{\sqrt{5}} d\theta \\ &= k \int_0^{2\pi} \frac{4}{3} \, d\theta = \frac{8}{3}\pi k \end{aligned}$$



9.15 Triple Integrals

58. The equation is $z = r$ and the density is $\rho = kr$.

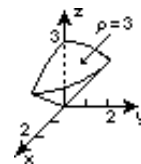
$$\begin{aligned} I_x &= \int_0^{2\pi} \int_0^1 \int_r^1 (y^2 + z^2)(kr)r \, dz \, dr \, d\theta = k \int_0^{2\pi} \int_0^1 \int_r^1 (r^4 \sin^2 \theta + r^2 z^2) \, dz \, dr \, d\theta \\ &= k \int_0^{2\pi} \int_0^1 \left[(r^4 \sin^2 \theta)z + \frac{1}{3}r^2 z^3 \right] \Big|_r^1 \, dr \, d\theta \\ &= k \int_0^{2\pi} \int_0^1 \left(r^4 \sin^2 \theta + \frac{1}{3}r^2 - r^5 \sin^2 \theta - \frac{1}{3}r^5 \right) \, dr \, d\theta \\ &= k \int_0^{2\pi} \left(\frac{1}{5}r^5 \sin^2 \theta + \frac{1}{9}r^3 - \frac{1}{6}r^6 \sin^2 \theta - \frac{1}{18}r^6 \right) \Big|_0^1 \, d\theta = k \int_0^{2\pi} \left(\frac{1}{30} \sin^2 \theta + \frac{1}{18} \right) \, d\theta \\ &= k \left(\frac{1}{60}\theta - \frac{1}{120} \sin 2\theta + \frac{1}{18}\theta \right) \Big|_0^{2\pi} = \frac{13}{90}\pi k \end{aligned}$$



59. (a) $x = (2/3) \sin(\pi/2) \cos(\pi/6) = \sqrt{3}/3$; $y = (2/3) \sin(\pi/2) \sin(\pi/6) = 1/3$;
 $z = (2/3) \cos(\pi/2) = 0$; $(\sqrt{3}/3, 1/3, 0)$
- (b) With $x = \sqrt{3}/3$ and $y = 1/3$ we have $r^2 = 4/9$ and $\tan \theta = \sqrt{3}/3$. The point is $(2/3, \pi/6, 0)$.
60. (a) $x = 5 \sin(5\pi/4) \cos(2\pi/3) = 5\sqrt{2}/4$; $y = 5 \sin(5\pi/4) \sin(2\pi/3) = -5\sqrt{6}/4$;
 $z = 5 \cos(5\pi/4) = -5\sqrt{2}/2$; $(5\sqrt{2}/4, -5\sqrt{6}/4, -5\sqrt{2}/2)$
- (b) With $x = 5\sqrt{2}/4$ and $y = -5\sqrt{6}/4$ we have $r^2 = 25/2$ and $\tan \theta = -\sqrt{3}$.
 The point is $(5/\sqrt{2}, 2\pi/3, -5\sqrt{2}/2)$.
61. (a) $x = 8 \sin(\pi/4) \cos(3\pi/4) = -4$; $y = 8 \sin(\pi/4) \sin(3\pi/4) = 4$; $z = 8 \cos(\pi/4) = 4\sqrt{2}$;
 $(-4, 4, 4\sqrt{2})$
- (b) With $x = -4$ and $y = 4$ we have $r^2 = 32$ and $\tan \theta = -1$. The point is $(4\sqrt{2}, 3\pi/4, 4\sqrt{2})$.
62. (a) $x = (1/3) \sin(5\pi/3) \cos(\pi/6) = -1/4$; $y = (1/3) \sin(5\pi/3) \sin(\pi/6) = -\sqrt{3}/12$;
 $z = (1/3) \cos(5\pi/3) = 1/6$; $(-1/4, -\sqrt{3}/12, 1/6)$
- (b) With $x = -1/4$ and $y = -\sqrt{3}/12$ we have $r^2 = 1/12$ and $\tan \theta = \sqrt{3}/3$. The point is $(1/2\sqrt{3}, \pi/6, 1/6)$.
63. With $x = -5$, $y = -5$, and $z = 0$, we have $\rho^2 = 50$, $\tan \theta = 1$, and $\cos \phi = 0$. The point is $(5\sqrt{2}, \pi/2, 5\pi/4)$.
64. With $x = 1$, $y = -\sqrt{3}$, and $z = 1$, we have $\rho^2 = 5$, $\tan \theta = -\sqrt{3}$, and $\cos \phi = 1/\sqrt{5}$. The point is $(\sqrt{5}, \cos^{-1} 1/\sqrt{5}, -\pi/3)$.
65. With $x = \sqrt{3}/2$, $y = 1/2$, and $z = 1$, we have $\rho^2 = 2$, $\tan \theta = 1/\sqrt{3}$, and $\cos \phi = 1/\sqrt{2}$. The point is $(\sqrt{2}, \pi/4, \pi/6)$.
66. With $x = -\sqrt{3}/2$, $y = 0$, and $z = -1/2$, we have $\rho^2 = 1$, $\tan \theta = 0$, and $\cos \phi = -1/2$. The point is $(1, 2\pi/3, 0)$.
67. $\rho = 8$
68. $\rho^2 = 4\rho \cos \phi$; $\rho = 4 \cos \phi$
69. $4z^2 = 3x^2 + 3y^2 + 3z^2$; $4\rho^2 \cos^2 \phi = 3\rho^2$; $\cos \phi = \pm\sqrt{3}/2$; $\phi = \pi/6$ or equivalently, $\phi = 5\pi/6$
70. $-x^2 - y^2 - z^2 = 1 - 2z^2$; $-\rho^2 = 1 - 2\rho^2 \cos^2 \phi$; $\rho^2(2 \cos^2 \phi - 1) = 1$
71. $x^2 + y^2 + z^2 = 100$
72. $\cos \phi = 1/2$; $\rho^2 \cos^2 \phi = \rho^2/4$; $4z^2 = x^2 + y^2 + z^2$; $x^2 + y^2 = 3z^2$
73. $\rho \cos \phi = 2$; $z = 2$
74. $\rho(1 - \cos^2 \phi) = \cos \phi$; $\rho^2 - \rho^2 \cos^2 \phi = \rho \cos \phi$; $x^2 + y^2 + z^2 - z^2 = z$; $z = x^2 + y^2$

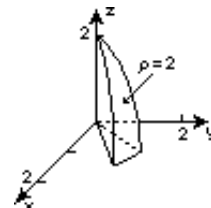
75. The equations are $\phi = \pi/4$ and $\rho = 3$.

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^{\pi/4} \int_0^3 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} \frac{1}{3} \rho^3 \sin \phi \Big|_0^3 \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} 9 \sin \phi \, d\phi \, d\theta \\ &= \int_0^{2\pi} -9 \cos \phi \Big|_0^{\pi/4} \, d\theta = -9 \int_0^{2\pi} \left(\frac{\sqrt{2}}{2} - 1 \right) d\theta = 9\pi(2 - \sqrt{2}) \end{aligned}$$



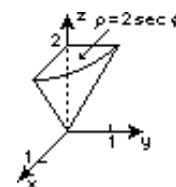
76. The equations are $\rho = 2$, $\theta = \pi/4$, and $\theta = \pi/3$.

$$\begin{aligned} \int_{\pi/4}^{\pi/3} \int_0^{\pi/2} \int_0^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta &= \int_{\pi/4}^{\pi/3} \int_0^{\pi/2} \frac{1}{3} \rho^3 \sin \phi \Big|_0^2 \, d\phi \, d\theta \\ &= \int_{\pi/4}^{\pi/3} \int_0^{\pi/2} \frac{8}{3} \sin \phi \, d\phi \, d\theta = \frac{8}{3} \int_{\pi/4}^{\pi/3} -\cos \phi \Big|_0^{\pi/2} d\theta \\ &= \frac{8}{3} \int_{\pi/4}^{\pi/3} (0 + 1) d\theta = \frac{2\pi}{9} \end{aligned}$$



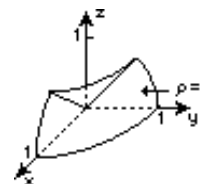
77. From Problem 69, we have $\phi = \pi/6$. Since the figure is in the first octant and $z = 2$ we also have $\theta = 0$, $\theta = \pi/2$, and $\rho \cos \phi = 2$.

$$\begin{aligned} V &= \int_0^{\pi/2} \int_0^{\pi/6} \int_0^{2 \sec \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{\pi/2} \int_0^{\pi/6} \frac{1}{3} \rho^3 \sin \phi \Big|_0^{2 \sec \phi} \, d\phi \, d\theta \\ &= \frac{8}{3} \int_0^{\pi/2} \int_0^{\pi/6} \sec^3 \phi \sin \phi \, d\phi \, d\theta = \frac{8}{3} \int_0^{\pi/2} \int_0^{\pi/6} \sec^2 \phi \tan \phi \, d\phi \, d\theta \\ &= \frac{8}{3} \int_0^{\pi/2} \frac{1}{2} \tan^2 \phi \Big|_0^{\pi/6} d\theta = \frac{4}{3} \int_0^{\pi/2} \frac{1}{3} d\theta = \frac{2}{9} \pi \end{aligned}$$



78. The equations are $\rho = 1$ and $\phi = \pi/4$. We find the volume above the xy -plane and double.

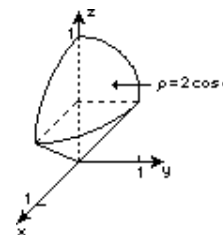
$$\begin{aligned} V &= 2 \int_0^{2\pi} \int_{\pi/4}^{\pi/2} \int_0^1 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = 2 \int_0^{2\pi} \int_{\pi/4}^{\pi/2} \frac{1}{3} \rho^3 \sin \phi \Big|_0^1 \, d\phi \, d\theta \\ &= \frac{2}{3} \int_0^{2\pi} \int_{\pi/4}^{\pi/2} \sin \phi \, d\phi \, d\theta = \frac{2}{3} \int_0^{2\pi} -\cos \phi \Big|_{\pi/4}^{\pi/2} d\theta = \frac{2}{3} \int_0^{2\pi} \frac{\sqrt{2}}{2} d\theta = \frac{2\pi\sqrt{2}}{3} \end{aligned}$$



79. By symmetry, $\bar{x} = \bar{y} = 0$. The equations are $\phi = \pi/4$ and $\rho = 2 \cos \phi$.

$$\begin{aligned} m &= \int_0^{2\pi} \int_0^{\pi/4} \int_0^{2 \cos \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} \frac{1}{3} \rho^3 \sin \phi \Big|_0^{2 \cos \phi} \, d\phi \, d\theta \\ &= \frac{8}{3} \int_0^{2\pi} \int_0^{\pi/4} \sin \phi \cos^3 \phi \, d\phi \, d\theta = \frac{8}{3} \int_0^{2\pi} -\frac{1}{4} \cos^4 \phi \Big|_0^{\pi/4} d\theta \\ &= -\frac{2}{3} \int_0^{2\pi} \left(\frac{1}{4} - 1 \right) d\theta = \pi \\ M_{xy} &= \int_0^{2\pi} \int_0^{\pi/4} \int_0^{2 \cos \phi} z \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} \int_0^{2 \cos \phi} \rho^3 \sin \phi \cos \phi \, d\rho \, d\phi \, d\theta \\ &= \int_0^{2\pi} \int_0^{\pi/4} \frac{1}{4} \rho^4 \sin \phi \cos \phi \Big|_0^{2 \cos \phi} \, d\phi \, d\theta = 4 \int_0^{2\pi} \int_0^{\pi/4} \cos^5 \phi \sin \phi \, d\phi \, d\theta \\ &= 4 \int_0^{2\pi} -\frac{1}{6} \cos^6 \phi \Big|_0^{\pi/4} d\theta = -\frac{2}{3} \int_0^{2\pi} \left(\frac{1}{8} - 1 \right) d\theta = \frac{7}{6} \pi \end{aligned}$$

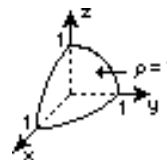
$$\bar{z} = M_{xy}/m = \frac{7\pi/6}{\pi} = 7/6. \text{ The centroid is } (0, 0, 7/6).$$



9.15 Triple Integrals

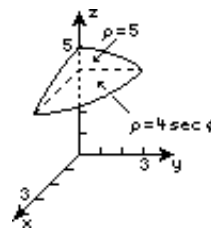
80. We are given density $= kz$. By symmetry, $\bar{x} = \bar{y} = 0$. The equation is $\rho = 1$.

$$\begin{aligned}
 m &= \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 kz \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = k \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \rho^3 \sin \phi \cos \phi \, d\rho \, d\phi \, d\theta \\
 &= k \int_0^{2\pi} \int_0^{\pi/2} \frac{1}{4} \rho^4 \sin \phi \cos \phi \Big|_0^1 \, d\phi \, d\theta = \frac{1}{4} k \int_0^{2\pi} \int_0^{\pi/2} \sin \phi \cos \phi \, d\phi \, d\theta \\
 &= \frac{1}{4} k \int_0^{2\pi} \frac{1}{2} \sin^2 \phi \Big|_0^{\pi/2} \, d\theta = \frac{1}{8} k \int_0^{2\pi} d\theta = \frac{k\pi}{4} \\
 M_{xy} &= \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 kz^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = k \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \rho^4 \cos^2 \phi \sin \phi \, d\rho \, d\phi \, d\theta \\
 &= k \int_0^{2\pi} \int_0^{\pi/2} \frac{1}{5} \rho^5 \cos^2 \phi \sin \phi \Big|_0^1 \, d\phi \, d\theta = \frac{1}{5} k \int_0^{2\pi} \int_0^{\pi/2} \cos^2 \phi \sin \phi \, d\phi \, d\theta \\
 &= \frac{1}{5} k \int_0^{2\pi} -\frac{1}{3} \cos^3 \phi \Big|_0^{\pi/2} \, d\theta = -\frac{1}{15} k \int_0^{2\pi} (0 - 1) \, d\theta = \frac{2}{15} k\pi \\
 \bar{z} = M_{xy}/m &= \frac{2k\pi/15}{k\pi/4} = 8/15. \text{ The center of mass is } (0, 0, 8/15).
 \end{aligned}$$



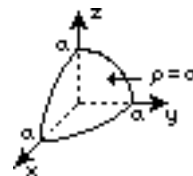
81. We are given density $= k/\rho$.

$$\begin{aligned}
 m &= \int_0^{2\pi} \int_0^{\cos^{-1} 4/5} \int_{4 \sec \phi}^5 \frac{k}{\rho} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = k \int_0^{2\pi} \int_0^{\cos^{-1} 4/5} \frac{1}{2} \rho^2 \sin \phi \Big|_{4 \sec \phi}^5 \, d\phi \, d\theta \\
 &= \frac{1}{2} k \int_0^{2\pi} \int_0^{\cos^{-1} 4/5} (25 \sin \phi - 16 \tan \phi \sec \phi) \, d\phi \, d\theta \\
 &= \frac{1}{2} k \int_0^{2\pi} (-25 \cos \phi - 16 \sec \phi) \Big|_0^{\cos^{-1} 4/5} \, d\theta = \frac{1}{2} k \int_0^{2\pi} [-25(4/5) - 16(5/4) - (-25 - 16)] \, d\theta \\
 &= \frac{1}{2} k \int_0^{2\pi} d\theta = k\pi
 \end{aligned}$$



82. We are given density $= k\rho$.

$$\begin{aligned}
 I_z &= \int_0^{2\pi} \int_0^\pi \int_0^a (x^2 + y^2)(k\rho) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\
 &= k \int_0^{2\pi} \int_0^\pi \int_0^a (\rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta) \rho^3 \sin \phi \, d\rho \, d\phi \, d\theta \\
 &= k \int_0^{2\pi} \int_0^\pi \int_0^a \rho^5 \sin^3 \phi \, d\rho \, d\phi \, d\theta = k \int_0^{2\pi} \int_0^\pi \frac{1}{6} \rho^6 \sin^3 \phi \Big|_0^a \, d\phi \, d\theta = \frac{1}{6} ka^6 \int_0^{2\pi} \int_0^\pi \sin^3 \phi \, d\phi \, d\theta \\
 &= \frac{1}{6} ka^3 \int_0^{2\pi} \int_0^\pi (1 - \cos^2 \phi) \sin \phi \, d\phi \, d\theta = \frac{1}{6} ka^3 \int_0^{2\pi} \left(-\cos \phi + \frac{1}{3} \cos^3 \phi \right) \Big|_0^\pi \, d\theta = \frac{1}{6} ka^3 \int_0^{2\pi} \frac{4}{3} \, d\theta = \frac{4\pi}{9} ka^6
 \end{aligned}$$



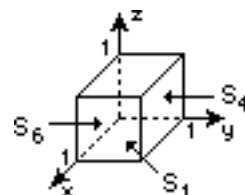
EXERCISES 9.16

Divergence Theorem

- 1.
- $\operatorname{div} \mathbf{F} = y + z + x$

The Triple Integral:

$$\begin{aligned}
 \iiint_D \operatorname{div} \mathbf{F} \, dV &= \int_0^1 \int_0^1 \int_0^1 (x + y + z) \, dx \, dy \, dz \\
 &= \int_0^1 \int_0^1 \left(\frac{1}{2}x^2 + xy + xz \right) \Big|_0^1 \, dy \, dz \\
 &= \int_0^1 \int_0^1 \left(\frac{1}{2} + y + z \right) \, dy \, dz = \int_0^1 \left(\frac{1}{2}y + \frac{1}{2}y^2 + yz \right) \Big|_0^1 \, dz \\
 &= \int_0^1 (1 + z) \, dz = \frac{1}{2}(1 + z^2) \Big|_0^1 = 2 - \frac{1}{2} = \frac{3}{2}
 \end{aligned}$$



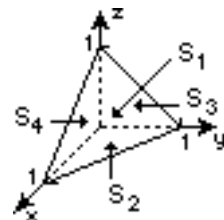
The Surface Integral: Let the surfaces be S_1 in $z = 0$, S_2 in $z = 1$, S_3 in $y = 0$, S_4 in $y = 1$, S_5 in $x = 0$, and S_6 in $x = 1$. The unit outward normal vectors are $-\mathbf{k}$, $\mathbf{k}$, $-\mathbf{j}$, $\mathbf{j}$, $-\mathbf{i}$ and $\mathbf{i}$, respectively. Then

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= \iint_{S_1} \mathbf{F} \cdot (-\mathbf{k}) \, dS_1 + \iint_{S_2} \mathbf{F} \cdot \mathbf{k} \, dS_2 + \iint_{S_3} \mathbf{F} \cdot (-\mathbf{j}) \, dS_3 + \iint_{S_4} \mathbf{F} \cdot \mathbf{j} \, dS_4 \\
 &\quad + \iint_{S_5} \mathbf{F} \cdot (-\mathbf{i}) \, dS_5 + \iint_{S_6} \mathbf{F} \cdot \mathbf{i} \, dS_6 \\
 &= \iint_{S_1} (-xz) \, dS_1 + \iint_{S_2} xz \, dS_2 + \iint_{S_3} (-yz) \, dS_3 + \iint_{S_4} yz \, dS_4 \\
 &\quad + \iint_{S_5} (-xy) \, dS_5 + \iint_{S_6} xy \, dS_6 \\
 &= \iint_{S_2} x \, dS_2 + \iint_{S_4} z \, dS_4 + \iint_{S_6} y \, dS_6 \\
 &= \int_0^1 \int_0^1 x \, dx \, dy + \int_0^1 \int_0^1 z \, dz \, dx + \int_0^1 \int_0^1 y \, dy \, dz \\
 &= \int_0^1 \frac{1}{2} \, dy + \int_0^1 \frac{1}{2} \, dx + \int_0^1 \frac{1}{2} \, dz = \frac{3}{2}.
 \end{aligned}$$

- 2.
- $\operatorname{div} \mathbf{F} = 6y + 4z$

The Triple Integral:

$$\begin{aligned}
 \iiint_D \operatorname{div} \mathbf{F} \, dV &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} (6y + 4z) \, dz \, dy \, dx \\
 &= \int_0^1 \int_0^{1-x} (6yz + 2z^2) \Big|_0^{1-x-y} \, dy \, dx \\
 &= \int_0^1 \int_0^{1-x} (-4y^2 + 2y - 2xy + 2x^2 - 4x + 2) \, dy \, dx
 \end{aligned}$$



9.16 Divergence Theorem

$$\begin{aligned}
 &= \int_0^1 \left(-\frac{4}{3}y^3 + y^2 - xy^2 + 2x^2y - 4xy + 2y \right) \Big|_0^{1-x} dx \\
 &= \int_0^1 \left(-\frac{5}{3}x^3 + 5x^2 - 5x + \frac{5}{3} \right) dx = \left(-\frac{5}{12}x^4 + \frac{5}{3}x^3 - \frac{5}{2}x^2 + \frac{5}{3}x \right) \Big|_0^1 = \frac{5}{12}
 \end{aligned}$$

The Surface Integral: Let the surfaces be S_1 in the plane $x + y + z = 1$, S_2 in $z = 0$, S_3 in $x = 0$, and S_4 in $y = 0$. The unit outward normal vectors are $\mathbf{n}_1 = (\mathbf{i} + \mathbf{j} + \mathbf{k})/\sqrt{3}$, $\mathbf{n}_2 = -\mathbf{k}$, $\mathbf{n}_3 = -\mathbf{i}$, and $\mathbf{n}_4 = -\mathbf{j}$, respectively. Now on S_1 , $dS_1 = \sqrt{3} dA_1$, on S_3 , $x = 0$, and on S_4 , $y = 0$, so

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iint_{S_1} \mathbf{F} \cdot \mathbf{n}_1 dS_1 + \iint_{S_2} \mathbf{F} \cdot (-\mathbf{k}) dS_2 + \iint_{S_3} \mathbf{F} \cdot (-\mathbf{i}) dS_3 + \iint_{S_4} \mathbf{F} \cdot (-\mathbf{j}) dS_4 \\
 &= \int_0^1 \int_0^{1-x} (6xy + 4y(1-x-y) + xe^{-y}) dy dx + \int_0^1 \int_0^{1-x} (-xe^{-y}) dy dx \\
 &\quad + \iint_{S_3} (-6xy) dS_3 + \iint_{S_4} (-4yz) dS_4 \\
 &= \int_0^1 \left(xy^2 + 2y^2 - \frac{4}{3}y^3 - xe^{-y} \right) \Big|_0^{1-x} dx + \int_0^1 xe^{-y} \Big|_0^{1-x} dx + 0 + 0 \\
 &= \int_0^1 [x(1-x)^2 + 2(1-x)^2 - \frac{4}{3}(1-x)^3 - xe^{x-1} + x] dx + \int_0^1 (xe^{x-1} - x) dx \\
 &= \left[\frac{1}{2}x^2 - \frac{2}{3}x^3 + \frac{1}{4}x^4 - \frac{2}{3}(1-x)^3 + \frac{1}{3}(1-x)^4 \right] \Big|_0^1 = \frac{5}{12}.
 \end{aligned}$$

3. $\operatorname{div} \mathbf{F} = 3x^2 + 3y^2 + 3z^2$. Using spherical coordinates,

$$\begin{aligned}
 \iiint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D 3(x^2 + y^2 + z^2) dV = \int_0^{2\pi} \int_0^\pi \int_0^a 3\rho^2 \rho^2 \sin \phi d\rho d\phi d\theta \\
 &= \int_0^{2\pi} \int_0^\pi \frac{3}{5} \rho^5 \sin \phi \Big|_0^a d\phi d\theta = \frac{3a^5}{5} \int_0^{2\pi} \int_0^\pi \sin \phi d\phi d\theta \\
 &= \frac{3a^5}{5} \int_0^{2\pi} -\cos \phi \Big|_0^\pi d\theta = \frac{6a^5}{5} \int_0^{2\pi} d\theta = \frac{12\pi a^5}{5}.
 \end{aligned}$$

4. $\operatorname{div} \mathbf{F} = 4 + 1 + 4 = 9$. Using the formula for the volume of a sphere,

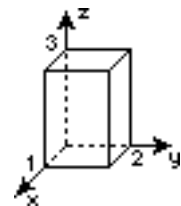
$$\iiint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_D 9 dV = 9 \left(\frac{4}{3} \pi 2^3 \right) = 96\pi.$$

5. $\operatorname{div} \mathbf{F} = 2(z-1)$. Using cylindrical coordinates,

$$\begin{aligned}
 \iiint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D 2(z-1) V = \int_0^{2\pi} \int_0^4 \int_1^5 2(z-1) dz r dr d\theta = \int_0^{2\pi} \int_0^4 (z-1)^2 \Big|_1^5 r dr d\theta \\
 &= \int_0^{2\pi} \int_0^4 16r dr d\theta = \int_0^{2\pi} 8r^2 \Big|_0^4 d\theta = 128 \int_0^{2\pi} d\theta = 256\pi.
 \end{aligned}$$

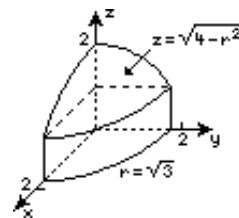
6. $\operatorname{div} \mathbf{F} = 2x + 2z + 12z^2$.

$$\begin{aligned}
 \iiint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} \mathbf{F} dV = \int_0^3 \int_0^2 \int_0^1 (2x + 2z + 12z^2) dx dy dz \\
 &= \int_0^3 \int_0^2 (x^2 + 2xz + 12xz^2) \Big|_0^1 dy dz = \int_0^3 \int_0^2 (1 + 2z + 12z^2) dy dz \\
 &= \int_0^3 2(1 + 2z + 12z^2) dz = (2z + 2z^2 + 8z^3) \Big|_0^3 = 240
 \end{aligned}$$



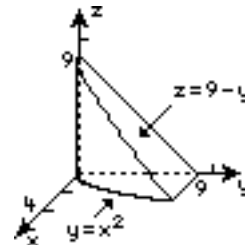
- 7.
- $\operatorname{div} \mathbf{F} = 3z^2$
- . Using cylindrical coordinates,

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} \mathbf{F} dV = \int_0^{2\pi} \int_0^{\sqrt{3}} \int_0^{\sqrt{4-r^2}} 3z^2 r dz dr d\theta \\
 &= \int_0^{2\pi} \int_0^{\sqrt{3}} r z^3 \Big|_0^{\sqrt{4-r^2}} dr d\theta = \int_0^{2\pi} \int_0^{\sqrt{3}} r(4-r^2)^{3/2} dr d\theta \\
 &= \int_0^{2\pi} -\frac{1}{5}(4-r^2)^{5/2} \Big|_0^{\sqrt{3}} d\theta = \int_0^{2\pi} -\frac{1}{5}(1-32) d\theta = \int_0^{2\pi} \frac{31}{5} d\theta = \frac{62\pi}{5}.
 \end{aligned}$$



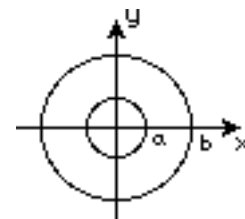
- 8.
- $\operatorname{div} \mathbf{F} = 2x$
- .

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} \mathbf{F} dV = \int_0^3 \int_{x^2}^9 \int_0^{9-y} 2x dz dy dx \\
 &= \int_0^3 \int_{x^2}^9 2x(9-y) dy dx = \int_0^3 -x(9-y)^2 \Big|_{x^2}^9 dx = \int_0^3 x(9-x)^2 dx \\
 &= \int_0^3 (x^3 - 18x^2 + 81x) dx = \left(\frac{1}{4}x^4 - 6x^3 + \frac{81}{2}x^2 \right) \Big|_0^3 = \frac{891}{4}
 \end{aligned}$$



- 9.
- $\operatorname{div} \mathbf{F} = \frac{1}{x^2 + y^2 + z^2}$
- . Using spherical coordinates,

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} \mathbf{F} dV = \int_0^{2\pi} \int_0^\pi \int_a^b \frac{1}{\rho^2} \rho^2 \sin \phi d\rho d\phi d\theta \\
 &= \int_0^{2\pi} \int_0^\pi (b-a) \sin \phi d\phi d\theta = (b-a) \int_0^{2\pi} -\cos \phi \Big|_0^\pi d\theta \\
 &= (b-a) \int_0^{2\pi} 2 d\theta = 4\pi(b-a).
 \end{aligned}$$



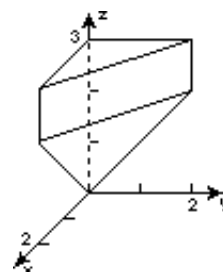
10. Since
- $\operatorname{div} \mathbf{F} = 0$
- ,
- $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_D 0 dV = 0$
- .

- 11.
- $\operatorname{div} \mathbf{F} = 2z + 10y - 2z = 10y$
- .

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D 10y dV = \int_0^2 \int_0^{2-x^2/2} \int_z^{4-z} 10y dy dz dx = \int_0^2 \int_0^{2-x^2/2} 5y^2 \Big|_z^{4-z} dz dx \\
 &= \int_0^2 \int_0^{2-x^2/2} (80 - 40z) dz dx = \int_0^2 (80z - 20z^2) \Big|_0^{2-x^2/2} dx = \int_0^2 (80 - 5x^4) dx = (80x - x^5) \Big|_0^2 = 128
 \end{aligned}$$

- 12.
- $\operatorname{div} \mathbf{F} = 30xy$
- .

$$\begin{aligned}
 \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D 30xy dV = \int_0^2 \int_0^{2-x} \int_{x+y}^3 30xy dz dy dx \\
 &= \int_0^2 \int_0^{2-x} 30xyz \Big|_{x+y}^3 dy dx \\
 &= \int_0^2 \int_0^{2-x} (90xy - 30x^2y - 30xy^2) dy dx \\
 &= \int_0^2 (45xy^2 - 15x^2y^2 - 10xy^3) \Big|_0^{2-x} dx \\
 &= \int_0^2 (-5x^4 + 45x^3 - 120x^2 + 100x) dx = \left(-x^5 + \frac{45}{4}x^4 - 40x^3 + 50x^2 \right) \Big|_0^2 = 28
 \end{aligned}$$



9.16 Divergence Theorem

13. $\operatorname{div} \mathbf{F} = 6xy^2 + 1 - 6xy^2 = 1$. Using cylindrical coordinates,

$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D dV = \int_0^\pi \int_0^{2\sin\theta} \int_{r^2}^{2r\sin\theta} dz r dr d\theta = \int_0^\pi \int_0^{2\sin\theta} (2r\sin\theta - r^2)r dr d\theta \\ &= \int_0^\pi \left(\frac{2}{3}r^3\sin\theta - \frac{1}{4}r^4 \right) \Big|_0^{2\sin\theta} d\theta = \int_0^\pi \left(\frac{16}{3}\sin^4\theta - 4\sin^4\theta \right) d\theta \\ &= \frac{4}{3} \int_0^\pi \sin^4\theta d\theta = \frac{4}{3} \left(\frac{3}{8}\theta - \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta \right) \Big|_0^\pi = \frac{\pi}{2}\end{aligned}$$

14. $\operatorname{div} \mathbf{F} = y^2 + x^2$. Using spherical coordinates, we have $x^2 + y^2 = \rho^2 \sin^2 \phi$ and $z = \rho \cos \phi$ or $\rho = z \sec \phi$. Then

$$\begin{aligned}\iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_D (x^2 + y^2) dS = \int_0^{2\pi} \int_0^{\pi/4} \int_{2\sec\phi}^{4\sec\phi} \rho^2 \sin^2 \phi \rho^2 \sin \phi d\rho d\phi d\theta \\ &= \int_0^{2\pi} \int_0^{\pi/4} \frac{1}{5} \rho^5 \sin^3 \phi \Big|_{2\sec\phi}^{4\sec\phi} d\phi d\theta = \int_0^{2\pi} \int_0^{\pi/4} \frac{992}{5} \sec^5 \phi \sin^3 \phi d\phi d\theta \\ &= \int_0^{2\pi} \int_0^{\pi/4} \frac{992}{5} \tan^3 \phi \sec^2 \phi d\phi d\theta = \frac{992}{5} \int_0^{2\pi} \frac{1}{4} \tan^4 \phi \Big|_0^{\pi/4} d\theta = \frac{992}{5} \int_0^{2\pi} \frac{1}{4} d\theta = \frac{496\pi}{5}.\end{aligned}$$

15. (a) $\operatorname{div} \mathbf{E} = q \left[\frac{-2x^2 + y^2 + z^2}{(x^2 + y^2 + z^2)^{5/2}} + \frac{x^2 - 2y^2 + z^2}{(x^2 + y^2 + z^2)^{5/2}} + \frac{x^2 + y^2 - 2z^2}{(x^2 + y^2 + z^2)^{5/2}} \right] = 0$

$$\iint_{S \cup S_a} (\mathbf{E} \cdot \mathbf{n}) dS = \iiint_D \operatorname{div} \mathbf{E} dV = \iiint_D 0 dV = 0$$

- (b) From (a), $\iint_S (\mathbf{E} \cdot \mathbf{n}) dS + \iint_{S_a} (\mathbf{E} \cdot \mathbf{n}) dS = 0$ and $\iint_S (\mathbf{E} \cdot \mathbf{n}) dS = -\iint_{S_a} (\mathbf{E} \cdot \mathbf{n}) dS$. On S_a ,

$|\mathbf{r}| = a$, $\mathbf{n} = -(\mathbf{x}\mathbf{i} + \mathbf{y}\mathbf{j} + \mathbf{z}\mathbf{k})/a = -\mathbf{r}/a$ and $\mathbf{E} \cdot \mathbf{n} = (q\mathbf{r}/a^3) \cdot (-\mathbf{r}/a) = -qa^2/a^4 = -q/a^2$. Thus

$$\iint_S (\mathbf{E} \cdot \mathbf{n}) dS = -\iint_{S_a} \left(-\frac{q}{a^2} \right) dS = \frac{q}{a^2} \iint_{S_a} dS = \frac{q}{a^2} \times (\text{area of } S_a) = \frac{q}{a^2} (4\pi a^2) = 4\pi q.$$

16. (a) By Gauss' Law $\iint_S (\mathbf{E} \cdot \mathbf{n}) dS = \iiint_D 4\pi\rho dV$, and by the Divergence Theorem

$$\iint_S (\mathbf{E} \cdot \mathbf{n}) dS = \iiint_D \operatorname{div} \mathbf{E} dV. \text{ Thus } \iiint_D 4\pi\rho dV = \iiint_D \operatorname{div} \mathbf{E} dV \text{ and } \iiint_D (4\pi\rho - \operatorname{div} \mathbf{E}) dV = 0.$$

Since this holds for all regions D , $4\pi\rho - \operatorname{div} \mathbf{E} = 0$ and $\operatorname{div} \mathbf{E} = 4\pi\rho$.

- (b) Since $\mathbf{E}$ is irrotational, $\mathbf{E} = \nabla\phi$ and $\nabla^2\phi = \nabla \cdot \nabla\phi = \nabla\mathbf{E} = \operatorname{div} \mathbf{E} = 4\pi\rho$.

17. Since $\operatorname{div} \mathbf{a} = 0$, by the Divergence Theorem

$$\iint_S (\mathbf{a} \cdot \mathbf{n}) dS = \iiint_D \operatorname{div} \mathbf{a} dV = \iiint_D 0 dV = 0.$$

18. By the Divergence Theorem and Problem 30 in Section 9.7,

$$\iint_S (\operatorname{curl} \mathbf{F} \cdot \mathbf{n}) dS = \iiint_D \operatorname{div} (\operatorname{curl} \mathbf{F}) dV = \iiint_D 0 dV = 0.$$

19. By the Divergence Theorem and Problem 27 in Section 9.7,

$$\begin{aligned}\iint_S (f\nabla g) \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} (f\nabla g) dV = \iiint_D \nabla \cdot (f\nabla g) dV = \iiint_D [f(\nabla \cdot \nabla g) + \nabla g \cdot \nabla f] dV \\ &= \iiint_D (f\nabla^2 g + \nabla g \cdot \nabla f) dV.\end{aligned}$$

20. By the Divergence Theorem and Problems 25 and 27 in Section 9.7,

$$\begin{aligned}\iint_S (f\nabla g - g\nabla f) \cdot \mathbf{n} dS &= \iiint_D \operatorname{div} (f\nabla g - g\nabla f) dV = \iiint_D \nabla \cdot (f\nabla g - g\nabla f) dV \\ &= \iiint_D [f(\nabla \cdot \nabla g) + \nabla g \cdot \nabla f - g(\nabla \cdot \nabla f) - \nabla f \cdot \nabla g] dV \\ &= \iiint_D (f\nabla^2 g - g\nabla^2 f) dV.\end{aligned}$$

21. If $G(x, y, z)$ is a vector valued function then we define surface integrals and triple integrals of $\mathbf{G}$ component-wise. In this case, if $\mathbf{a}$ is a constant vector it is easily shown that

$$\iint_S \mathbf{a} \cdot \mathbf{G} dS = \mathbf{a} \cdot \iint_S \mathbf{G} dS \quad \text{and} \quad \iiint_D \mathbf{a} \cdot \mathbf{G} dV = \mathbf{a} \cdot \iiint_D \mathbf{G} dV.$$

Now let $\mathbf{F} = f\mathbf{a}$. Then

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_S (f\mathbf{a}) \cdot \mathbf{n} dS = \iint_S \mathbf{a} \cdot (f\mathbf{n}) dS$$

and, using Problem 27 in Section 9.7 and the fact that $\nabla \cdot \mathbf{a} = 0$, we have

$$\iiint_D \operatorname{div} \mathbf{F} dV = \iiint_D \nabla \cdot (f\mathbf{a}) dV = \iiint_D [f(\nabla \cdot \mathbf{a}) + \mathbf{a} \cdot \nabla f] dV = \iiint_D \mathbf{a} \cdot \nabla f dV.$$

By the Divergence Theorem,

$$\iint_S \mathbf{a} \cdot (f\mathbf{n}) dS = \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_D \operatorname{div} \mathbf{F} dV = \iiint_D \mathbf{a} \cdot \nabla f dV$$

and

$$\mathbf{a} \cdot \left(\iint_S f\mathbf{n} dS \right) = \mathbf{a} \cdot \left(\iiint_D \nabla f dV \right) \quad \text{or} \quad \mathbf{a} \cdot \left(\iint_S f\mathbf{n} dS - \iiint_D \nabla f dV \right) = 0.$$

Since $\mathbf{a}$ is arbitrary,

$$\iint_S f\mathbf{n} dS - \iiint_D \nabla f dV = 0 \quad \text{and} \quad \iint_S f\mathbf{n} dS = \iiint_D \nabla f dV.$$

$$\begin{aligned}22. \quad \mathbf{B} + \mathbf{W} &= - \iint_S p\mathbf{n} dS + m\mathbf{g} = m\mathbf{g} - \iiint_D \nabla p dV = m\mathbf{g} - \iiint_D \rho\mathbf{g} dV = m\mathbf{g} - \left(\iiint_D \rho dV \right) \mathbf{g} \\ &= m\mathbf{g} - m\mathbf{g} = \mathbf{0}\end{aligned}$$

EXERCISES 9.17

Change of Variables in Multiple Integrals

1. $T: (0, 0) \rightarrow (0, 0); (0, 2) \rightarrow (-2, 8); (4, 0) \rightarrow (16, 20); (4, 2) \rightarrow (14, 28)$
2. Writing $x^2 = v - u$ and $y = v + u$ and solving for u and v , we obtain $u = (y - x^2)/2$ and $v = (x^2 + y)/2$. Then the images under T^{-1} are $(1, 1) \rightarrow (0, 1); (1, 3) \rightarrow (1, 2); (\sqrt{2}, 2) \rightarrow (0, 2)$.

9.17 Change of Variables in Multiple Integrals

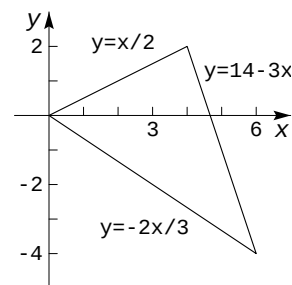
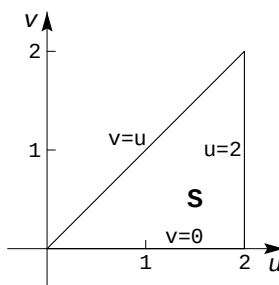
3. The uv -corner points $(0,0)$, $(2,0)$, $(2,2)$ correspond to xy -points $(0,0)$, $(4,2)$, $(6,-4)$.

$$v=0: x=2u, y=u \implies y=x/2$$

$$u=2: x=4+v, y=2-3v \implies$$

$$y=2-3(x-4)=-3x+14$$

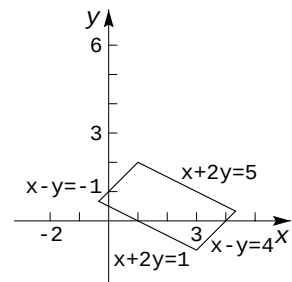
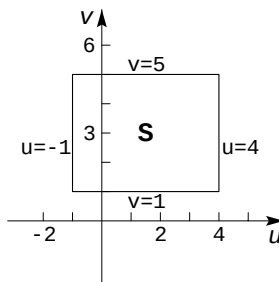
$$v=u: x=3u, y=-2u \implies y=-2x/3$$



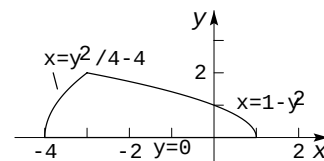
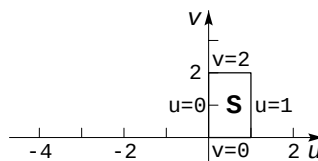
4. Solving for x and y we see that the transformation is $x=2u/3+v/3$, $y=-u/3+v/3$. The uv -corner points $(-1,1)$, $(4,1)$, $(4,5)$, $(-1,5)$ correspond to the xy -points $(-1/3, 2/3)$, $(3,-1)$, $(13/3, 1/3)$, $(1,2)$.

$$v=1: x+2y=1; \quad v=5: x+2y=5;$$

$$u=-1: x-y=-1; \quad u=4: x-y=4$$



5. The uv -corner points $(0,0)$, $(1,0)$, $(1,2)$, $(0,2)$ correspond to the xy -points $(0,0)$, $(1,0)$, $(-3,2)$, $(-4,0)$.



$$v=0: x=u^2, y=0 \implies y=0 \text{ and } 0 \leq x \leq 1$$

$$u=1: x=1-v^2, y=v \implies x=1-y^2$$

$$v=2: x=u^2-4, y=2u \implies x=y^2/4-4$$

$$u=0: x=-v^2, y=0 \implies y=0 \text{ and } -4 \leq x \leq 0$$

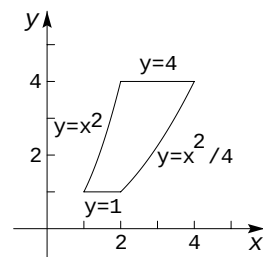
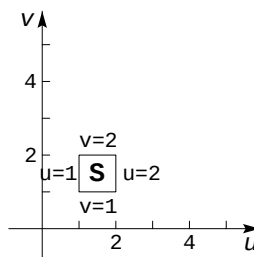
6. The uv -corner points $(1,1)$, $(2,1)$, $(2,2)$, $(1,2)$ correspond to the xy -points $(1,1)$, $(2,1)$, $(4,4)$, $(2,4)$.

$$v=1: x=u, y=1 \implies y=1, 1 \leq x \leq 2$$

$$u=2: x=2v, y=v^2 \implies y=x^2/4$$

$$v=2: x=2u, y=4 \implies y=4, 2 \leq x \leq 4$$

$$u=1: x=v, y=v^2 \implies y=x^2$$



$$7. \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} -ve^{-u} & e^{-u} \\ ve^u & e^u \end{vmatrix} = -2v$$

$$8. \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} 3e^{3u} \sin v & e^{3u} \cos v \\ 3e^{3u} \cos v & -e^{3u} \sin v \end{vmatrix} = -3e^{6u}$$

$$9. \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} -2y/x^3 & 1/x^2 \\ -y^2/x^2 & 2y/x \end{vmatrix} = -\frac{3y^2}{x^4} = -3\left(\frac{y}{x^2}\right)^2 = -3u^2; \quad \frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{-3u^2} = -\frac{1}{3u^2}$$

$$10. \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{2(y^2-x^2)}{(x^2+y^2)^2} & \frac{-4xy}{(x^2+y^2)^2} \\ \frac{4xy}{(x^2+y^2)^2} & \frac{2(y^2-x^2)}{(x^2+y^2)^2} \end{vmatrix} = \frac{4}{(x^2+y^2)^2}$$

9.17 Change of Variables in Multiple Integrals

From $u = 2x/(x^2 + y^2)$ and $v = -2y/(x^2 + y^2)$ we obtain $u^2 + v^2 = 4/(x^2 + y^2)$. Then $x^2 + y^2 = 4/(u^2 + v^2)$ and $\partial(x, y)/\partial(u, v) = (x^2 + y^2)^2/4 = 4/(u^2 + v^2)^2$.

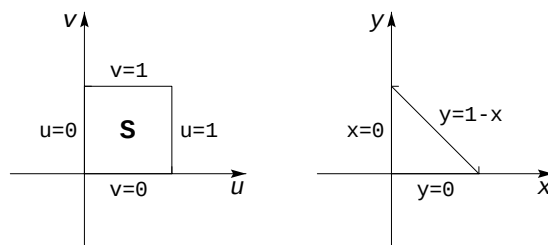
11. (a) The uv -corner points $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$ correspond to the xy -points $(0, 0)$, $(1, 0)$, $(0, 1)$, $(0, 0)$.

$$v = 0: x = u, y = 0 \implies y = 0, 0 \leq x \leq 1$$

$$u = 1: x = 1 - v, y = v \implies y = 1 - x$$

$$v = 1: x = 0, y = u \implies x = 0, 0 \leq y \leq 1$$

$$u = 0: x = 0, y = 0$$



- (b) Since the segment $u = 0, 0 \leq v \leq 1$ in the uv -plane maps to the origin in the xy -plane, the transformation is not one-to-one.

12. $\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 1-v & v \\ -u & u \end{vmatrix} = u$. The transformation is 0 when u is 0, for $0 \leq v \leq 1$.

13. R1: $x + y = -1 \implies v = -1$

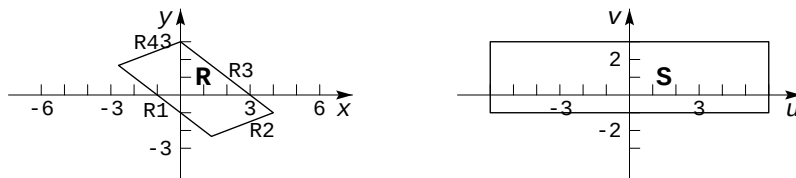
$$R2: x - 2y = 6 \implies u = 6$$

$$R3: x + y = 3 \implies v = 3$$

$$R4: x - 2y = -6 \implies u = -6$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & -2 \\ 1 & 1 \end{vmatrix} = 3 \implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{3}$$

$$\iint_R (x + y) dA = \iint_S v \left(\frac{1}{3}\right) dA' = \frac{1}{3} \int_{-1}^3 \int_{-6}^6 v du dv = \frac{1}{3} (12) \int_{-1}^3 v dv = 4 \left(\frac{1}{2}\right) v^2 \Big|_{-1}^3 = 16$$



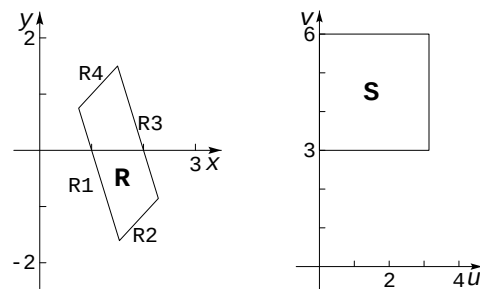
14. R1: $y = -3x + 3 \implies v = 3$

$$R2: y = x - \pi \implies u = \pi$$

$$R3: y = -3x + 6 \implies v = 6$$

$$R4: y = x \implies u = 0$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & -1 \\ 3 & 1 \end{vmatrix} = 4 \implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{4}$$



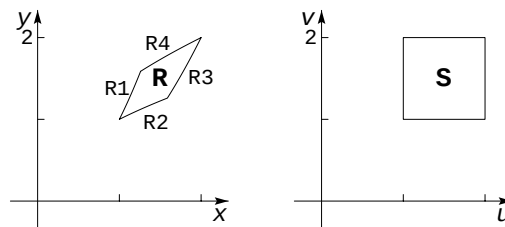
$$\begin{aligned} \iint_R \frac{\cos \frac{1}{2}(x - y)}{3x + y} dA &= \iint_S \frac{\cos u/2}{v} \left(\frac{1}{4}\right) dA' = \frac{1}{4} \int_3^6 \int_0^\pi \frac{\cos u/2}{v} du dv = \frac{1}{4} \int_3^6 \frac{2 \sin u/2}{v} \Big|_0^\pi dv \\ &= \frac{1}{2} \int_3^6 \frac{dv}{v} = \frac{1}{2} \ln v \Big|_3^6 = \frac{1}{2} \ln 2 \end{aligned}$$

15. R1: $y = x^2 \implies u = 1$

$$R2: x = y^2 \implies v = 1$$

$$R3: y = \frac{1}{2}x^2 \implies u = 2$$

$$R4: x = \frac{1}{2}y^2 \implies v = 2$$



$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 2x/y & -x^2/y^2 \\ -y^2/x^2 & 2y/x \end{vmatrix} = 3 \implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{3}$$

9.17 Change of Variables in Multiple Integrals

$$\iint_R \frac{y^2}{x} dA = \iint_S v \left(\frac{1}{3} \right) dA' = \frac{1}{3} \int_1^2 \int_1^2 v du dv = \frac{1}{3} \int_1^2 v dv = \frac{1}{6} v^2 \Big|_1^2 = \frac{1}{2}$$

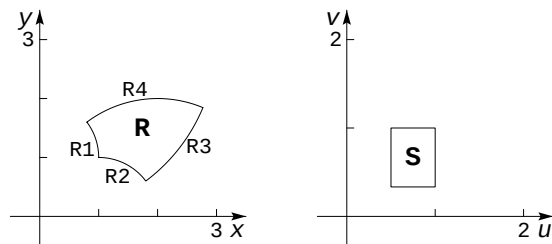
16. $R1: x^2 + y^2 = 2y \implies v = 1$

$R2: x^2 + y^2 = 2x \implies u = 1$

$R3: x^2 + y^2 = 6y \implies v = 1/3$

$R4: x^2 + y^2 = 4x \implies u = 1/2$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{2(y^2 - x^2)}{(x^2 + y^2)^2} & \frac{-4xy}{(x^2 + y^2)^2} \\ \frac{-4xy}{(x^2 + y^2)^2} & \frac{2(x^2 - y^2)}{(x^2 + y^2)^2} \end{vmatrix} = \frac{-4}{(x^2 + y^2)^2}$$



Using $u^2 + v^2 = 4/(x^2 + y^2)$ we see that $\partial(x, y)/\partial(u, v) = -4/(u^2 + v^2)^2$.

$$\iint_R (x^2 + y^2)^{-3} dA = \iint_S \left(\frac{4}{u^2 + v^2} \right)^{-3} \left| \frac{-4}{(u^2 + v^2)^2} \right| dA' = \frac{1}{16} \int_{1/3}^1 \int_{1/2}^1 (u^2 + v^2) du dv = \frac{115}{5184}$$

17. $R1: 2xy = c \implies v = c$

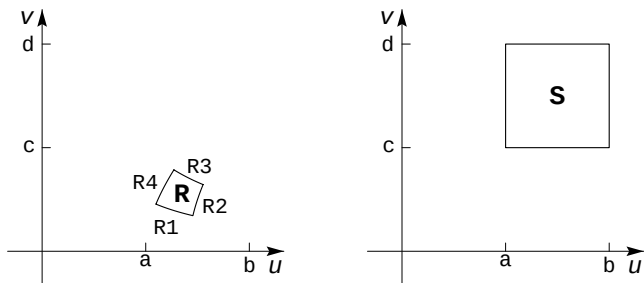
$R2: x^2 - y^2 = b \implies u = b$

$R3: 2xy = d \implies v = d$

$R4: x^2 - y^2 = a \implies u = a$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4(x^2 + y^2)$$

$$\implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{4(x^2 + y^2)}$$



$$\iint_R (x^2 + y^2) dA = \iint_S (x^2 + y^2) \frac{1}{4(x^2 + y^2)} dA' = \frac{1}{4} \int_c^d \int_a^b du dv = \frac{1}{4} (b - a)(d - c)$$

18. $R1: xy = -2 \implies v = -2$

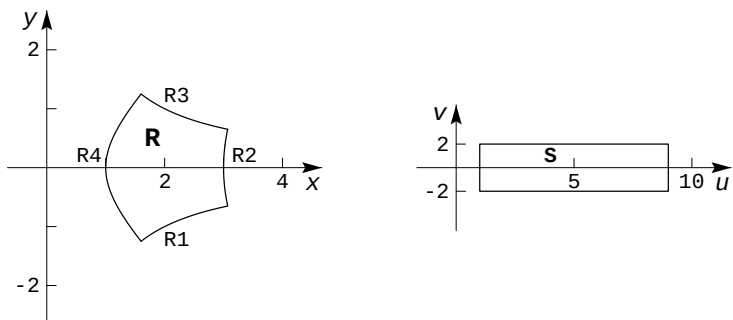
$R2: x^2 - y^2 = 9 \implies u = 9$

$R3: xy = 2 \implies v = 2$

$R4: x^2 - y^2 = 1 \implies u = 1$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 2x & -2y \\ y & x \end{vmatrix} = 2(x^2 + y^2)$$

$$\implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{2(x^2 + y^2)}$$



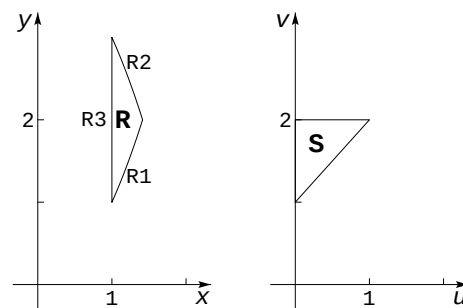
$$\iint_R (x^2 + y^2) \sin xy dA = \iint_S (x^2 + y^2) \sin v \left(\frac{1}{2(x^2 + y^2)} \right) dA' = \frac{1}{2} \int_{-2}^2 \int_1^9 \sin v du dv = \frac{1}{2} \int_{-2}^2 8 \sin v dv = 0$$

19. $R1: y = x^2 \implies v + u = v - u \implies u = 0$

$R2: y = 4 - x^2 \implies v + u = 4 - (v - u)$
 $\implies v + u = 4 - v + u \implies v = 2$

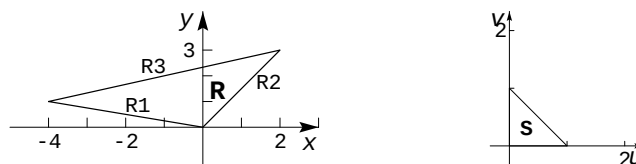
$R3: x = 1 \implies v - u = 1 \implies v = 1 + u$

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} -\frac{1}{2\sqrt{v-u}} & \frac{1}{2\sqrt{v-u}} \\ 1 & 1 \end{vmatrix} = -\frac{1}{\sqrt{v-u}}$$



$$\begin{aligned} \iint_R \frac{x}{y + x^2} dA &= \iint_S \frac{\sqrt{v-u}}{2v} \left| -\frac{1}{\sqrt{v-u}} \right| dA' = \frac{1}{2} \int_0^1 \int_{1+u}^2 \frac{1}{v} dv du = \frac{1}{2} \int_0^1 [\ln 2 - \ln(1+u)] du \\ &= \frac{1}{2} \ln 2 - \frac{1}{2} [(1+u) \ln(1+u) - (1+u)] \Big|_0^1 = \frac{1}{2} \ln 2 - \frac{1}{2} [2 \ln 2 - 2 - (0 - 1)] = \frac{1}{2} - \frac{1}{2} \ln 2 \end{aligned}$$

20. Solving $x = 2u - 4v$, $y = 3u + v$ for u and v we obtain $u = \frac{1}{14}x + \frac{2}{7}y$, $v = -\frac{3}{14}x + \frac{1}{7}y$. The xy -corner points $(-4, 1)$, $(0, 0)$, $(2, 3)$ correspond to the uv -points $(0, 1)$, $(0, 0)$, $(1, 0)$.



$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 2 & -4 \\ 3 & 1 \end{vmatrix} = 14$$

$$\begin{aligned} \iint_R y dA &= \iint_S (3u + v)(14) dA' = 14 \int_0^1 \int_0^{1-u} (3u + v) dv du = 14 \int_0^1 \left(3uv + \frac{1}{2}v^2 \right) \Big|_0^{1-u} du \\ &= 14 \int_0^1 \left(\frac{1}{2} + 2u - \frac{5}{2}u^2 \right) du = \left(7u + 14u^2 - \frac{35}{3}u^3 \right) \Big|_0^1 = \frac{28}{3} \end{aligned}$$

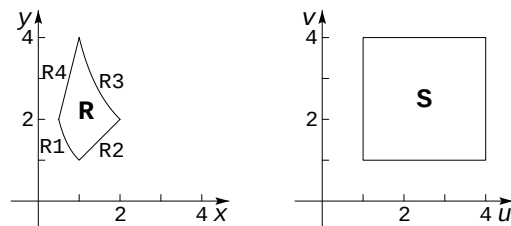
21. $R1: y = 1/x \implies u = 1$

$R2: y = x \implies v = 1$

$R3: y = 4/x \implies u = 4$

$R4: y = 4x \implies v = 4$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} y & x \\ -y/x^2 & 1/x \end{vmatrix} = \frac{2y}{x} \implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{x}{2y}$$



$$8 \iint_R y^4 dA = \iint_S u^2 v^2 \left(\frac{1}{2v} \right) du dv = \frac{1}{2} \int_1^4 u^2 v du dv = \frac{1}{2} \int_1^4 \frac{1}{3} u^3 v \Big|_1^4 dv = \frac{1}{6} \int_1^4 63v dv = \frac{21}{4} v^2 \Big|_1^4 = \frac{315}{4}$$

22. Under the transformation $u = y + z$, $v = -y + z$, $w = x - y$ the parallelepiped D is mapped to the parallelepiped E : $1 \leq u \leq 3$, $-1 \leq v \leq 1$, $0 \leq w \leq 3$.

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} 0 & 1 & 1 \\ 0 & -1 & 1 \\ 1 & -1 & 0 \end{vmatrix} = 2 \implies \frac{\partial(x, y, z)}{\partial(u, v, w)} = \frac{1}{2}$$

9.17 Change of Variables in Multiple Integrals

$$\begin{aligned}
 \iint_D (4z + 2x - 2y) dV &= \iint_E (2u + 2v + 2w) \frac{1}{2} dV' = \frac{1}{2} \int_0^3 \int_{-1}^1 \int_1^3 (2u + 2v + 2w) du dv dw \\
 &= \frac{1}{2} \int_0^3 \int_{-1}^1 (u^2 + 2uv + 2uw) \Big|_1^3 dv dw = \frac{1}{2} \int_0^3 \int_{-1}^1 (8 + 4v + 4w) dv dw \\
 &= \int_0^3 (4v + v^2 + 2vw) \Big|_{-1}^1 dw = \int_0^3 (8 + 4w) dw = (8w + 2w^2) \Big|_0^3 = 42
 \end{aligned}$$

23. We let $u = y - x$ and $v = y + x$.

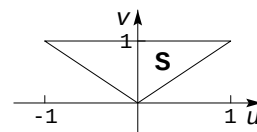
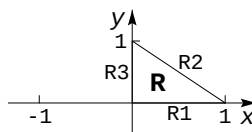
$$R1: y = 0 \implies u = -x, v = x \implies v = -u$$

$$R2: x + y = 1 \implies v = 1$$

$$R3: x = 0 \implies u = y, v = y \implies v = u$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -2 \implies \frac{\partial(x, y)}{\partial(u, v)} = -\frac{1}{2}$$

$$\begin{aligned}
 \iint_R e^{(y-x)/(y+x)} dA &= \iint_S e^{u/v} \left| -\frac{1}{2} \right| dA' = \frac{1}{2} \int_0^1 \int_{-v}^v e^{u/v} du dv = \frac{1}{2} \int_0^1 v e^{u/v} \Big|_{-v}^v dv \\
 &= \frac{1}{2} \int_0^1 v(e - e^{-1}) dv = \frac{1}{2}(e - e^{-1}) \frac{1}{2} v^2 \Big|_0^1 = \frac{1}{4}(e - e^{-1})
 \end{aligned}$$



24. We let $u = y - x$ and $v = y$.

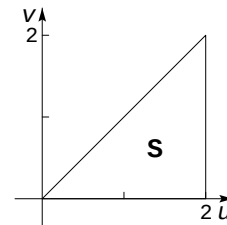
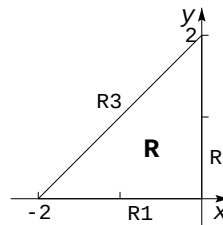
$$R1: y = 0 \implies v = 0, u = -x \implies v = 0, 0 \leq u \leq 2$$

$$R2: x = 0 \implies v = u$$

$$R3: y = x + 2 \implies u = 2$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} -1 & 1 \\ 0 & 1 \end{vmatrix} = -1 \implies \frac{\partial(x, y)}{\partial(u, v)} = -1$$

$$\iint_R e^{y^2 - 2xy + x^2} dA = \iint_S e^{u^2} | -1 | dA' = \int_0^2 \int_0^u e^{u^2} dv du = \int_0^2 u e^{u^2} du = \frac{1}{2} e^{u^2} \Big|_0^2 = \frac{1}{2}(e^4 - 1)$$



25. Noting that $R2$, $R3$, and $R4$ have equations $y + 2x = 8$, $y - 2x = 0$, and $y + 2x = 2$, we let $u = y/x$ and $v = y + 2x$.

$$R1: y = 0 \implies u = 0, v = 2x \implies u = 0, 2 \leq v \leq 8$$

$$R2: y + 2x = 8 \implies v = 8$$

$$R3: y - 2x = 0 \implies u = 2$$

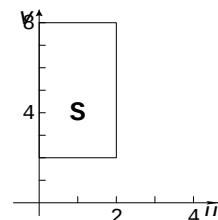
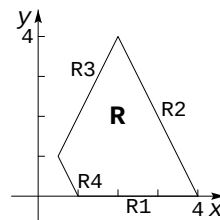
$$R4: y + 2x = 2 \implies v = 2$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} -y/x^2 & 1/x \\ 2 & 1 \end{vmatrix} = -\frac{y + 2x}{x^2} \implies \frac{\partial(x, y)}{\partial(u, v)} = -\frac{x^2}{y + 2x}$$

$$\iint_R (6x + 3y) dA = 3 \iint_S (y + 2x) \left| -\frac{x^2}{y + 2x} \right| dA' = 3 \iint_S x^2 dA'$$

From $y = ux$ we see that $v = ux + 2x$ and $x = v/(u + 2)$. Then

$$3 \iint_S x^2 dA' = 3 \int_0^2 \int_2^8 v^2 (u + 2)^2 dv du = \int_0^2 \frac{v^3}{(u + 2)^2} \Big|_2^8 du = 504 \int_0^2 \frac{du}{(u + 2)^2} = -\frac{504}{u + 2} \Big|_0^2 = 126.$$



26. We let $u = x + y$ and $v = x - y$.

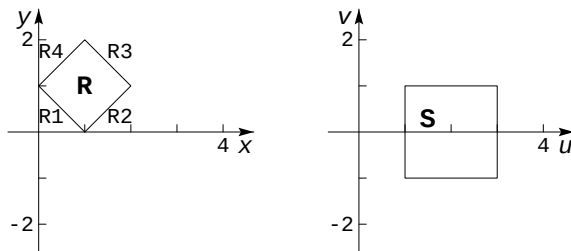
$$R1: x + y = 1 \implies u = 1$$

$$R2: x - y = 1 \implies v = 1$$

$$R3: x + y = 3 \implies u = 3$$

$$R4: x - y = -1 \implies v = -1$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2 \implies \frac{\partial(x, y)}{\partial(u, v)} = -\frac{1}{2}$$



$$\begin{aligned} \iint_R (x+y)^4 e^{x-y} dA &= \iint_S u^4 e^v \left| -\frac{1}{2} \right| dA' = \frac{1}{2} \int_1^3 \int_{-1}^1 u^4 e^v dv du = \frac{1}{2} \int_1^3 u^4 e^v \Big|_{-1}^1 du \\ &= \frac{e - e^{-1}}{2} \int_1^3 u^4 du = \frac{e - e^{-1}}{10} u^5 \Big|_1^3 = \frac{242(e - e^{-1})}{10} = \frac{121}{5}(e - e^{-1}) \end{aligned}$$

27. Let $u = xy$ and $v = xy^{1.4}$. Then $xy^{1.4} = c \implies v = c$; $xy = b \implies u = b$; $xy^{1.4} = d \implies v = d$; $xy = a \implies u = a$.

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} y & x \\ y^{1.4} & 1.4xy^{0.4} \end{vmatrix} = 0.4xy^{1.4} = 0.4v \implies \frac{\partial(x, y)}{\partial(u, v)} = \frac{5}{2v}$$

$$\iint_R dA = \iint_S \frac{5}{2v} dA' = \int_c^d \int_a^b \frac{5}{2v} du dv = \frac{5}{2}(b-a) \int_c^d \frac{dv}{v} = \frac{5}{2}(b-a)(\ln d - \ln c)$$

28. The image of the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ under the transformation $u = x/a$, $v = y/b$, $w = z/c$, is the unit sphere $u^2 + v^2 + w^2 = 1$. The volume of this sphere is $\frac{4}{3}\pi$. Now

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = abc$$

and

$$\iiint_D dV = \iiint_E abc dV' = abc \iiint_E dV' = abc \left(\frac{4}{3}\pi \right) = \frac{4}{3}\pi abc.$$

29. The image of the ellipse is the unit circle $x^2 + y^2 = 1$. From $\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 5 & 0 \\ 0 & 3 \end{vmatrix} = 15$ we obtain

$$\begin{aligned} \iint_R \left(\frac{x^2}{25} + \frac{y^2}{9} \right) dA &= \iint_S (u^2 + v^2) 15 dA' = 15 \int_0^{2\pi} \int_0^1 r^2 r dr d\theta = \frac{15}{4} \int_0^{2\pi} r^4 \Big|_0^1 d\theta \\ &= \frac{15}{4} \int_0^{2\pi} d\theta = \frac{15\pi}{2}. \end{aligned}$$

$$30. \frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \begin{vmatrix} \sin \phi \cos \theta & \rho \cos \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \cos \phi \sin \theta & \rho \sin \phi \cos \theta \\ \cos \phi & -\rho \sin \phi & 0 \end{vmatrix}$$

$$\begin{aligned} &= \cos \phi (\rho^2 \sin \phi \cos \phi \cos^2 \theta + \rho^2 \sin \phi \cos \phi \sin^2 \theta) + \rho \sin \phi (\rho \sin^2 \phi \cos^2 \theta + \rho \sin^2 \phi \sin^2 \theta) \\ &= \rho^2 \sin \phi \cos^2 \phi (\cos^2 \theta + \sin^2 \theta) + \rho^2 \sin^3 \phi (\cos^2 \theta + \sin^2 \theta) = \rho^2 \sin \phi (\cos^2 \phi + \sin^2 \phi) = \rho^2 \sin \phi \end{aligned}$$

CHAPTER 9 REVIEW EXERCISES

1. True; $|\mathbf{v}(t)| = \sqrt{2}$
2. True; for all t , $y = 4$.
3. True
4. False; consider $\mathbf{r}(t) = t^2\mathbf{i}$. In this case, $\mathbf{v}(t) = 2t\mathbf{i}$ and $\mathbf{a}(t) = 2\mathbf{i}$. Since $\mathbf{v} \cdot \mathbf{a} = 4t$, the velocity and acceleration vectors are not orthogonal for $t \neq 0$.
5. False; ∇f is perpendicular to the level curve $f(x, y) = c$.
6. False; consider $f(x, y) = xy$ at $(0, 0)$.
7. True; the value is $4/3$.
8. True; since $2xy \, dx - x^2 \, dy$ is not exact.
9. False; $\int_C x \, dx + x^2 \, dy = 0$ from $(-1, 0)$ to $(1, 0)$ along the x -axis and along the semicircle $y = \sqrt{1 - x^2}$, but since $x \, dx + x^2 \, dy$ is not exact, the integral is not independent of path.
10. True
11. False; unless the first partial derivatives are continuous.
12. True
13. True
14. True; since $\text{curl } \mathbf{F} = \mathbf{0}$ when $\mathbf{F}$ is a conservative vector field.
15. True
16. True
17. True
18. True
19. $\mathbf{F} = \nabla\phi = -x(x^2 + y^2)^{-3/2}\mathbf{i} - y(x^2 + y^2)^{-3/2}\mathbf{j}$
20. $\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f(x) & g(y) & h(z) \end{vmatrix} = \mathbf{0}$
21. $\mathbf{v}(t) = 6\mathbf{i} + \mathbf{j} + 2t\mathbf{k}$; $\mathbf{a}(t) = 2\mathbf{k}$. To find when the particle passes through the plane, we solve $-6t + t + t^2 = -4$ or $t^2 - 5t + 4 = 0$. This gives $t = 1$ and $t = 4$. $\mathbf{v}(1) = 6\mathbf{i} + \mathbf{j} + 2\mathbf{k}$, $\mathbf{a}(1) = 2\mathbf{k}$; $\mathbf{v}(4) = 6\mathbf{i} + \mathbf{j} + 8\mathbf{k}$, $\mathbf{a}(4) = 2\mathbf{k}$
22. We are given $\mathbf{r}(0) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$.

$$\mathbf{r}(t) = \int \mathbf{v}(t) \, dt = \int (-10t\mathbf{i} + (3t^2 - 4t)\mathbf{j} + \mathbf{k}) \, dt = -5t^2\mathbf{i} + (t^3 - 2t^2)\mathbf{j} + t\mathbf{k} + \mathbf{c}$$

$$\mathbf{i} + 2\mathbf{j} + 3\mathbf{k} = \mathbf{r}(0) = \mathbf{c}$$

$$\mathbf{r}(t) = (1 - 5t^2)\mathbf{i} + (t^3 - 2t^2 + 2)\mathbf{j} + (t + 3)\mathbf{k}$$

$$\mathbf{r}(2) = -19\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$$

23. $\mathbf{v}(t) = \int \mathbf{a}(t) \, dt = \int (\sqrt{2} \sin t \mathbf{i} + \sqrt{2} \cos t \mathbf{j}) \, dt = -\sqrt{2} \cos t \mathbf{i} + \sqrt{2} \sin t \mathbf{j} + \mathbf{c}$;
 $-\mathbf{i} + \mathbf{j} + \mathbf{k} = \mathbf{v}(\pi/4) = -\mathbf{i} + \mathbf{j} + \mathbf{c}$, $\mathbf{c} = \mathbf{k}$; $\mathbf{v}(t) = -\sqrt{2} \cos t \mathbf{i} + \sqrt{2} \sin t \mathbf{j} + \mathbf{k}$;

$$\mathbf{r}(t) = -\sqrt{2} \sin t \mathbf{i} - \sqrt{2} \cos t \mathbf{j} + t \mathbf{k} + \mathbf{b}; \quad \mathbf{i} + 2\mathbf{j} + (\pi/4)\mathbf{k} = \mathbf{r}(\pi/4) = -\mathbf{i} - \mathbf{j} + (\pi/4)\mathbf{k} + \mathbf{b}, \quad \mathbf{b} = 2\mathbf{i} + 3\mathbf{j};$$

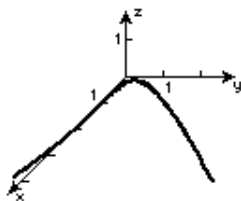
$$\mathbf{r}(t) = (2 - \sqrt{2} \sin t)\mathbf{i} + (3 - \sqrt{2} \cos t)\mathbf{j} + t\mathbf{k}; \quad \mathbf{r}(3\pi/4) = \mathbf{i} + 4\mathbf{j} + (3\pi/4)\mathbf{k}$$

$$24. \quad \mathbf{v}(t) = t\mathbf{i} + t^2\mathbf{j} - t\mathbf{k}; \quad |\mathbf{v}| = t\sqrt{t^2 + 2}, \quad t > 0; \quad \mathbf{a}(t) = \mathbf{i} + 2t\mathbf{j} - \mathbf{k}; \quad \mathbf{v} \cdot \mathbf{a} = t + 2t^3 + t = 2t + 2t^3;$$

$$\mathbf{v} \times \mathbf{a} = t^2\mathbf{i} + t^2\mathbf{k}, \quad |\mathbf{v} \times \mathbf{a}| = t^2\sqrt{2}; \quad a_T = \frac{2t + 2t^3}{t\sqrt{t^2 + 2}} = \frac{2 + 2t^2}{\sqrt{t^2 + 2}}, \quad a_N = \frac{t^2\sqrt{2}}{t\sqrt{t^2 + 2}} = \frac{\sqrt{2}t}{\sqrt{t^2 + 2}};$$

$$\kappa = \frac{t^2\sqrt{2}}{t^3(t^2 + 2)^{3/2}} = \frac{\sqrt{2}}{t(t^2 + 2)^{3/2}}$$

25.



$$26. \quad \mathbf{r}'(t) = \sinh t \mathbf{i} + \cosh t \mathbf{j} + \mathbf{k}, \quad \mathbf{r}'(1) = \sinh 1 \mathbf{i} + \cosh 1 \mathbf{j} + \mathbf{k};$$

$$|\mathbf{r}'(t)| = \sqrt{\sinh^2 t + \cosh^2 t + 1} = \sqrt{2 \cosh^2 t} = \sqrt{2} \cosh t; \quad |\mathbf{r}'(1)| = \sqrt{2} \cosh 1;$$

$$\mathbf{T}(t) = \frac{1}{\sqrt{2}} \tanh t \mathbf{i} + \frac{1}{\sqrt{2}} \mathbf{j} + \frac{1}{\sqrt{2}} \operatorname{sech} t \mathbf{k}, \quad \mathbf{T}(1) = \frac{1}{\sqrt{2}} (\tanh 1 \mathbf{i} + \mathbf{j} + \operatorname{sech} 1 \mathbf{k});$$

$$\frac{d\mathbf{T}}{dt} = \frac{1}{\sqrt{2}} \operatorname{sech}^2 t \mathbf{i} - \frac{1}{\sqrt{2}} \operatorname{sech} t \tanh t \mathbf{k}; \quad \frac{d\mathbf{T}}{dt}(\mathbf{T}(1)) = \frac{1}{\sqrt{2}} \operatorname{sech}^2 1 \mathbf{i} - \frac{1}{\sqrt{2}} \operatorname{sech} 1 \tanh 1 \mathbf{k},$$

$$\left| \frac{d\mathbf{T}}{dt}(\mathbf{T}(1)) \right| = \frac{\operatorname{sech} 1}{\sqrt{2}} \sqrt{\operatorname{sech}^2 1 + \tanh^2 1} = \frac{1}{\sqrt{2}} \operatorname{sech} 1; \quad \mathbf{N}(1) = \operatorname{sech} 1 \mathbf{i} - \tanh 1 \mathbf{k};$$

$$\mathbf{B}(1) = \mathbf{T}(1) \times \mathbf{N}(1) = -\frac{1}{\sqrt{2}} \tanh 1 \mathbf{i} + \frac{1}{\sqrt{2}} (\tanh^2 1 + \operatorname{sech}^2 1) \mathbf{j} - \frac{1}{\sqrt{2}} \operatorname{sech} 1 \mathbf{k}$$

$$= \frac{1}{\sqrt{2}} (-\tanh 1 \mathbf{i} + \mathbf{j} - \operatorname{sech} 1 \mathbf{k})$$

$$\kappa = \left| \frac{d\mathbf{T}}{dt}(\mathbf{T}(1)) \right| / |\mathbf{r}'(1)| = \frac{(\operatorname{sech} 1)/\sqrt{2}}{\sqrt{2} \cosh 1} = \frac{1}{2} \operatorname{sech}^2 1$$

$$27. \quad \nabla f = (2xy - y^2)\mathbf{i} + (x^2 - 2xy)\mathbf{j}; \quad \mathbf{u} = \frac{2}{\sqrt{40}} \mathbf{i} + \frac{6}{\sqrt{40}} \mathbf{j} = \frac{1}{\sqrt{10}} (\mathbf{i} + 3\mathbf{j});$$

$$D_{\mathbf{u}}f = \frac{1}{\sqrt{10}} (2xy - y^2 + 3x^2 - 6xy) = \frac{1}{\sqrt{10}} (3x^2 - 4xy - y^2)$$

$$28. \quad \nabla F = \frac{2x}{x^2 + y^2 + z^2} \mathbf{i} + \frac{2y}{x^2 + y^2 + z^2} \mathbf{j} + \frac{2z}{x^2 + y^2 + z^2} \mathbf{k}; \quad \mathbf{u} = -\frac{2}{3} \mathbf{i} + \frac{1}{3} \mathbf{j} + \frac{2}{3} \mathbf{k}; \quad D_{\mathbf{u}}F = \frac{-4x + 2y + 4z}{3(x^2 + y^2 + z^2)}$$

$$29. \quad f_x = 2xy^4, \quad f_y = 4x^2y^3.$$

$$(a) \quad \mathbf{u} = \mathbf{i}, \quad D_{\mathbf{u}}(1, 1) = f_x(1, 1) = 2$$

$$(b) \quad \mathbf{u} = (\mathbf{i} - \mathbf{j})/\sqrt{2}, \quad D_{\mathbf{u}}(1, 1) = (2 - 4)/\sqrt{2} = -2/\sqrt{2}$$

$$(c) \quad \mathbf{u} = \mathbf{j}, \quad D_{\mathbf{u}}(1, 1) = f_y(1, 1) = 4$$

$$30. \quad (a) \quad \frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}$$

$$= \frac{x}{\sqrt{x^2 + y^2 + z^2}} 6 \cos 2t + \frac{y}{\sqrt{x^2 + y^2 + z^2}} (-8 \sin 2t) + \frac{z}{\sqrt{x^2 + y^2 + z^2}} 15t^2$$

$$= \frac{(6x \cos 2t - 8y \sin 2t + 15zt^2)}{\sqrt{x^2 + y^2 + z^2}}$$

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$$\begin{aligned}
 \text{(b)} \quad \frac{\partial w}{\partial t} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial t} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial t} \\
 &= \frac{x}{\sqrt{x^2 + y^2 + z^2}} \frac{6}{r} \cos \frac{2t}{r} + \frac{y}{\sqrt{x^2 + y^2 + z^2}} \left(\frac{8r}{t^2} \sin \frac{2r}{t} \right) + \frac{z}{\sqrt{x^2 + y^2 + z^2}} 15t^2 r^3 \\
 &= \frac{\left(\frac{6x}{r} \cos \frac{2t}{r} + \frac{8yr}{t^2} \sin \frac{2r}{t} + 15zt^2 r^3 \right)}{\sqrt{x^2 + y^2 + z^2}}
 \end{aligned}$$

31. $F(x, y, z) = \sin xy - z$; $\nabla F = y \cos xy \mathbf{i} + x \cos xy \mathbf{j} - \mathbf{k}$; $\nabla F(1/2, 2\pi/3, \sqrt{3}/2) = \frac{\pi}{3} \mathbf{i} + \frac{1}{4} \mathbf{j} - \mathbf{k}$. The equation of the tangent plane is

$$\frac{\pi}{3} \left(x - \frac{1}{2} \right) + \frac{1}{4} \left(y - \frac{2\pi}{3} \right) - \left(z - \frac{\sqrt{3}}{2} \right) = 0$$

$$\text{or } 4\pi x + 3y - 12z = 4\pi - 6\sqrt{3}.$$

32. We want to find a normal to the surface that is parallel to $\mathbf{k}$. $\nabla F = (y-2)\mathbf{i} + (x-2y)\mathbf{j} + 2z\mathbf{k}$. We need $y-2=0$ and $x-2y=0$. The tangent plane is parallel to $z=2$ when $y=2$ and $x=4$. In this case $z^2=5$. The points are $(4, 2, \sqrt{5})$ and $(4, 2, -\sqrt{5})$.

$$33. \text{ (a) } V = \int_0^1 \int_x^{2x} \sqrt{1-x^2} dy dx = \int_0^1 y \sqrt{1-x^2} \Big|_x^{2x} dx = \int_0^1 x \sqrt{1-x^2} dx = -\frac{1}{3} (1-x^2)^{3/2} \Big|_0^1 = \frac{1}{3}$$

$$\text{(b) } V = \int_0^1 \int_{y/2}^y \sqrt{1-x^2} dx dy + \int_1^2 \int_{y/2}^1 \sqrt{1-x^2} dx dy$$

34. We are given $\rho = k(x^2 + y^2)$.

$$\begin{aligned}
 m &= \int_0^1 \int_{x^3}^{x^2} k(x^2 + y^2) dy dx = k \int_0^1 \left(x^2 y + \frac{1}{3} y^3 \right) \Big|_{x^3}^{x^2} dx \\
 &= k \int_0^1 \left(x^4 + \frac{1}{3} x^6 - x^5 - \frac{1}{3} x^9 \right) dx = k \left(\frac{1}{5} x^5 + \frac{1}{21} x^7 - \frac{1}{6} x^6 - \frac{1}{30} x^{10} \right) \Big|_0^1 = \frac{k}{21}
 \end{aligned}$$

$$\begin{aligned}
 M_y &= \int_0^1 \int_{x^3}^{x^2} k(x^3 + xy^2) dy dx = k \int_0^1 \left(x^3 y + \frac{1}{3} xy^3 \right) \Big|_{x^3}^{x^2} dx = k \int_0^1 \left(x^5 + \frac{1}{3} x^7 - x^6 - \frac{1}{3} x^{10} \right) dx \\
 &= k \left(\frac{1}{6} x^6 + \frac{1}{24} x^8 - \frac{1}{7} x^7 - \frac{1}{33} x^{11} \right) \Big|_0^1 = \frac{65k}{1848}
 \end{aligned}$$

$$\begin{aligned}
 M_x &= \int_0^1 \int_{x^3}^{x^2} k(x^2 y + y^3) dy dx = k \int_0^1 \left(\frac{1}{2} x^2 y^2 + \frac{1}{4} y^4 \right) \Big|_{x^3}^{x^2} dx = k \int_0^1 \left(\frac{1}{2} x^6 + \frac{1}{4} x^8 - \frac{1}{2} x^8 - \frac{1}{4} x^{12} \right) dx \\
 &= k \left(\frac{1}{14} x^7 - \frac{1}{36} x^9 - \frac{1}{52} x^{13} \right) \Big|_0^1 = \frac{20k}{819}
 \end{aligned}$$

$$\bar{x} = M_y/m = \frac{65k/1848}{k/21} = 65/88; \quad \bar{y} = M_x/m = \frac{20k/819}{k/21} = 20/39 \quad \text{The center of mass is } (65/88, 20/39).$$

$$\begin{aligned}
 35. \quad I_y &= \int_0^1 \int_{x^3}^{x^2} k(x^4 + x^2 y^2) dy dx = k \int_0^1 \left(x^4 y + \frac{1}{3} x^2 y^3 \right) \Big|_{x^3}^{x^2} dx = k \int_0^1 \left(x^6 + \frac{1}{3} x^8 - x^7 - \frac{1}{3} x^{11} \right) dx \\
 &= k \left(\frac{1}{7} x^7 + \frac{1}{27} x^9 - \frac{1}{8} x^8 - \frac{1}{36} x^{12} \right) \Big|_0^1 = \frac{41}{1512} k
 \end{aligned}$$

36. (a) Using symmetry,

$$\begin{aligned}
 V &= 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} dz \, dy \, dx = 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{a^2-x^2-y^2} \, dy \, dx \\
 &\quad \boxed{\text{Trig substitution}} \\
 &= 8 \int_0^a \left(\frac{y}{2} \sqrt{a^2-x^2-y^2} + \frac{a^2-x^2}{2} \sin^{-1} \frac{y}{\sqrt{a^2-x^2}} \right) \bigg|_0^{\sqrt{a^2-x^2}} dx = 8 \int_0^a \frac{\pi}{2} \frac{a^2-x^2}{2} dx \\
 &= 2\pi \left(a^2x - \frac{1}{3}x^3 \right) \bigg|_0^a = \frac{4}{3}\pi a^3
 \end{aligned}$$

(b) Using symmetry,

$$\begin{aligned}
 V &= 2 \int_0^{2\pi} \int_0^a \int_0^{\sqrt{a^2-r^2}} r \, dz \, dr \, d\theta = 2 \int_0^{2\pi} \int_0^a r \sqrt{a^2-r^2} \, dr \, d\theta \\
 &= 2 \int_0^{2\pi} \left(-\frac{1}{3}(a^2-r^2)^{3/2} \right) \bigg|_0^a d\theta = \frac{2}{3} \int_0^{2\pi} a^3 \, d\theta = \frac{4}{3}\pi a^3
 \end{aligned}$$

(c)
$$\begin{aligned}
 V &= \int_0^{2\pi} \int_0^\pi \int_0^a \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^\pi \frac{1}{3} \rho^3 \sin \phi \bigg|_0^a d\phi \, d\theta \\
 &= \frac{1}{3} \int_0^{2\pi} \int_0^\pi a^3 \sin \phi \, d\phi \, d\theta = \frac{1}{3} \int_0^{2\pi} -a^3 \cos \phi \bigg|_0^\pi d\theta = \frac{1}{3} \int_0^{2\pi} 2a^3 \, d\theta = \frac{4}{3}\pi a^3
 \end{aligned}$$

37. We use spherical coordinates.

$$\begin{aligned}
 V &= \int_0^{2\pi} \int_{\tan^{-1} 1/3}^{\pi/4} \int_0^{3 \sec \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_{\tan^{-1} 1/3}^{\pi/4} \frac{1}{3} \rho^3 \sin \phi \bigg|_0^{3 \sec \phi} d\phi \, d\theta \\
 &= \frac{1}{3} \int_0^{2\pi} \int_{\tan^{-1} 1/3}^{\pi/4} 27 \sec^3 \phi \sin \phi \, d\phi \, d\theta = 9 \int_0^{2\pi} \int_{\tan^{-1} 1/3}^{\pi/4} \tan \phi \sec^2 \phi \, d\phi \, d\theta \\
 &= 9 \int_0^{2\pi} \frac{1}{2} \tan^2 \phi \bigg|_{\tan^{-1} 1/3}^{\pi/4} d\theta = \frac{9}{2} \int_0^{2\pi} \left(1 - \frac{1}{9} \right) d\theta = 8\pi
 \end{aligned}$$

38.
$$\begin{aligned}
 V &= \int_0^{2\pi} \int_0^{\pi/6} \int_1^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/6} \frac{1}{3} \rho^3 \sin \phi \bigg|_1^2 d\phi \, d\theta \\
 &= \int_0^{2\pi} \int_0^{\pi/6} \left(\frac{8}{3} \sin \phi - \frac{1}{3} \sin \phi \right) d\phi \, d\theta = \frac{7}{3} \int_0^{2\pi} \int_0^{\pi/6} \sin \phi \, d\phi \, d\theta = \frac{7}{3} \int_0^{2\pi} -\cos \phi \bigg|_0^{\pi/6} d\theta \\
 &= \frac{7}{3} \int_0^{2\pi} \left[-\frac{\sqrt{3}}{2} - (-1) \right] d\theta = \frac{7}{3} \left(1 - \frac{\sqrt{3}}{2} \right) 2\pi = \frac{7\pi}{3} (2 - \sqrt{3})
 \end{aligned}$$

39. $2xy + 2xy + 2xy = 6xy$

40.
$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2y & xy^2 & 2xyz \end{vmatrix} = 2xz\mathbf{i} - 2yz\mathbf{j} + (y^2 - x^2)\mathbf{k}$$

41.
$$\frac{\partial}{\partial x}(2xz) - \frac{\partial}{\partial y}(2yz) + \frac{\partial}{\partial z}(y^2 - x^2) = 0$$

42. $\nabla(6xy) = 6y\mathbf{i} + 6x\mathbf{j}$

43.
$$\int_C \frac{z^2}{x^2 + y^2} ds = \int_\pi^{2\pi} \frac{4t^2}{\cos^2 2t + \sin^2 2t} \sqrt{4\sin^2 2t + 4\cos^2 2t + 4} \, dt = \int_\pi^{2\pi} 8\sqrt{2} t^2 \, dt = \frac{8\sqrt{2}}{3} t^3 \bigg|_\pi^{2\pi} = \frac{56\sqrt{2}\pi^3}{3}$$

44.
$$\int_C (xy + 4x) ds = \int_1^0 [x(2-2x) + 4x] \sqrt{1+4} \, dx = \sqrt{5} \int_1^0 (6x - 2x^2) \, dx = \sqrt{5} \left(3x^2 - \frac{2}{3}x^3 \right) \bigg|_1^0 = -\frac{7\sqrt{5}}{3}$$

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45. Since $P_y = 6x^2y = Q_x$, the integral is independent of path.

$$\phi_x = 3x^2y^2, \quad \phi = x^3y^2 + g(y), \quad \phi_y = 2x^3y + g'(y) = 2x^3y - 3y^2; \quad g(y) = -y^3; \quad \phi = x^3y^2 - y^3;$$

$$\int_{(0,0)}^{(1,-2)} 3x^2y^2 dx + (2x^3y - 3y^2) dy = (x^3y^2 - y^3) \Big|_{(0,0)}^{(1,-2)} = 12$$

46. Let $x = a \cos t$, $y = a \sin t$, $0 \leq t \leq 2\pi$. Then using $dx = -a \sin t dt$, $dy = a \cos t dt$, $x^2 + y^2 = a^2$ we have

$$\oint_C \frac{-y dx + x dy}{x^2 + y^2} = \int_0^{2\pi} \frac{1}{a^2} [-a \sin t(-a \sin t) + a \cos t(a \cos t)] dt = \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt = \int_0^{2\pi} dt = 2\pi.$$

47. $\int_C y \sin \pi z dx + x^2 e^y dy + 3xyz dz$

$$\begin{aligned} &= \int_0^1 [t^2 \sin \pi t^3 + t^2 e^{t^2} (2t) + 3tt^2 t^3 (3t^2)] dt = \int_0^1 (t^2 \sin \pi t^3 + 2t^3 e^{t^2} + 9t^8) dt \\ &= \left(-\frac{1}{3\pi} \cos \pi t^3 + t^9 \right) \Big|_0^1 + 2 \int_0^1 t^3 e^{t^2} dt \quad \boxed{\text{Integration by parts}} \\ &= \frac{2}{3\pi} + 1 + (t^2 e^{t^2} - e^{t^2}) \Big|_0^1 = \frac{2}{3\pi} + 2 \end{aligned}$$

48. Parameterize C by $x = \cos t$, $y = \sin t$; $0 \leq t \leq 2\pi$. Then

$$\begin{aligned} \oint_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{2\pi} [4 \sin t(-\sin t dt) + 6 \cos t(\cos t) dt] = \int_0^{2\pi} (6 \cos^2 t - 4 \sin^2 t) dt \\ &= \int_0^{2\pi} (10 \cos^2 t - 4) dt = \left(5t + \frac{5}{2} \sin 2t - 4t \right) \Big|_0^{2\pi} = 2\pi. \end{aligned}$$

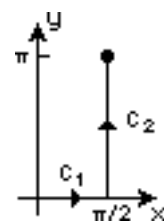
Using Green's Theorem, $Q_x - P_y = 6 - 4 = 2$ and $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R 2 dA = 2(\pi \cdot 1^2) = 2\pi$.

49. Let $\mathbf{r}_1 = \frac{\pi}{2} t \mathbf{i}$ and $\mathbf{r}_2 = \frac{\pi}{2} \mathbf{i} + \pi t \mathbf{j}$ for $0 \leq t \leq 1$. Then $d\mathbf{r}_1 = \frac{\pi}{2} \mathbf{i}$, $d\mathbf{r}_2 = \pi \mathbf{j}$, $\mathbf{F}_1 = \mathbf{0}$,

$$\mathbf{F}_2 = \frac{\pi}{2} \sin \pi t \mathbf{i} + \pi t \sin \frac{\pi}{2} \mathbf{j} = \frac{\pi}{2} \sin \pi t \mathbf{i} + \pi t \mathbf{j},$$

and

$$W = \int_{C_1} \mathbf{F}_1 \cdot d\mathbf{r}_1 + \int_{C_2} \mathbf{F}_2 \cdot d\mathbf{r}_2 = \int_0^1 \pi^2 t dt = \frac{1}{2} \pi^2 t^2 \Big|_0^1 = \frac{\pi^2}{2}.$$

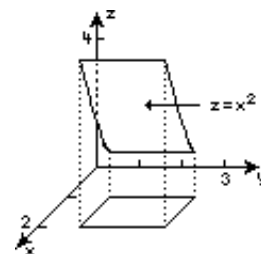


50. Parameterize the line segment from $(-1/2, 1/2)$ to $(-1, 1)$ using $y = -x$ as x goes from $-1/2$ to -1 . Parameterize the line segment from $(-1, 1)$ to $(1, 1)$ using $y = 1$ as x goes from -1 to 1 . Parameterize the line segment from $(1, 1)$ to $(1, \sqrt{3})$ using $x = 1$ as y goes from 1 to $\sqrt{3}$. Then

$$\begin{aligned} W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_{-1/2}^{-1} \mathbf{F} \cdot (dx \mathbf{i} - dx \mathbf{j}) + \int_{-1}^1 \mathbf{F} \cdot (dx \mathbf{i}) + \int_1^{\sqrt{3}} \mathbf{F} \cdot (dy \mathbf{j}) \\ &= \int_{-1/2}^{-1} \left(\frac{2}{x^2 + (-x)^2} - \frac{1}{x^2 + (-x)^2} \right) dx + \int_{-1}^1 \frac{2}{x^2 + 1} dx + \int_1^{\sqrt{3}} \frac{1}{1 + y^2} dy \\ &= \int_{-1/2}^{-1} \frac{1}{2x^2} dx + \int_{-1}^1 \frac{2}{1 + x^2} dx + \int_1^{\sqrt{3}} \frac{1}{1 + y^2} dy \\ &= -\frac{1}{2x} \Big|_{-1/2}^{-1} + 2 \tan^{-1} x \Big|_{-1}^1 + \tan^{-1} y \Big|_1^{\sqrt{3}} = -\frac{1}{2} + 2 \left(\frac{\pi}{2} \right) + \frac{\pi}{12} = \frac{13\pi - 6}{12}. \end{aligned}$$

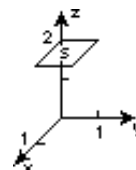
51. $z_x = 2x$, $z_y = 0$; $dS = \sqrt{1 + 4x^2} dA$

$$\begin{aligned} \iint_S \frac{z}{xy} dS &= \int_1^3 \int_1^2 \frac{x^2}{xy} \sqrt{1 + 4x^2} dx dy = \int_1^3 \frac{1}{y} \left[\frac{1}{12} (1 + 4x^2)^{3/2} \right] \bigg|_1^2 dy \\ &= \frac{1}{12} \int_1^3 \frac{17^{3/2} - 5^{3/2}}{y} dy = \frac{17\sqrt{17} - 5\sqrt{5}}{12} \ln y \bigg|_1^3 \\ &= \frac{17\sqrt{17} - 5\sqrt{5}}{12} \ln 3 \end{aligned}$$



52. $\mathbf{n} = \mathbf{k}$, $\mathbf{F} \cdot \mathbf{n} = 3$;

$$\text{flux} = \iint_S \mathbf{F} \cdot \mathbf{n} dS = 3 \iint_S dS = 3 \times (\text{area of } S) = 3(1) = 3$$



53. The surface is $g(x, y, z) = x^2 + y^2 + z^2 - a^2 = 0$. $\nabla g = 2(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = 2\mathbf{r}$, $\mathbf{n} = \mathbf{r}/|\mathbf{r}|$,

$$\mathbf{F} = c\nabla(1/|\mathbf{r}|) + c\nabla(x^2 + y^2 + z^2)^{-1/2} = c \frac{-x\mathbf{i} - y\mathbf{j} - z\mathbf{k}}{(x^2 + y^2 + z^2)^{3/2}} = -c\mathbf{r}/|\mathbf{r}|^3$$

$$\mathbf{F} \cdot \mathbf{n} = -\frac{\mathbf{r}}{|\mathbf{r}|^3} \cdot \frac{\mathbf{r}}{|\mathbf{r}|} = -c \frac{\mathbf{r} \cdot \mathbf{r}}{|\mathbf{r}|^4} = -c \frac{|\mathbf{r}|^2}{|\mathbf{r}|^4} = -\frac{c}{|\mathbf{r}|^2} = -\frac{c}{a^2}$$

$$\text{flux} = \iint_S \mathbf{F} \cdot \mathbf{n} dS = -\frac{c}{a^2} \iint_S dS = -\frac{c}{a^2} \times (\text{area of } S) = -\frac{c}{a^2} (4\pi a^2) = -4\pi c$$

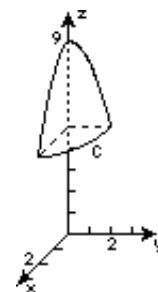
54. In Problem 53, $\mathbf{F}$ is not continuous at $(0, 0, 0)$ which is in any acceptable region containing the sphere.

55. Since $\mathbf{F} = c\nabla(1/r)$, $\text{div } \mathbf{F} = \nabla \cdot (c\nabla(1/r)) = c\nabla^2(1/r) = c\nabla^2[(x^2 + y^2 + z^2)^{-1/2}] = 0$ by Problem 37 in Section 9.7. Then, by the Divergence Theorem,

$$\text{flux } \mathbf{F} = \iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_D \text{div } \mathbf{F} dV = \iiint_D 0 dV = 0.$$

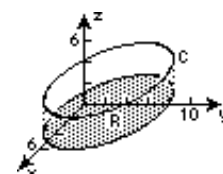
56. Parameterize C by $x = 2 \cos t$, $y = 2 \sin t$, $z = 5$, for $0 \leq t \leq 2\pi$. Then

$$\begin{aligned} \iint_S (\text{curl } \mathbf{F} \cdot \mathbf{n}) dS &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C 6x dx + 7z dy + 8y dz \\ &= \int_0^{2\pi} [12 \cos t (-2 \sin t) + 35(2 \cos t)] dt \\ &= \int_0^{2\pi} (70 \cos t - 24 \sin t \cos t) dt = (70 \sin t - 12 \sin^2 t) \bigg|_0^{2\pi} = 0. \end{aligned}$$



57. Identify $\mathbf{F} = -2y\mathbf{i} + 3x\mathbf{j} + 10z\mathbf{k}$. Then $\text{curl } \mathbf{F} = 5\mathbf{k}$. The curve C lies in the plane $z = 3$, so $\mathbf{n} = \mathbf{k}$ and $dS = dA$. Thus,

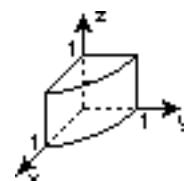
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = \iint_R 5 dA = 5 \times (\text{area of } R) = 5(25\pi) = 125\pi.$$



58. Since $\text{curl } \mathbf{F} = 0$, $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F} \cdot \mathbf{n}) dS = \iint_S 0 dS = 0$.

59. $\text{div } \mathbf{f} = 1 + 1 = 2$;

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_D \text{div } \mathbf{F} dV = \iiint_D 3 dV = 3 \times (\text{volume of } D) = 3\pi$$



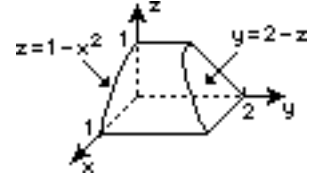
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60. $\operatorname{div} \mathbf{F} = x^2 + y^2 + z^2$. Using cylindrical coordinates,

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= \iiint_D \operatorname{div} \mathbf{F} \, dV = \iiint_D (x^2 + y^2 + z^2) \, dV = \int_0^{2\pi} \int_0^1 \int_0^1 (r^2 + z^2) r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^1 \left(r^3 z + \frac{1}{3} r z^3 \right) \Big|_0^1 \, dr \, d\theta = \int_0^{2\pi} \int_0^1 \left(r^3 + \frac{1}{3} r \right) \, dr \, d\theta \\ &= \int_0^{2\pi} \left(\frac{1}{4} r^4 + \frac{1}{6} r^2 \right) \Big|_0^1 \, d\theta = \int_0^{2\pi} \frac{5}{12} \, d\theta = \frac{5\pi}{6}. \end{aligned}$$

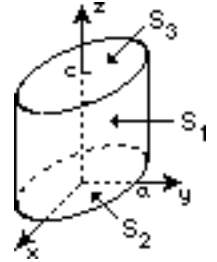
61. $\operatorname{div} \mathbf{F} = 2x + 2(x + y) - 2y = 4x$

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= \iiint_D \operatorname{div} \mathbf{F} \, dV = \iiint_D 4x \, dV = \int_0^1 \int_0^{1-x^2} \int_0^{2-z} 4x \, dy \, dz \, dx \\ &= \int_0^1 \int_0^{1-x^2} 4x(2-z) \, dz \, dx = \int_0^1 \int_0^{1-x^2} (8x - 4xz) \, dz \, dx \\ &= \int_0^1 (8xz - 2xz^2) \Big|_0^{1-x^2} \, dx = \int_0^1 [8x(1-x^2) - 2x(1-x^2)^2] \, dx \\ &= \left[-2(1-x^2)^2 + \frac{1}{3}(1-x^2)^3 \right] \Big|_0^1 = \frac{5}{3} \end{aligned}$$



62. For S_1 , $\mathbf{n} = (x\mathbf{i} + y\mathbf{j})/\sqrt{x^2 + y^2}$; for S_2 , $\mathbf{n}_2 = -\mathbf{k}$ and $z = 0$; and for S_3 , $\mathbf{n}_3 = \mathbf{k}$ and $z = c$. Then

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} \, dS &= \iint_{S_1} \mathbf{F} \cdot \mathbf{n}_1 \, dS_1 + \iint_{S_2} \mathbf{F} \cdot \mathbf{n}_2 \, dS_2 + \iint_{S_3} \mathbf{F} \cdot \mathbf{n}_3 \, dS_3 \\ &= \iint_{S_1} \frac{x^2 + y^2}{\sqrt{x^2 + y^2}} \, dS_1 + \iint_{S_2} (-z^2 - 1) \, dS_2 + \iint_{S_3} (z^2 + 1) \, dS_3 \\ &= \iint_{S_1} \sqrt{x^2 + y^2} \, dS_1 + \iint_{S_2} (-1) \, dS_2 + \iint_{S_3} (c^2 + 1) \, dS_3 \\ &= a \iint_{S_1} dS_1 - \iint_{S_2} dS_2 + (c^2 + 1) \iint_{S_3} dS_3 \\ &= a(2\pi ac) - \pi a^2 + (c^2 + 1)\pi a^2 = 2\pi a^2 c + \pi a^2 c^2. \end{aligned}$$



63. $x = 0 \implies u = 0, v = -y^2 \implies u = 0, -1 \leq v \leq 0$

$$x = 1 \implies u = 2y, v = 1 - y^2 = 1 - u^2/4$$

$$y = 0 \implies u = 0, v = x^2 \implies u = 0, 0 \leq v \leq 1$$

$$y = 1 \implies u = 2x, v = x^2 - 1 = u^2/4 - 1$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 2y & 2x \\ 2x & -2y \end{vmatrix} = -4(x^2 + y^2) \implies \frac{\partial(x, y)}{\partial(u, v)} = -\frac{1}{4(x^2 + y^2)}$$

$$\begin{aligned} \iint_R (x^2 + y^2)^{3/2} \sqrt{x^2 + y^2} \, dA &= \iint_S (x^2 + y^2)^{3/2} \sqrt{v} \left| -\frac{1}{4(x^2 + y^2)} \right| dA' = \frac{1}{4} \int_0^2 \int_{u^2/4-1}^{1-u^2/4} v^{1/3} \, dv \, du \\ &= \frac{1}{4} \int_0^2 \frac{3}{4} v^{4/3} \Big|_{u^2/4-1}^{1-u^2/4} \, du = \frac{3}{16} \int_0^2 \left[(1 - u^2/4)^{4/3} - (u^2/4 - 1)^{4/3} \right] \, du \\ &= \frac{3}{16} \int_0^2 \left[(1 - u^2/4)^{4/3} - (1 - u^2/4)^{4/3} \right] \, du = 0 \end{aligned}$$

$$\begin{aligned}
 64. \quad y = x &\implies u + uv = v + uv \implies v = u \\
 x = 2 &\implies u + uv = 2 \implies v = (2 - u)/u \\
 y = 0 &\implies v + uv = 0 \implies v = 0 \text{ or } u = -1 \\
 &\quad (\text{we take } v = 0)
 \end{aligned}$$

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 1 + w & u \\ v & 1 + u \end{vmatrix} = 1 + u + v$$

Using $x = u + uv$ and $y = v + uv$ we find

$$\begin{aligned}
 (x - y)^2 &= (u + uv - v - uv)^2 = (u - v)^2 = u^2 - 2uv + v^2 \\
 x + y &= u + uv + v + uv = u + v + 2uv \\
 (x - y)^2 + 2(x + y) + 1 &= u^2 + 2uv + v^2 + 2(u + v) + 1 = (u + v)^2 + 2(u + v) + 1 = (u + v + 1)^2.
 \end{aligned}$$

Then

$$\begin{aligned}
 \iint_R \frac{1}{\sqrt{(x - y)^2 + 2(x + y) + 1}} dA &= \iint_S \frac{1}{u + v + 1} (u + v + 1) dA' = \int_0^1 \int_v^{2/(1+v)} du dv \\
 &= \int_0^1 \left(\frac{2}{1 + v} - v \right) dv = \left[2 \ln(1 + v) - \frac{1}{2} v^2 \right] \Big|_0^1 = 2 \ln 2 - \frac{1}{2}.
 \end{aligned}$$

65. The equations of the spheres are $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 + (z - a)^2 = 1$. Subtracting these equations, we obtain $(z - a)^2 - z^2 = 1 - a^2$ or $-2az + a^2 = 1 - a^2$. Thus, the spheres intersect on the plane $z = a - 1/2a$. The region of integration is $x^2 + y^2 + (a - 1/2a)^2 = a^2$ or $r^2 = 1 - 1/4a^2$. The area is

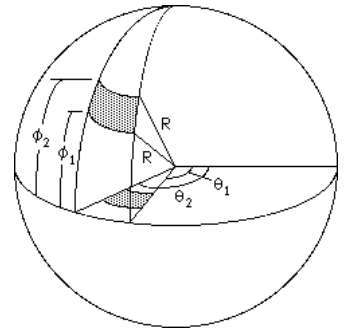
$$\begin{aligned}
 A &= a \int_0^{2\pi} \int_0^{\sqrt{1-1/4a^2}} (a^2 - r^2)^{-1/2} r dr d\theta = 2\pi a \left[-(a^2 - r^2)^{1/2} \right] \Big|_0^{\sqrt{1-1/4a^2}} \\
 &= 2\pi a \left(a - \left[a^2 - \left(1 - \frac{1}{4a^2} \right) \right]^{1/2} \right) = 2\pi a \left(a - \left[\left(a - \frac{1}{2a} \right)^2 \right]^{1/2} \right) = \pi.
 \end{aligned}$$

66. (a) Both states span 7 degrees of longitude and 4 degrees of latitude, but Colorado is larger because it lies to the south of Wyoming. Lines of longitude converge as they go north, so the east-west dimensions of Wyoming are shorter than those of Colorado.
- (b) We use the function $f(x, y) = \sqrt{R^2 - x^2 - y^2}$ to describe the northern hemisphere, where $R \approx 3960$ miles is the radius of the Earth. We need to compute the surface area over a polar rectangle P of the form $\theta_1 \leq \theta \leq \theta_2$, $R \cos \phi_2 \leq r \leq R \cos \phi_1$. We have

$$f_x = \frac{-x}{\sqrt{R^2 - x^2 - y^2}} \quad \text{and} \quad f_y = \frac{-y}{\sqrt{R^2 - x^2 - y^2}}$$

so that

$$\sqrt{1 + f_x^2 + f_y^2} = \sqrt{1 + \frac{x^2 + y^2}{R^2 - x^2 - y^2}} = \frac{R}{\sqrt{R^2 - r^2}}.$$



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Thus

$$\begin{aligned} A &= \iint_P \sqrt{1 + f_x^2 + f_y^2} \, dA = \int_{\theta_1}^{\theta_2} \int_{R \cos \phi_2}^{R \cos \phi_1} \frac{R}{\sqrt{R^2 - r^2}} \, r \, dr \, d\theta \\ &= (\theta_2 - \theta_1) R \sqrt{R^2 - r^2} \Big|_{R \cos \phi_1}^{R \cos \phi_2} = (\theta_2 - \theta_1) R^2 (\sin \phi_2 - \sin \phi_1). \end{aligned}$$

The ratio of Wyoming to Colorado is then $\frac{\sin 45^\circ - \sin 41^\circ}{\sin 41^\circ - \sin 37^\circ} \approx 0.941$. Thus Wyoming is about 6% smaller than Colorado.

- (c) $97,914/104,247 \approx 0.939$, which is close to the theoretical value of 0.941. (Our formula for the area says that the area of Colorado is approximately 103,924 square miles, while the area of Wyoming is approximately 97,801 square miles.)